

6.24 Cryogenic Coil System Operation

A cryogenic coil system is built into this machine using a cryogenerator unit to provide the cooling medium. The coolant will cool the coil down to the temperature of -114°C under normal circumstances.

The system is used to condense moisture on to the coils rather than letting it vaporise and create a large gas load for the pumping system.

	Caution !
	Low Temperature Hazard. Do not touch coils when being cooled



Figure 6.24 - 1 Photograph Of Cryogenic Coils In A Machine

The cryogenerator unit is controlled from the computer to switch into the following modes;

Vacuum System Mode	Evaporation Zone Cryogenerator Operation Mode
Vacuum	Cool Mode
Hold	Defrost for three minutes
Vent	Standby
Vent & chamber open	Standby

Table 6.24 - 1 Cryogenerator Operation Table



Figure 6.24 - 2 Photograph of Free Standing Cryogenerator Unit

6.25 Machine Diagnostics

The user of the machine can check on the status of quite a number of machine sub sections through the MMI screen diagnostics.

Network Diagnostics Screen

Main ASI - A nodes			Main ASI - B nodes			Profibus		
Node	Fault	Device	Node	Fault	Device	Node	Fault	Device
1A	☐	VV1	1B	☐	VV1B	2	☐	Driver PLC
2A	☐	VV2	2B	☐	VV2B	3	☐	ASI gateway
3A	☐	VV3	3B	☐	Mech closed safety	4	☐	Thyristor 1
4A	☐	VV4	4B	☐	VV5	5	☐	Thyristor 2
5A	☐	VV5	5B	☐	VV6	6	☐	Thyristor 3
6A	☐	VV5A	6B	☐	VV6B	7	☐	Thyristor 4
7A	☐	VV7	7B	☐	VV7B	8	☐	Thyristor 5
8A	☐	Drive emergency stop	8B	☐	VV8	9	☐	Thyristor 6
9A	☐	Drive guard stop	9B	☐	VV9	10	☐	Thyristor 7
10A	☐	VV10A	10B	☐	VV10B	11	☐	Chiller
11A	☐	Vessel emergency stop	11B	☐	VV11B	29	☒	Booster pump 1
12A	☐	Vessel viewing flap	12B	☐	VV12B	27	☒	Booster pump 2
13A	☐	Source viewing flap	13B	☐	VV13B	28	☒	Booster pump 3
14A	☐	VV14C	14B	☐	VV14B	29	☒	Booster pump 4
15A	☐	Shutter	15B	☐	VV15B	32	☐	Vacuum control
16A	☐	Water temperature	16B	☐	VV16B	47	☐	Wire feed 1
17A	☐	Water temperature	17B	☐	VV17B	54	☐	Monitor 1
18A	☐	Pendant emergency stop	18B	☐	VV18B	59	☐	Drive control
19A	☐	Vessel guard stop	19B	☐	VV19B	94	☐	Source viewing strobe
20A	☐	Mech open/close safety	20B	☐	VV20B			
21A	☐	Mech enable	21B	☐	VV21B			
22A	☐	Nip solenoid	22B	☐	VV22B			
23A	☐	Hinged shield pushbuttons	23B	☐	VV23B			
24A	☐	Emergency stop & guard stop reset	24B	☐	VV24B			
25A	☐	Mech isolator	25B	☐	VV25B			
26A	☐	Mech seal	26B	☐	VV26B			
27A	☐	Mech closed limit switch	27B	☐	VV27B			
28A	☐	Mech door switch	28B	☐	VV28B			
29A	☐	Mech open/close pushbutton	29B	☐	VV29B			
30A	☐	Mech two hand operation safety	30B	☐	VV30B			
31A	☐		31B	☐	VV31B			

This screen shows the status of the control items located around the machine that are linked on Asi and profibus networks.

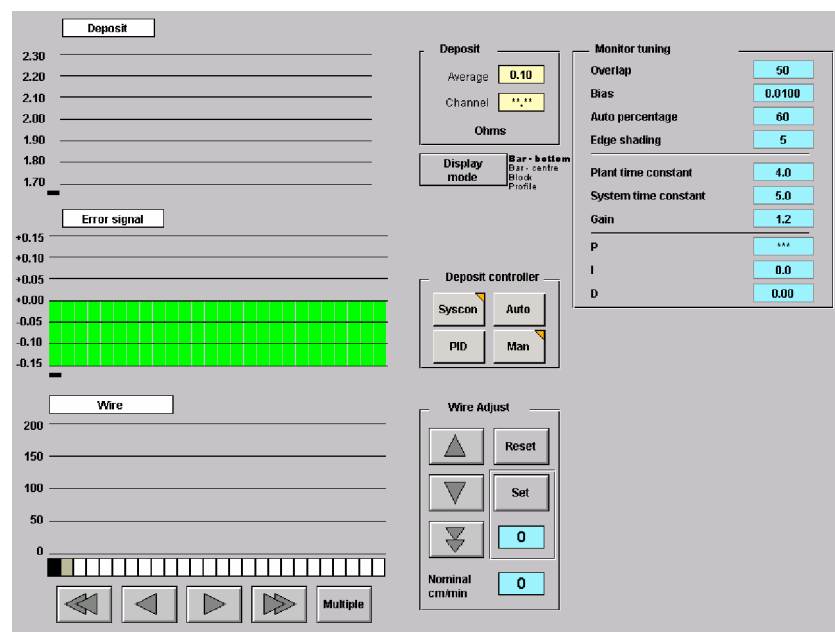
Figure 6.25 - 1 Network Diagnostic Screen

The status of the individual nodes are colour coded;-

Grey: - Network element/node is monitored and is operational mode.
Red: - Fault or not connected, or powered up.

In the situation where some units are powered down in the case of a E-stop being activated, once power has been reapplied, the 'RESET I/O' button has to be touched

Deposition Monitor Diagnostic Screen



This screen is available at the supervisor level, and is used to troubleshoot/set up the deposition monitor system.

Use the help button for information

Figure 6.25 - 2 Deposition Monitor Diagnostic Screen

Interlocks Diagnostic Screen

This screen provided the user with the ability to check on which interlock in a string is not in the correct state to allow the function of a piece of equipment or operation.

Touching the relevant title in the upper section will show the interlock string. The elements are colour coded to signify their state. The colour red signifies that the element (interlock) is in a condition that will not allow the operation to take place. The colour green signifies that the interlock is in the correct state.

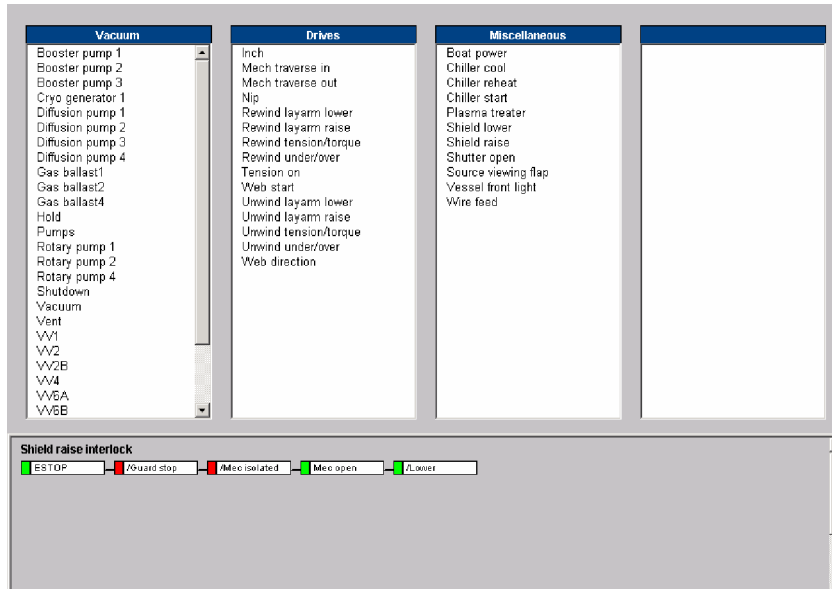
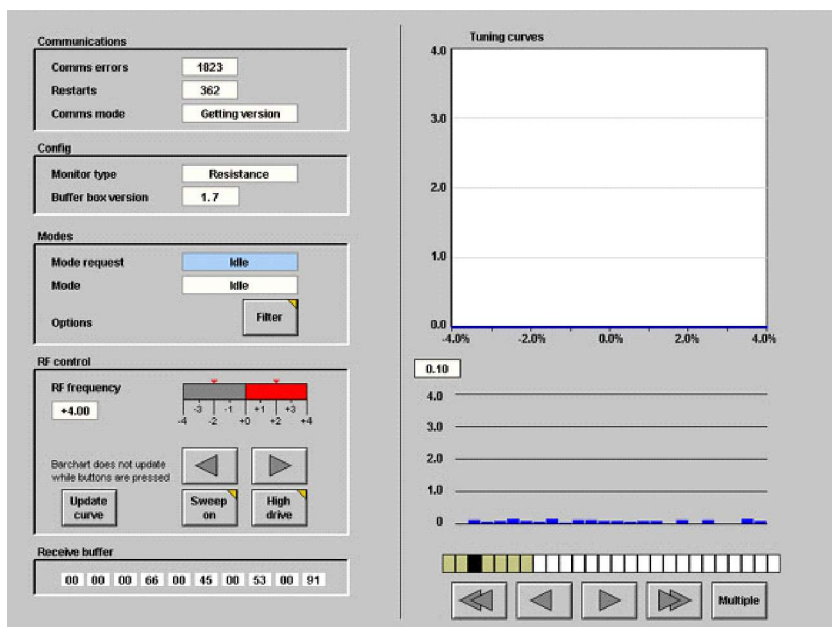


Figure 6.25 - 3
Interlocks Diagnostic
Screen

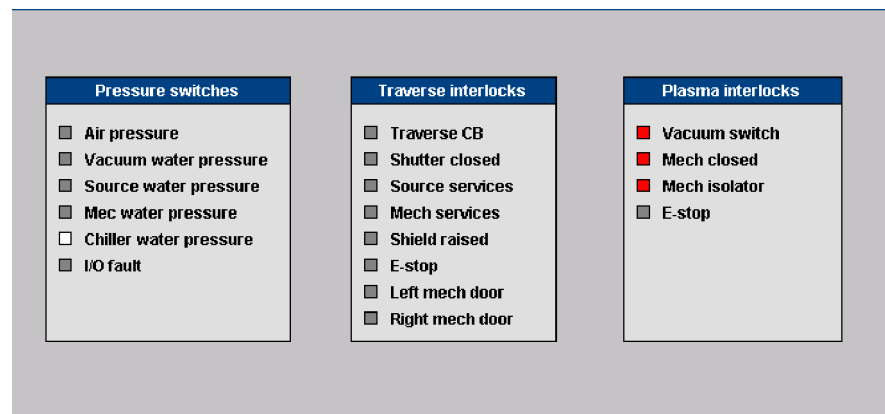
Deposition Monitor Control System



This screen is provided for investigating potential issues with the monitor system supplied with the machine. This operation can only be carried out at a supervisor/ maintenance level.

Figure 6.25 - 4
Deposition Monitor
Diagnostic Screen

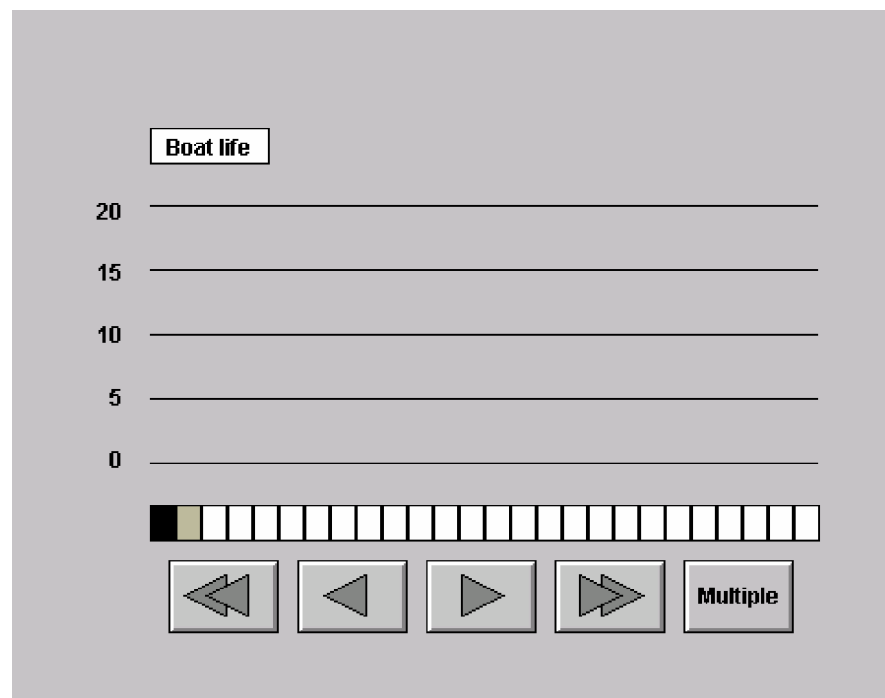
Machine Services Diagnostic Screen



This screen is used to check on the status of some of the water and air interlocks of the machine

Figure 6.25 - 5
Services Diagnostic
Screen

Evaporation Source Diagnostic Screen

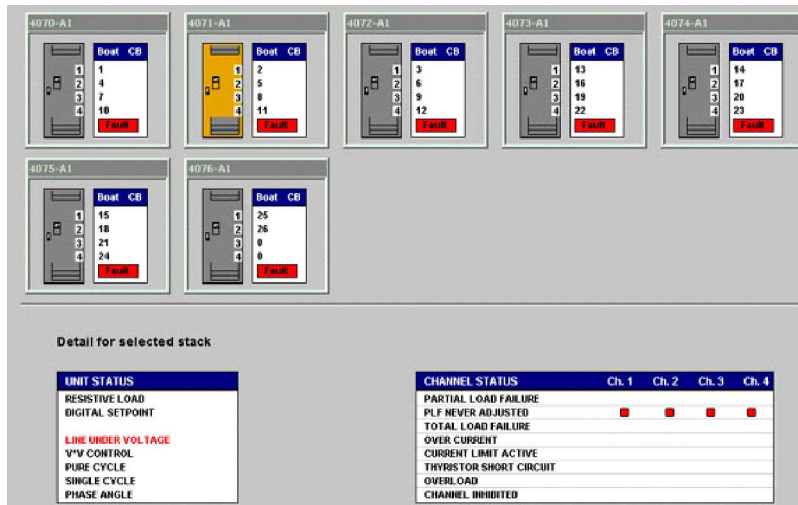


This screen has been designed to allow the user to check on evaporator life.

Figure 6.25 - 6
Evaporation Source
Diagnostic Screen

Thyristor Stack Diagnostic Screen

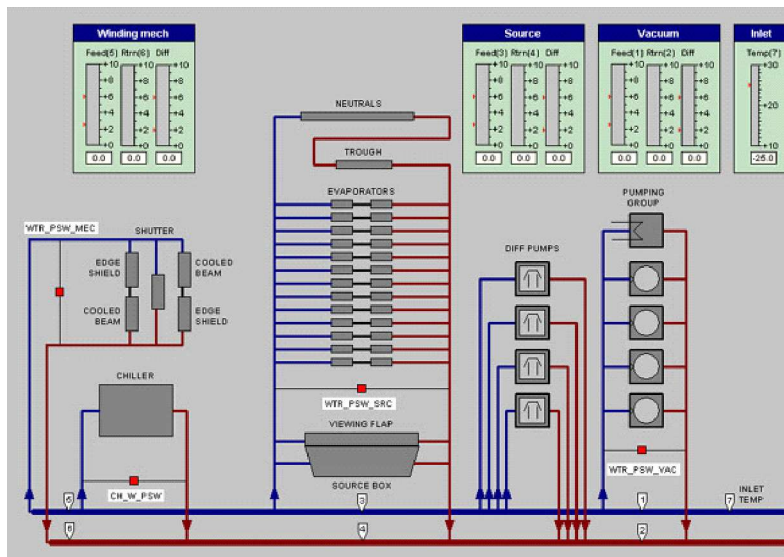
This screen is provided to allow monitoring of the evaporator power supply thyristor units.



It is normal that when the 'BOATS OFF' condition is monitored that a 'Line under voltage' message is displayed.

Figure 6.25 - 7 Thyristor Stack Diagnostic Screen

Water Circuits Diagnostic Screen



This screen is provided to allow the user to check on the status of the water system on the machine.

There are a number of sensors monitoring the feed and return pressure levels to sections of the machine.

Figure 6.25 - 8 Water Circuits Diagnostic Screen

The incoming supply temperature is also monitored and in cases when the temperature rises above the alarm set point, an alarm will be given.

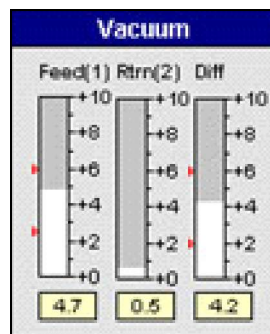


Figure 6.25 - 9 Water Pressure Diagnostic Section of Screen

The pressure monitoring part of the screen has three indicators.

- Inlet (Feed) pressure
- Outlet (Return) pressure
- Differential.

The red arrow heads on the outside of the columns indicates the preset operating levels that the system is to work within.

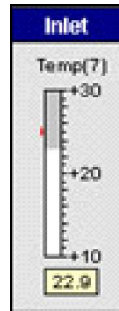


Figure 6.25 - 10 Water Pressure Diagnostic Section of Screen

The inlet temperature is also monitored not to rise above a predetermined level, as indicated by the red arrow head.

Drives Diagnostic Screen

Strobe motor	
☐ Ready to switch on ☐ Switched on ☐ Operation enabled ☐ Fault ☐ Setpoint acknowledge ☐ Quick stop ☐ Switch on disabled ☐ Warning ☐ Handshake ☐ Remote ☐ Target reached ☐ Internal limit active ☐ Homing attained ☐ Following error ☐ Accelerating phase ☐ Constant phase	Status
☐ Limit position ☐ Over temperature ☐ Under voltage ☐ Internal position limit ☐ Error message generated ☐ Error control of load angle ☐ Position setpoint out of range ☐ Program error	Warning
☐ Over temperature ☐ Error control of load angle ☐ Short circuit ☐ Under voltage ☐ Error motion control ☐ Error position limit control	Fault

BP1	
☐ Ready for ON ☐ Ready to run ☐ Enabled ☐ Fault ☐ Off2 applied ☐ Off3 applied ☐ Start lockout ☐ Alarm	☐ Setpoint OK ☐ Master control ☐ f reached ☐ Current limit ☐ Brake ☐ Motor overload ☐ CW rotation ☐ Inverter overload

BP2	
☐ Ready for ON ☐ Ready to run ☐ Enabled ☐ Fault ☐ Off2 applied ☐ Off3 applied ☐ Start lockout ☐ Alarm	☐ Setpoint OK ☐ Master control ☐ f reached ☐ Current limit ☐ Brake ☐ Motor overload ☐ CW rotation ☐ Inverter overload

BP3	
☐ Ready for ON ☐ Ready to run ☐ Enabled ☐ Fault ☐ Off2 applied ☐ Off3 applied ☐ Start lockout ☐ Alarm	☐ Setpoint OK ☐ Master control ☐ f reached ☐ Current limit ☐ Brake ☐ Motor overload ☐ CW rotation ☐ Inverter overload

This screen gives the status of various drive components on the machine.

Figure 6.25 - 11
Drives Diagnostic
Screen

6.26 Plasma Treater System

The plasma treater is supplied to enhance the coating of the substrate with the aluminium.

In simple terms plasma is similar to the fluorescent tube used to light the chamber. A gas mixture is fed into the plasma treater and an electrical charge is placed across its internal surfaces and as in the fluorescent tube, a plasma is “struck”.

The benefits of the plasma treater will vary depending on the substrate being processed. Benefits recorded are better adhesion levels and barrier properties.

Plasma Treater System Components

There are three main components to the General Vacuum Equipment's plasma treater System, namely:

- The Heat Exchanger Section (Contained in one unit with the generator)
- The Generator, or Power Source Section (Contained in one unit with the heat exchanger)
- The Plasma Treater Assembly
- Process Gas Control Equipment



Figure 6.26 - 1 Photograph showing Plasma electrical panel containing the power supply

Plasma Treater Assembly

The plasma treater assembly is installed within the vacuum chamber and is positioned to allow the film to pass through it for treatment when it leaves the unwind roll.



Figure 6.26 - 2 Photograph showing a plasma treater.

Gas Connection and Distribution

In order to generate a plasma glow both power and a sufficient gas environment (or pressure regime) is required. As the plasma is located in a vacuum chamber where the pressure regime is below a satisfactory level for plasma generation, we need to feed the process gas into the associated local atmosphere to ignite and maintain plasma. It has been found various gas mixtures can enhance certain physical properties of a passing substrate surface.

In order to feed the necessary gas mixtures in specific ratios, it is necessary to incorporate a suitable gas control system into the machine.



Figure 6.26 - 3 Photograph of Mass Flow Control Equipment

Baratron Pressure Gauge

In order to measure the vacuum level of inside the plasma treater zone accurately when using a number of different process gases, a cold cathode Baratron gauge is utilised.

Although the gauge is located on the outside of the chamber, a internal flexible tube runs between the gauge connection and the treater assembly.

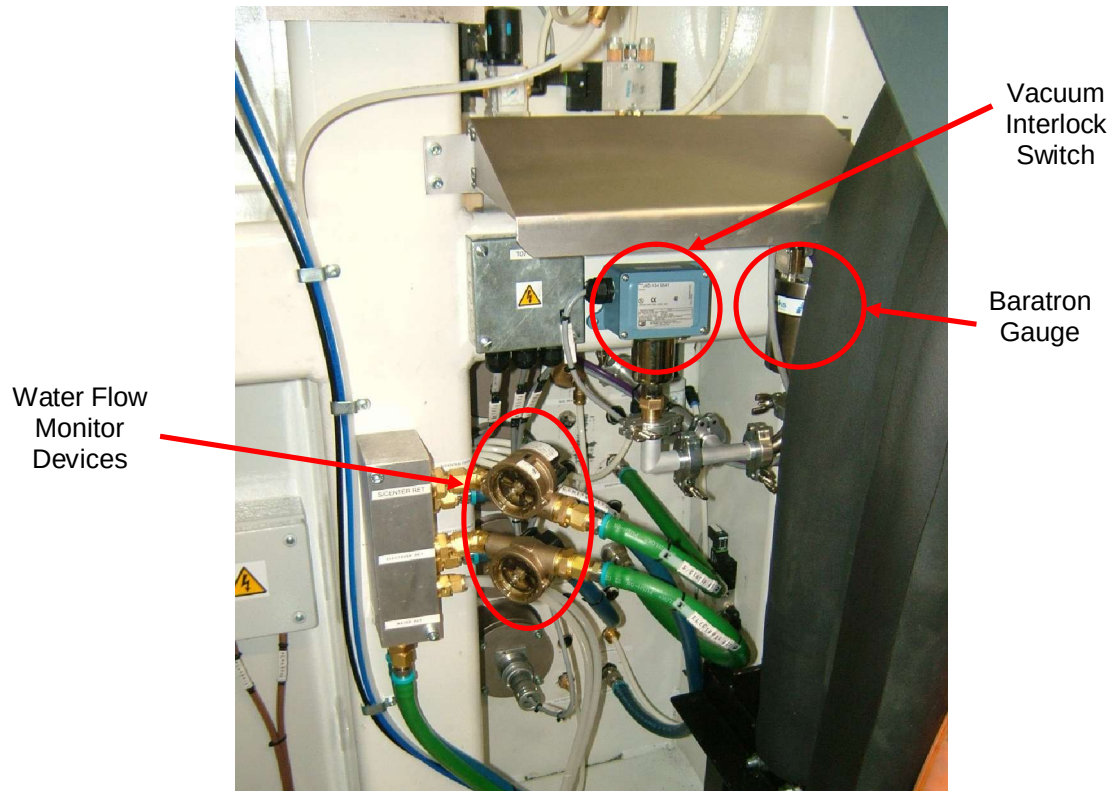


Figure 6.26 - 4 Photograph of Location of Baratron Gauge

6.27 Plasma Treater Interface MMI Screen

Plasma System Set Up Interface Screen

The plasma treater unit operational parameters are set up at the time of setting up the rest of the machine in the SETUP screen.

The screenshot displays the 'Setup' screen for a K4000 machine, specifically for a '12 mic pet' recipe. The interface is divided into several sections for parameter configuration:

- Recipe name:** 12 mic pet
- Roll details:**
 - Roll ID: Auto-000088
 - Customer: (empty)
 - Web type: PET
 - Roll length: 32000 m
 - Thickness: 12 um
 - Web width: 2200 mm
 - Roll weight: 1360
 - Target: 2.10
 - Target units: OD
 - Upper limit: 2.00 OD
 - Lower limit: 1.90 OD
 - Boats: Old
 - Web speed: 630.0 m/min
 - Wire speed: 90 cm/min
 - Chiller temp: -10
- Unwind:**
 - Tension: 400.0 N
 - Over: (selected)
 - Start diameter: 654.2 mm
 - Final diameter: 190.0 mm
 - Gain at min diameter: 750
 - Gain at max diameter: 1000
 - Contact: 50.0 N
- Rewind:**
 - Tension: 200.0 N
 - Under: (selected)
 - Start diameter: 373.3 mm
 - Gain at min diameter: 500
 - Gain at max diameter: 1000
 - Gap: 15.0 mm
- Miscellaneous:**
 - DTR1: +20.0 Torque
 - DTR2: +35.0 Torque
 - Web acceleration: 60 s
 - Web deceleration: 60 s
 - Web density: 1.40 g/cm3
 - PET=1.4 BOPP=0.9 CAVI=0.85
 - Gas wedge: On
- Plasma treater:**
 - Gas 1: Pre-mix
 - Power control: Fixed
 - Power setpoint: 5.0 kW
 - Plasma treater: On

On the right side, there are buttons for 'Recipes...', 'Load', 'Curve...', 'Update all', 'Main', 'Old boats', 'New boats', 'Rewind Profile', 'OD Curve', and 'Modes'. The bottom status bar shows '11', 'Vac: 8.82e+2', 'Length: 0', 'Drum: +22', 'Unwind: 654.2', 'MET CARE', 'OK', and the date/time '9/12/2007 11:45:07 AM'.

Figure 6.27 - 1 Setup Screen

In this screen, the following parameters can be selected:

Power Control Operation

Select the operation mode of control. Selecting RAMPED will allow the user to set up a set point that is increased automatically at the speed the machine is increased. Selecting FIXED will allow the total set point in KW to be set. This power level will be applied from switching on the plasma power supply.

Power Set point

The required plasma power supply required set point in kilowatts.

Plasma Treater

Whether the plasma treater is ON or OFF in the run. Has to be set as on and loaded from the recipe. Cannot be over-riden on control interface if said OFF.

Plasma System Operational Interface

To adjust the different system controls and to monitor the performance of the plasma system, an interface is provided. If selected as on in the recipe the plasma treater will ramp and turn on the water pump automatically, although these can also be controlled manually using this control

interface. The power supply can also be controlled manually by deselecting remote and controlling from the panel on the front.

The interface is accessed from the PROCESS screen and is selected by touching the treater button at the bottom right of the screen. It is possible to alternate between source and plasma control.

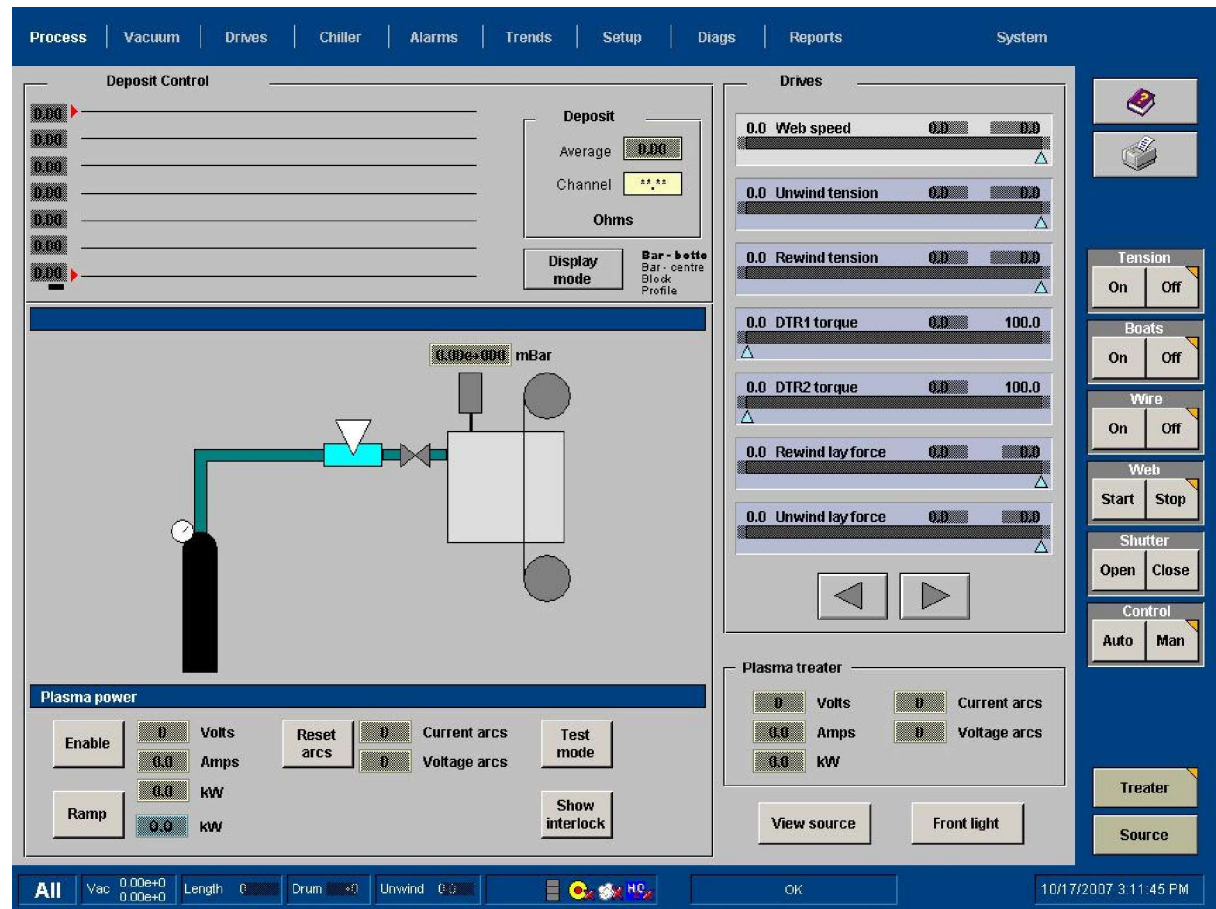


Figure 6.27 - 2 PROCESS Screen Showing Plasma Treater Control



Figure 6.27 - 3 Treater Screen Selection button on the PROCESS screen

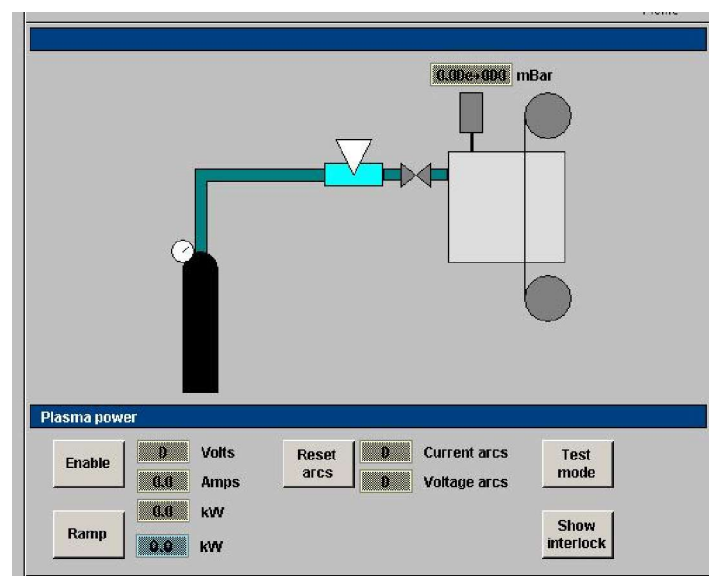


Figure 6.27 - 4 Plasma Treater Control Section of PROCESS Screen

This screen allows the following parameters to be changed:

Gas Control

Enable Button

There are electrically operated isolation valves located at each gas controller, which are energised from the ENABLE button. Grey indicates closed, white indicates open.

The amount of gas flowing will be indicated on the screen.

Plasma Power Supply Controls

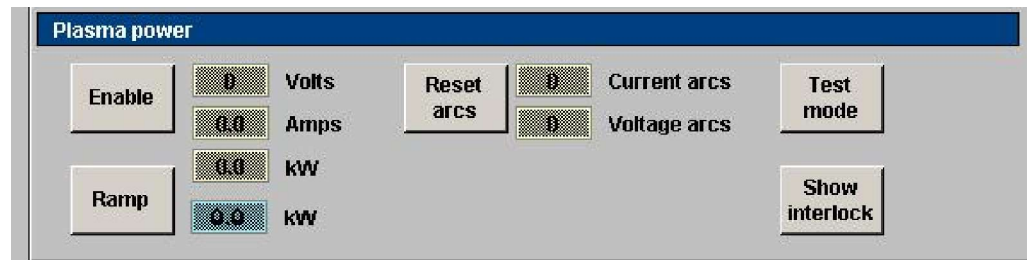


Figure 6.27 - 5 Plasma Power Supply Controls On Plasma Interface

Set-point Control

The power supply operation power set-point level can be set by pressing the blue box.

When the unit is operating, the power level is indicated in KW (yellow box), and the volts and amps drawn are indicated.

Enable Button

This button is used to energise the plasma power supply.

Ramp

This button ramps the power proportional to the speed as the shutter opens, to the value displayed

Show Interlock Button

This button will open the Plasma Treater Interlock section on the DIAGNOSTICS screen.

Alarms

- Over Temperature– This alarm is generated within the power supply, refer to the supplier manual for further information.
- Out of Regulation – This alarm is generated within the power supply, refer to the supplier manual for further information.

Reset Arc Button

This button is used to reset the current and the voltage arc counters.

Test Mode Button

There is a requirement when initially powering up the plasma treater to allow its operation to be checked out. Pressing the 'test button' allows the user to power up the plasma treater without any web in the winding mechanism, to allow clean up of the target plates. There is an interlock indication panel to allow the user to ensure that all necessary interlocks are made.

If this mode is to be used, remove the substrate from the machine, unless machine closed and running substrate.

Selecting the test button will allow the operation of the plasma treater.

All the interlocks have to be present for the plasma treater to be powered up.

Note: If the 'Test button' is left selected, tension cannot be applied to the substrate.

6.28 Threading the Substrate Through The Plasma Treater Unit

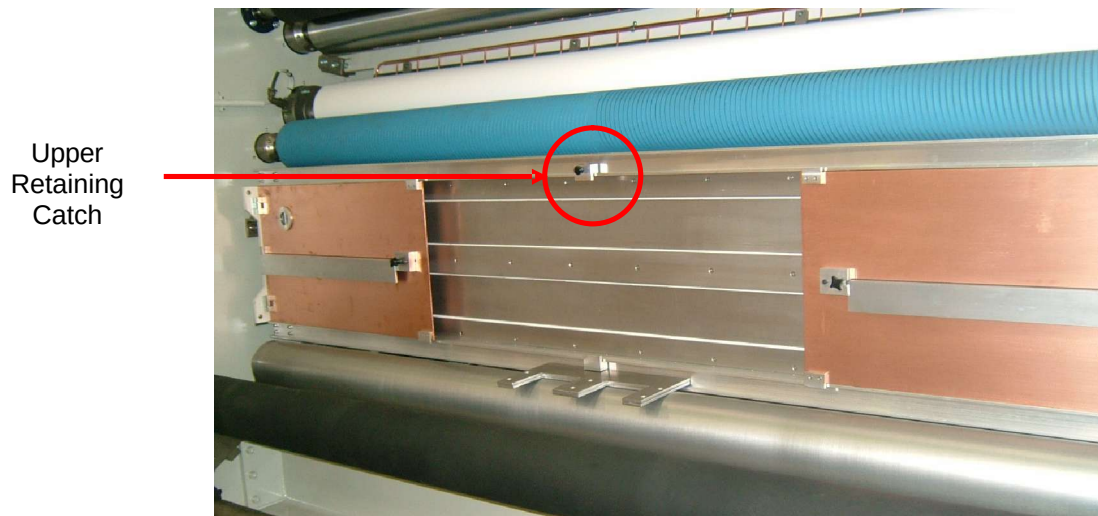


Figure 6.28 - 1 photograph to show the plasma treater with its front central cover removed

The front central cover is removed by lifting the upper retaining catch.

	Caution !
	Take care not to damage the roller located below the plasma treater, when removing the central cover.

In order to web substrate through the plasma treater, it is generally only required to remove this panel. This provides enough space to access the substrate as it is fed through the lower idler roller and low conductance seal arrangement. It is essential not to touch the electrode plates during this operation, or, drag contamination or debris with the substrate into the plasma zone, as arcing will result. With the film routed past the electrodes the operator can then pass the substrate through the upper zone slot.

6.29 Setting the Back-Side Plasma Shields and Focusing Shields

In order to prevent backside treatment of the substrates, which can later result in blocking problem, a pair of backside treater or plasma shields has been incorporated into the design.

There are also sets of impedance shields supplied to ensure that the plasma is focused at the location of the substrate.

These can be accessed from the front of the treater without opening the doors. There are adjustment blocks for the backside shields, and to clamp the impedance shields.

Sections of aluminium angle extrusion are supplied to be used for the impedance shields. One side is longer than the other, and this is the side that fits into the slot in the shield.

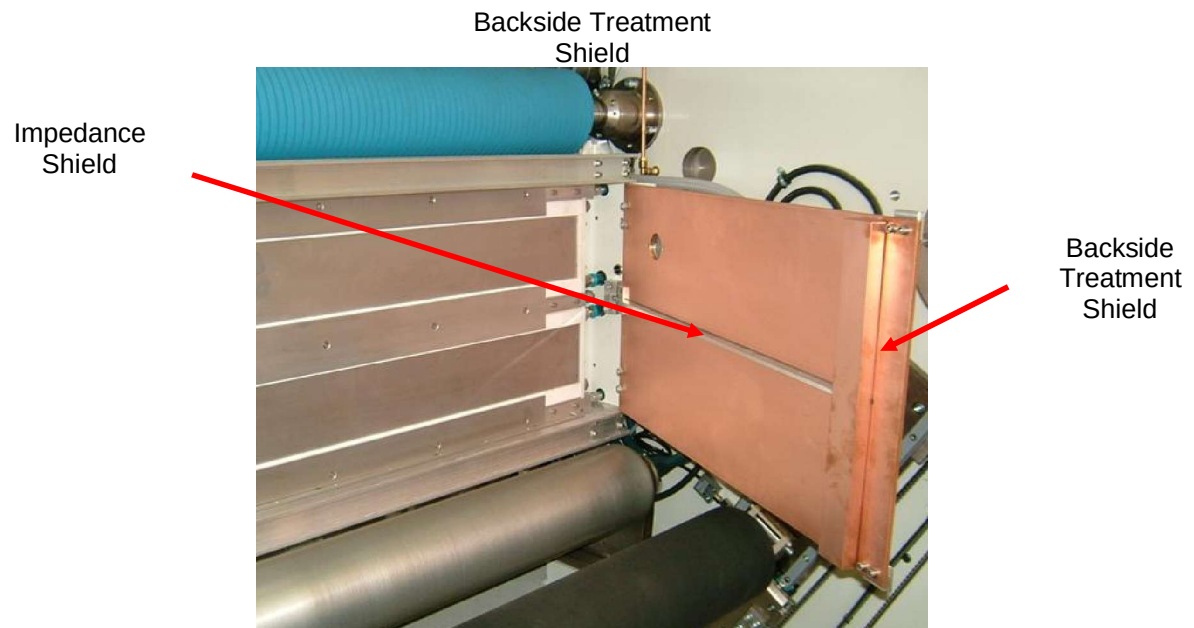


Figure 6.29 - 1 A photograph to showing the internal view of backside treatment and impedance shield



Figure 6.29 - 2 A photograph to showing the external view of backside treatment and impedance shield

Sufficient aluminium strips have been supplied to allow the backside treatment block to be moved in the treater.

To modify the width of the shields, the following instructions should be used.

1. Measure the width of the substrate.
2. Loosen the clamping blocks on the external section of the cover.
3. Slide the other clamp away and release the aluminium angle shield.

Carry out the same procedure on the other side

With the measurement of the width of the substrate known. Move the backside treatment blocks (clamp) until they are an equal distance way from the centre of the treater that equates to the width of the film. The fixed screw in each block should be used as a datum, and the block should be moved in 6mm from the edge of the substrate width. This will ensure that the blocks cover the edge of the substrate to stop backside treatment.

Locate the aluminium angle shield to fit the gap that exists outside the substrate.

6.30 Adjustable Source Height

This machine has been designed to allow the evaporator to drum height to be varied.

This is accomplished by rotating the threaded rods located at four points of the source enclosure.



Figure 6.30 - 1 Photograph of Operator Adjusting the Source Height

The closer the evaporators are to the drum, the more heat is transferred onto the web, and the collection efficiency is better than when it is lowered.

In the lower position, the possibility of any spitting occurring is greatly reduced.

	Note!
	Ensure that both sides of the source are raised evenly. Use a spirit level to check for correct levels.

SECTION 7

METALLIZER OPERATIONAL SEQUENCES

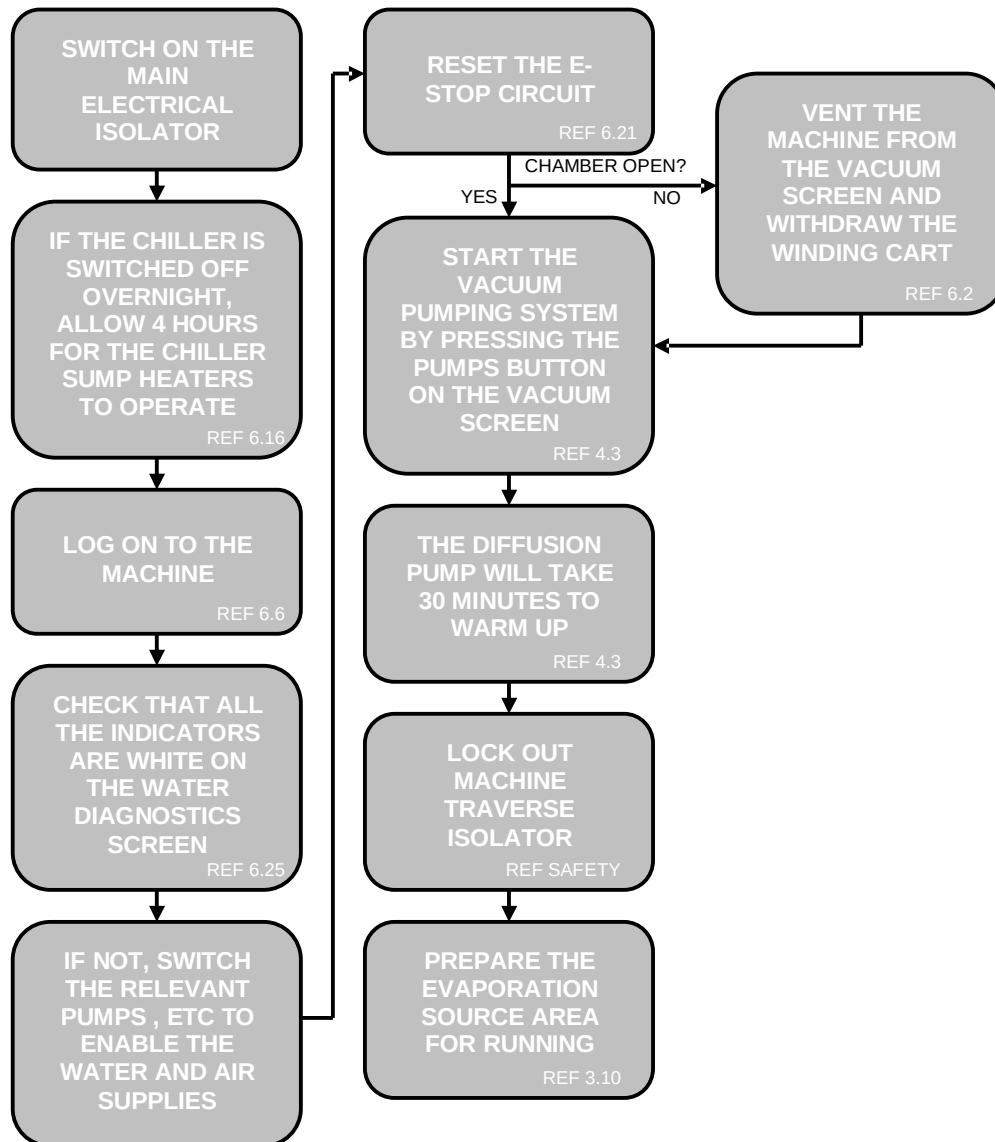
Section 7.1 Operational Sequence of Aluminium Metallizing

7.1 Operational Sequence of Aluminium Metallizing

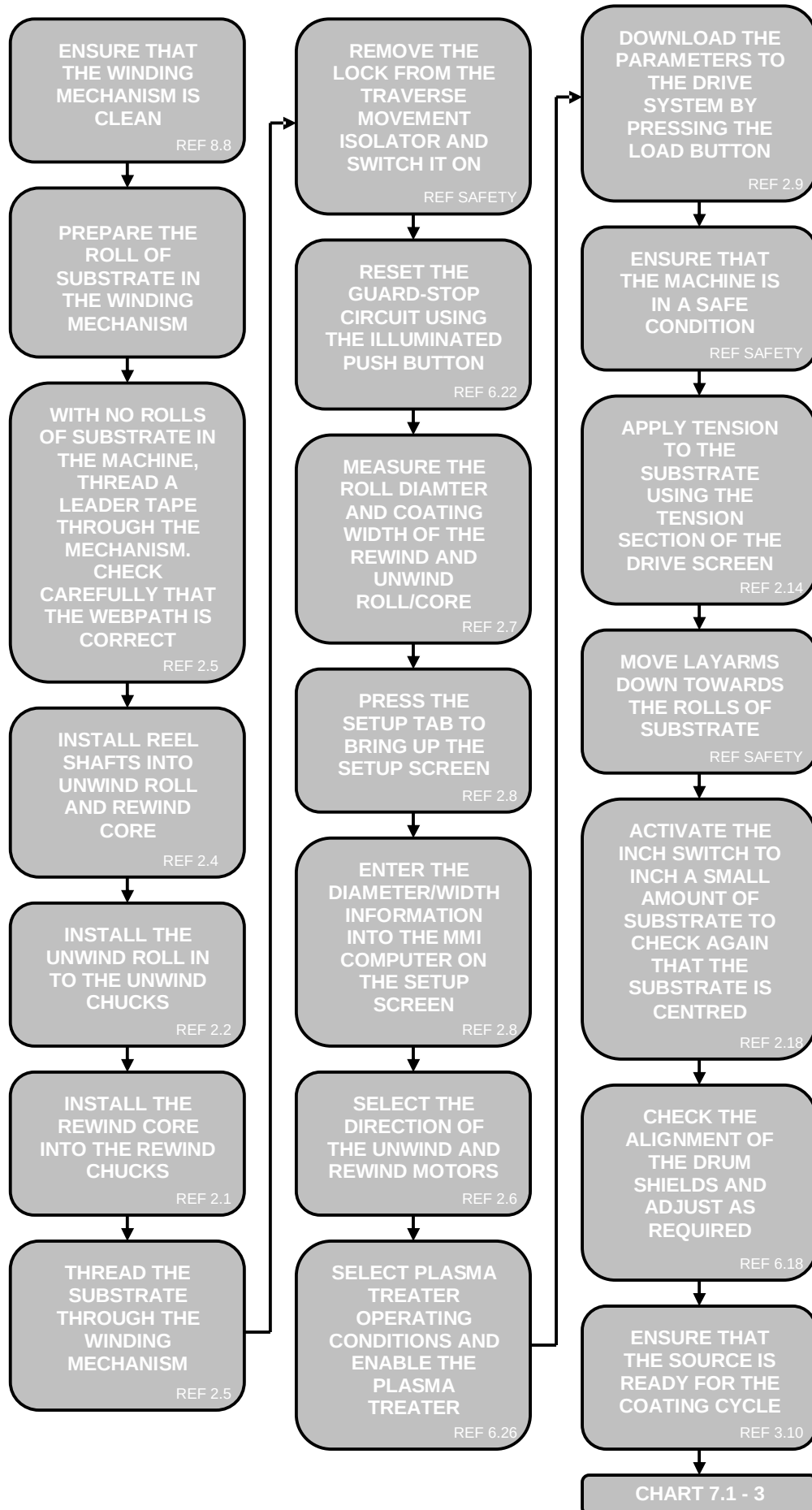
Rather than trying to explain the sequence of operating the machine for metallizing with in-depth written descriptions, flow charts are provided, to help the operator of the machine follow the correct operational sequences.

It has been found that operators of this type of equipment prefer this type of information, while operating the machine.

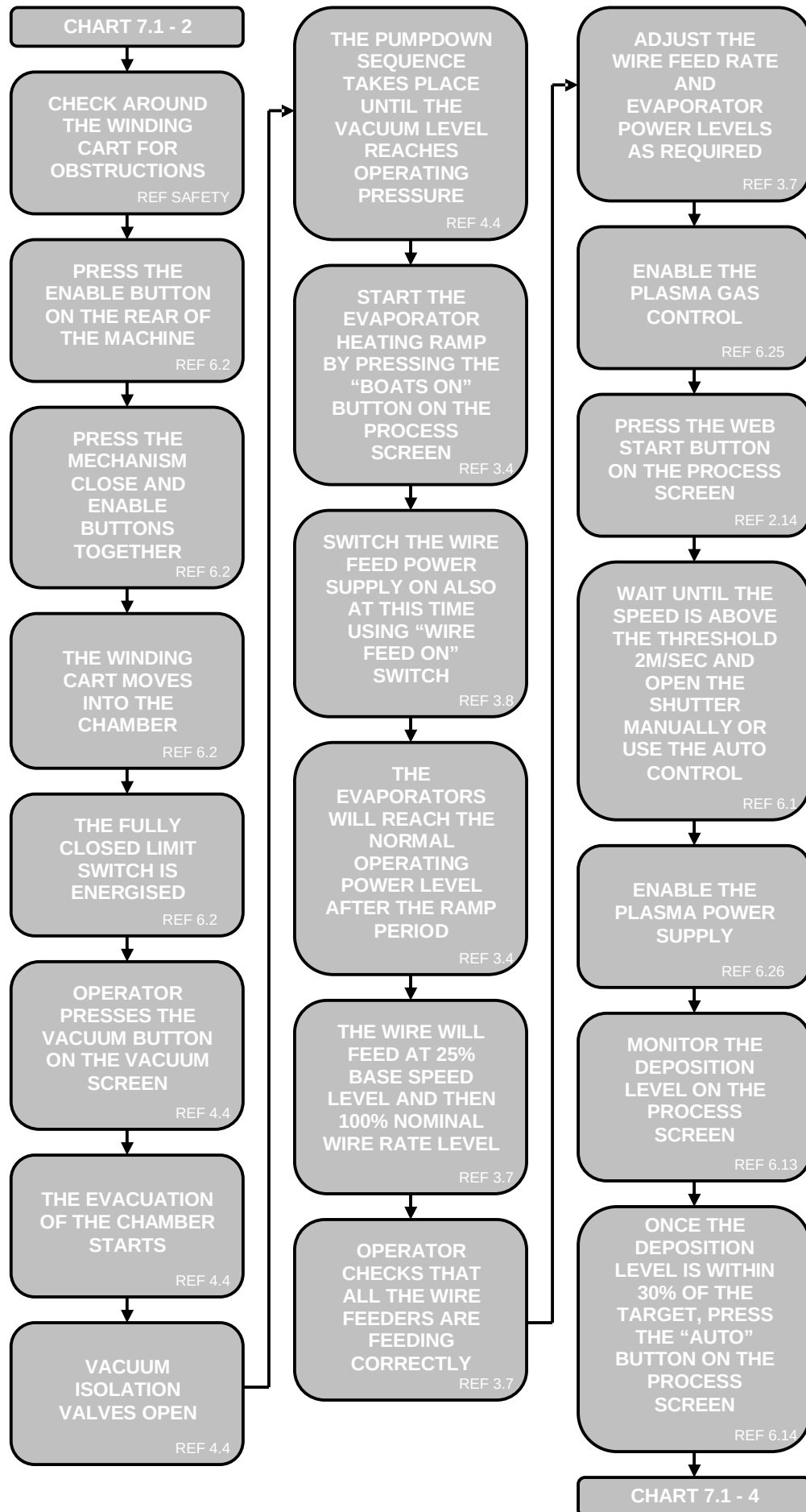
At relevant sections throughout the charts, reference is made to other sections in this manual (section numbers), where further information is provided about any relevant subject.



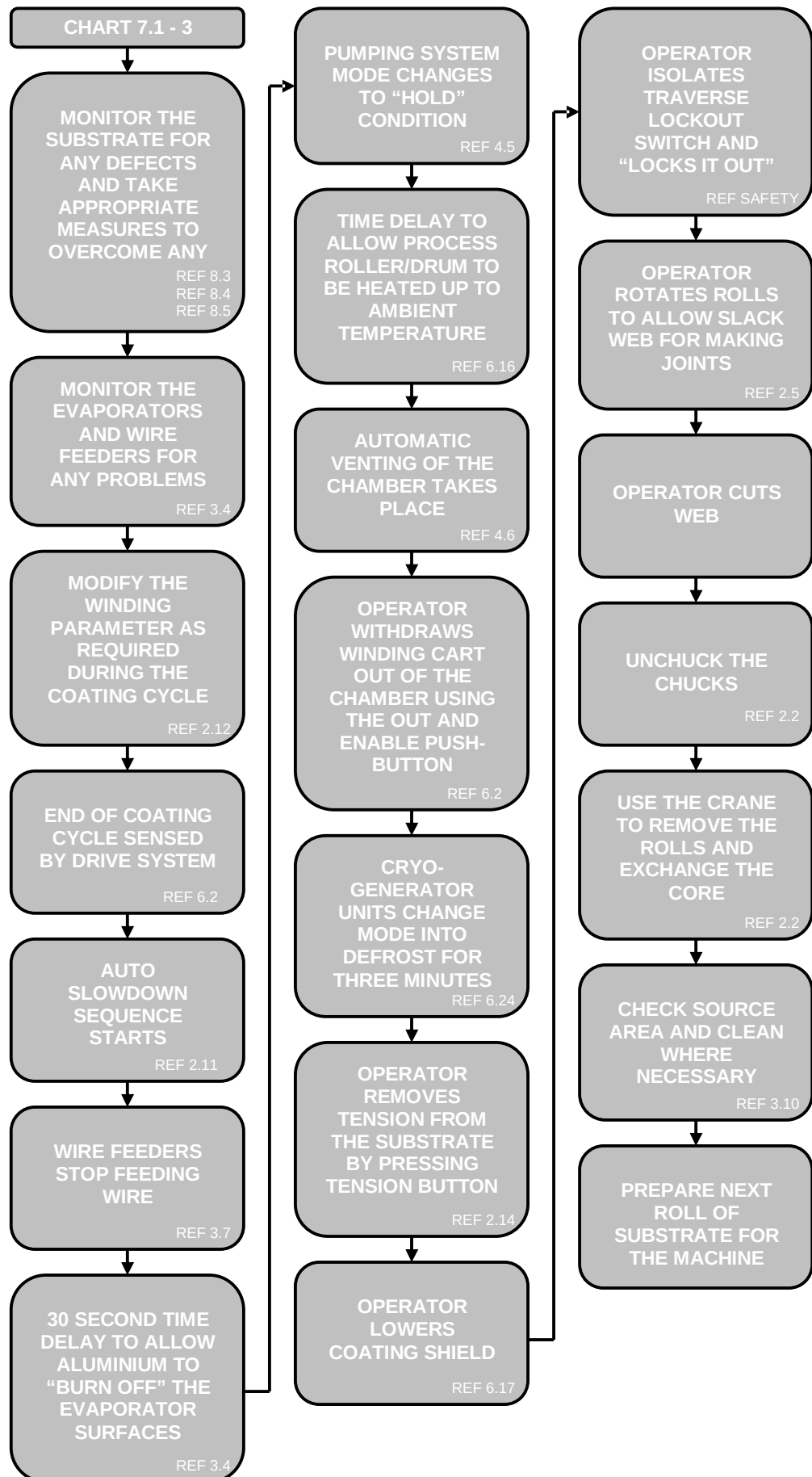
Flow Chart 7.1 - 1 Starting The Machine Flow Chart



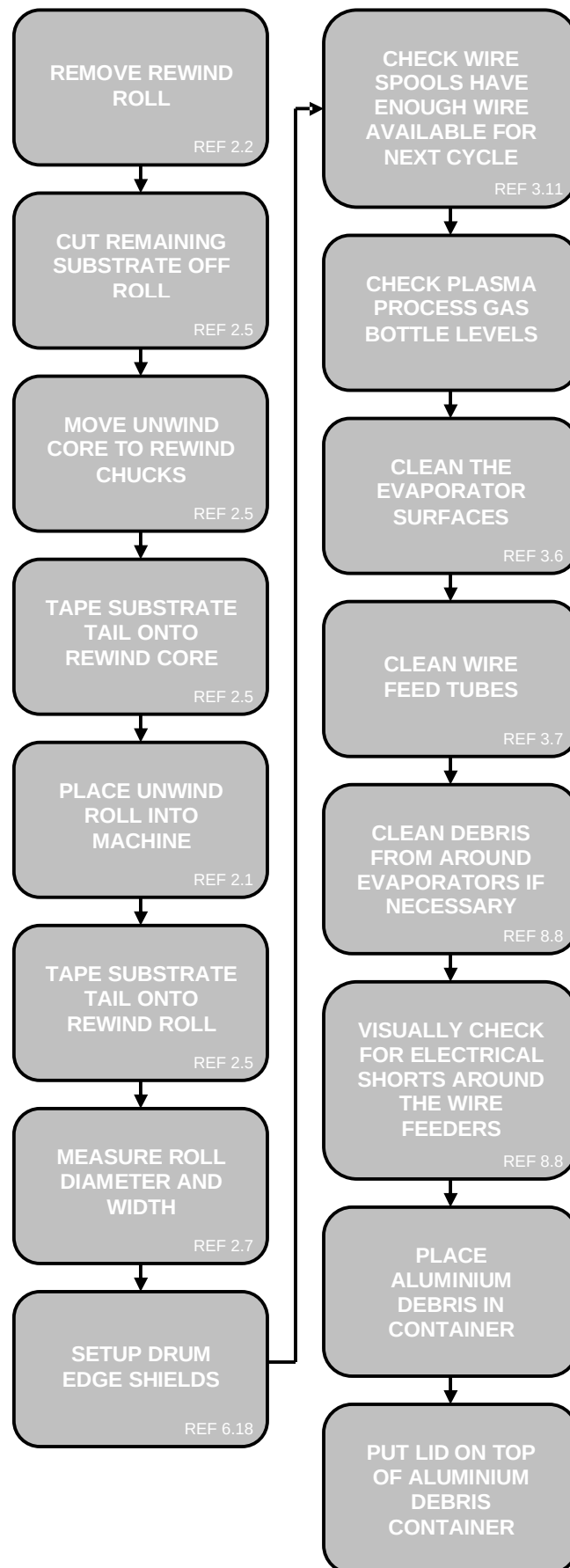
Flow Chart 7.1 - 2 Normal Operation Flow Chart 1



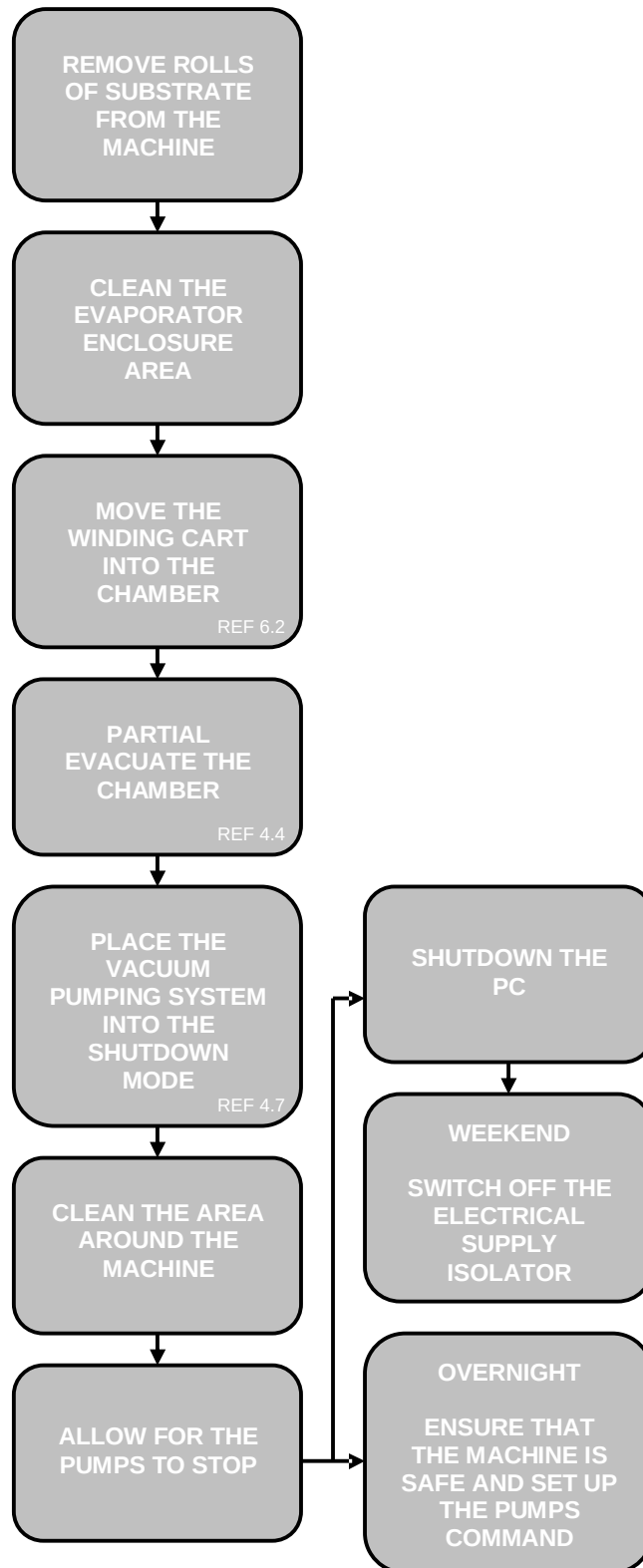
Flow Chart 7.1 - 3 Normal Operation Flow Chart 2



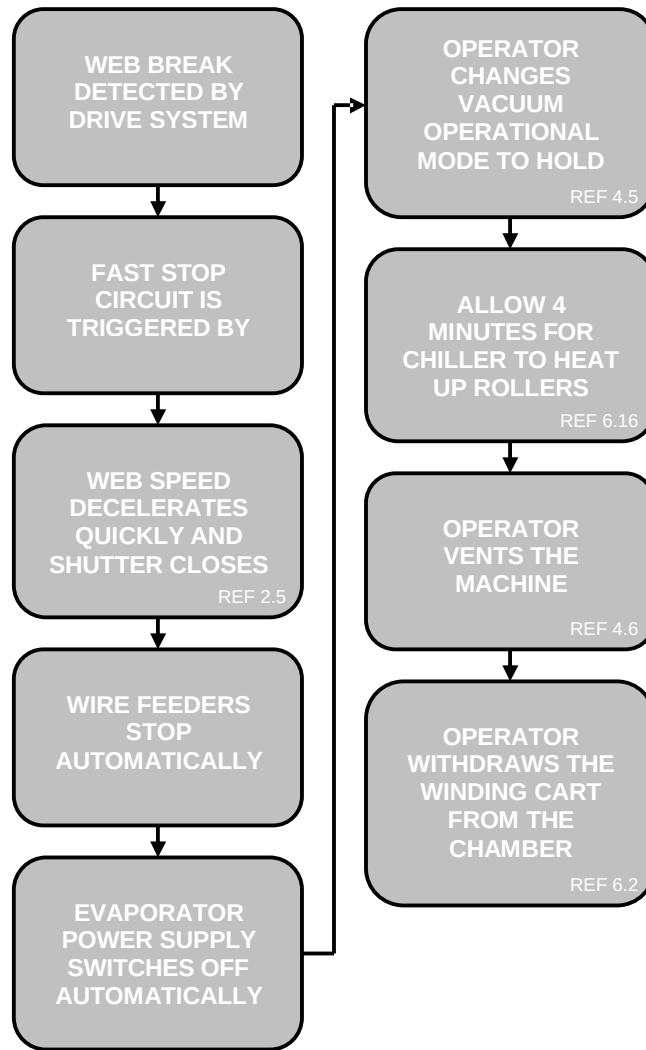
Flow Chart 7.1 - 4 Normal Operation Flow Chart 3



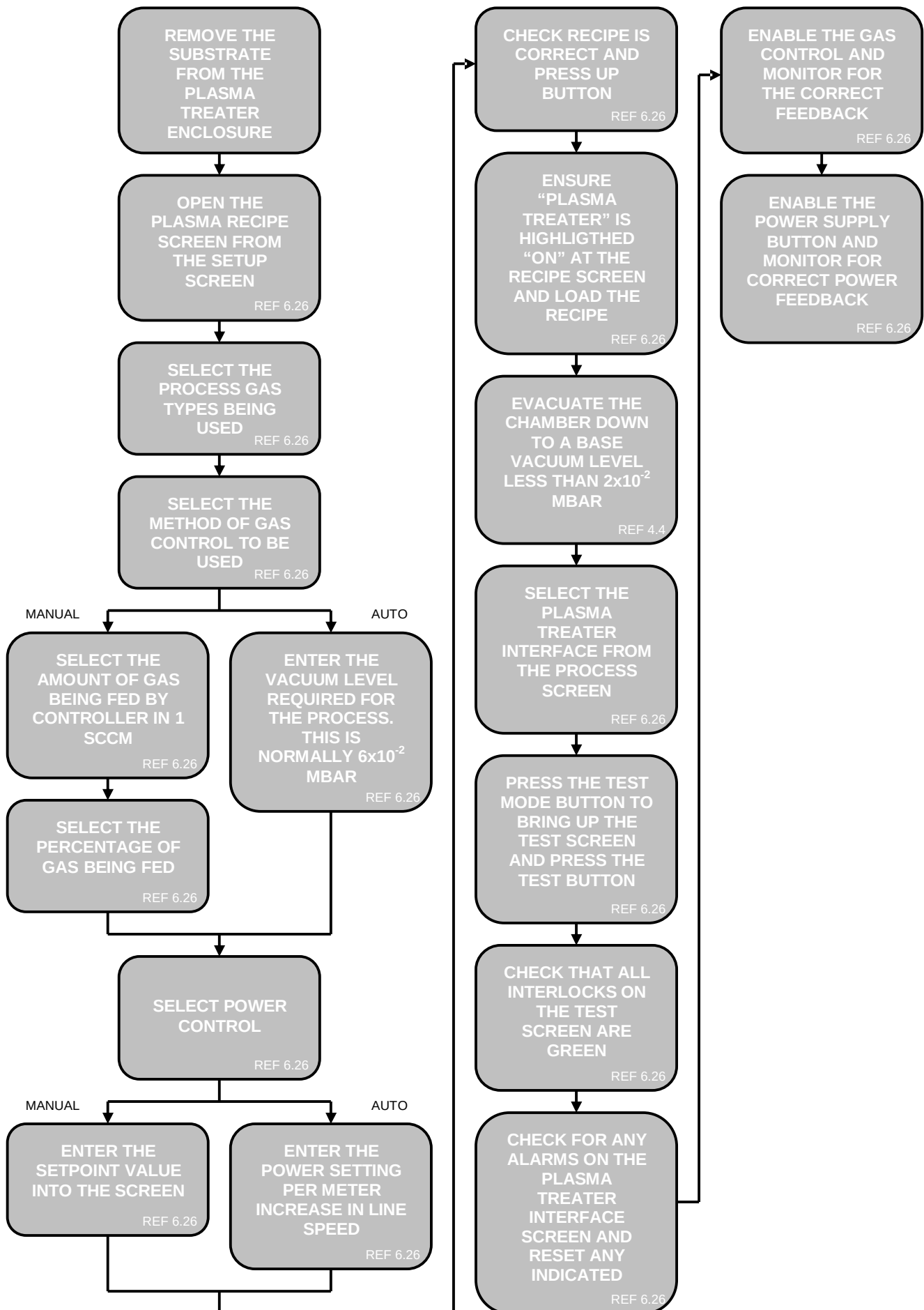
Flow Chart 7.1 - 5 Work Carried Out Between Cycles



Flow Chart 7.1 - 6 Shutdown Flow Chart



Flow Chart 7.1 - 7 Action By The Operator And Machine When A Web Break Sensed



Flow Chart 7.1 - 8 Actions Required By Operator When Checking Operation Of Plasma Treater In Test Mode

SECTION 8 MACHINE TROUBLESHOOTING AND CLEANING PROCESSES

Section 8.1	Process Trouble Shooting
Section 8.2	Vacuum Process & System Problems
Section 8.3	Winding Problems
Section 8.4	Source Problems (Evaporators & Wire Feeding)
Section 8.5	Deposition Control Problems
Section 8.6	Plasma Treater Process Problems (Optional Process)
Section 8.7	Zinc Sulphide Process Issues (Optional Process)
Section 8.8	Cleaning Vacuum Roll Coaters
Section 8.9	Metallizer Cleaning Schedule
Section 8.10	Disposal

8.1 Process Trouble Shooting

The following chapters are to be used as a means of recognizing basic process faults. Further information is given in other sections of this manual and training or vendor manuals.

8.2 Vacuum Process & System Problems

The level of the vacuum pressure inside the machine is a very important process variable. When the vacuum level required is not reached or it takes an extended time to reach, we must investigate why?

What is Wrong?

There can be many reasons why the vacuum level within a chamber may not reach the desired pressure. The reasons can singular or a combination of the following;

- An air leak anywhere, Usually at a seal, feed-through, rotary lead-through, valve, pump or bellows.
- A water leak, located internally.
- Contamination from improper cleaning, lubrication materials or substrates (solvents)
- Malfunctioning gauges, valves or pumps.
- Accidental damage to the machine.

The user must develop a diagnostic method for solving the potential vacuum problems. The problem must be firstly defined.

- Are the valves opening in the correct sequence?
- Are the pumps making abnormal noises?
- When does the problem occur in the pumpdown cycle?

General Vacuum Equipment provides a diagnostic tool on the machine. It is the pumpdown curve. When the machine is delivered. The commissioning engineer will set up the pumpdown screen to show guide line curves of the pumpdown cycle of the machine. Comparison should be made with this curve for the above problems.

Comparisons can be made with the curves shown below;

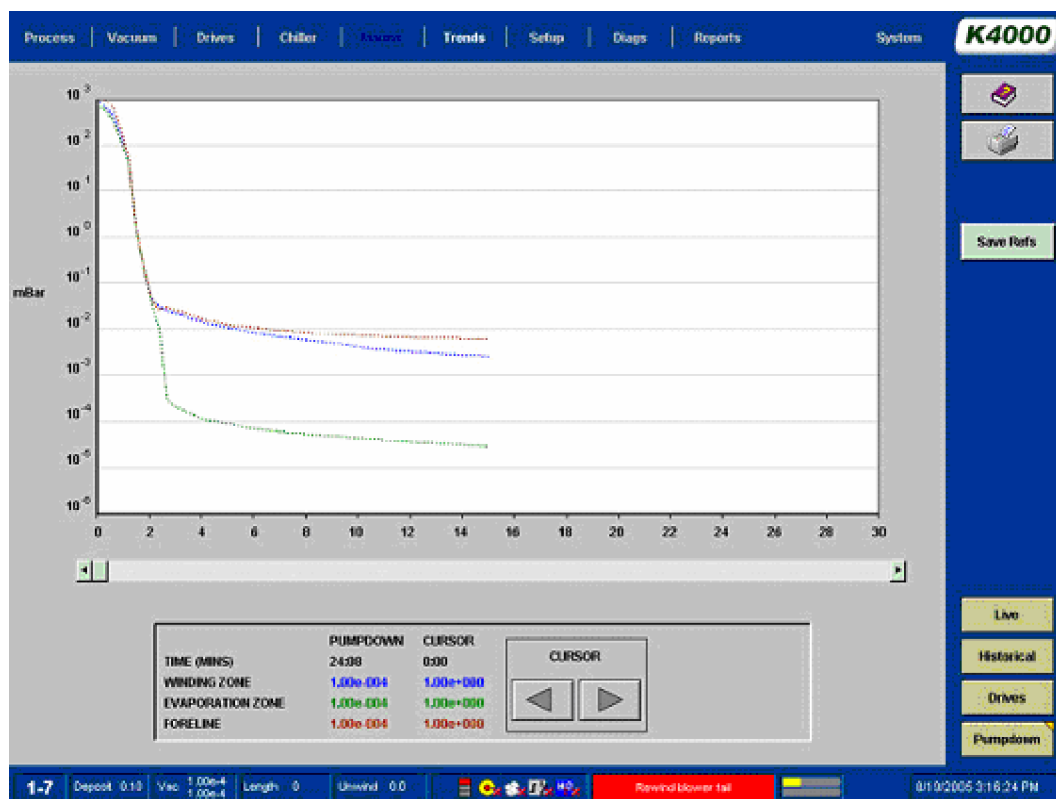


Figure 8.2 - 1 Vacuum Pump Trend Screen

Normal Machine Pumpdown Curve

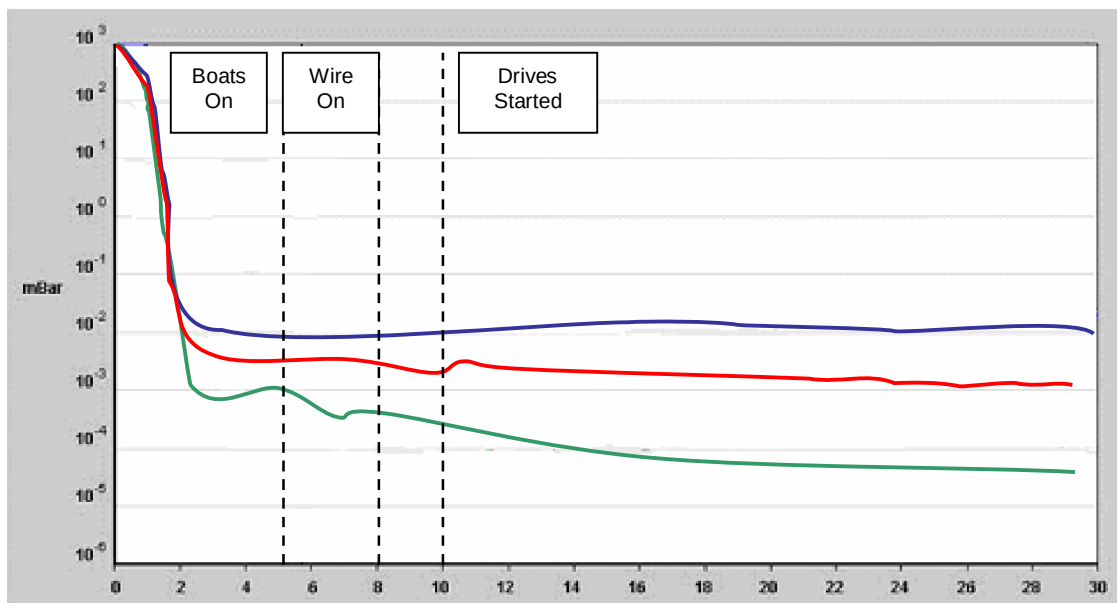


Figure 8.2 - 2 Typical Normal Pumpdown Cycle

It is normal for a small peak to be seen when the following occur;

Evaporator Power on (Boats On)	The radiant heat generated from the evaporators heats up the local surfaces, which will release gas molecules.
Wire Feeders On	The wire is evaporated and the hot evaporant will reach the shutter. The surface will "give up" gas molecules
Rotating Substrate	The substrate can "outgas" and give up gas molecules Also as the drive system accelerates, the shutter will be withdrawn. This allows radiant heat from the evaporators to heat the drum shield area, causing further "outgassing"

Outgassing

This is a term used in the vacuum industry to describe the slow release of gas molecules from all surfaces within the chamber. This includes the substrate, aluminium oxide, evaporators, etc.

Applying heat to these surfaces will increase the release rate from the surfaces.

Machine Cleanliness

During the normal coating process, stray aluminium attaches to all parts around the source box. Also, aluminium oxide is produced, in a dust form. While the chamber is open, the oxide and dust will absorb moisture from the surrounding atmosphere. This absorption will slow down the pumpdown times. The chamber should be cleaned as described in a later section of this instruction manual.

It will be noticeable, that as the machine becomes dirty the longer the pumpdowns will take.

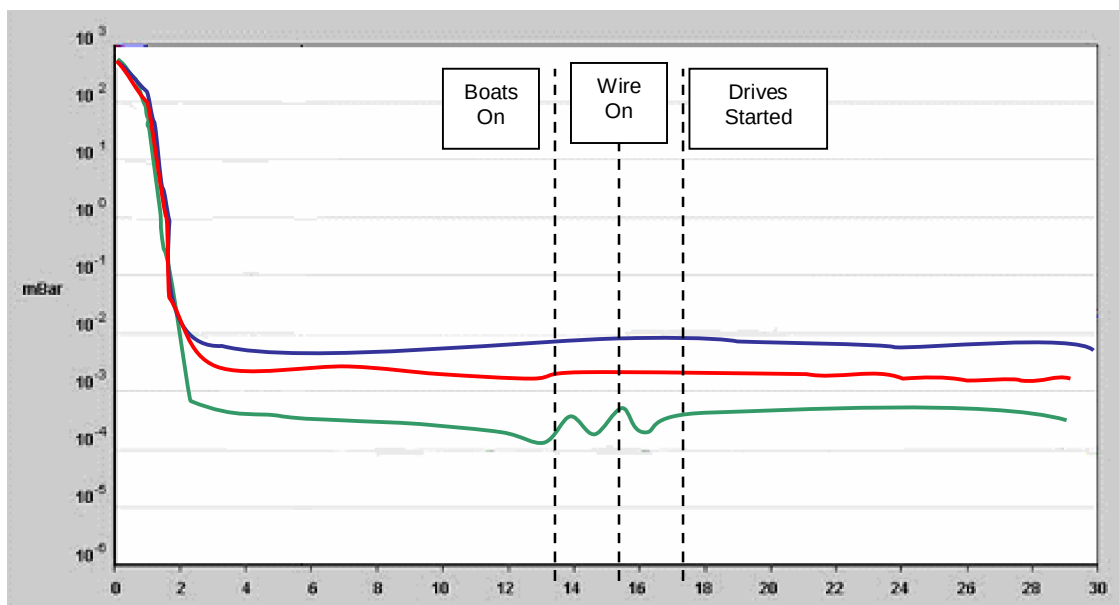


Figure 8.2 - 3 Typical Pumpdown Curve Indicating Problems with A Contaminated Machine

Material Problems

Some substrates have a tendency to give off gases (i.e. "outgas"). If this is suspected, remove the material from the chamber and pump out the empty chamber. If the pumpdown time and vacuum level returns to normal, then the material is to be suspected.

It may be necessary to accept longer pumpdown times when this material is being processed. Alternatively, the line speed may have to be reduced.

Typical reasons for outgassing are;

High moisture	Substrates stored in high humidity or moist conditions.
Solvents	Solvent release from the surface of the substrate
Coatings	Coatings on the substrate surface can be the source of outgassing.

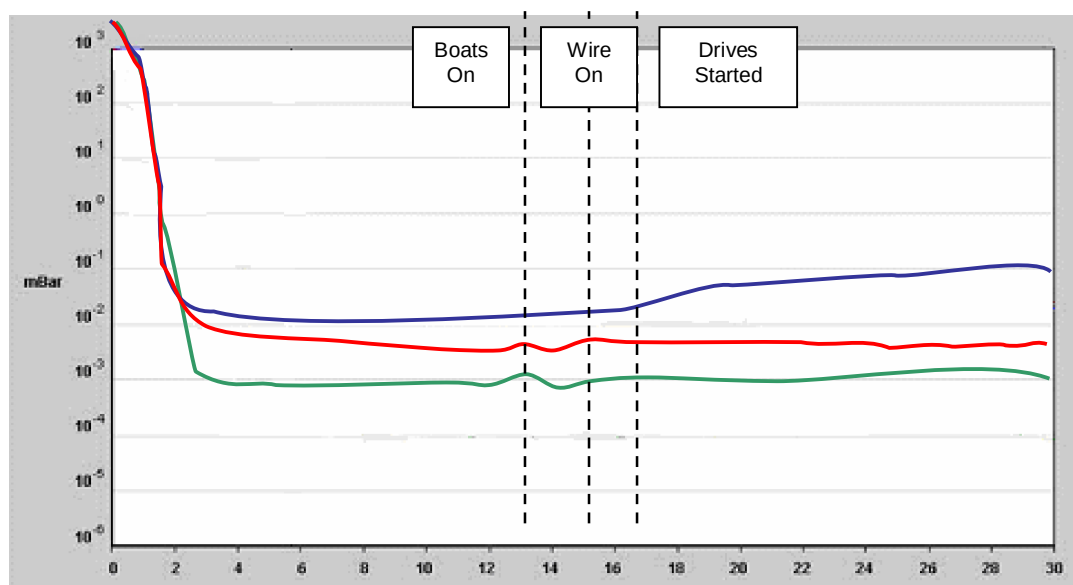


Figure 8.2 - 4 Typical Pump down With Substrate Outgassing.

Air Leaks

Check the main vessel joint band, all sight-glasses, rotary leadthroughs etc. for the location of air leaks. This is described in the leak detection section of the maintenance manual.

Performance of Pumps

Check the ultimate vacuum levels of the pumps to the values given in individual proprietary manuals using the recommended equipment.

Water Leaks

Check all water hoses, pipes, and shields for signs of water leaks. These usually appear as a rust or dark stain on the Aluminium oxide coatings. A method of checking for leaks is to isolate the individual water manifold circuit, open the feed and monitor for rises on the vacuum gauges.

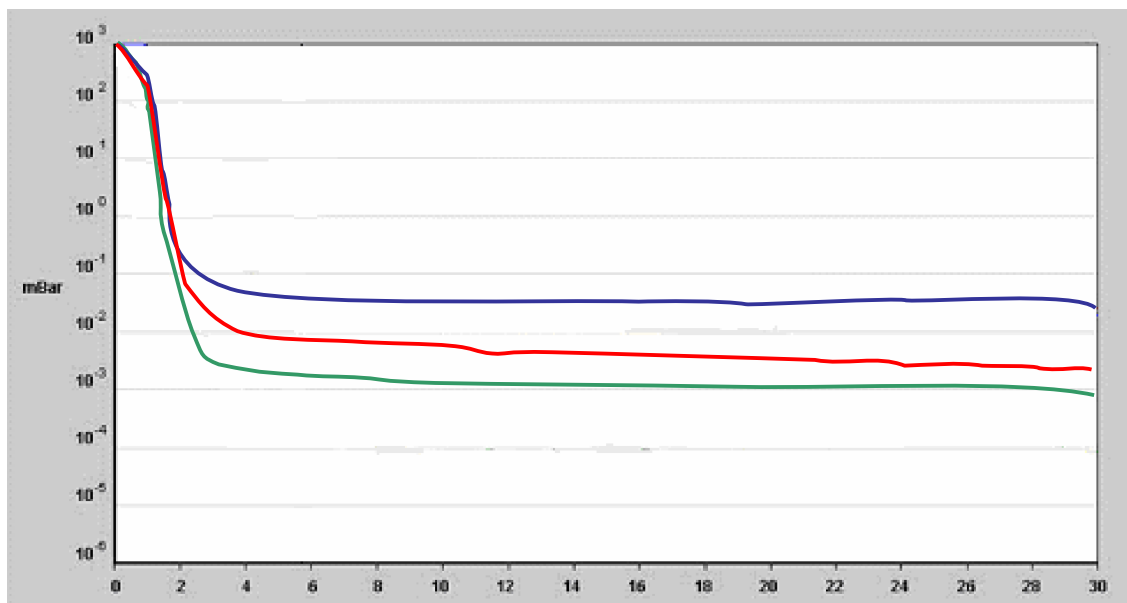


Figure 8.2 - 5 Typical Pump down Curve of Machine with Water Leak Problem

Diffusion Pump Problems

One of the most common problems that are seen with the vacuum system is the shutting down of the diffusion pumps. The pumps are interlocked to flash the mimic screen between red and green at the relevant pump. As the mimic starts to flash the pump's heaters are also de-energized. The reasons that the pump will be shutdown are;

- Overheating - lack of cooling water- generally caused by blockage of the water manifold's strainer. Generally indicated as "Diff Overtemp" Alarm
- Overheating - cooling water being too warm - caused by problems with the customers cooling system or closing of the valves on the return manifold.
- Loss of vacuum level because of debris located between the high vacuum valve (VV7) and the chamber. This can be detected by the vacuum level in the pump becoming lower when the chamber is evacuated.
- Loss of water pressure at the vacuum pump manifold.

Cryo-generator Unit Problems

If the Cryo-generator unit is not operating correctly due to loss of the refrigeration gas for example, the pumpdown will be affected.

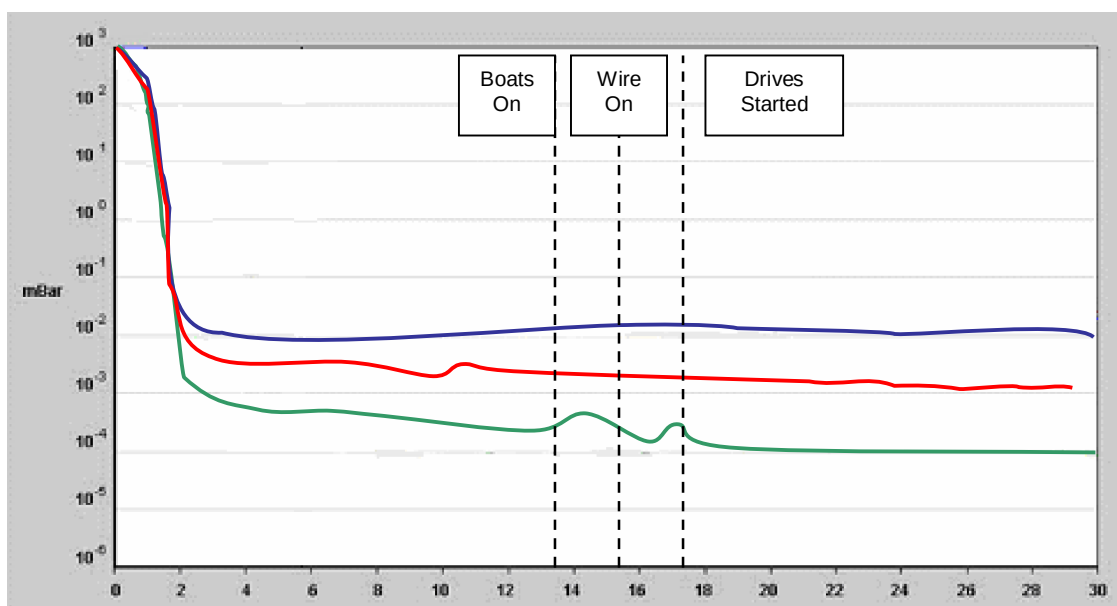


Figure 8.2 - 6 Typical Pumpdown Curve of Machine with Faulty Cryo-generator/Cryogenic System

Other Vacuum Problems

Other Problems that have been monitored in the vacuum system control re described below;

Problem	Probable Cause	Suggested Remedy
Foreline valves (VV6) do not open	- Major air leak - Faulty Gauge in foreline - Poorly performing pump(s) - Incorrect interlocking pressure switch points	- Leak check the machine for leaks. - Check calibration of gauge in foreline pipe work. - Check the performance of the pumps against the information supplied by the manufacturer - Set-points are set by General Vacuum Equipment.
Foreline valves (VV6) close after high-vacuum valve open (VV7)	- Zone separation not completed - Vacuum zone breakdown between winding and evaporation zones - High vapour pressure source located in evaporation zone - Poor Pump Performance	- Check that the relevant valves are closing correctly i.e.; VV 8 and 4. - Check zone seals for correct performance. Replace or re-align as necessary. - Check evaporation area for incorrect greases, etc. - Check the performance of the pumps against the information supplied by the manufacturer.
Fore-line valve(s) (VV6) opens after a long period	Poorly performing pump	Check the performance of the pumps against the information supplied by the manufacturer

Problem	Probable Cause	Suggested Remedy
High Vacuum valve (VV7) does not open	- Machine Interlocking stopping valves opening. - Major air leak - Faulty Gauge in chamber - Poorly performing pump - Incorrect interlocking pressure switch points	- Check the machine plc program for interlocks required for valve/s to open - Leak check the machine - Check calibration of gauge in the top of the chamber. - Check the performance of the pumps against the information supplied by the manufacturer. - Set-points are set by General Vacuum Equipment.
High vacuum valve (VV7) cycles open and closed	- Diffusion Pumps below operating temperature. - Diffusion pump cooling water too cold	- Check that the power being taken by the pumps is correct. Check with the manufacturer's information. - Check the return temperature from the diffusion pumps is at 67 °C
Coating of oil on cooled process roller	Contamination by oil/grease which produces a high vapour at low pressures.	Clean the contamination and remove the item that is producing the vapour.
Diffusion pumps pressure rise to atmosphere when chamber is vented	- Debris on valve seal - Valve not being closed	- Lift the valve as described in the maintenance manual and inspect. - Check the operation of the related spool valve
Poor Ultimate pressure in the 10^{-2} mbar range	- Small air Leak - Small Water Leak - Pump Fault	- Leak check the chamber - Check all the water hoses - Check the oil level and quality - Compare the ultimate pressure of the pumps against the manufacturer's data
Vacuum Mimic Screen does not update when actions are taking place	- Communications problem - Loose communication wiring/termination	Check the communications check screen to check that correct communications is taking place.
Poor Ultimate pressure in the 10^{-1} mbar range	- Large air Leak - Large Water Leak - Pump failure	- Isolate the chamber from the pumping group and monitor if the pressure rises in the chamber. If it does, switch the pumps off and listen for an air leak. - If the chamber pressure does not rise. Listen to the area around the pumping group for leaks. Use the gauge in different positions on the pipe work to identify the leak. - Check all internal hoses for ice build up. - Ensure all the pumps are rotating. Check the pumping performance.

Problem	Probable Cause	Suggested Remedy
High pressure rise in chamber when drive system starts to rotate.	- Rotary lead-through Leak - Outgassing Substrate	- Remove the substrate, close the chamber and evacuate it. Inch the drive system and monitor the pressure levels on the gauges in the chamber. This will determine if the problem is on a lead-through. Loss of oil from the gravity bottle will indicate seal wear. - See section above.
Booster Pump does not start	- Pump Seized - Inverter Unit not operating - Pump circuit breaker tripped	- Check that the pump rotates. - Check the operation of the inverter and it's associated wiring - Reset the circuit breaker.
Isolation valve indicator on vacuum mimic flashes when valve is supposed to be open	- Valve not opened or closed - Position sensor not being activated	- Check position of manual indicator. Strip the valve down to investigate a problem. - Check that the sensor is being activated
High pressure rise in evaporation zone when shutter opens	- Rotary lead-through Leak - Outgassing Substrate - Outgassing of aluminium oxide - Too much gas flow to Gas wedge	- With the evaporators powered down, operate the drive at a low speed and open the shutter to see if the pressure rises on the gauge in the evaporation zone. - Substitute the substrate for another. If this product does not cause a pressure rise, the substrate is suspect. - Ensure that the machine has been cleaned and that large build ups of aluminium are not present. - Normal level less than 2 litres/sec

Table 8.2 - 1 Other Vacuum Problems

8.3 Winding Problems

Introduction

This section of the manual deals with substrate and winding problems that may be encountered when winding substrates in the machine.

Terminology

There are a number of different terms to describe the same problem on a roll of substrate.

Reference should be made to this section when describing a problem to anybody.

The following listing is of all possible winding defects that have been identified.

Fibre Core Problems

Problem	Paper or Film Defect	Problem
Core length too short	Film & Paper	Core length too long
Core offset to substrate	Film & Paper	Raised edges on core ends
Substrate not square	Film & Paper	Crushed core
Debris on core	Film & Paper	Plastic Core Problems
Thin walled core	Film & Paper	Thin walls
Core not engaged on reel-shaft	Film & Paper	Score marks

Unwind Roll Problems

Problem	Paper or Film Defect	Problem
Blocking of Substrate	Film	Backside Corona Treatment
Telescoping	Film & Paper	Eccentric Cores
Dishing	Film & Paper	Out Of Round Cores
Feathering	Film & Paper	Even Roll Firmness
Core Movement	Film & Paper	High Edges
Soft Edges	Film & Paper	Starring
Droop/Gullswings/Bird Wings	Film	Chicken Wire Pattern
Incorrect Treatment Side	Film	Surface Friction Level
Contamination	Paper & Film	Creases
Wedge Profile	Paper & Film	Excessive Splices
Excessive Gauge Bands	Paper & Film	Poor Splices
Tin Canning	Paper & Film	Buckles
Blade Streaks	Paper	Wandering
Lacquer Line	Paper	Base Paper Line
Fish Eyes	Paper	Calendar Cut
Clay Skip	Paper	Ridges
Ballooning	Film	Drips
Die lines	Film	Gels
Flat Spots	Film & Paper	

Processing Winding Problem

Problem	Paper or Film Defect	Problem
Screwing	Film & Paper	Creases - Wrinkles • Pipe Creases • Pinch Creases
Roll Too Soft - Troughing	Film & Paper	
Flapping Edges	Film	
Curl	Paper	Roping
Outgassing	Film & Paper	Web breaks
Roller Marks	Film	
Baggy Film	Film	Roller Marks

Rewind Roll Problem

Problem	Paper or Film Defect	Problem
Cracking of Coatings	Film & Paper	Dishing of Roll Edge
Telescoping	Film & Paper	Lipped Edge
Starring of Roll Edge	Film & Paper	Hard Rolls
Blocking of Substrate	Film	Wrinkles • Unwind Area • Evaporation Area • Post Draw Roller Area • Rewind/Layarm Area
Roll Contraction	Film	
Width Reduction	Film	
High Temperature Rolls	Film	
Tin Can Effect	Film	Curvature /skew - measurement
Offset Movement	Film & Paper	Soft Rolls
Horned Edge	Film & Paper	Stickouts

Coating Process Problems

Problem	Paper or Film Defect	Problem
Tramlines	Film	Railroad Tracks/ Tramlines
Wide Margins	Film & Paper	Scratches • Small Diagonal After Metallizing • Small Diagonal Before Metallizing • Winding Mechanism Deflection • Repeated Pattern • Continuous Single Scratch • Speed Differences
Uneven Deposition- Uneven Lacquer	Paper	
Non Metallized Lanes	Film	

Coating Process Problems

Problem	Paper or Film Defect	Problem
Low Surface Tension - post Coating	Film	Low Adhesion Levels
Coating drum Contamination	Film & Paper	Debris in Edge of Roll
Blue Tint	Film & Paper	Brown Tint
Gold/Green Tint	Film & Paper	Metal Pick Off
Spitting	Film & Paper	Poor Uniformity
Spattering	Film & Paper	Light Lane
Blisters	Paper	Fish Eyes
Low Gloss	Paper	Uncoated Spots
Banding	Film & Paper	Dark Lanes
Jail Bars	Film & Paper	
Blooming	Film	Pin Holes
Reflected heat lines	Film	

Table 8.3 - 1 Winding Problems

Roll Storage

The method used for storing the rolls of substrate will differ from location to location. Some materials such as thick paper can be stored on the floor on their end, while others have to be stored in racks. Care must be taken not to damage the substrate while it is in storage.

Generally plastic film is supported in a rack with a pipe placed through the core. However, if the roll is soft, stresses can be induced as shown below;

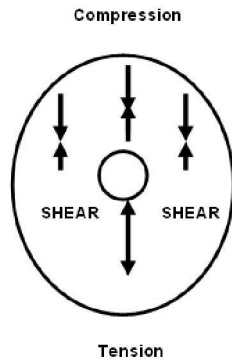


Figure 8.3 - 1 Roll Damage Caused By Storage Problems

If the roll is very heavy and a cardboard core is used, the core can become distorted, as shown below;

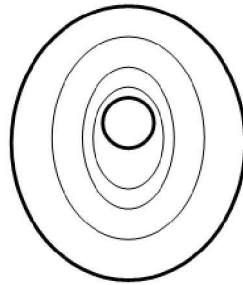


Figure 8.3 - 2 Core Support Deformation

If rolls are stored on a flat floor for long periods of time, a flat on it can be formed. This will lead to a "lobbing" roll or even "blocking" of the substrate.

Care must be taken not to damage the outer edges of the roll when moving into to the machine. Damage on the ends can lead to web breaks (snap outs).

Storage Environment

If a roll of substrate is stored in a high humidity area prior to being metallized, the resulting rewind roll has been monitored to be hotter than rolls stored in a correctly air conditioned area. Another aspect is that the substrate will be expelling moisture as it is unwound.

Cores

All substrate is wound off and onto cores. These cores may be manufactured from the following materials;

- Steel
- Plastic
- Aluminium
- Composites
- Cardboard Fibre

The above cores may be fitted onto a shaft or be run in the machine in a "shaftless mode". The figure below demonstrates the difference between "with shaft" and "shaftless".

Some companies have a common core that is used on all the various equipment in their factory, While others will have large cores at a unwind station and rewind onto smaller cores.

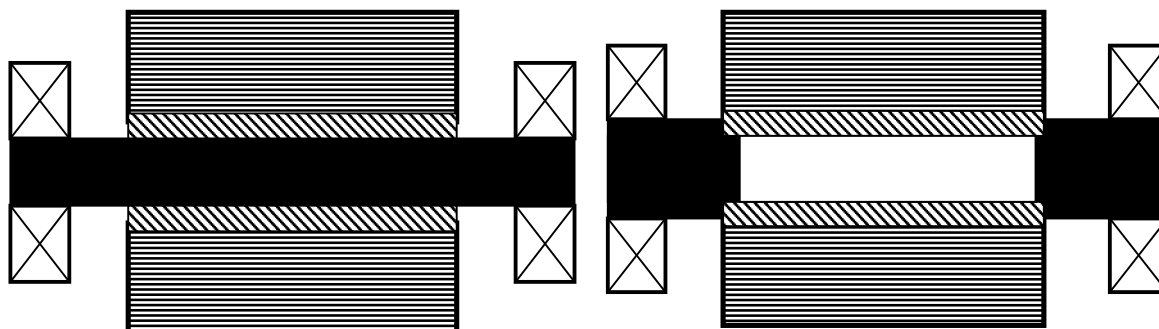


Figure 8.3 - 3 Rolls with Shafts & Without Shafts

The common core sizes used in machines are;

- 76mm - usually fibreboard
- 152mm - usually fibreboard or plastic
- 304mm - usually steel or aluminium

If the material is not concentric on the core, vibration can be induced within the winding mechanism and further winding problems can occur.

Fibre Core Problems

Core Length Too Short

If the core used is too short, the substrate can droop over the ends. This results in soft edges on the completed roll. The substrate can show marks on the metallized surface at the position where the core ends.

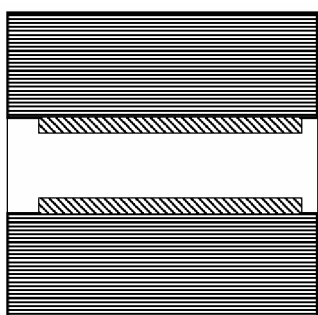


Figure 8.3 - 4 Core Too Short Sketch

Core Length Too Long

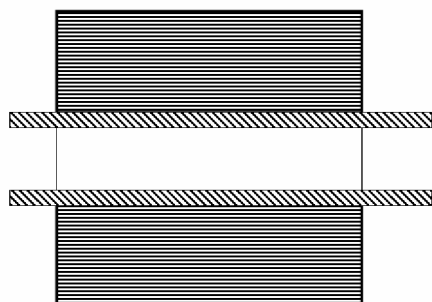


Figure 8.3 - 5 Core Too Long Sketch

This situation can cause problems in handling the finished roll after completion of the process.

It is an undesirable situation and with care can be easily overcome.

Core Offset to Substrate

The unwind and rewind cores should be the same length and be placed even to each other.

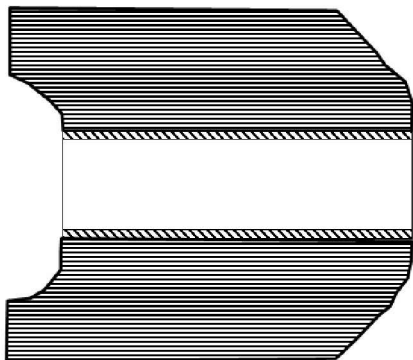


Figure 8.3 - 6 Core Offset To Roll Sketch

If they are not positioned even, but offset, the roll shown below will be manufactured. It is undesirable for this type of roll to be produced. It will produce major problems down stream on progressive processes.

Substrate Not Square

If the substrate is not aligned within the winding mechanism and attached to the core in an offset position, possible winding defects such as wrinkling may occur. A profile of this type of roll can be seen below.

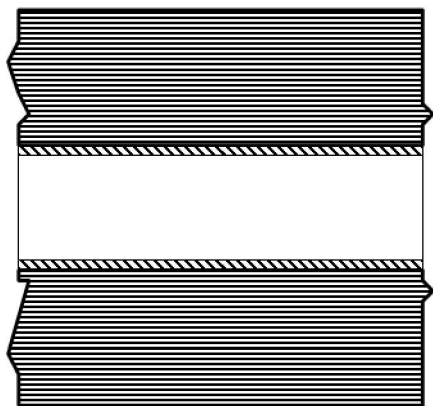


Figure 8.3 - 7 Substrate Not Located On Core Squarely Diagram Sketch

Debris on Core

The cardboard cores used should be new for every roll that is wound at the rewind position.

However, some users recycle the cores throughout the factory and they can become damaged with dents. Another problem is the remnants of adhesive tape and substrate material. This can lead to hollows in the finished roll and lumps. Ensure that the core is clean and free from defects.

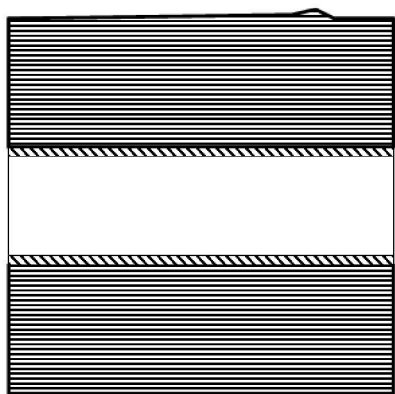


Figure 8.3 - 8 Debris on Core Sketch

Thin Walled Core

If the wall on the cardboard cores is thin, it can be deflected by the expanded segments of the mechanical reel shaft. The pattern created looks like a clam, see the figure below.

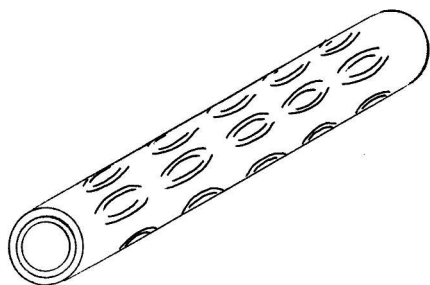


Figure 8.3 - 9 Thin Walled Core Showing Defect Marks

To overcome this problem, use cardboard cores of a thicker wall.

Core Not Engaged on Reel Shaft.

The core can be seen to move on the shaft and show results similar to when the unwind and rewind cores are not aligned.

Check that the shaft is engaged correctly in the core, either mechanically or pneumatically. A side product of this slippage can be cardboard dust.

The recommended torque level to be applied to the mechanical shafts is 100 Newton metres.

When moving unwind cores to the rewind position, ensure that the core is still engaged correctly with the torque wrench.

Crushed Core

If a roll is wound too tight on a core, it can cause the core wall to collapse under the pressure. The first symptom is that the reel shaft will be difficult to remove from the core. The deformation can be seen when the shaft has been removed.

Place wooden stoppers (bungs) in the end until the roll is moved to the next process.

Ensure that the situation is not repeated on the next roll.

In very bad cases a commercial mechanical expander can be used to press the inner wall of the core out.

Raised Edges on Core Ends

If the cores have not been cut with a core cutter, but by some other means, the edge will sometimes be "lipped up". This will induce a lip build up on the roll being produced.

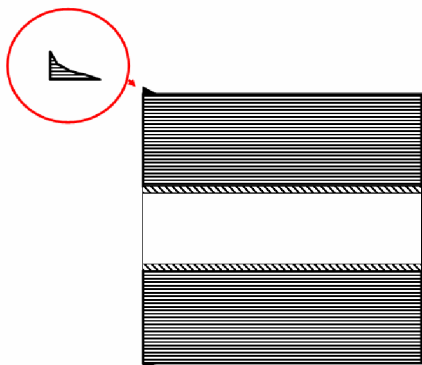


Figure 8.3 - 10 Edge Build Up Diagram

Ensure that the edges of the core is not "lipped up". If it is, smooth it down using a file.

Plastic Core Problems

Basically plastic cores can suffer similar problems to those of fibre cores, as far as alignment is concerned. Plastic cores are used in the narrow width machine applications with thinner substrates.

Thin walls

If the core walls are too thin, they will deflect with large roll build ups and the core becomes "egg shaped". Use thicker walled cores to overcome this problem.

Score Marks

If the cores are re-used often, they must be inspected for score damage when the previous rolls remnants have been removed. The knife scores can cause ridges on the finished roll.

Metal Cores Problems

Metal cores once again can suffer the same debris and alignment problems. They can also suffer locator damage on the ends of the cores. Usually, the metal cores are located between chucks without a reel shaft.

Unwind Roll Problems

Blocking of Substrate

A blocked roll is one where the layers do not separate easily on unwinding. A characteristic of this defect is retarded unwinding tangent as shown below. Often, the tangent will move between two positions in a saw tooth fashion and is sometimes accompanied by a ripping or bellowing noise (like a sail boat sail in wind).

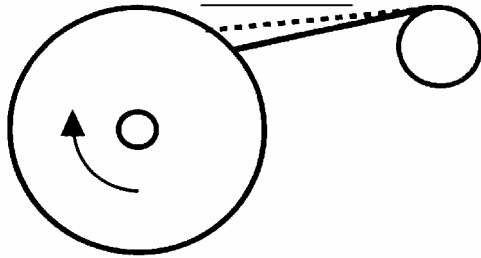


Figure 8.3 - 11 Diagram of "Blocked" Substrate on Unwind Roll

The reasons for the "blocking" could be one of the following;

- Mechanical Interlocking - If the surfaces of the layers of substrate are rough or fibrous, the layers can interlock and stick together.
- Adhesion: - If there is an adhesive action due to coatings, treatments etc., the layers may stick together.

It has been found in cases of severe blocking that the substrate will rip and cause a web-break situation.

It is very difficult to overcome this problem and generally the roll is rejected.

Backside Corona Treatment

A number of types of plastic film are corona treated to increase the dyne level to the required level. The method of treating the plastic film requires the film to be flat on a ceramic backing roll. If the film is not flat on the roller, the treatment can pass through the film and cause backside treatment.

It has been found that this backside treatment will cause blocking of the unwind roll. This has been monitored in thinner gauge OPP, LDPE and CPP films.

This is one of the main reasons for blocking of unwind rolls.

Telescoping

The term telescoping is used to describe the condition when a roll of material moves when it is being unwound. A profile is shown below.

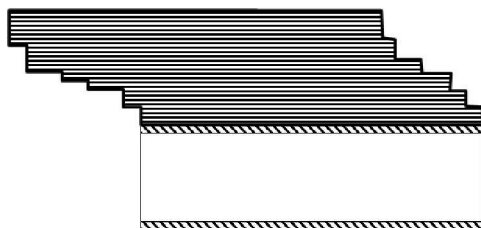


Figure 8.3 - 12 Diagram Showing Telescoped Roll

The profile shows a telescoping action that the substrate's edge profile has moved a large distance over a few layers. Obviously there was interlayer slippage.

This problem is most likely on types of substrate that have a low surface tension level, for example, cellophane.

Other cases occur with rolls of substrate that have been poorly wound with non constant tension profile. This typically can be monitored when multiple starts and stops of the winder has occurred.

If a roll has a soft hardness profile, throughout the roll, air is trapped within it and this can also lead to the problem.

To overcome the problem, a device for holding the edges together will have to be used. The devices used can be side cheeks or edge profile rollers.

Dishing

The term "dishing" refers to the roll, whose profile is shown below;



Figure 8.3 - 13 Diagram of Dished Roll

The reasons for the profile being generated could be;

- Core movement side-wards gradually on the shaft
- Core bellied or crushed
- Interlayer slipping

The movement will always be in the same direction

It has been known that some users have process rolls that are presented to the machine on a regular basis with "dished" profiles.

Some machines drive systems have been designed to allow gradual automatic movement of the unwind roll to overcome the "dishing". Unfortunately standard machines do not have any facility to overcome this problem. The user has the choice of not being able to coat the full width of the substrate or have the roll rewound with straight edges.

Feathering

This is the term to describe a condition when just a few layers of the substrate move sideways at the surface of the roll. This condition can be caused by the following reasons;

- Material Outgassing as it unwinds
- Soft Rolls
- Low surface friction of material (Common with cellophane).

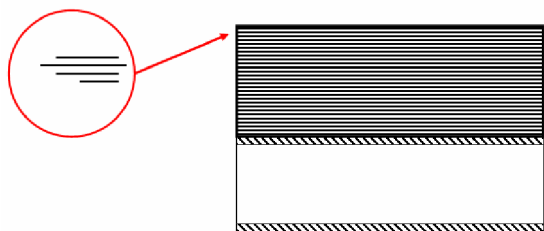


Figure 8.3 - 14 Feathering Of Unwind Roll

This problem will cause the substrate to move across on the drum and therefore the drum will become coated with aluminium. The other undesirable effect is that one edge of the substrate will not be coated.

To overcome the problem, the followings steps can be taken;

- Increase the unwind tension slightly
- Fit side plates (cheeks) to the roll
- Use side guide rollers
- Use a layon roller (if provided)

Core Movement

As already mentioned, core movement will cause various winding problems and possible contamination of the winding mechanism with core dust.

Eccentric Cores

Eccentric cores will cause vibration and wrinkling from the unwind roll. It is unlikely that a quality roll of metallized product can be achieved with this type of defect.

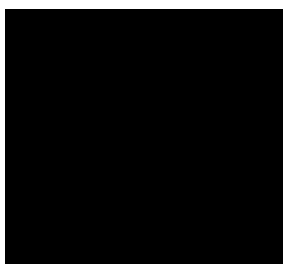


Figure 8.3 - 15 Eccentric Roll

Remove the roll and rewind the material onto a new core.

Out of Round Roll

If out of round rolls of substrate are presented to the machine, the following problems can occur;

- Vibration within the machine.
- Poor tension control
- Wrinkles

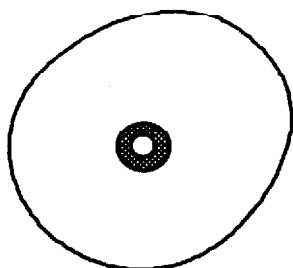


Figure 8.3 - 16 Out Of Round Roll Defect

The roll could have been formed in this manner by allowing it to be stored incorrectly and inducing a flat. Another reason could be the use of a poorly manufactured core.

In the case of any of the previously mentioned problems, the reason for the defect must be found.

Even Roll Firmness

The roll should be checked for firmness using a commercial hardness tester, such as a Rho-meter.

The hardness profile of the roll will effect the winding of the substrate through the winding mechanism. Soft and hard areas will indicate problems within the roll, such as gauge bands.

As a general guide, the rolls should be between 30 - 40 Rho with no single point variation of more than 10 Rho. Some rolls may be wound softer to overcome possible blocking problems.

A roll may be presented to the metallizer, which looks perfect to the user's eye. However, the roll may contain gauge bands for example that will be hidden by air layers between the layers of the substrate. Once the substrate has been processed in the metallizer, the defects may be seen in the finished roll. This is due to the fact that the roll is wound in a vacuum environment with no air entrapment.

High Edges

If the roll has been slit at a previous process with a blunt knife, the usual result is that localized heat is produced and the slit edge becomes rough.

This rough edge can lead to a high edge similar to that seen in the paragraph on "raised edges on core ends".

Once metallized the edge profile becomes more exaggerated and will contact the rewind layon roller, even if it is in gap wind.

To overcome the problem, the layarm can be operated in the layon mode with a light layon force.

Or the rewind roll can be split before the raised edge becomes too large.

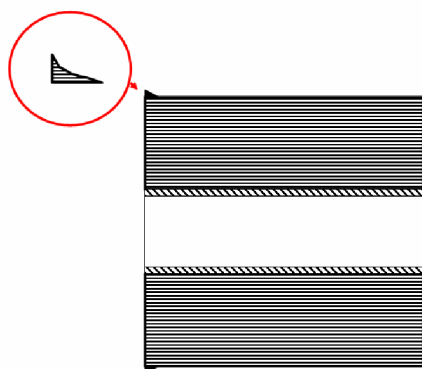


Figure 8.3 - 17 High Edge Defect

The raised edge will also induce wrinkles and eventually hard creases on the rewind roll.

Where plastic film has been slit with a poorly adjusted or blunt knife, the slit edge can be seen to be rough and have a serrated pattern. This fault can lead to raised edges.

Soft Edges

If the profile of the substrate is checked with a hardness tester and the outer edges are very soft (i.e.; 10 Rho's lower), the following problems can be expected;

- Hotter rewind rolls in these areas, as the substrate was not in good contact with the cooled rollers/drums.
- "Flappy Edges" throughout the winding mechanism that ride over spreader rollers and induce wrinkles.

Poor deposition control in these areas as the substrate moves towards and then away from the deposition monitor probes. This in turn gives false signals to the deposition control system.



Figure 8.3 - 18 Photograph of Soft Roll

Staring

Staring is the descriptive term given to rolls that have spoke like patterns on the side of the rolls, which characterize the defect. There may be only a single radial spoke or multiple spokes.

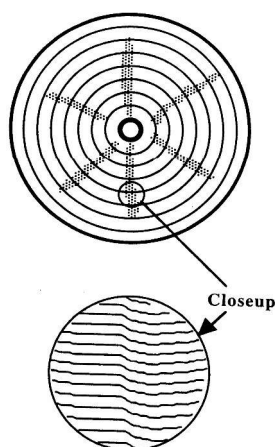


Figure 8.3 - 19 Staring of Roll layers

The previous figure shows an example of a "starring defect".

The spokes will extend outwards towards the surface of the roll.

The fault is generally caused by compression of softer wound layers by tighter layers further away from the core. This can occur if major changes in the desired tension level are made throughout the coating cycle.

Ideally this defect is not desired and modifications to the winding parameters should be made to overcome the defect.

Droop/Gulls wings, Birds wings

This is a descriptive name to a condition where the edges of the substrate fall away at either edge, as shown below. This is an undesirable parameter within the film. It can lead to hot rewind rolls on the edges and possible wrinkling.

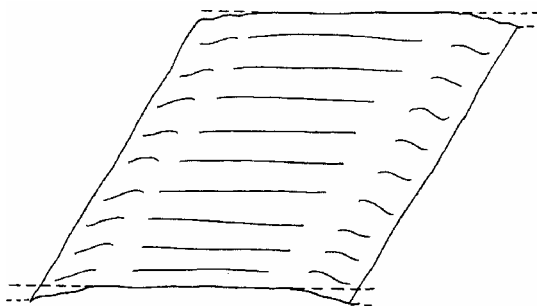


Figure 8.3 - 20 Sketch Showing Gull Wing Defect

To overcome the problem, while winding the material through the machine, the rollers may have to be taped with small amounts of tape in the location of the edges of the film.

Chicken wire pattern

This is the description of a pattern on a roll of film that has a surface that is partially collapsed area that resembles "diamond shaped marks" or "cells of a honeycomb"

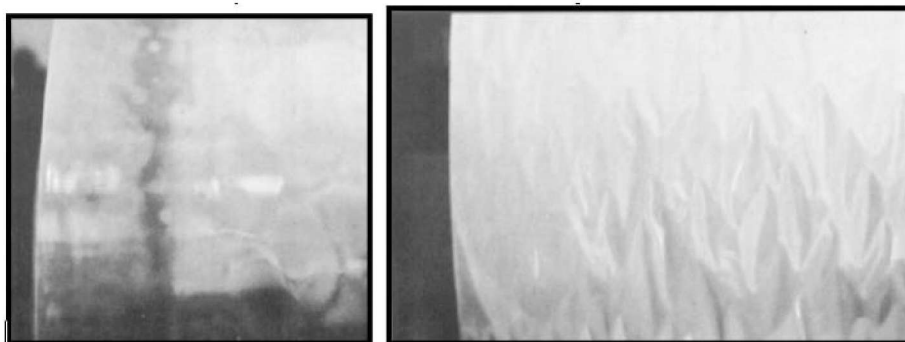


Figure 8.3 - 21 Photographs of Typical Chicken Wire Defect Pattern

Contamination

If there is some contamination of the substrate, such as oil, the downstream processes will be effected and the roll maybe deemed scrap depending upon the contamination.

The source of contamination should be investigated and overcome.

Surface Friction Level

The surface friction level on a film can be measured using special fluids. The fluid is manufactured to indicate the "wettability" of the surface of the substrate. Test results are based on how the various surface tension solutions react when applied to a non absorbent surface. If a given solution wets the surface, its dyne level is lower than the substrate's surface friction. If it forms beads, its dyne level exceeds that of the substrate. There is a standard test method (ASTM D-2578)

There is a minimum dyne level required to achieve good adhesion between the evaporated aluminium and the substrate surface.

The minimum level will vary as specified by different film manufacturers. The minimum for all plastic substrates should be > 38 dyne. A quick method of determining the level is to use special dyne level pens. The pen appears to have a felt "nib", which contains a solution, which when rubbed on the film, will indicate if the dyne level is above the required level. Indication of the dyne level will be that the coloured solution will stay on the film surface. If the level is lower than 38 dyne, the solution will dissipate, i.e., bead.

Pens are available from a number of manufacturers.

Dyne level testing kits can also be bought from these manufacturers to find the exact dyne level of the treatment.

Incorrect Treatment Side

Generally with most plastic films, such as BOPP, OPP and CPP, they are either flame or corona treated on one side. It is important that this side is the one that is opposite the coating source when the substrate passes through the evaporation zone.

To identify the treated side, normally film suppliers will identify the treated side with a label or piece of paper stating "inside or outside treated".

Failure to coat the treated side will be later identified by poor adhesion.

Excessive Gauge Bands

When the film is manufactured, there are variations in the thickness across its width. The thickness is generally called the gauge.

When there are lanes of thicker gauge within the overall width of the substrate, they are described as gauge bands.

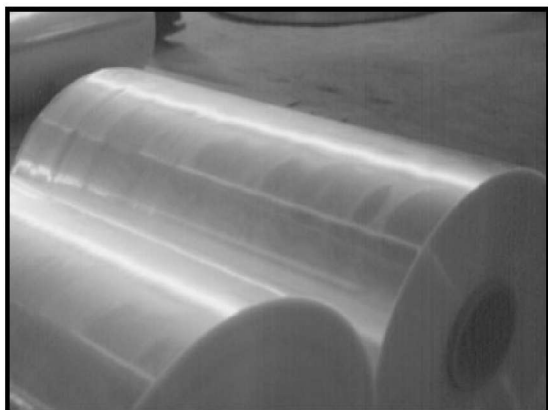


Figure 8.3 - 22 Photograph of a Roll Showing Heavy Gauge Banding

Gauge bands can be detected sometimes when hardness profiles are made of the rolls. However, some may be only present down within a roll, for example, when a joint is present.

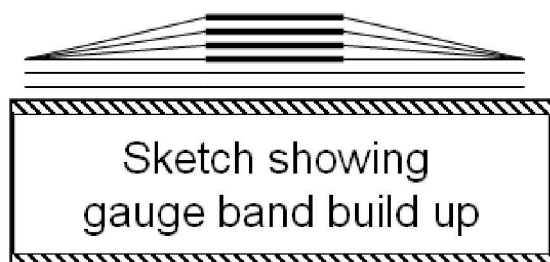


Figure 8.3 - 23 Gauge Band Build Up of Layers Defect

Excessive gauge variations can cause uncontrollable wrinkles when running the substrate through the winding mechanism.

As previously mentioned a roll of material when presented to the machine may hide the defect visually to the eye. When the roll has been processed, the defect will be enhanced by the lack of air.

Creases

Any creases, either soft or hard in the base roll cannot be removed from the substrate by the machine. It is desirable that the roll does not have any creases present.

Soft creases can be pulled out of the substrate by pulling the web in a sideways manner. The substrate is acceptable for most applications.

Hard creases can not be pulled out as described above and will cause the substrate to be rejected.

Wedge Profile

If the unwind roll substrate has been manufactured in a manner, which allows the thickness/gauge to be in a wedge profile from one edge to the other. This profile will cause wrinkles to be produced as the roll is unwound. The figure below shows a cross section of a roll, indicating an excessive wedge profile.

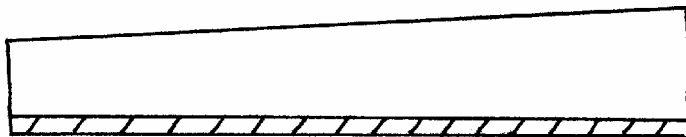


Figure 8.3 - 24 Roll Shape- Wedge Defect

Generally the winding system will have a spreader roller located soon after the unwind roll to remove any wrinkles from the unwind roll. However, if the wedge profile is excessive, the wrinkles will not be removed completely.

A quick method of checking rolls of substrate for this defect, the circumference of the roll can be checked on both edges using a diameter tape ruler.

Excessive Splices

Most suppliers of rolls of substrate will have an in-house quality requirement for maximum number of splices. Excessive numbers of splices are likely to cause wrinkles, as different base roll material has been used.

At low processing speeds, this may not be a problem, but at the high metallizing speeds, they will cause wrinkles and tension control problems. Also, every splice is a potential web break situation, see next item.

Poor Splices

A major problem with any machine processing substrates in a web form is web breaks. One of the major sources of web breaks is poor splicing joints.

The manufacture of the splices is generally to a standard procedure, written by the substrate manufacturer. A simple guide follows;

- The two pieces of substrate should either be butted together or overlapped.
- There should be no loose flapping material.
- There should be no adhesive material left face up that can stick to another layer of the roll or a roller within the mechanism.
- The joint should be flat and free from creases

Buckles

Buckles are closely related in both mechanical and in appearance to "Starring", but tend to occur over a few layers rather than the whole roll.

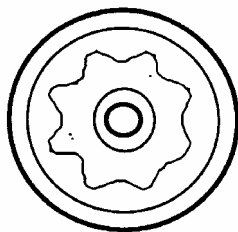


Figure 8.3 - 25 Buckle Defect in Roll

Tin canning

This defect has been also called "ribbing" or corduroy

On a base roll of substrate, the defect may not be seen on the outer surface, as the defect causing the problem can be hidden by air entrapment within the roll. However, once the roll is being unwound in a vacuum environment, the lines can be seen.

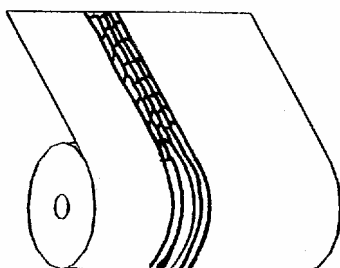


Figure 8.3 - 26 Unwind Roll Showing "Tin Can" Pattern Defect

The defect can be described as light or heavy. The light wrinkles can be pulled out by applying tension across the web in a cross machine direction.

The source of the defect can be;

Problem	Description	Suggested Remedy
Core deflection	The walls of the core are thin and are pressed away from the reel shaft when the shaft is engaged into the core	Use cores with a thicker wall
Core debris	Debris such as tape maybe on the core.	Remove the debris
Uneven core surface	Check the core surface for uneven areas	Replace the core
Excessive tension	The tension levels used to wind the substrate were too high	

Table 8.3 - 2 Tin Canning Effect Problem

The defect will be seen both at the unwind roll and the rewind roll as it is processed through the machine.

Wandering

The same condition as feathering. This term is usually used in the paper metallizing industry.

Ballooning

This is a term that relates to when a roll has a soft centre section and either side is hard. Air can be entrapped within the roll that leads to the middle of the roll appearing to inflate when the roll is placed inside a vacuum environment. This defect is very rare.

Fisheye

This is a term used to describe a coating defect. The defect is caused by a foreign substance on a coater backing roller on the coating machine. The fault is shown below.

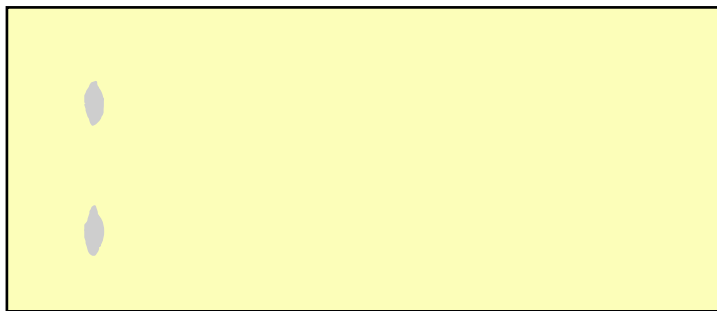


Figure 8.3 - 27 Substrate Showing "Fish Eye" Damage

Base Paper Line

This is a line in the base material and looks like a continuous scratch in the machine direction (MD). It is created at the paper manufacturer's plant where the clay coating on the paper's surface has been scratched off.

It can be detected by using a bright light shone across the substrate.

Blade Streak

This defect is similar to a base paper line, but is generally narrower. It is caused by debris being located under the coating blade on the coating machine.

Lacquer Line

Same as "Blade streak"

Calendar Cut

Paper manufacturer fault. Appears as diagonal wrinkles of less than 150mm in length.

Clay Skip

Paper manufacturer fault. There is an absence of clay coating on the paper and it is normally seen as a blotchy pattern. The coating medium will not adhere to this area of the substrate.

Die Lines

This defect is usually only found in plastic films that are manufactured by a casting process on a drum. The defect can be described as a continuous scratch, but with raised edges.



Figure 8.3 - 28 Substrate Showing "Die Lines"

The reason for this fault is that the die has become contaminated.

Gels

This is a description given to a cast plastic defect, when the polymer granules are not melted fully and pass through the extruder to be located in the substrate surface.

Ridges

This is a term used in the paper industry to describe a defect, the same as that as described as gauge bands in film. The thickness (calendar) of the substrate is thicker in one area than others.

This defect will therefore cause ridges to be formed where the substrate is thicker. If excessive, ridges will be seen.

Flat Spots

This defect describes an area on a roll of substrate which is flat, while the rest of the circumference is round. This is generally caused by the storage of the roll on a flat surface. Care should be taken to store the roll of substrate in a suitable manner.

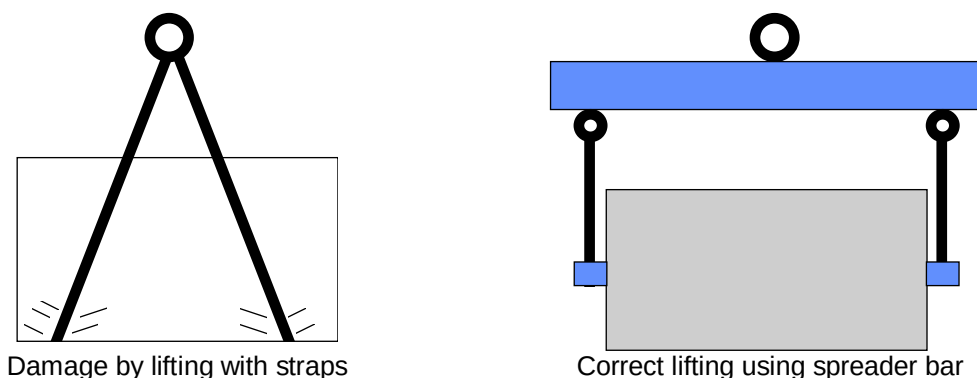


Figure 8.3 - 29 Correct & Incorrect Lifting Of Rolls

This fault can also be caused by using lifting straps to pick the roll up. See the previous drawing to explain the best method of lifting the rolls.

Rewind Roll Problems

Blocking

As previously mentioned, a blocked roll is one where the layers do not separate easily on unwinding. A characteristic of this defect is retarded unwinding tangent as shown below. Often, the tangent will move between two positions in a saw-tooth fashion and is sometimes accompanied by a ripping or bellowing noise (like a sail boat sail in wind).

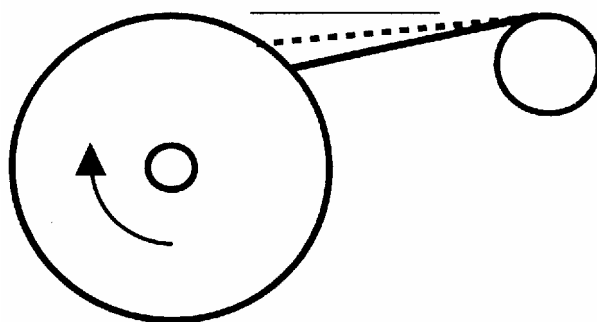


Figure 8.3 - 30 Blocking of Substrate on Rewind Roll

The reasons for the "blocking" of the rewind roll are as follows.

Problem	Description	Suggested Remedy
Mechanical Interlocking	If the surfaces of the layers of substrate are rough or fibrous, the layers can interlock and stick together. This can be the case when the roll is wound hard and hard wrinkles are formed.	Operate the machine's winding system at a lower tension setting. Ensure that it is not too low to create other problems described later.

Problem	Description	Suggested Remedy
Adhesion	If there is a adhesive action due to coatings, treatments etc., the layers may stick together. This case when the roll is too hot on removal from the machine.	See the section of "High temperature rolls"
Backside Treatment	If the rear (non coated side) of the substrate has been accidentally treated to a high level > 30 dyne, it is likely to stick to the layer above it after processing.	Check the substrate for high surface tension levels on the non coated side before the unwind roll is placed in the machine.

Table 8.3 - 3 Blocking Defects

It has been found in cases of severe blocking that the substrate will rip and cause a web-break situation.

Roll Contraction

Some substrates will contract on the rewind roll after being processed through the winding mechanism. This is generally restricted to very extensible substrates like CPP and LPDE.

If excessive tension is applied to the substrate as it passes through the winding mechanism and it is relaxed on the rewind roll, the layers will contract. This is especially true, if too much draw tension is used as the substrate is wound through the evaporation zone.

To overcome this problem, ensure that the tension in the substrate is kept to a minimum, which allows good rolls to be formed without stretching the substrate.

Width Reduction

Once again, if the substrate is stretched through the winding mechanism, it can become narrower in width. As it has already been stated, extensible substrates will result in contracting on the rewind roll, where less extensible substrates will be deformed permanently by a reduction in width.

To overcome this problem, the tensioning of the substrate must be reduced to an acceptable level to allow the building of a good roll.

Cracking of Coatings

With some coated substrates, over tensioning of the them, will lead to physical damage to the coating. This can be the case in stamping foils, where the damage will not be seen until the foil has been stamped onto the final product.

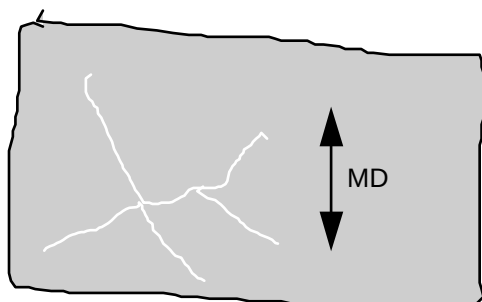


Figure 8.3 - 31 Cracking Of Coatings on Substrate

Care must be taken not to place too much stress on the coating by applying too much tension.

Telescoping

The term telescoping is used to describe the condition when a roll of material moves when it is being wound onto the rewind roll. A profile is shown below.



Figure 8.3 - 32 Photograph Showing Rewind Roll That Has Telescoped

The profile shows a telescoping action that the substrate's edge profile has moved a large distance over a few layers. Obviously there was interlayer slippage.

The problem is generally caused on the rewind section of the machine by a lack of tension at the beginning of the roll formation. The tension maybe increased as the roll becomes larger and then the roll moves. This is a non desirable defect and will cause problems downstream on further equipment. It can also lead to defects in the barrier properties of the metallized layer.

To overcome the problem, the tension setting and taper function should be used to form a firm, but not hard finished roll.

Starring of Edge

If the tension setting for the rewind roll has been set low and the taper setting was set to increase the tension throughout the roll, it is possible that this defect will be seen.

It is not a serious defect in most cases. It is a indication that the roll is becoming hard.

The tension settings should be modified to stop this defect occurring. Set the starting rewind tension a little higher and reduce the taper setting.

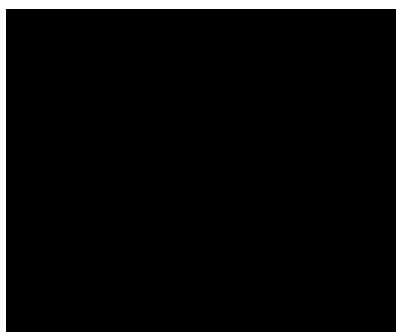


Figure 8.3 - 33 Star Pattern on Roll Of Substrate

Dishing of Roll Edge

As described in the unwind section of this booklet, the term dishing refers to the roll profile below

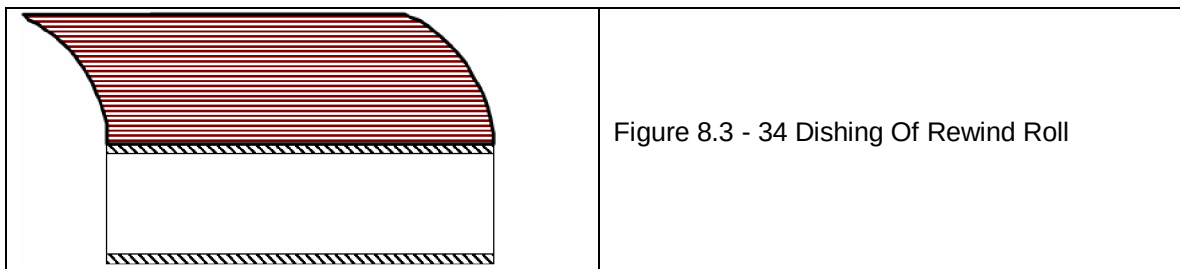


Figure 8.3 - 34 Dishing Of Rewind Roll

The reasons for the profile being generated could be;

- Core movement sideways gradually on the shaft
- Rewind core belled or crushed
- Interlayer slipping

The movement will be in one direction

Check the above for the source of the fault

High Temperature Rolls

High temperature rewind rolls can be a major defect with some rolls being scrapped.

In the case of stamping foils, where heat and force are used to transfer the coating off the carrier substrate onto the product, this defect is disastrous. The force of the layers wound together and the heat contained within the roll generate the same effect as the stamping machine. Therefore all the coating can be transferred.

In the case of heat sealable plastic films, such as BOPP and Nylon, excessive heat will ensure that the layers stick together and give a "blocking" defect.

To ensure that the substrate is metallized at the lowest possible temperature possible without physical damage, the following rules should be followed;

- Store the substrate in an environmental controlled room before being metallized. If the roll being presented to the machine is already hot/warm, it will get even warmer through the process.
- Ensure that the cooled rollers/drum/s are being cooled to the correct process temperature.
- Ensure that the cooled roller/drum surface is clean and smooth, i.e., no debris upon it.
- Ensure that there is no debris that can fall in behind the substrate while it is being coated.
- Operate the coating sources at the lowest required temperature level to reduce the radiant heat load from them.
- Ensure that the substrate is held onto the cooling rollers/drum with the required tension, without stretching it.
- Ensure that the substrate is spread on to the cooling rollers/drum to provide good surface contact.

Tin Can Effect

This defect has been also called "ribbing"

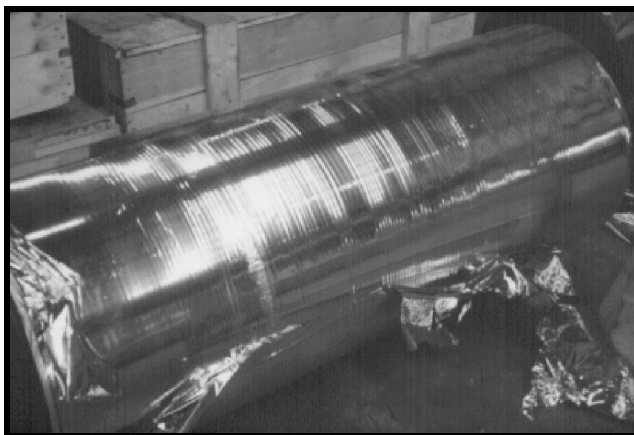


Figure 8.3 - 35 "Tin can" Effect on Rewind Roll

The defect can be described as light or heavy. The light wrinkles can be pulled out by applying tension across the web in a cross machine direction.

The source of the defect can be;

Problem	Description	Suggested Remedy
Core deflection	The walls of the core are thin and are pressed away from the reel shaft when the shaft is engaged into the core	Use cores with a thicker wall
Core debris	Debris such as tape maybe on the core.	Remove the debris
Uneven core surface	Check the core surface for uneven areas	Replace the core
Excessive tension	The tension levels used to wind the substrate were too high	Reduce the tension set-point
Base material on unwind roll	Check the base substrate on the unwind roll	

Table 8.3 - 4 Tin Canning – Rewind

On the unwind roll, the defect may not be seen on the outer surface, as the defect causing the problem can be hidden by air entrapment within the roll. However, once the roll is being unwound in a vacuum environment, the lines can be seen.

The defect can be seen both at the unwind and the rewind as it is processed through the machine, if it is on the base roll.

To overcome, check all of the above items; reduce the tension to a level that allows a good quality roll to be formed without any other defects.

Lipped Edge

This is a fault described previously as a "high edge" in the unwind section of this booklet.

It will not be created in the machine, but reducing the heat-load in the substrate will reduce the problem.

Wrinkles

Light wrinkles will be formed in the machine as the substrate is wound through the winding mechanism.

- Unwind

As the substrate is unwound from the roll, light wrinkles can be expected to be seen. One of the first rollers in a modern machine will be a spreader roller. Its purpose is to remove any light wrinkles that are present.

- Evaporation Area

It is desired that wrinkles are not present as the substrate passes through the evaporation zone. If they occur, there will be uneven coating by the aluminium and possible heat damage. Therefore it is usual that a spreader roller is used prior to the drum to spread the substrate onto it.

- Post Draw roller/s

As the substrate passes from the draw section of the machine, it will relax and wrinkles will appear. These are removed by the spreader above the light.

- Rewind/layarm area

The layarm assembly contains a spreader roller. Its purpose is to spread the substrate as it reaches the rewind roll.

Why wrinkles are formed is discussed under the section titled "creases"

Off Set Movement

If a roll of substrate moves over in the machine as it is wound through the winding mechanism, misalignment may be suspected. The alignment for all the rollers should be checked. Another situation that has been found to cause this type of fault is when a finger type spreader roller has not been manufactured correctly.



Figure 8.3 - 36 Sketch of Spreader Roller That Can Cause Offset Defect

Soft Rolls

Soft rewind rolls usually produce problems with telescoping, dishing, troughing flat spots (storage) defects as described in other sections of this booklet.

Possible reasons could be;

Problem	Description	Suggested Remedy
Rewind Tension too low	The tension set-point is too low for the substrate being processed	Increase tension set-point settings
Tension Transducer calibration	If the calibration of the load cell system is incorrect, reduced tension may be applied.	Check calibration of transducers
Low Tension range selected	Low tension range selected when user thinks high range has been selected.	Check tension range is correct

Problem	Description	Suggested Remedy
Tension roller isolation	If one of the rollers between the rewind load cell roller and the rewind roll is mechanically binding and allowing tension isolation	Check all rollers to be free running, Free up as required
Low unwind tension	The unwind roll tension is too low and the substrate is being pulled off the roll.	Increase the unwind tension setting

Table 8.3 - 5 Soft Roll Defects

Hard Rolls

The target hardness of the roll will be generally the same as the roll being presented to the unwind or slightly softer.

The rolls can be tested using the Rhometer to allow recordings for quality control and process data.

Hard rolls will lead to potential problems such as;

- Hard creases
- Blocking of layers
- Transfer of coatings
- Substrate damage
- Potential web breaks.

The general reason for hard rolls could be.

Problem	Description	Suggested Remedy
Rewind Tension too high	The tension set-point is too high for the substrate being processed	Reduce settings
Tension Transducer calibration	If the calibration of the load cell system is incorrect, excess tension may be applied.	Check calibration of transducers
High Tension range selected	High tension range selected when user thinks low range has been selected.	Check tension range is correct
Tension roller isolation	If one of the rollers between the rewind load cell roller and the rewind roll is mechanically binding and allowing tension isolation	Check all rollers to be free running, Free up as required
Contraction of substrate on rewind roll	Substrate shrinks, due to heat contraction	Check for reasons of excessive roll temperature.
Excess Draw Tension	Stretching of substrate around evaporation area	Reduce draw levels
Unwind Tension High	The unwind section of the machine is applying excessive tension to the substrate.	Reduce the tension setting to the unwind section of the machine.
Layarm Laying on with excess pressure	The layarm roller presses onto the rewind roll with excessive pressure causing compression of the roll. Can also sometimes lead to inverse "high edge" as described previously	Check that the layarm is operating correctly in the layon mode

Table 8.3 - 6 Hard Roll Defects

Horned Edge

This is another term that describes the defect previously described as a lipped edge or high edge.

Stick Outs

When a few layers protrude from the side wall of the roll, they can be termed "stick outs". This defect is also called a "shiners" or "Stickouts"

They are usually the result in tension or line speed changes while the roll is being formed.

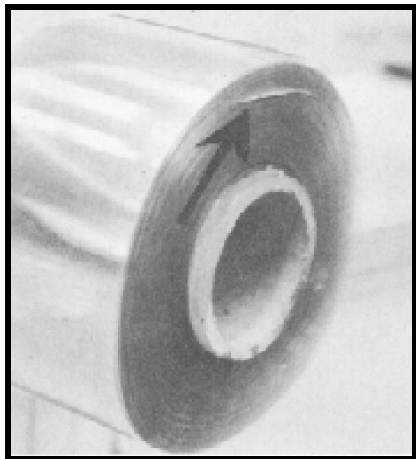


Figure 8.3 - 37 Rewind Roll Showing Substrate "Stickouts"

This is not a serious defect, but should be avoided, as it will cause problems with other process equipment.

Increases in tension should be controlled using taper tension.

Increases in the line speed set-point should be gradual.

Curvature/Skew - measurement

Curvature of the roll will produce wrinkles and creases as shown below. Curvature occurs generally in BOPP films where one side of the film is longer than the other. To check, remove 20 foot of film from the roll, place the two ends together and fold the web in two. The edges should be parallel. A slight difference is acceptable, but if the difference is more than 1/2", problems will occur.

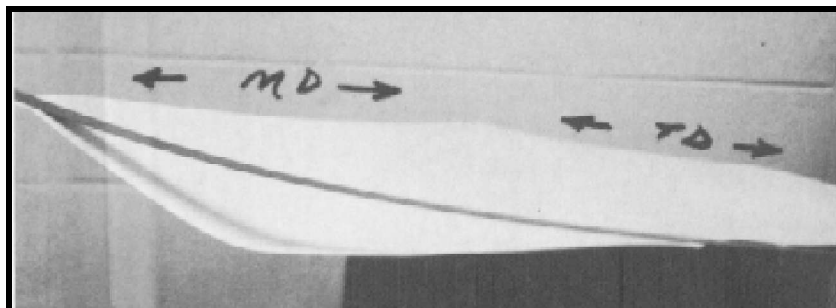


Figure 8.3 - 38 How to Measure the Skew

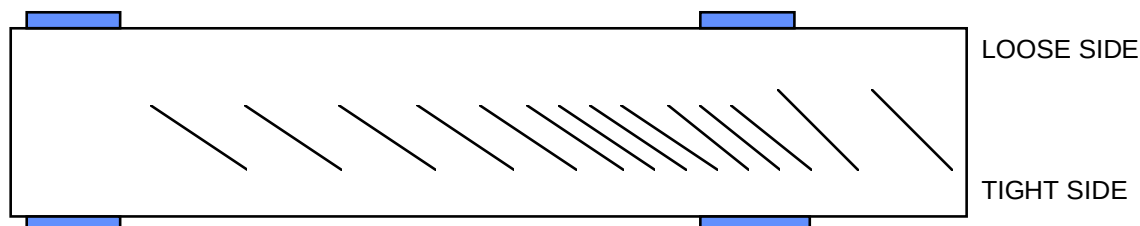


Figure 8.3 - 39 Diagram of Substrate Showing Loose Edge Defect

With one side of the roll tight and the other loose, all the tension in the substrate will be located in the tighter side. Diagonal creases and wrinkles will be seen within the machine

Processing Winding Problems

Creases - Wrinkles

Creasing can be described where the substrate is folded or pinched and therefore the rewind roll is then uneven. The creasing is normally formed by wrinkles in the machine direction (MD).

Alternatively they can be diagonal, but not normally across the width the substrate.

- Pipe Wrinkles

Problem	Description	Suggested Remedy
Roller Misalignment ①	Diagonal wrinkles start between certain rollers	Check for roller alignment at entry and exit of area of source
Base Roll Defect ②	Already in base roll	Check the unwind roll for sources of the defect, including blocking, "piping" etc.
Baggy Film ③	Film profile not flat. Wrinkles start and stop within a short distance, Run diagonally	Check the roll flatness as described in a later section
Poor nipping action ④	Surging wrinkles generally in MD direction	Check by removing nip roller while running. Check nip roller operation for parallel contact. Check the roller for wear.
Bow roller Setting ⑤	Wrinkles tend to be slightly diagonal. Generally will point to tighter part of substrate. Will be in middle of roll or both edges. Single running wrinkle that is present at all times will indicate bow type spreader is 180 degrees from correct position	Check the roller for correct alignment and apex position
Roller Drag ⑥	Wrinkles surge in and out after a roller	Check roller to be free running
Roller Deflection ⑦	Amount of wrinkles can be increased and decreased if the tension setting is increased or decreased	Check that the minimum roll width is not being exceeded. Has a roller been substituted for a smaller roller?
Winding mechanism twist ⑧	Wrinkles will appear diagonally between a number of rollers	Check the alignment as per the manufacturer's instructions.
Gauge Bands ⑨	Gauge bands will also cause diagonal wrinkles to be generated as the rewind roll is formed. The wrinkling will become worse as the diameter builds.	Check for hard spots on the roll before processing. Check the thickness of the film across the substrate.
Layarm setting ① ⑨	If the layarm assembly with the rollers mounted upon it is too far away from the rewind roll, wrinkles can be induced by the tension in the substrate.	Ensure that the assembly is tracking the rewind roll at the desired gap, i.e. less than 20mm (3/4") in layoff mode.

Table 8.3 - 7 Pipe Wrinkles In Web

Pipe creases are soft and will pull out under tension. When pipe creasing occurs, it should be determined if the machine is inducing it, or if it is in the base roll (unwind).

- Pinch Creasing

These are small hard creases that will not pull out under tension on the slitter. It can be caused by the following and the above reasons (worst cases);

Problem	Description	Suggested Remedy
Base material fault ① ①	Wrinkles/creases in base roll at unwind position.	
Winding too hard ① ②	Wrinkles can be pulled into the substrate that cause creases when pulled over rollers within the mechanism.	Reducing the tension will normally overcome this problem
Inability to remove wrinkles ① ③	If the film is very "baggy", the spreader rollers in the machine may not be able to remove the wrinkles	Try to increase the circumference of the lead in roller with a wrap of tape on the area where the film is "baggy"
Heat damage ① ④	If the radiant heat to the plastic film surface is excessive. The substrate will tend to lift of the surface of the drum as the substrate shrinks.	Check the operation of the chiller etc. Reduce the heat source.

Table 8.3 - 8 Pinch Creases in Web

Examples of Wrinkle Patterns

①

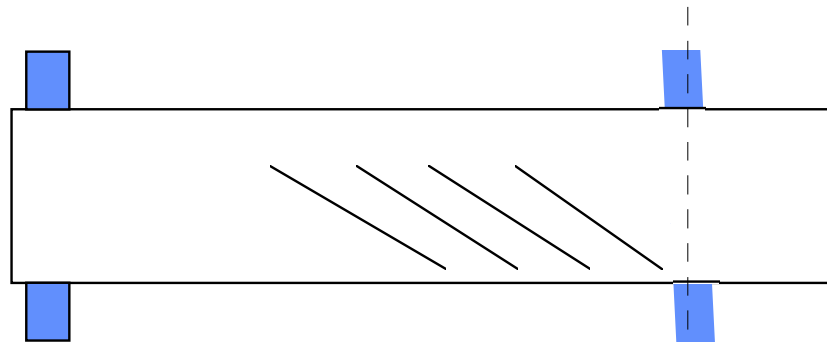


Figure 8.3 - 40 Roller Misalignment

②

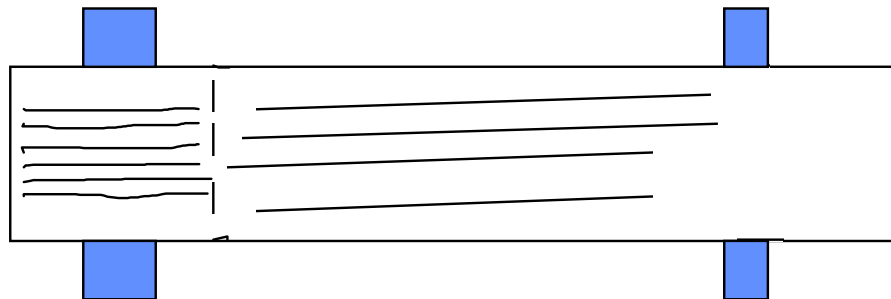


Figure 8.3 - 41 Base Roll Defect

③

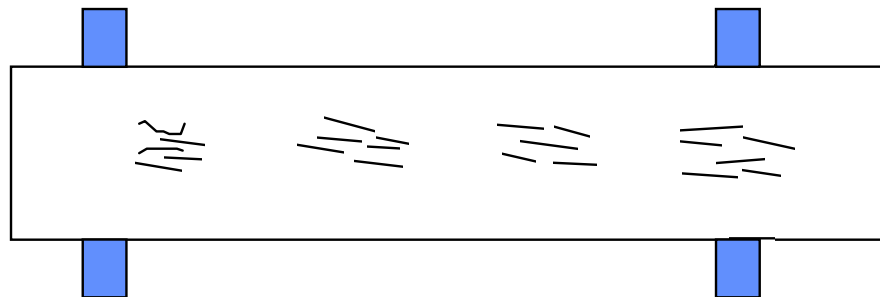


Figure 8.3 - 42 Baggy Film

④

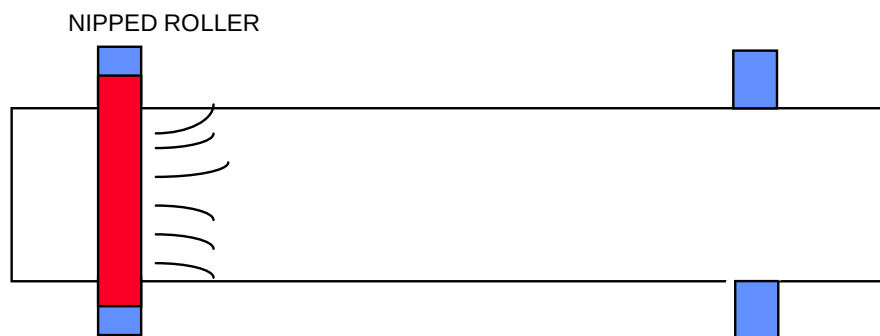


Figure 8.3 - 43 Poor Nip Action - Drum Machines Only

⑤

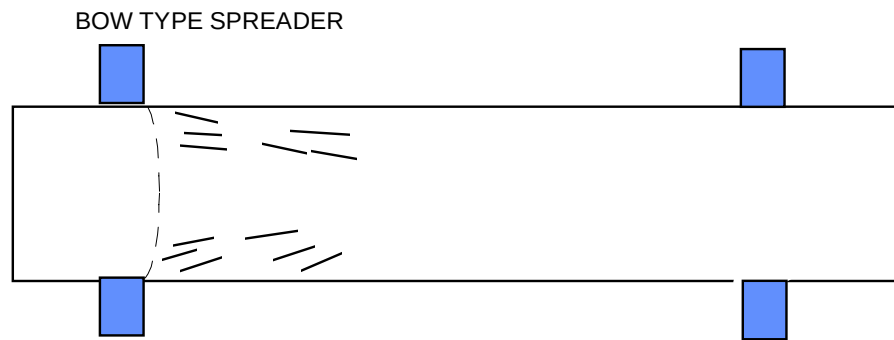


Figure 8.3 - 44 Bow Roller Setting Incorrect

⑥

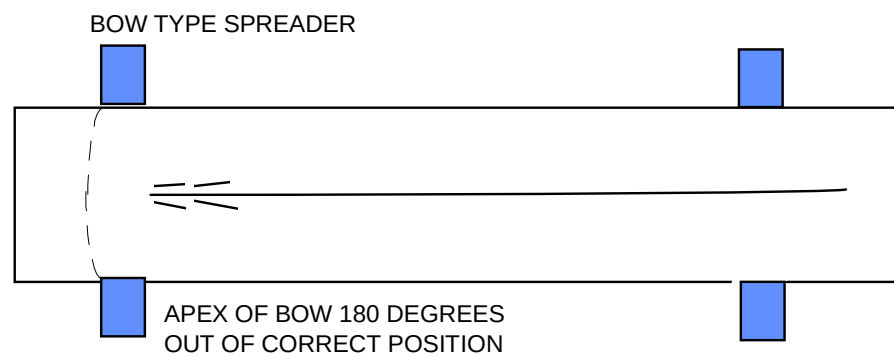


Figure 8.3 - 45 Bow Roller Setting Incorrect

⑦

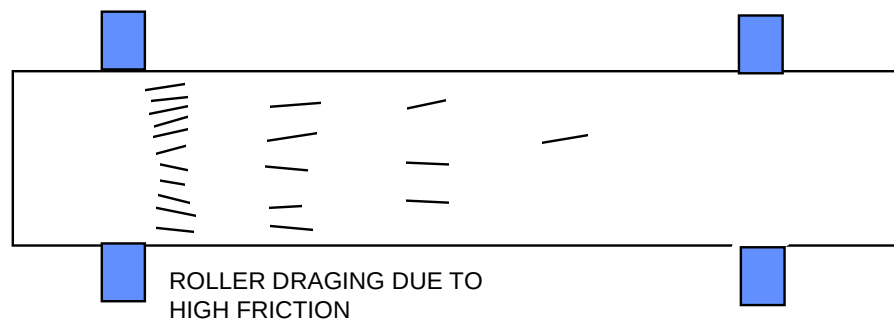


Figure 8.3 - 46 Roller Drag

⑧

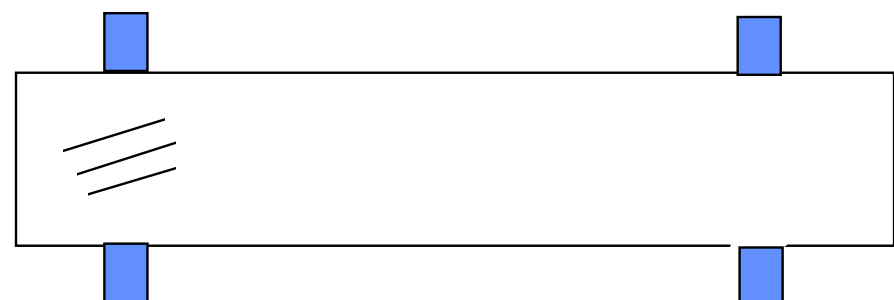


Figure 8.3 - 47 Roller Deflection

⑨

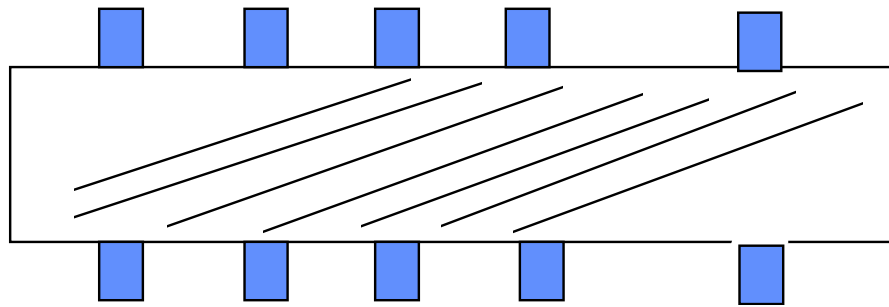


Figure 8.3 - 48 Winding Mechanism Twist

① ①

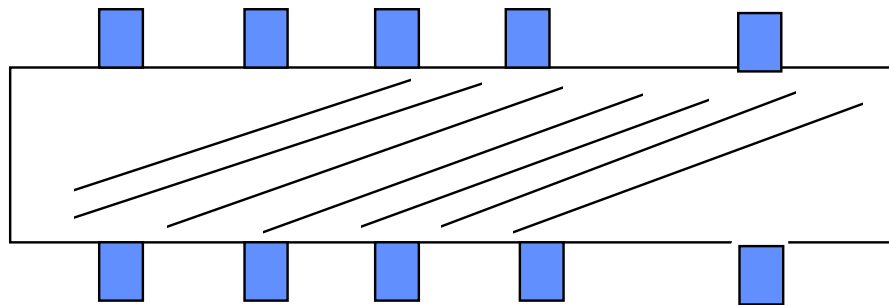


Figure 8.3 - 49 Gauge Bands on Unwind

① ①

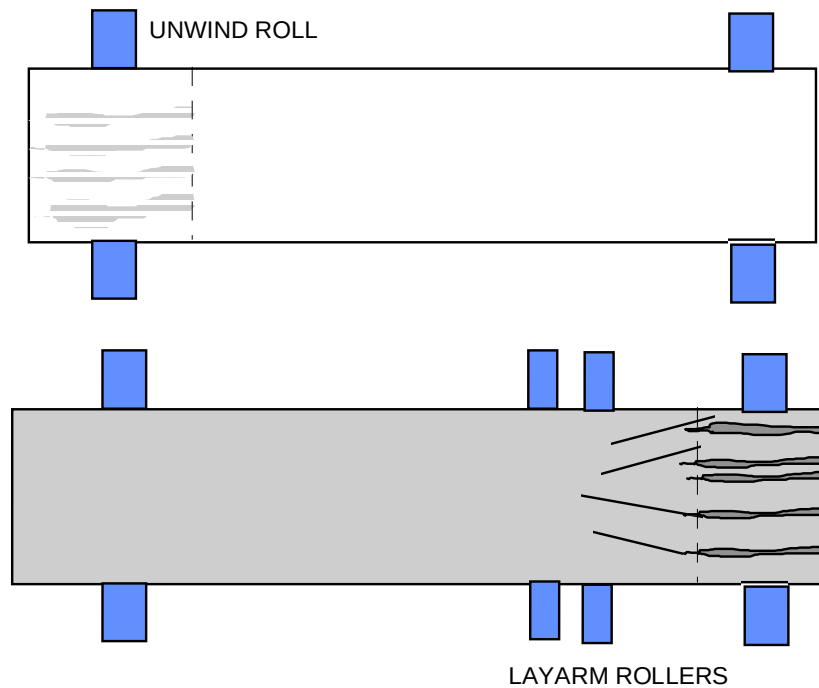


Figure 8.3 - 50 Layarm Setting Too Far Away

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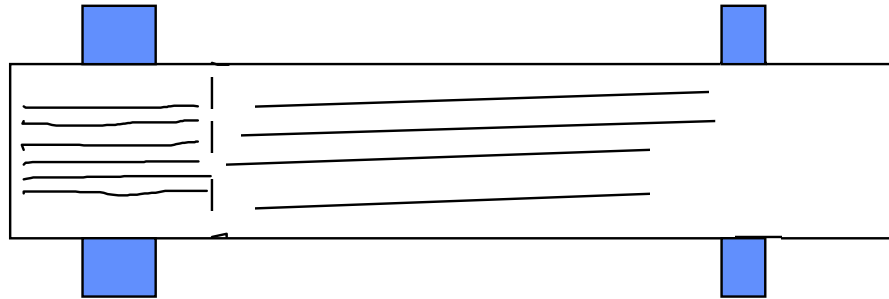


Figure 8.3 - 51 Base Roll Defect

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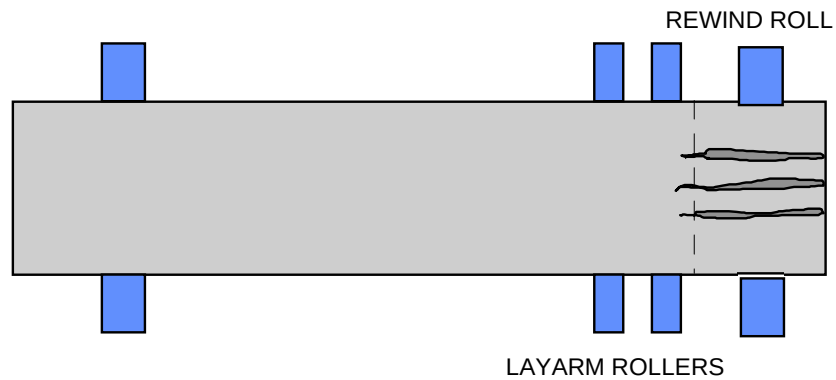


Figure 8.3 - 52 Winding Rewind Roll with Too Much Tension

① ④

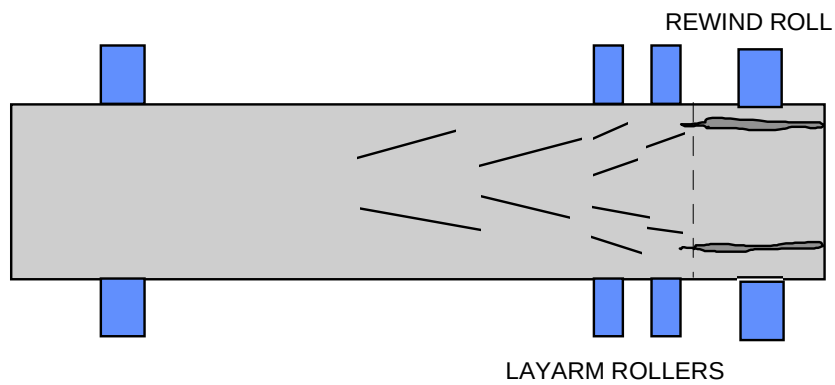


Figure 8.3 - 53 Inability of Spreaders to Remove Excessive Wrinkles

① ⑤

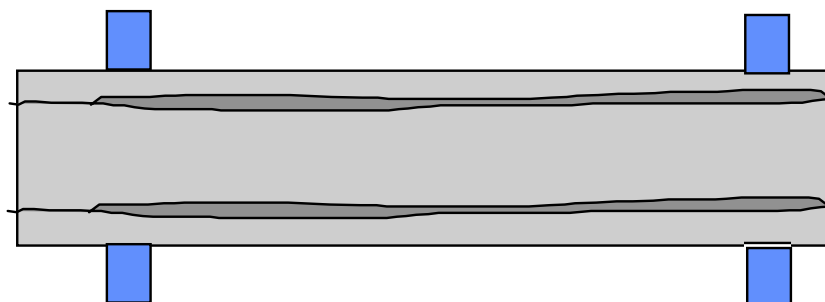


Figure 8.3 - 54 Heat Damage Caused By Radiant Heat

Screwing

This term describes a defect when one side of the rewind roll is wound tighter than the other.

Another example is when the centre is wound tighter or looser than the centre.

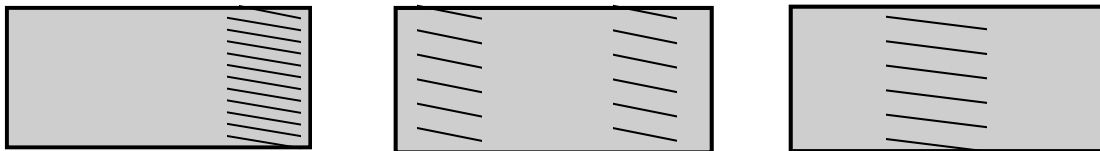


Figure 8.3 - 55 Roll Showing Screwing patterns

The reason this type of defect occurs can be;

- Layarm roller/assembly out of alignment
- Layarm roller laying on with high level of force
- Very bad gauge bands within film
- Incorrect bow roller spreading action producing stretching of substrate.

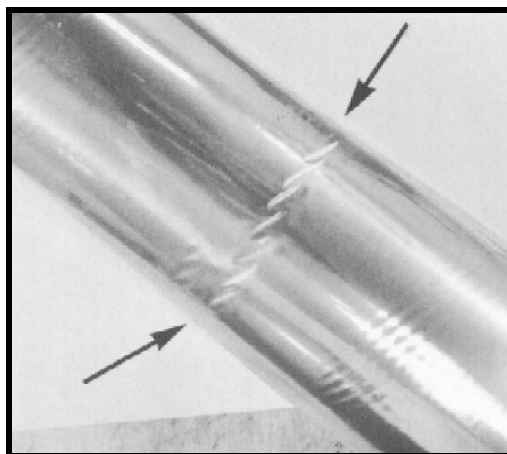


Figure 8.3 - 56 Photograph of Screwing Effect on Film

Roll Too Soft -Troughing

If a roll is wound at a low tension, there is potentially a problem created when the chamber is vented. The air will enter the roll of substrate and cause a condition called "Troughing", as shown in the profile below.



Figure 8.3 - 57 Roll of Substrate Indicating "Troughing" Problem

The solution is to wind the roll with the correct amount of tension and use the taper control to ensure that a firm roll is produced.

If the roll of material is also hot, the hard creases can block and therefore produce a scrap roll.

Another reason that this defect can be produced is that a poor joint is present in the roll that produces a "lump"

Flapping Edges

This name also describes the same defect as gulls wings etc. this defect can be formed in the winding mechanism if excessive tension is applied to the substrate while it passes over a bow type spreader roller that is inverted.

Curl

This term is used to describe a defect when the edges of a paper based substrate curls within itself due to excessive moisture removal within the machine or on a previous process. If within the metallizer, it will usually accompanied by dis-colourment.

Outgassing

This is a term used to describe the condition when the vacuum level in the chamber is affected by gas molecules being given off by a item. In the case of rolls of substrate, the substrate is the item. All rolls of substrate will "outgas" to certain level depending upon the gas molecules trapped within it's layers.

If a roll of substrate contains large amounts of gas molecules, for example moisture, problems can occur when the unwind roll is unwound.

This will be seen in a large rise in the pressure level in the winding zone of the machine. This in turn will affect the evaporation zone and it's pressure will rise. The overall effect of this out gassing will be less efficient metallizing and potential adhesion and colour problems.

If a roll is suspected to be out gassing, one possible means of overcoming the problem is to wind it through the metallizer once without actually coating it. Then wind it through once again and coat it.

Roller Marks

With some older machines, there is no idler roller after a spreader roller on the layarm assembly. This arrangement can lead to the defects shown below;

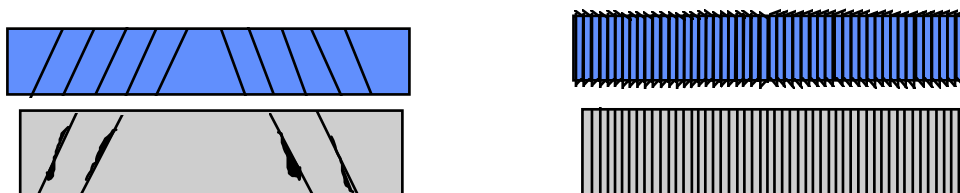


Figure 8.3 - 58 Roller Damage Induced Marks

The roll on the left has had excessive tension applied and the substrate has been deformed.

The roll on the right has been formed into a grooved pattern as the thin material drops into the grooves on the straight cut spreader roller.

Roping

Roping is a term used in the paper industry to describe a fault that looks as if a rope had been pressed on the surface of the substrate.



Figure 8.3 - 59 "Roping" Pattern on Paper Substrate

Roping can be present in a roll of paper as it is brought to the machine, but can not be seen without thorough examination.

The defect is caused by high tension at a previous piece of equipment. The defect is highlighted when the paper is dried as it passes through the evaporation zone.

If the substrate is wound under excessive tension levels within the winding mechanism, this fault can occur.

Another source of this defect can be the layarm spreader roller. If it is worn, there will be localized excessive tension applied on the areas that are not worn.

Web Breaks

The reasons that web breaks can occur are listed below;

Problem	Description	Suggested Remedy
Splices	Poorly manufactured splices are one of the major causes of web breaks	Splices should be manufactured with no trailing edges using quality tape.
Nicks	Nicks and damage to the edge of the roll can be a major problem	Check the edges of rolls at the unwind to ensure that the rolls are not damaged.
Heat	If the coating shutter is not located correctly or opens too early, excessive heat will be received by the substrate and if it is plastic, it will melt or burn	Ensure that the shutter is closed correctly. Do not open the shutter too early.
Tension	Excessive tension levels in the substrate will snap the substrate.	Check that the tension levels set for the substrate are not excessive.
Holes	Any holes in the layers of the substrate will cause the substrate to "snap out"	Check the substrate for obvious defects.
Mechanical	Any mechanical drags, etc. within the machine will place mechanical stress on the substrate.	Check the winding mechanism for free running rollers and mechanical damage.

Table 8.3 - 9 Web Breaks Defects

Web break are the source of major lost production output in both substrate and time. All precautions should be taken to guard against conditions that cause them.

Coating Process Defects

Tramlines

As the substrate is drawn tight against a coating drum through the evaporation zone, this defect can be generated. If the substrate enters the coating area with loose areas within its width, the film is pulled taught and the loose area of film are pulled flat. In the time between the substrate being loose and then flat, wrinkles can be pulled into the substrate. In these areas, the substrate will act as shown below, with a certain area being shadowed by the wrinkle.

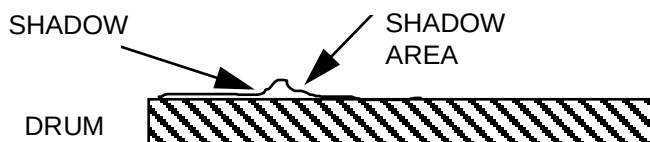


Figure 8.3 - 60 Tramline Formation In Substrate

Generally these wrinkles are not continuous and run in a diagonal direction as shown below.

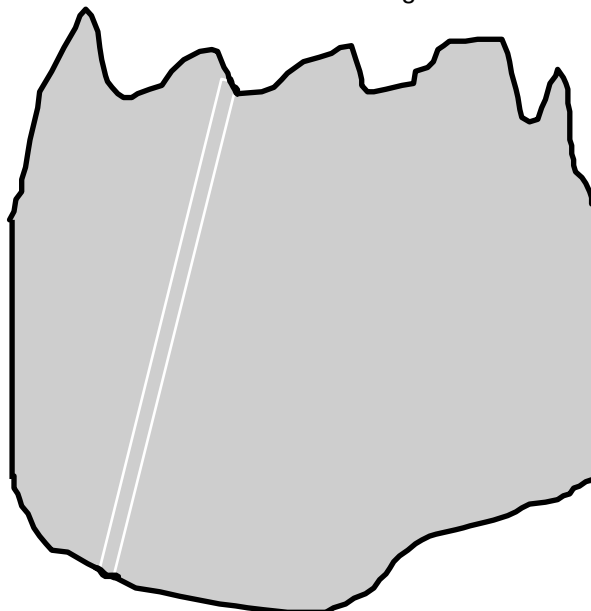


Figure 8.3 - 61 Tramline Defect Pattern on Substrate

The wrinkles, if any, can be seen through the source viewing window.

The reasons that the wrinkles are generated can be;

- Incorrect positioning of pre-drum bow type spreader roller apex.
- Gauge bands in substrate.
- Misalignment of rollers in the unwind area.
- Incorrect tensioning of substrate over drum.

To overcome the majority of these wrinkles, the draw level can be increased to stretch the substrate slightly. However with non extensible substrates, care must be taken not to stretch them.

If the problem occurs in all the rolls of substrates being produced, check the other items listed above.

Heat Tunnelling

Another defect that is similar to "Tramlines". However, in this situation, the wrinkles will be generally present across the full width of the substrate.

The reasons for this type of fault can be;

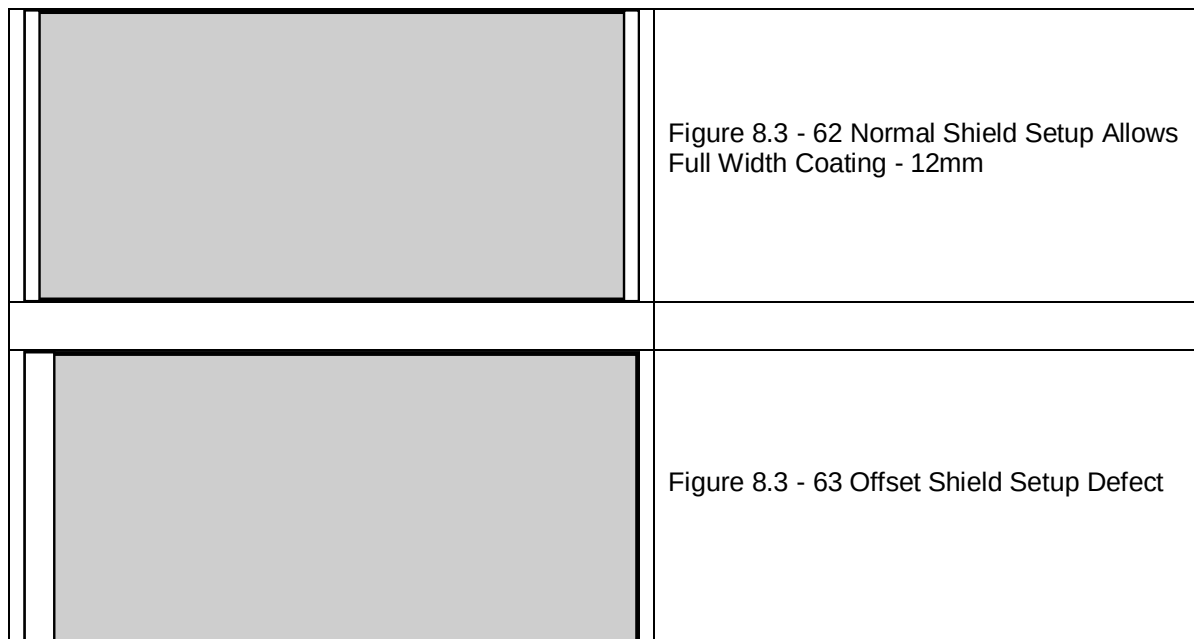
- No or very little tension created by the draw roller
- Cooled drum not at correct temperature
- Outgassing of substrate, which lifts the substrate from the drum's surface.

Check that the correct tension is being applied to the substrate and the chiller is at the required process temperature. Try a different substrate to check for an out gassing problem.

Wide Margins

Drum Machines Only:

If the drum shields have not been set correctly to cover only 6mm, wide uncoated edges can be created. This will cause the yield of finished product to fall, and may even cause the waste of the roll of material.



Care must be taken to ensure that the shields are set correctly.

Another reason for clear edges is telescoping or excessive feathering of the unwind roll.

Precautions should be taken with known problematic substrates or on-line modifications to the process parameters must be made.

Uneven Deposition due to uneven lacquer Coating

If the lacquer coating on paper substrate is not uniform, the inline monitor system will find it more difficult to control the deposition levels.

This can be detected by monitoring the output (wire feed) display of the different wire feeders and therefore evaporation levels to different parts of the substrate.

In most plants, the user does not have monitoring equipment to check the uniformity of the lacquer coating.

To check if this is the reason for erratic deposition readings, replace the roll with a different one from another process batch

Hot Rewind Roll

This subject is covered by the defect listed as "high temperature rolls"

Non Metallized lanes

If there are non metallized lanes passing the post coating viewing window, the cause is generally due to very poor quality substrate. There will be situations when the "baggy" substrate folds over on itself as it passes through the evaporation zone.

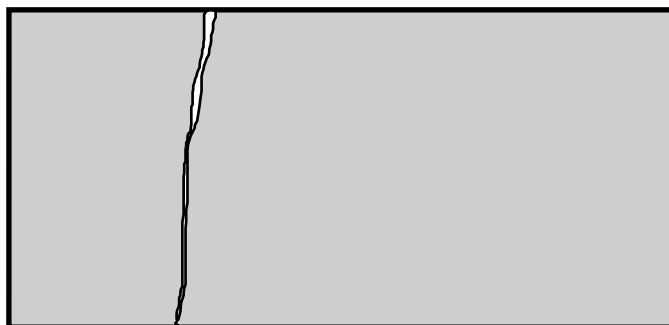


Figure 8.3 - 64 Substrate Showing Non Coated Area.

The word "baggy" is used to describe a length of substrate that has been manufactured with poor flatness.

Scratches

To determine if scratches are located below or in the coating medium, a light table (box) should be used.



Figure 8.3 - 65 Diagram Showing Differences Between Different Scratches In Metallized Substrate.

Wiping the coating medium off the surface of the substrate where the scratch is will highlight if the scratch is in the base substrate or the coating, or not. The scratch will no longer be present, if it is in the coating medium.

If scratches are found in the substrate before metallizing, the source will be somewhere prior to the evaporation zone.

- Small diagonal After Metallizing



Figure 8.3 - 66 Small Diagonal Scratches after Metallizing

Check the rollers that touch the coated side of the substrate for small defects on the outer edges.

This defect can also occur when the coated side of the substrate is wound over a bow type spreader roller.

- Small Diagonal before Metallizing



Figure 8.3 - 67 Small Diagonal Scratches before Metallizing

This type of defect has been seen on precoated substrates and is caused by slight movement of the cover on bow type spreader rollers.

- Winding mechanism deflection



Figure 8.3 - 68 Scratches Caused By Roller Deflection

If a roller in the mechanism deflects, it will cause wrinkles to be created which in turn can lead to the type of scratches shown above. Another reason for this type of scratch is the winding of very "baggy" substrate. As it passes through the winding mechanism, the coated surface may touch the zone gaps or monitor system probes, causing the coating to be scratched.

- Repeated pattern

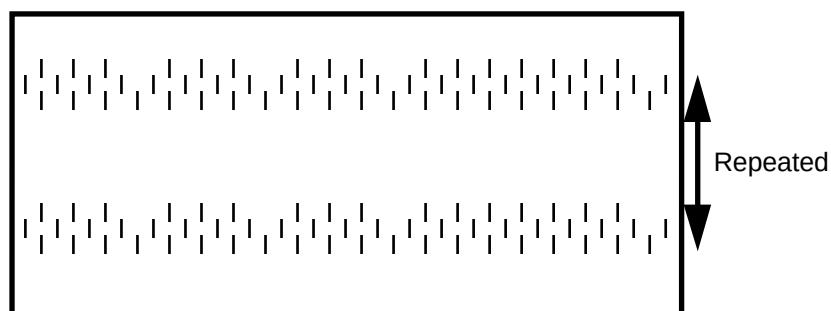


Figure 8.3 - 69 Repeated Scratch Defect

In the above situation, it can be seen that the scratching pattern is repeated. Check for vibration problems that may produce a resonant pattern when the substrate touches a certain part of the machine. The pattern will generally indicate the location of the touching medium by the distance between the scratches.

- Continuous Single



Figure 8.3 - 70 Continuous Scratch I Substrate

In the defect shown above, the normal cause is a piece of debris touching the substrate surface.

Check the area around the resistance system probes and the gap between the drum and the zone gaps for debris/coating medium build up.

- Speed Differences

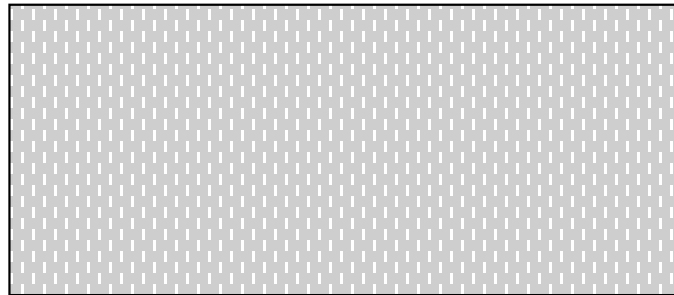


Figure 8.3 - 71 Scratch Pattern Caused By Roller Speed Differences

If there is a speed difference between the substrate and a roller that it is in contact with, there is a good chance that the above scratch pattern will be seen. Check the surface speed of all the rollers within the machine.

Coating the Process Drum

An undesirable effect of telescoping and feathering of the unwind rolls is the coating of the drum in the evaporation zone. This leads to the following;

- Contamination of the rollers
- Indentation marks (Operator)
- Aluminium flakes in the sides of the rolls
- Web breaks
- Torn edges - paper

Adjust the running parameters to stop the unwind roll feathering, or stop the process and remove a badly telescoped roll.

Another reason for this problem is the incorrect width setting of the drum shields. These should be corrected.

Stretch Lines on Film

If the tension applied to the substrate while it is being coated is excessive, stretch lanes can be produced. This excess tension will cause the film to form ridges where heat will be concentrated. This will cause the type of defect shown below.



Figure 8.3 - 72 Photographs Showing Examples of Stretch Lanes

Blue Tint

A grey or bluish tint to the coating, which has been seen to be caused by old or damaged evaporators.

Low Surface Tension of Substrate

The dyne level of the substrate can be negatively effected by the following situations;

Problem	Description	Suggested Remedy
Base film low tension levels.	The substrate surface treatment level will decrease over long time periods	Ensure the base film tension level meets the minimum specification
Substrate stored for excessive periods between production and metallizing	The substrate surface treatment level will decrease over long time periods	Ensure the substrate is used within a reasonable time period
Substrate too hot while processing		Ensure the chiller is operating at the required process temperature. The radiant heat being given by the evaporators should be minimized by correct operation of them. The substrate is held against the cooling rollers/drum by the draw rollers.

Table 8.3 - 10 Low Surface Tension on Substrate Surface

Debris in edge of Roll

Flakes of aluminium can be sometimes be found in the edges of the roll. These flakes are formed on the drum when it is coated with the aluminium. The surface breaks off in the winding zone and the flakes can stick to the substrate and become wound into the edges. This can lead to contamination of the roll and possible web breaks.

The drum should be protected using drum shields.

Alternatively, the drum should be taped and this tape should be removed on a regular basis.

The zoning shields that the are located around the drum should not be opened while coating the substrate, as this condition will lead to higher pressures in the evaporation zone and further contamination of the winding zone.

Brown Tint

Dis-colourment of the aluminium coating from a silver to a brown/gold colour. This dis-colourment can be caused by excess deposition with the burning of the substrate or a poor vacuum level in the evaporation zone.

This fault can be seen across the full width of the substrate or just in localized areas.

Ensure that the pressure level in the evaporation area is low enough to provide good metallizing conditions.

Low Adhesion levels

The adhesion of the aluminium coating medium to the carrier substrate is effected by the following factors;

Problem	Description	Suggested Remedy
Surface roughness	If the surface of the substrate is very smooth, such as glassine paper, the coating medium will not stick to the surface very well	Modify the surface by pre metallizing treatment, if possible.
Surface treatment levels	If the substrate has not had a surface treatment process applied to it, such as flame or corona treatment. The coating medium will not adhere to the surface very well	Check the surface treatment level of the substrate.
Coating medium contamination	If the coating medium is contaminated, it can effect the adhesion of itself to the substrate.	Check that the coating medium matches the specification of that purchased
Poor vacuum levels	If the vacuum level in the evaporation zone is not at the correct level, poor adhesion will take place	Ensure that the vacuum level in the evaporation zone is below 1×10^{-3} mbar (torr)

Table 8.3 - 11 Low Adhesion Problems

Some substrates such as BOPP have to be treated to a high dyne level by corona or flame treaters to allow for good adhesion. If there is no treatment or low levels, the coating medium will not stick to the substrate well.

The other factors are mentioned elsewhere in this manual.

Gold/green Tint

This defect in the colour of the coated substrate has been monitored when the incorrect grade of aluminium wire has been used for metallizing.

If the wire has not been chemically cleaned or contaminated will also cause this colour.

Blisters

This defect is caused by pockets of air trapped between the lacquer coat on paper and the paper itself. After metallizing, there will be a area, the size of the "blister" that will be uncoated.

This fault cannot be generally be seen at the metallizer and efforts must be made to determine its source in prior process equipment.

Pick Off

This is the term given to a defect when the coating medium is removed from the carrier substrate in a manner described by the term. The removal of the coating medium is not constant and can be one of many shapes. The reasons for "pick off" are usually;

Problem	Description	Suggested Remedy
Slight blocking of the layers	As the layers stick together and then part, the coating medium can be partly transferred from the carrier layer to the backside of the other layer	
Roller defects	Check any rollers that touch the coating for damage.	Remove damage from rollers
Roller drag	A roller on the metal side maybe rotating slowly with a mechanical force holding it back.	Check all rollers spin freely. maintain any that do not rotate freely
Slight scratching	Scratching from small foreign items may cause this defect "look a like"	Check for possible foreign objects in close proximity to the substrate.

Table 8.3 - 12 Coating Pick Off Defects

Poor Uniformity

When the term uniformity is used, we are describing the level of metal coating across the width of the substrate. Ideally, a perfect uniform layer is desired. However in the real world, a +/- 3% uniformity is the target. There are a number of factors that affect the uniformity of the coating;

Problem	Description	Suggested Remedy
Vacuum Level	The vacuum level in the evaporation zone should be in a region suitable for normal operation, i.e. $< 8 \times 10^{-4}$ mbar	Check the vacuum level in the evaporation zone is at the correct level.
Surface treatment level	Surface friction level should be above 40 dyne	Check the surface friction to be higher than 40 dyne using the appropriate test equipment.
Evaporator Age	As the evaporator becomes old, the evaporation pool becomes smaller, which in turn effects the uniformity.	Replace worn and old evaporators. See the booklet on evaporators for these conditions.
Calibration of monitor System	Calibration of the system should be checked on a regular basis.	Check the calibration of the monitor system as described in the maintenance manual or the training booklet on this subject.
High Wire feed speeds	High wire speeds will require higher power levels from the evaporators. This in turn will effect the uniformity.	If it is thought that there is a problem, reduce the line speed and therefore the evaporation rates.
Spacing of coating sources	Dependent upon the spacing of the evaporation sources is the uniformity pattern.	On older machines, the evaporation sources were further away from the substrate and the spacing between them was wider. Do not try to run at high line speeds
High evaporation rates	High rates will effect the "evaporation plume" and will concentrate it over the evaporator.	See the training booklet on evaporators for further information.
Substrate outgassing	Certain substrates will outgas as the metal coating coats the surface. This can give local pressure rises that affects the uniformity.	Remove the roll of substrate and substitute it for another.
Uniform coating (paper)	The coating placed on the paper before metallizing is not even.	Try to check if possible and makes changes to the running parameters on the coating machine.

Table 8.3 - 13 Poor Uniformity Defects

Spitting

Spitting is a term used to describe a defect, but also the reason for the defect. When the coating medium, usually aluminium is projected up to the substrate

Spattering

Another name describing "spitting"

Light Lane

This is a defect name that relates to lanes above evaporators where the metal deposition is low.

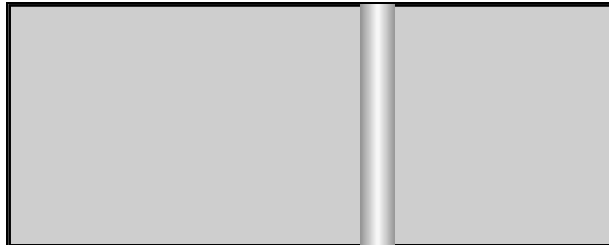


Figure 8.3 - 73 Section of Substrate Showing A Light Lane

Dark Lane

This is a name given to a defect that relates to one or more lanes above the evaporators where the metal deposition is high.



Figure 8.3 - 74 Section of Substrate Showing A Dark Lane

Uncoated spots - dust

Very small areas of the substrate can have a speckled appearance, when substrate is looked at off the machine (i.e., on a light table/box). Clean all dust from the machine that will potentially land on the substrates surface.

Banding

This is a defect term given when there is a pattern as shown below on the substrate as it passes the back light.

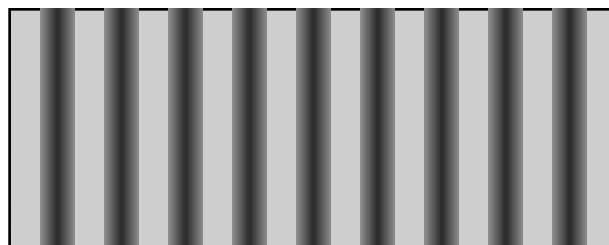


Figure 8.3 - 75 Area of Substrate Showing Heavy Banding

It can be reduced by slowing down the line speed, and therefore reducing the evaporation rate and evaporator temperature.

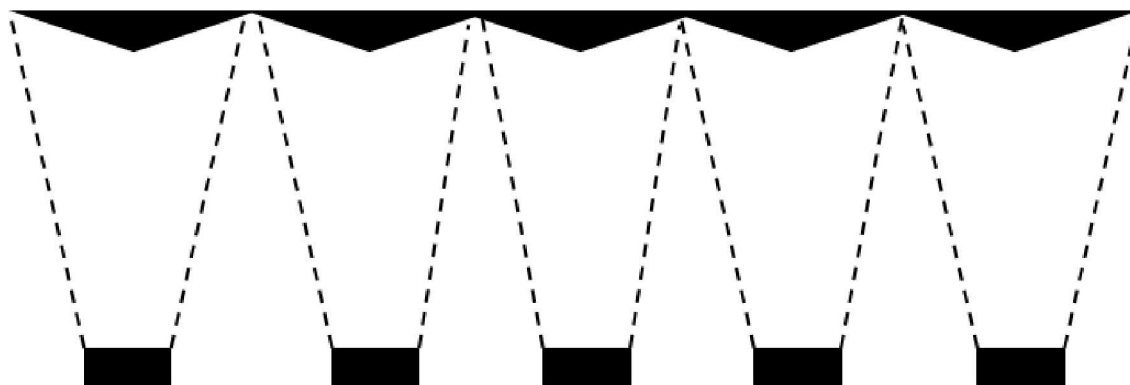


Figure 8.3 - 76 Figure Showing Very High Rate of Evaporation That Causes Banding

Distribution of coating when evaporators are operated at higher operating temperatures or as the evaporator's age.

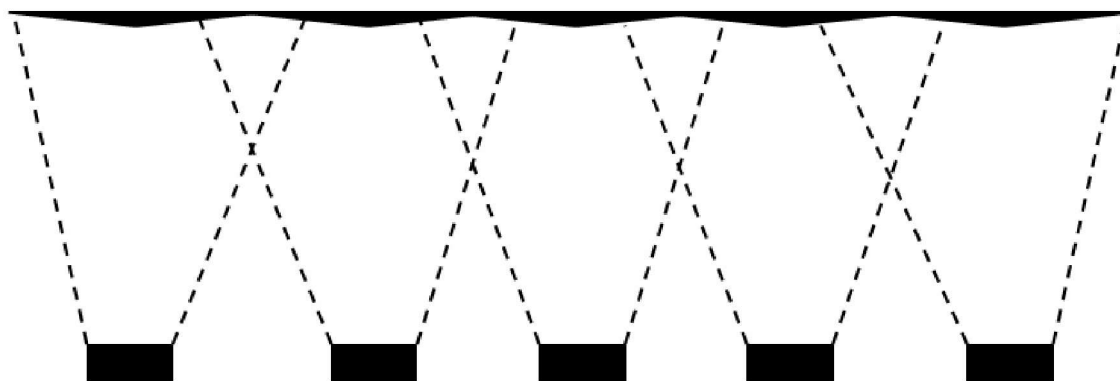


Figure 8.3 - 77 Figure Showing Normal Rate of Evaporation

The best evaporation conditions are when the evaporators are operated at 1450°C

Jail Bars

Another term to describe "banding"

Blooming

A defect term that is used to describe a fault in the manufacture of plastic films. When tension is applied to the film and it is wound through the winding mechanism, a deposit can sometimes be left on the rollers. It is said that the film has bloomed to leave the deposit.

The deposit can be a chemical deposit such as slip agent that has not been processed correctly.

Low Gloss

A term used by paper metallizing companies to describe a dull finish after the paper has been metallized.

Usually caused by an incorrect or low coating application.

Pin Holes

This term is used to describe small holes in the metallized surface. It is usually thought that they are caused by "spitting". However, some have been found to be in the base substrate.

The defect will cause failures of packaging product substrate's barrier properties.

8.4 Source Problems (Evaporators & Wire Feeding)

Evaporator "Spitting"

This name is given when molten aluminium is ejected from the evaporator towards the substrate from the evaporator surface. Various reasons are listed below, investigate and repair.

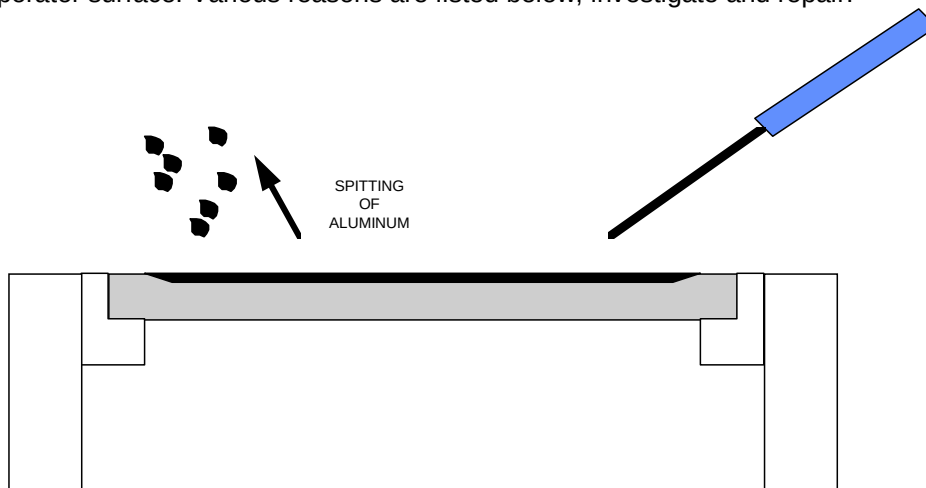


Figure 8.4 - 1 Sketch of Evaporator in a Spitting Condition

Problem	Description	Suggested Remedy
Wire feed tube or wire at earth potential	- Shorting to source enclosure. - Shorting to surrounding steelwork.	Check for electrical "Arc" marks on steelwork
Evaporators not installed	Evaporators not installed correctly, leading to build up of aluminium	Check that the evaporators have been installed correctly
Evaporators being over fed with aluminium.	Excess wire being fed to the evaporators.	Reduce the wire feed rate.
Excess of aluminium debris falling onto evaporators	- Excessive aluminium build up on coating shields - Aluminium debris falling off the shutter when it is moved	- Clean coating shields more frequently. - Reduce the thickness of release paint being used on the shutter
Contaminated aluminium wire	Aluminium used has not been chemically cleaned.	Ensure that the wire is chemically cleaned
Disturbance of Aluminium pool	- Aluminium "bombing" onto evaporator. - Wire Vibration "stirring" the pool. - Opening in guide tube too large	- Reduce the temperature of the evaporator. - Ensure that wire feed tube is long enough. - Check wire feeder for blockages that cause vibration to be induced - Check correct guide tube material is being used

Problem	Description	Suggested Remedy
Dirty wire feed assemblies.	Wire feeders over greased, which contaminates the wire	Reduce the amount of grease being applied and clean the wire-feeders.
Outgassing From Evaporators	New evaporators sometimes give off gases which cause spitting	Do not allow evaporators to absorb moisture from the surrounding atmosphere
Incorrect Grade of aluminium wire being used.	Contaminates inside wire cause outgassing problems, etc.	Ensure specified wire is supplied.

Table 8.4 - 1 "Spitting" Defects

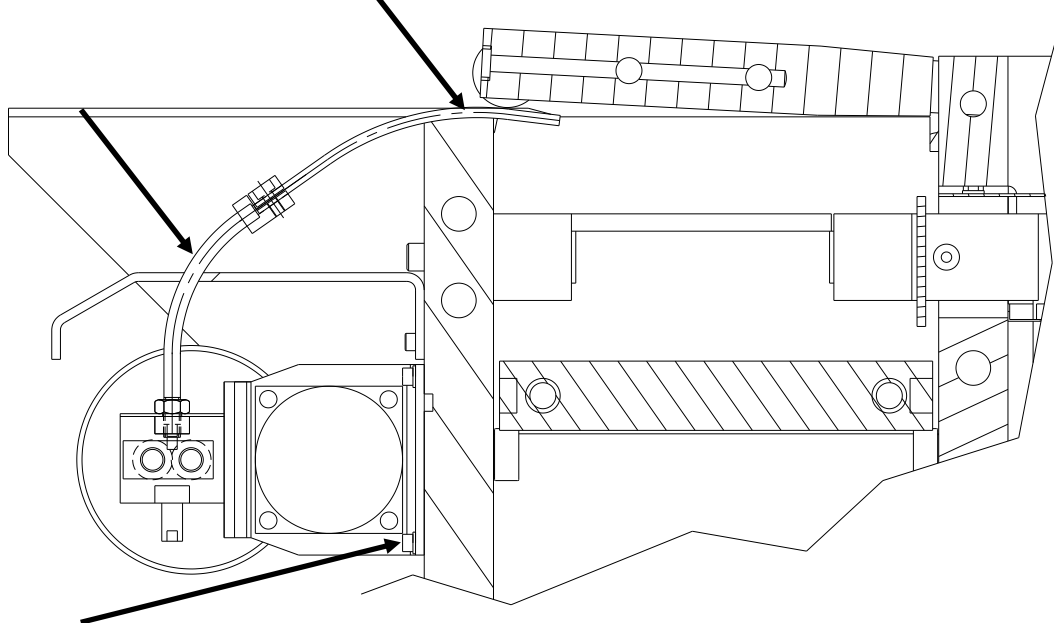


Figure 8.4 - 2 Checking Wire-Feeder Assembly for Electrical Shorts

Aluminium Wire "Bombing"

This term is used to describe a situation when the aluminium starts to melt before it reaches the evaporator surface. The liquid aluminium will form a build up on the end of the aluminium wire, until it becomes so heavy that it falls off the wire.

A direct effect of this situation is poor deposition uniformity caused by the sudden evaporation of the aluminium and then a lack of it.

The reasons that this problem occurs are listed below with the remedies.

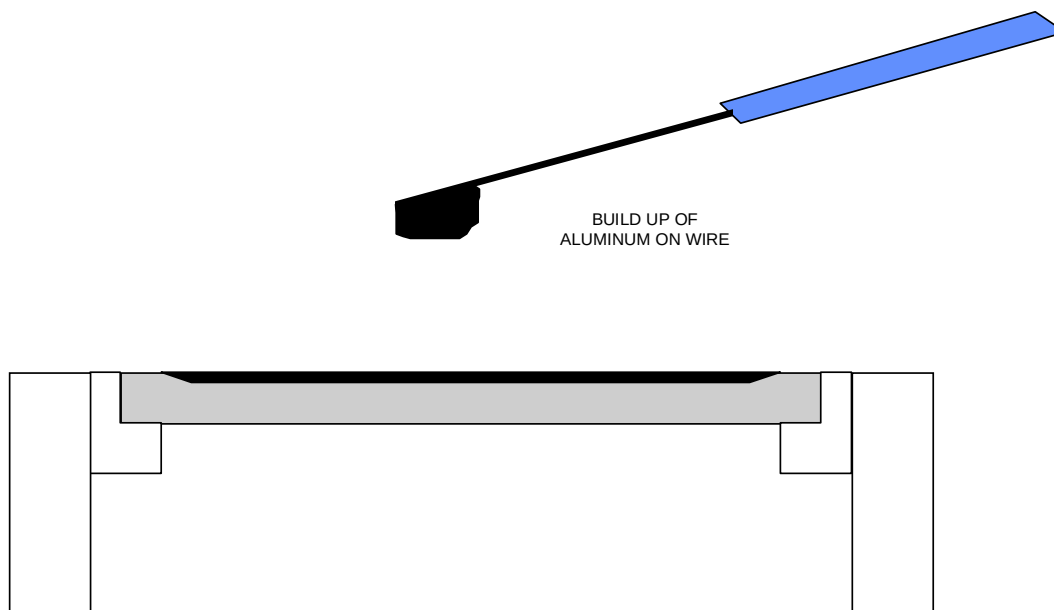


Figure 8.4 - 3 Sketch Of Evaporator With Wire In A "Bombing" Condition.

Problem	Description	Suggested Remedy
Evaporator too Hot	The excess heat melts the wire.	Reduce the power supply to the evaporators.
Wire Too Far Away	The wire is being guided too far over the evaporator.	Ensure the wire touches the evaporator in the correct position.
Low Wire Speed	Wire feed speed is too low causing the problem	Increase the wire feed speed
Non - Constant feed	Wire is not being fed constantly onto the evaporator.	Check the wire-feeder operation with the chamber open.

Table 8.4 - 2 Wire "Bombing" Defects

Evaporator Life

The normal average evaporator life should be approximately 15 hours. The following table relates known reasons that shorten the evaporator life.

Problem	Description	Suggested Remedy
Low MOC Values	The hot resistivity of the evaporators should be between 2300 - 2800 MOC	Ensure Evaporators are in the recommended range.
Incorrect Use of Ramp Cycles	- The "New Evaporator" ramp Cycle is not being used when new evaporators are being heated for the first cycle. "New Boat "Ramp Curve is incorrect.	Ensure that the "New Boat" ramp is being selected in the ROLL 'setup screen.
Evaporators Ran Too Hot	Excess power applied to the evaporators will reduce the life of the evaporators	Ensure that the power to the evaporators is kept to a minimum level to allow the evaporation of the aluminium. There should be full surface usage with no hot spots

Problem	Description	Suggested Remedy
Poor Contact	The contact surface between the evaporator and the contact block is poor	Ensure that the evaporator is located with good contact with the copper contact posts
Non Cleaning of Evaporator Surface	Contamination of oxides and debris will limit the surface area of the evaporator that can be used. This leads to thermal damage of areas of the evaporator.	Clean evaporators with a scraping tool between cycles to keep the surface area clean.
Over feeding of Evaporators	There is a maximum amount of wire that can be evaporated. When running higher levels, there is a tendency to increase the power level.	Do not feed more than the recommended amount of wire for the surface area of the evaporator.
Inappropriate Venting of Machine	The machine has been vented to atmosphere while the evaporators are still hot.	Allow the colour of the evaporators to go black before venting the chamber. This normally takes 3 minutes from the power being removed from them

Table 8.4 - 3 Evaporator Life Defects

Evaporator Contact Blocks

Damage is known to occur to the copper blocks when:

Problem	Description	Suggested Remedy
Over-temperature	Water cooling is not sufficient.	Ensure that the water supplied to the source water manifold is at the correct level, i.e., check against water schematic drawing provided.
Contact between post and contact block poor	Fixing bolt is not tightened sufficiently.	Ensure that the bolt that holds the copper contact block has been tightened
Poor contact between Evaporator and contact block	Evaporators not fitted correctly.	Ensure that the evaporators are located squarely in between the contact posts
Non use of graphite tape	Graphite tape is NOT USED between the evaporator and terminal block.	Always use one layer of graphite tape between the evaporator and the contact block.
Debris on contact block	The blocks are not kept clean	Ensure that the contact block is cleaned of all debris before a new evaporator is installed.

Table 8.4 - 4 Contact Block Defects

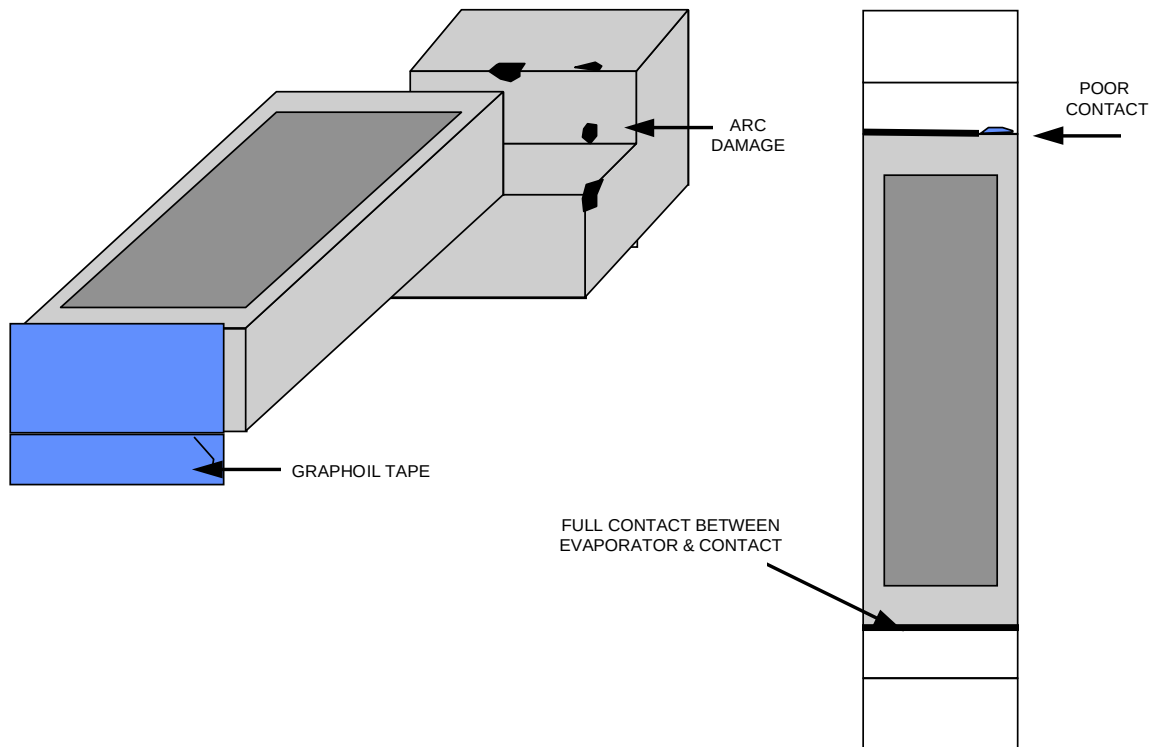


Figure 8.4 - 4 Copper Contact Block Damage

Uneven Heating of Evaporator

The evaporator will glow unevenly in the following situations. This will lead to thermal stress and a limited life span.

Problem	Description	Suggested Remedy
Excess Cooling of contact block.	If the cooling to evaporator power supply posts is very cold, it will cool down the ends of the evaporators.	Ensure the cooling water to the source supply manifold meets the specification on the water schematic.
Poor use of evaporator surface area.	The surface area of the evaporator is contaminated and limited area.	Ensure the evaporator is cleaned correctly with a scraper.
Wire fed in one area	If the wire is fed onto the evaporator in the incorrect area, it will not allow the whole of the evaporator surface to be used.	Ensure that the wire feeds onto the evaporator at a position between 1/3 and 1/2 of the length of it from the wire feeder end.

Table 8.4 - 5 Uneven Heating Of Evaporators Defect Table

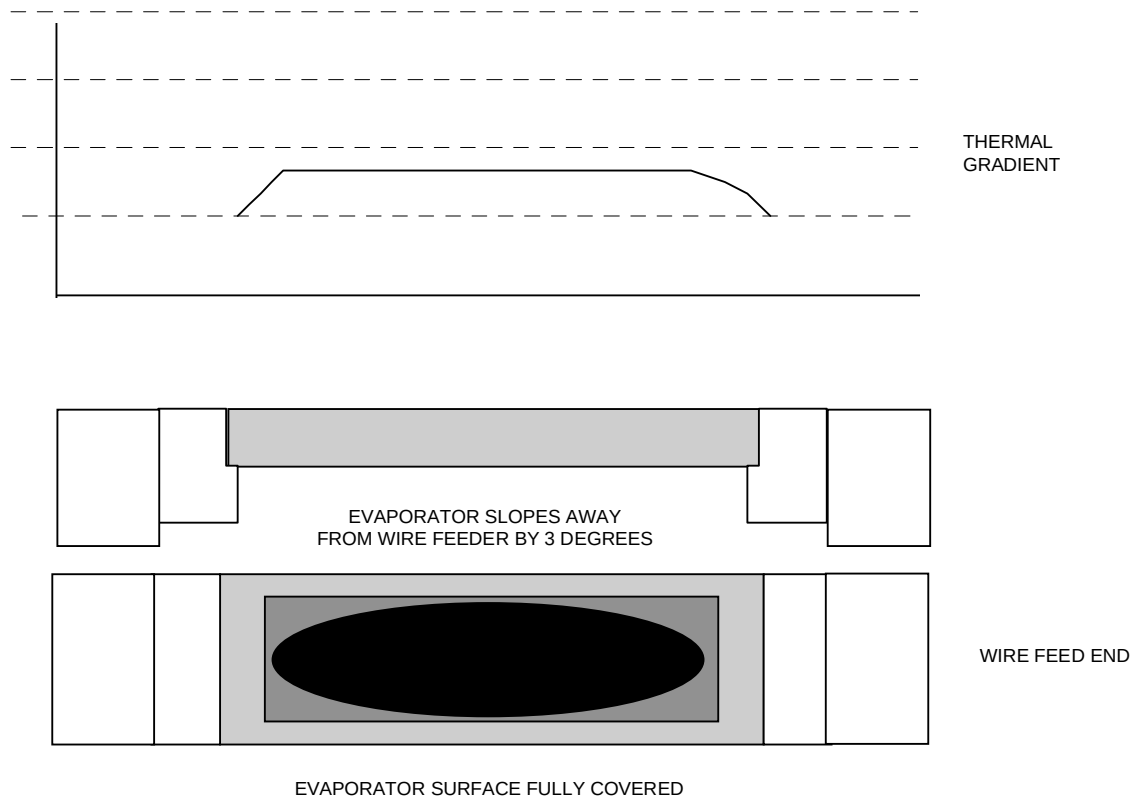


Figure 8.4 - 5 Thermal Radiant on a Correctly Installed Evaporator

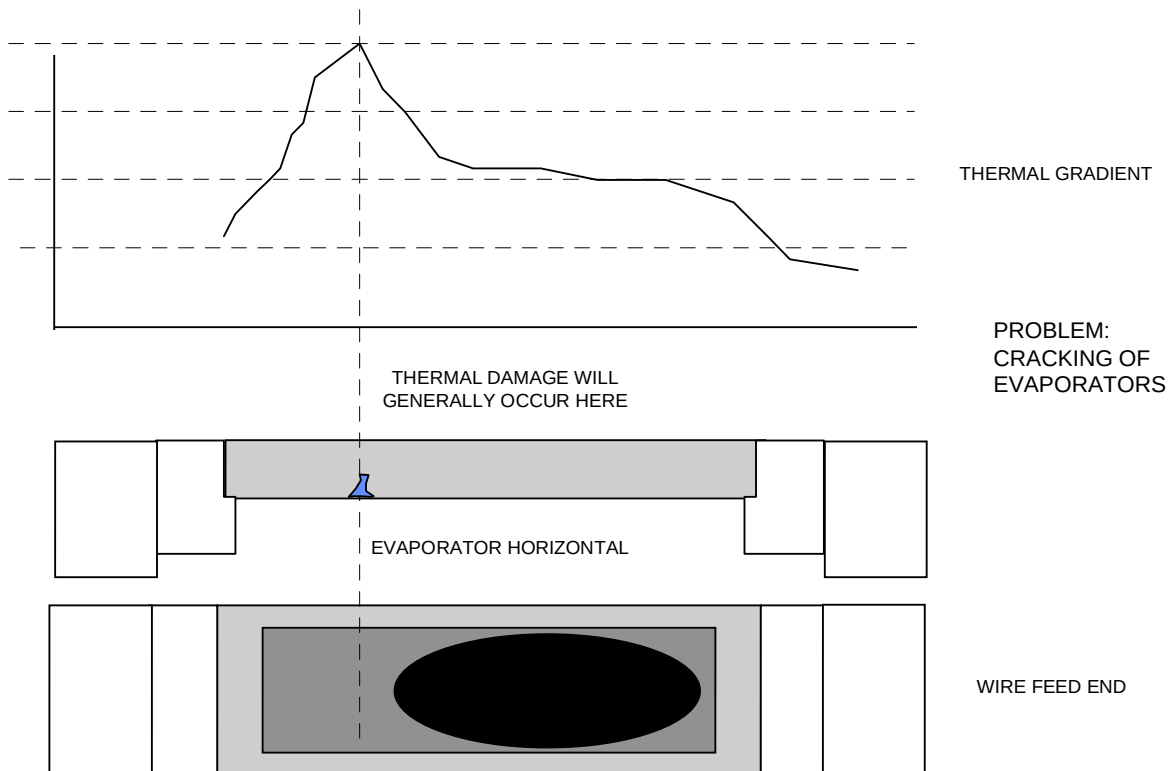


Figure 8.4 - 6 Thermal Gradient on a Correctly Horizontally Installed Evaporator

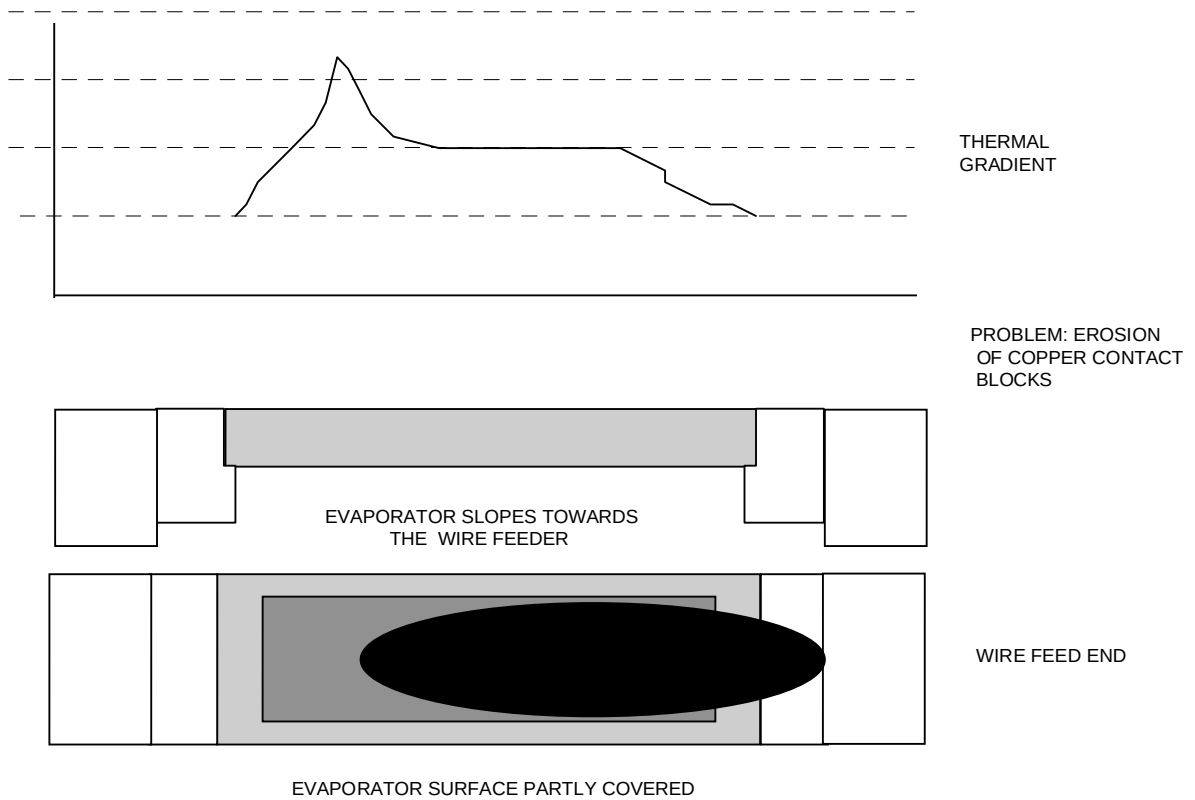


Figure 8.4 - 7 Thermal Gradient on an Incorrectly Installed Evaporator

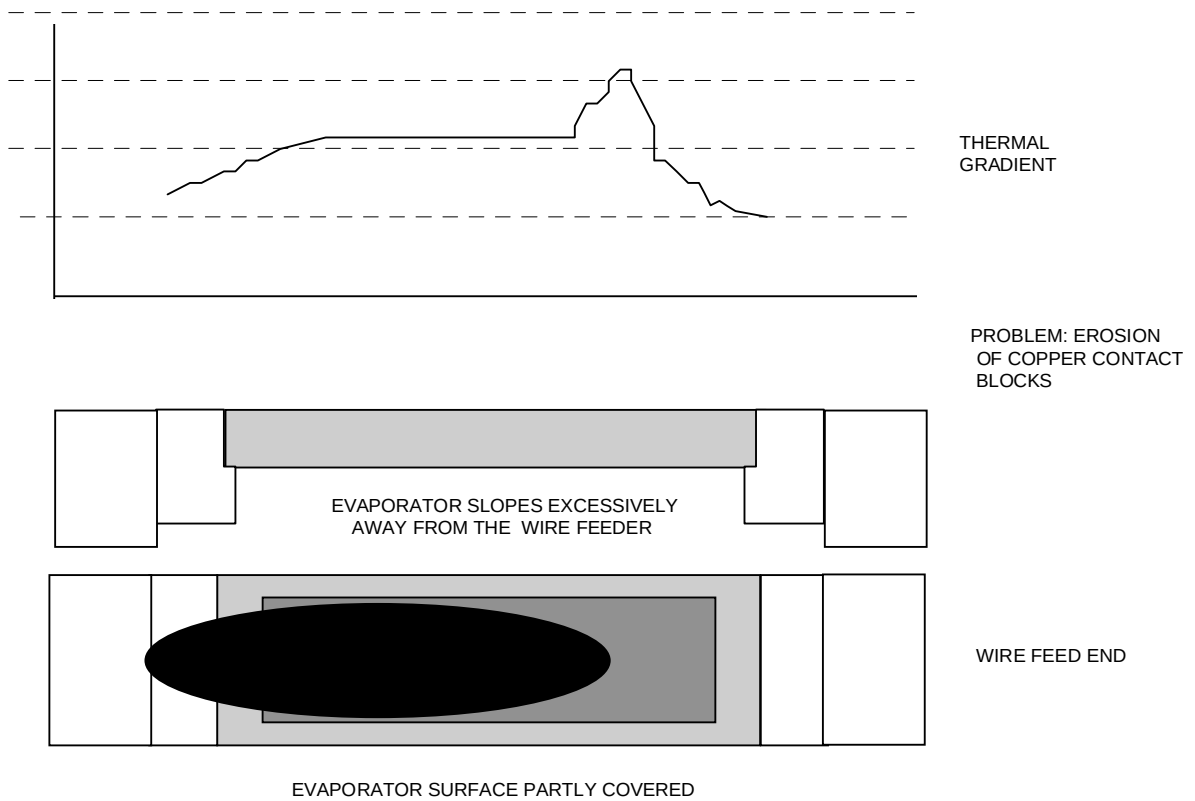


Figure 8.4 - 8 Thermal Gradient on an Incorrectly Installed Evaporator

Wire Alignment onto Evaporators

If the aluminium wire is not fed onto the evaporator in the correct area, there can be problems with it not melting and poor uniformity control. The wire should always be fed down the centre line of the evaporator and a third of the way along.

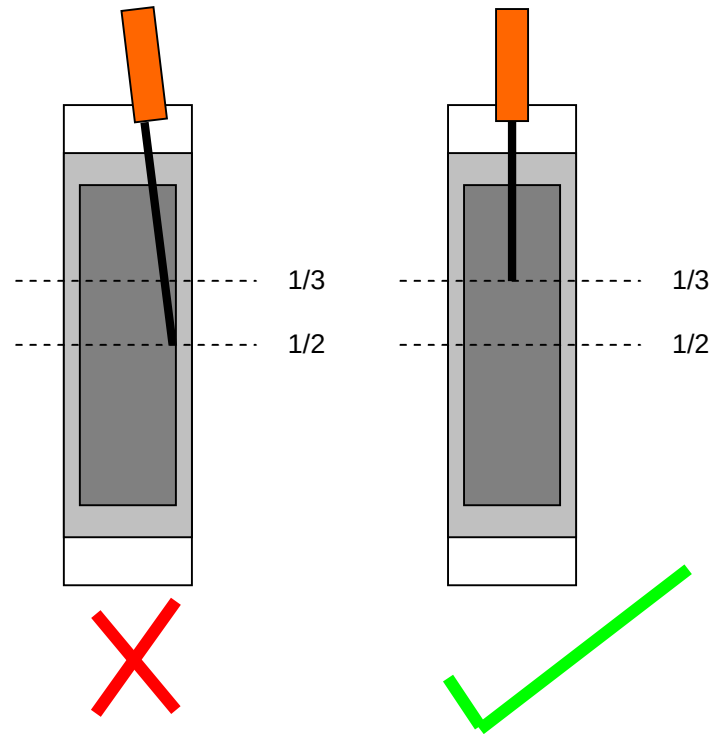
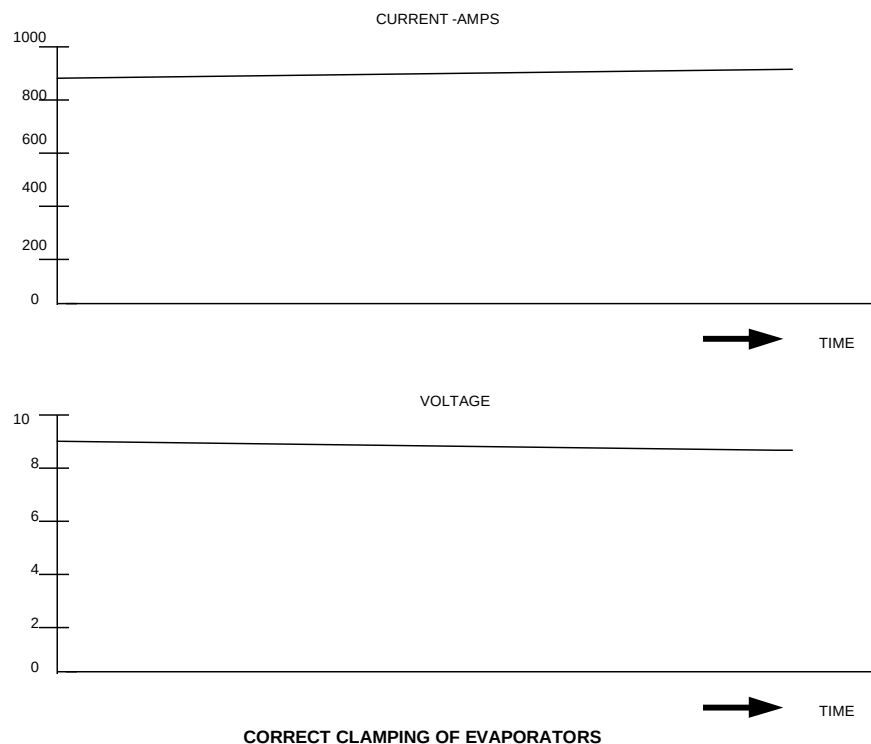


Figure 8.4 - 9 Feeding Wire Correctly Onto Evaporator Surface

Evaporator will not Heat Up

If the evaporator will not heat up to a temperature where the aluminium wire melts, there are a number of possible reasons;



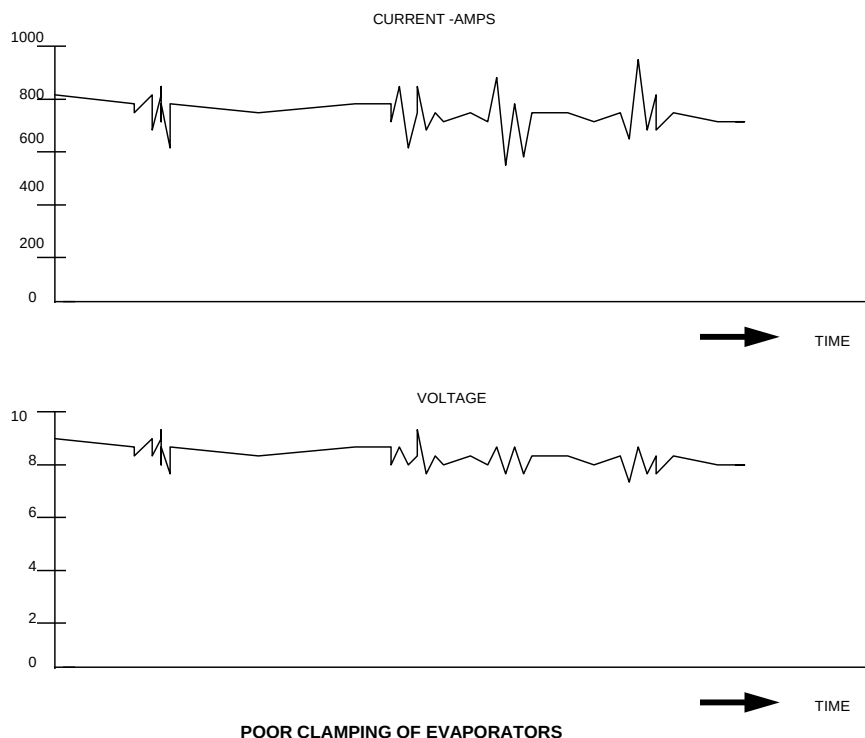


Figure 8.4 - 10 Graph Showing Differences in Electrical Power Supply In Poor Contact Situation.

Reasons for poor contact between the evaporators and the contact blocks are as follows;

Problem	Description	Suggested Remedy
Incorrect Resistivity	Higher MOC values of a value more than 3000 MOC will require more than the designed power supply	Ensure that evaporators used are within the specified MOC range
No power Supply	The power is not being supplied to the evaporator	Check the control screen to see if the evaporator is being supplied. Reference should be made to the maintenance manual.
Poor Contact (Most likely)	Poor contact between the evaporator and the contact block	Ensure the contact between the evaporator and contact post is correct, as recommended.

Table 8.4 - 6 Poor Evaporator Contact Defect Table

Wire Stoppages

The following chart lists possible reasons for wire-feeder stoppage during the coating cycle;

Problem	Description	Suggested Remedy
Wire Tube Blockage	- Tube damaged, which reduced area for wire to pass through. - Tube blocked by aluminium.	Ensure the tube opening is OK. Clean the tube of any aluminium build up or replace it.
Aluminium Wire Diameter	Aluminium wire diameter not constant	Ensure that the diameter is constant and meets the specification given.

Problem	Description	Suggested Remedy
Mechanical Fault on Wire Feeder	Faulty wire feeder assembly -- gears not meshing together	Check the operation of the wire-feeder, monitoring the engagement of the gears.
Failure of Stepper Motor	Failure of stepper motor to run correctly	Check operation of stepper motor
Incorrect installation of consumable wire feed tube	Tube not installed correctly, causes wire to drag on wall.	Ensure that the tube is installed correctly.
Incorrect Tube Used	Tube used has a small inside diameter.	Ensure that any replacement tube is the same as the original item.
Bend In Tube incorrect	Guide tube has sharp bends in it that drag on the wire.	Ensure that any bends formed in the guide tubes are smooth
Wire Feeder Movement Stiff	If the feeder has not been greased, the drag will be high	Ensure that the feeders are greased regularly with the correct grease.

Table 8.4 - 7 Wire Feeder Stoppage Defects Table

Erosion of Evaporator Surface

The surface of the evaporator will be eroded over a period of operating time.

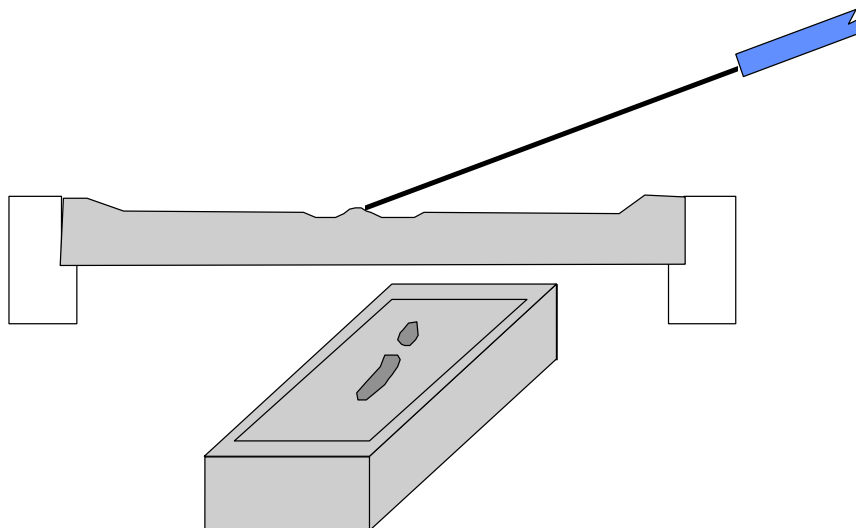


Figure 8.4 - 11 Sketch Showing Chemical Erosion of the Evaporator Surface

The erosion of the surface of the evaporator is caused by a chemical reaction between the aluminium wire and the ceramic components.

To reduce the effect of this erosion, the wire feed rate should be reduced.

It has been noticed that this problem occurs more often on certain types of evaporators than others. Contact your evaporator supplier for advice on the chemical composition of the evaporators.

Contamination of Evaporator Surface

Contamination of the evaporator surface will cause a potential number of problems, such as

- Dis-colourment of the aluminium coating
- Vacuum Problems
- Poor Uniformity

Problem	Description	Suggested Remedy
Release Paint	Evaporator is accidentally painted with release paint	Ensure that surface is not painted
Dirty Wire	- Wire not chemically cleaned correctly - Incorrect grade of wire being used	- Ensure that the aluminium wire has been chemically cleaned. - Ensure correct grade of wire is being used.
Contaminant	Contamination from external source	Ensure that no oil, etc. is contaminating the evaporators
Debris	Aluminium Debris falling off surrounding shields	Clean shields to a higher standard between cycles

Table 8.4 - 8 Contamination of Evaporator Surface Defect Table

Flooding of Evaporators

This is a term given to the situation when liquid aluminium will flow over the sides of the evaporator onto the contact blocks, or over the sides of the evaporators.

Problem	Description	Suggested Remedy
Evaporator Too Cold	The evaporator is at too low a temperature to vaporize the aluminium being fed upon it.	Increase the power applied to the evaporator when possible.
Excess Wire Feed	Wire feed rate exceeds the maximum evaporation rate.	Reduce the wire feed rate.
Wire Feed Position Incorrect	The wire feed is being fed onto the evaporator at an incorrect position	Ensure that the wire is fed onto the evaporator at the correct position.
Incorrect Installation of evaporator	Evaporator not leaning away from wire feeder end to encourage molten aluminium to flow down the evaporator.	Ensure that the evaporator has been installed correctly.
Contamination of surface area.	Oxide build up restricts surface usage.	Clean the oxide build up from the evaporators.
Graphite tape	Excessive graphite tape acts as a condensation point	Trim the graphite tape to remove this possible cause.

Table 8.4 - 9 Flooding of Evaporator Surface Defect Table

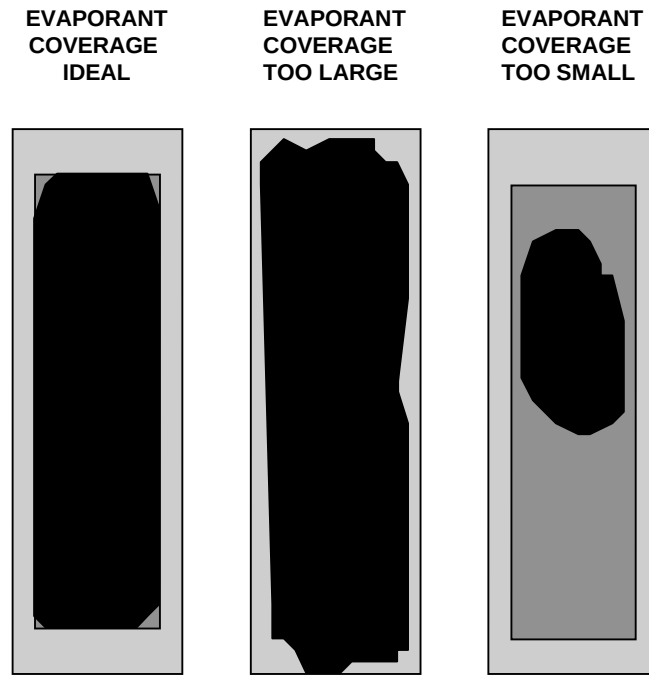


Figure 8.4 - 12 Evaporant Coverage of Evaporator Surface

8.5 Deposition Control Problems

This is a brief summary of potential problems that may occur during the first few weeks of production, mainly because of the lack of experience.

Problems that occur on heating up the evaporators

Display Symptom	Reasons	Suggested Remedy
After the heater main power supply is switched on, the power ramp does not start.	No values being sent from computer	Download SETUP Screen with correct information
Edge channel deposit is light and the controller will not switch into the AUTO mode on starting the coating cycle	- The roll of substrate has not been placed in the centre of the winding mechanism. - The incorrect width has been entered into the initialize screen	- Stop the cycle and align the roll correctly - Stop the cycle and re-initialize with the machine open.
Light deposit channel in the middle of the substrate and the controller will not go into the AUTO mode on starting the coating cycle	Wire feeder stops feeding wire to the evaporator	Stop the cycle and check why the problem has occurred
Heavy deposit channel in the middle of the substrate and the controller will not go into the AUTO mode	Aluminium Debris fell onto an evaporator from the shutter as it opened.	Allow it to vaporize off the evaporator. If it does not clear, increase the power to the evaporator. If it still does not clear, stop the cycle and remove.
All channels have a light deposit on starting the run	- Wire Feed rate is too low - Vacuum level is not correct - Substrate will not accept coating	- Increase the base wire feed rate slowly. - Check that the vacuum is below 8×10^{-4} mbar - Check that the correct treated side is facing the evaporators.
No wire feeds onto the evaporators after the evaporators are heated up	Wire feed contactor not engaged.	Reduce the wire feed base speed rate to 20% of set-point and press the wire on push-button. Ensure that the wire is feeding and melting. Then increase the base speed back to its previous level
The controller will not switch into AUTO mode from WARM UP mode on start up	- Erratic Wire feed rates - Variable evaporation rates due to excessive temperature variations of evaporators. - Untreated substrate being coated	- Use the RESET function to bring all the outputs even again. - Lower or raise the power levels to allow even evaporation and full cavity coverage. - Ensure that the treated side of the substrate faces the evaporators

Display Symptom	Reasons	Suggested Remedy
Wire feeder goes backwards on start up	Faulty stepper unit controller or wiring	Check the control circuit as described in the maintenance manual.
Evaporators do not heat up enough to melt wire	- Incorrect wire feed setting - Low temperature due to allowing the evaporators to power down on the previous coating cycle.	- Ensure that the wire feed rate is at the normal running rate in the initialize page before attempting to heat the evaporators. - Increase the evaporator power levels to increase the evaporators.

Table 8.5 - 1 Deposition Control Problems -1

Problems that occur on when the coating cycle is taking place.

Display Symptom	Reasons	Suggested Remedy
Edge channel deposit goes light and controller goes into WARM UP mode from AUTO mode	- Substrate has moved sideways and no longer covers the probe - Unwind roll has moved on shaft	- Increase the unwind tension, if possible to stop slippage on unwind roll. - Stop machine, remove the roll and centralize.
Light deposit channel in the middle of the substrate and the controller goes into the WARM UP mode during the run	Wire feeder stops feeding wire to the evaporator	Stop the coating cycle and check why the problem has occurred
Heavy deposit channel in the middle of the substrate and the controller goes into the WARM UP mode during the run	Aluminium Debris fell onto an evaporator from shields.	Allow it to vaporize off the evaporator. If it does not clear, increase the power to the evaporator. If it still does not clear, stop the cycle and remove.
All channels indicate a light deposit on the substrate and the controller goes into the WARM UP mode during the run	- Poor treatment on substrate - Patchy treatment or coatings	Check age of substrate and ensure the quality of the coating or treatment.
All channels indicate a heavy deposit on the substrate and the controller goes into the WARM UP mode during the run	Patchy treatment levels or coatings	Check age of substrate and ensure the quality of the coating or treatment.
All wire feeders stop	- Communications Problem - No main power supply - Faulty wiring	- Check communications screen. - Check for the correct voltage at the controllers - Check the wiring to the stepper controller

Display Symptom	Reasons	Suggested Remedy
MMI screen "Freezes"-- sticks -- cannot be changed	Internal Software fault	Note the steps taken prior to the screen "freezing". Also steps taken to clear the fault. Please report to General Vacuum Equipment
Wire feeder number x stops at the same length every coating cycle	Overheating of the controller unit	Check the heat sink part of the unit is in good contact with the background plate.
A section of the wire-feeders stop	- Stepper controller fault - Faulty wiring	- Check for fault as described in the maintenance manual on the stepper at the highest position that has stopped.(i.e.; evaporator 14) - Check the wiring to the stepper controller.

Table 8.5 - 2 Deposition Control Problems -2

8.6 Plasma Treater Process Problems

The equipment supplied should provide the user with a reliable process tool. However, there maybe some problems during its operational life cycle. In this section of the manual, potential problems are listed for the users information.

Display Symptom	Reasons	Suggested Remedy
Scratching of Substrate	• Roller not turning	• Check that the entry roller is free to turn, by hand
	• Substrate touch exit slot	• Adjust slot gap
	•	• Ensure that film is reasonably flat and does not have very bad slack areas.
No Gas Flow	• Gas supply empty	• Check gas supply.
	• Gas being produced by outgasing of substrate.	• If plasma does not ignite, substrate may not be suitable
	• Gas controllers not switch on or set-up correctly.	• Check
	• Isolation valves closed	• Check and open
No Display on Power Supply	• No main power supply	• Check that the main power supply is switched on.
	• Internal fault	• Refer to the manual supplied by the vendor
No Plasma Created	• Poor background vacuum level	• Check that the chamber background vacuum level will allow vacuum level in enclosure to be set to 5×10^{-2} torr.
Arcing & Power Supply Instability	• Contaminated electrode area	• Clean the electrode area and burn in the unit.
	• Incorrect vacuum level.	• Check
Excessive Electrode Cover Plate Erosion	• Vacuum level too low, causing 'sputtering' process to take place.	• Check calibration of baratron gauge.

Table 8.6 - 1 Plasma Treater Problems

8.7 Zinc Sulphide Process Problems (Not applicable)

8.8 Cleaning Vacuum Roll Coaters

Regular cleaning of the machine is essential if the machine is to maintain a high standard of finished product, pumping speed and maximum processing speed.

	Warning !
	Do not attempt to clean the winding mechanism or enter the chamber without rotating the traverse lockout isolator and locking it out.

Personal Protection Equipment.

It is advised that operators use the relative personal protection equipment recommended on MSDS sheets provided with the machine and suppliers of any cleaning substance.

As a general guideline, for dealing with Aluminium oxide/dust.

- Skin Contact -Protective gloves.
- Breathing - Local exhaust or breathing protection.
- Eyes - Safety goggles, or eye protection in combination with breathing protection.

Tools

It is recommended that good quality cleaning tools are used in the cleaning procedures described in this section of the manual. The use of inappropriate or damaged tools will lead to machine damage.

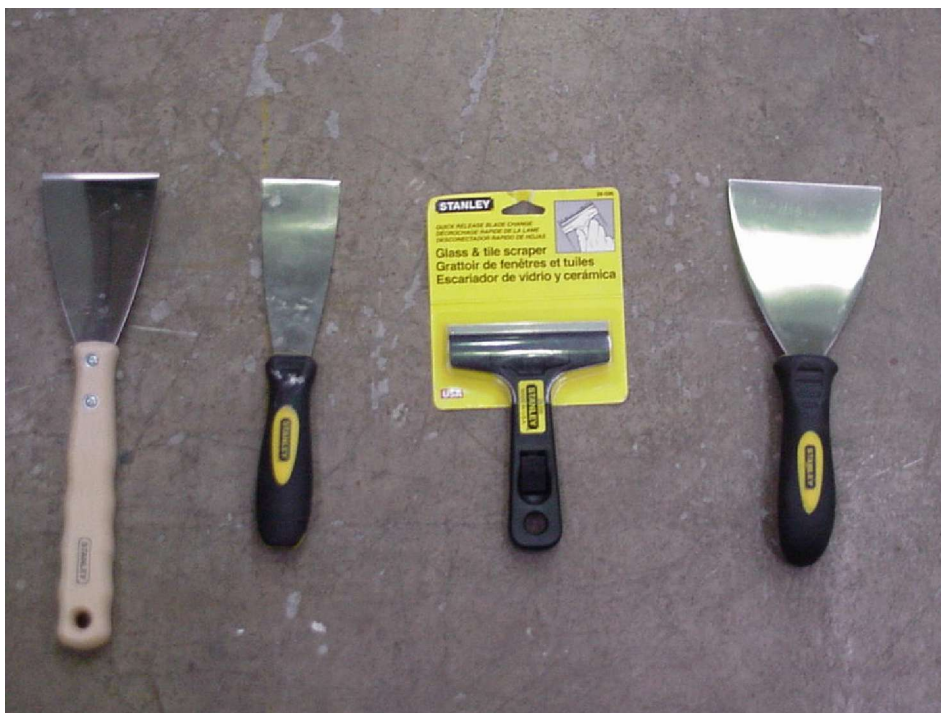


Figure 8.8 - 1 Photograph of Selection of Scrapers Recommended for Cleaning the Machine

Vacuum Cleaner

It is recommended that an industrial rated vacuum cleaner be supplied to allow a high level of aluminium dust collection from the machine, when cleaning takes place.

Aluminium Debris Storage Container

Aluminum dust should be specifically stored, prior to disposure in steel sealable containers. For example and oil drum type container with a removable lid that can be closed to stop the contents becoming damp

Cleaning Procedures

A layer of aluminium or oxide inside the machine will increase the pump down time and decrease the ultimate pressure attainable in the vessel. This in time will affect the quality of the product.

Also, if dust is disturbed and falls onto the substrate during the operation of machine, it will cause pin holes in the coating. It is strongly suggested that the use of compressed air to "blow" the dust is discouraged, as the dust will only land on another surface.

	Warning ! The rollers must only be rotated by hand while cleaning takes place and the power should be switched off and locked out.
	Caution ! Do not use steel scrapers or chisels, etc. When cleaning chrome or idler rollers, this can lead to damage the surface of the rollers store aluminium debris in sealed steel containers

Rubber Sleeved "Bow Type " Spreaders

The manufacturer recommends that the sleeve be cleaned with water or a cloth dampened with kerosene or stoddart solvent. Do not use an excessive amount of solvent.

Cork Covered Rollers

The cork cannot be cleaned perfectly. Use an adhesive backed tape to lift dust from the surface of the roller. Eventually the cork will need to be replaced. Use adhesive backed cork for this purpose. This material is stocked by General Vacuum Equipment.

Do not attempt to use a piece of cork without an adhesive backing. It has been found that using a non-backed cork and glue causes problems through contamination.

Rubber Straight Cut or Triple Cut Spreader Rollers

The manufacturer recommends the use of "mild white spirit" or "any proprietary rubber roller cleaning solution, as supplied to the printing industry".

	Caution !
	Do not use any gasoline derivatives or nylon or stiff brushes, as damage will occur to the roller surface.

Chrome Idler Rollers

Clean the rollers with a metal polish, such as "Brasso" and the residue should be removed with clean cloth. Then wipe the whole surface with a cloth damped with "Acetone"

Chrome Cooling Roller

Clean deposits with a metal polish, such as "Brasso" and lint free cloth. Then wipe the whole surface with a cloth damped with "Acetone".

In the case where the drum has been coated with a light to medium coating of aluminium, in cases, for example, where a web break has occurred, use a scraper of the type shown below. Care must be taken not to scrape the surface of the drum.



Figure 8.8 - 2 Photograph of Operator Cleaning the Drum.

Note. The cleaning motion should be in the direction around the drum, not across it's surface.

Non-Contact Resistance Monitor Beam

Ensure that there is no film trapped between the roller and monitoring heads. Aluminium dust may settle in parts of the beam. This should be vacuumed from the machine on a regular basis.

Cleaning source box (enclosure)

The source box and shields are subject to heavy coatings of aluminium and must be examined and cleaned at regular intervals.

The source box can be cleaned in its normal operating position by using a flat scraper.

Remove the larger pieces of aluminium debris and evacuate the smaller pieces and dust using a vacuum cleaner.

If the instructions concerning the release paint have been followed, the aluminium deposits will be easily be removed.

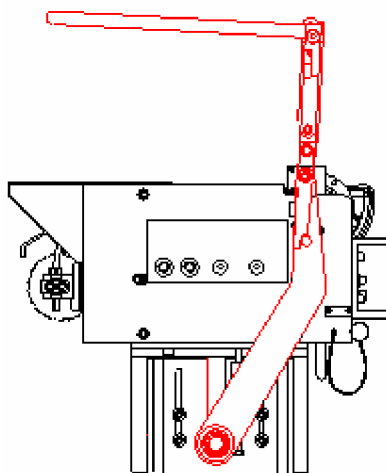


Figure 8.8 - 3 Source Enclosure Shown With Shutter raised to Allow Cleaning

Do not use hammers and chisels to remove the aluminium debris as this can lead to "pits" in the source box surface.

The source box base can be removed from the machine to allow the base to be cleaned away from the chamber. This can be accomplished by following the sequence below;

1. Close the feed and return valves to the source manifold
2. Drain the water from the cooling circuits
3. Disconnect the hoses inside the chamber
4. Pull the source base out, support it and clean it
5. Return the source base and reconnect the hoses.
6. Switch the water on and check for leaks



Caution !

Ensure that the hoses are reconnected to the source box (enclosure), so that cooling is present during heating of evaporators.

Cleaning stray aluminium collection shields

There are a number of shields provided around the source enclosure. These are provided to collect the stray aluminium evaporation. These are manufactured out of stainless steel and can be cleaned using scrapers and wire brushes, outside the chamber.

General Vacuum Equipment recommends the user must wear protective goggles and dust mask when cleaning with such tools and care should be taken not to damage the collection shields

These shields can be cleaned by placing them in a mild caustic bath away from the chamber.



Figure 8.8 - 4 Removing Collection Shields From Within The Chamber

Evaporator Posts/Contact Blocks

To extend the life of the contact blocks, it is suggested that all debris be removed from them when replacing the evaporators. It has been found that cleaning the contact area as the evaporators are changed, with a "pam" sanding machine, extends the life.

Cleaning of other exposed surfaces subject to aluminium coating

The cleaning of exposed surfaces, such as the drum shields, or shutter, can be simplified if they are painted with a release agent such as Boron Nitride suspension. Only a thin coating is necessary, if too thick a coat is used, the aluminium coating will flake off and possibly fall onto or across the evaporators causing electrical short circuits. This will cause patches on the film being coated. If correctly applied this will not occur and its use is highly recommended.



Figure 8.8 - 5 Cleaning The Source Enclosure Shutter



Figure 8.8 - 6 Photograph Of User Cleaning Coating Shield In The Lowered Position

There are a number of ways of cleaning the aluminium coating off the shield surfaces; scraper, wire brush, etc. The user must wear protective eye goggles and dust mask when cleaning. Care should be taken not to damage the shields.

If possible remove large pieces of the coating first and then remove remainder with tooling. Finally clean all areas with an industrial vacuum cleaner.

Cleaning Of Wire-Feeder Units

The wire-feeder units should be removed from the machine and cleaned in a degreasing bath using commonly used degreasing solvents used in maintenance workshops. Some wire feeders have been manufactured from Nylon and checks must be made that solvents to be used will not attack the nylon. Consult with General Vacuum Equipment, if there is any doubts.

Once the excess debris have been removed the complete wire-feeder should be submerged in a container of Acetone to remove any remaining solvents used for the degreasing stage.

They should be then removed and placed in a further separate container containing acetone before being reinstalled on the machine.

Cleaning Of Wire Feed Spouts

The piece of tube that guides the aluminium wire onto the evaporator is located in the vapour cloud and becomes heavily coated with aluminium debris. If not cleaned on a regular basis, the tube can become blocked and the wire will not feed.

The tubes should be painted with release agent before being used and this coating should be repeated as the debris is removed.

The debris can be removed by twisting its outer surface lightly with a pair of pliers.

To clean the inner part of the tube, the tubes can be placed in a weak caustic solution, e.g. commercial drain cleaner. Take all the precautions as instructed on the cleaner's container.

It is suggested that aluminium foil be placed around the clamp section of the feed tube to keep it clean.



Figure 8.8 - 7 Cleaning Wire Feeder Spouts Of Aluminium Debris

Cleaning Of Contaminated Bearings

Some bearings on the machine are not fully sealed, such as those used in the load cell transducers, because of high frictional losses. Depending upon the cleanliness of the machine during normal operating cycles, reduced contamination of the bearings occur.

If the winding area of the machine is kept clean of dust, excessive grease can be removed by wiping the outside of the bearing with a cloth.

However to clean the bearings of the lubricating grease and aluminium debris, the bearings should be removed from the machine for cleaning.

The suggested method of cleaning off the machine is;

- The bearings should be removed from the machine and cleaned in a degreasing bath or cleaner using degreasing solvents used in maintenance workshops.
- Once the excess grease and debris have been removed, the bearings should be submerged in a container of Acetone to remove any remaining solvents used for the degreasing stage.
- They should be then removed and placed in a further separate container containing acetone before being reinstalled on the machine and greased with the recommended grease.

It may not be easy to remove some components from the machine for this purpose. A temporary measure is to wipe the grease out of the bearings with a lint free cloth and then re-grease them.

Cleaning of Layarm slides

The layarm assemblies slide on linear bearings that travel along the slide rods. Over a period of time, debris can be seen on the rods. It is suggested that these should be cleaned by wiping them with a lint free cloth dampened with a lightweight oil (use the same oil as in the gravity filled bottles on the rotary leadthroughs).

Do not use release agents such as WD40, as they have a high outgassing rate.



Figure 8.8 - 8 Wiping The Layarm Slide Units

It may become necessary to remove the assembly to replace the bearings or the rod if it becomes damaged and pitted. Please contact GENERAL VACUUM EQUIPMENT for information if required.

Cleaning Concave Zoning Seals

The concave steel zoning strips provide a low conductance seal between the winding and evaporation zones of the chamber.

To remove the debris, it is suggested that a scraper be used to remove heavy deposits, while light build ups, can be removed with a slightly abrasive cloth.

Cleaning Of Silicone Zoning Seals

The suggested method of cleaning is to use a slightly damp cloth to wipe away any debris.



Figure 8.8 - 9 Photograph Showing User Wiping the Seals Down

General Cleaning of Machine

	Warning !
	Wear protective gloves and eye protection when using caustic solutions.

If the machine is being operated correctly the remaining stray aluminium deposits on the vessel wall and mechanisms will be small but should be cleaned once a week to prevent a build-up. The main vessel flange and seals should be examined on a regular basis and kept clean.

The chamber walls are painted a yellow colour to highlight the oxide coatings.

Care should be taken not to inadvertently introduce any high vapour pressure substances into the vessel during cleaning, e.g., certain solvents.

Internal Viewing Windows

Use a dry cloth to remove all dust that has built up upon the window.

Use a plexiglass/perspex cleaner as recommended by your local plastic distributor. They will normally stock a cleaning solution for this type of application.

Ensure that you do not use an abrasive cleaner.

Cleaning Source Viewing Window -- Internally

The internal surface of this window will become coated with aluminium over a period of time.

The light coating can be removed by using a razor blade type scraper and scraping the surface of the glass.

An alternative is to use a lint free cloth damped with a weak caustic acid solution.

	Warning !
	Wear protective gloves and eye protection when using caustic solutions.

Do not use wire wool, sand paper or abrasive cleaners as the window will become scratched and difficult to view through.

Cleaning the Cryogenic Coils

Due to the fact that the cryogenic coils collect water on them, they also sometimes collect aluminium dust.

The coils must be wiped clean of this aluminium debris. If not the collection efficiency of the coils will decrease and lead to longer pump down periods.

A lint free cloth can be used to carry out this operation.

Do not use any abrasive material to "polish" the coils as excess use could lead to reduced wall thickness and potential leaks of the cooling medium.

Cleaning Plasma Treaters

The electrode plates have to be inspected and cleaned as necessary.

Film debris should be removed after any web break situation.

The cleaning interval is somewhat of an unknown. This is because it is very dependent upon the types of substrates being pre-treated. Many substrates have coatings, which can deposit on the electrode plates and plasma surfaces during treatment and require regular cleaning prior to running the unit again.

Aligners are particularly problematic and can initiate arcing events on the plasma treater electrode plates. The general rule is if unacceptable levels of arcs occur, the plates require cleaning.

Depending on the deposit or level of contamination on the plates the level of cleaning can differ dramatically. Again, the general rule is, use the least abrasive material possible to remove the source of contamination responsible for causing the arc events.

To inspect the electrodes and insulators in this equipment, the front plates can be hinged back to give access to inspect the treater plates.



Figure 8.8 - 10 Photograph of Plasma treater

The two water-cooled copper sections can be released to allow inspection without the water having to be removed.



Figure 8.8 - 11 Photograph of Plasma treater hinged Doors Opened

Basic Clean

If the contamination is found to be loose process particles deposited during a venting cycle or film pieces deposited during a web break. A IPA/acetone clean with a lint free cloth is sufficient.

Definition: A basic clean refers to a clean up procedure which removes any oxide build-up, general contamination (dust/dirt) and/or arc marks on any visible electrodes surface, without the need to remove any edge shielding. A basic clean is required if arcs are being generated/observed on the electrode in racetrack, or the area inside the racetrack.

Procedure: In order to perform a 'basic clean' of the plasma treater, it is a requirement to seal the circumference of the electrodes to prevent potential contamination/ oxide build-up from being transferred from the electrode surface to the area underneath the edge shielding. Sealing can be achieved with the aid of appropriate masking tape. The actual removal of contamination/ oxide build-up from the electrode surface can be achieved with an appropriate abrasive material. In order to avoid unnecessary wear of the electrode surface, a minimum grit size of 600 is recommended. Following the removal of the contamination it is advised to thoroughly vacuum the entire exposed surfaces of the plasma treater, followed by a final clean with alcohol or IPA solution.

Note: The use of protective gloves is essential throughout any contact with the plasma treater, particularly the electrode surfaces.

Thorough Clean

If however a hard, abrasive deposit is found adhered to the plates from a deposit from the film, or Al Ox formed at the edge of the racetrack during prolonged process running of the plasma treater, a through clean is required.

Definition: A thorough clean refers to a complete strip down of the copper edge cover and ceramics in order to remove the electrodes for cleaning. A thorough clean is required if arcing events are being generated/observed in the area of the electrode outside the racetrack.

Procedure: Remove the copper front covers and seal the outer edges of the plates as shown below



Figure 8.8 - 12 Electrode Plates With Masking Around The Edges

It is recommended to use a form of masking tape to seal around the electrode plates to prevent debris and dust particles falling down the edges of the plasma treater.

It is recommended to seal the electrode plate to cover/ protect the dielectric coating around the circumference

In order to perform a 'through clean' of the plasma treater, it is required to use a rotary sanding wheel which has been found to be an effective way of removing the deposit. After using the abrasive sanding wheel, a finer grade abrasive material is required to remove any scratch marks evident on the plates.

Note: please use a minimum of 250 grade abrasive pads, ideally 1000 grade.

Following the removal of the contamination it is advised to thoroughly vacuum all surfaces of the electrode, followed by a final full clean with alcohol or IPA solution.

	Note!
	The use of protective gloves is essential throughout any contact with the plasma treater, particularly the electrode surfaces.

8.9 Metallizer Cleaning Schedule

The following is intended as a guide only and generally users will find their own sequence of cleaning based upon their particular operating methods and time periods.

Cleaning Between Runs

1. Visually check source area and remove loose debris from coating shield.
2. Visually check wire feed area for any potential wire feed snags.
3. Check that there is sufficient Aluminium wire on the reels for the next run.
4. Use small brush or a scraper to clean oxide off the evaporators.
5. Check and clean wire feed spouts as required.
6. Check vessel joint band and 'O' Ring for obstructing debris.
7. Dry the cryogenic coils and walls with a lint free cloth.
8. Remove any loose substrate debris

Shift Clean Up (To be performed at the end of each shift)

1. All the above plus the following.
2. Inspect the shields, remove any build up and paint with Boron Nitride.
3. Check the rubber covered rollers for aluminium build up and surface slickness.
4. Remove any substrate debris in the mechanism.

Weekly Clean Up

Source Area

1. Clean source box to bare metal.
2. Clean copper evaporator posts and blocks to bare copper.
3. Remove wall shields (if fitted) and clean.
4. Clean top and bottom of movable shutter.
5. Wipe all supports down.
6. Wire feed tubes to be cleaned or replaced.
7. Check the condition of the flexible cloth

Coating Area

1. Clean all shielding down to original surface.
2. Wipe all surface areas.
3. Wipe zone sealing strip with clean cloth and regrease it.
4. Wipe main chamber 'O' Ring and regrease it
5. Check zone separation gap around the rollers is clear of debris.
6. Remove all aluminium debris from above the zone separation seal.

Winding Mechanism Area

1. Clean all rollers as described in the previous section
2. Remove any debris build up.
3. Clean any oil spillage etc.
4. Check for substrate trapped between the monitor beam and the ebonite roller.

Viewing Windows

1. Clean all viewing windows inside using dry cloth.

Plasma Unit

1. Clean electrode plates and remove any debris

General Cleaning

1. Clean desk and cabinet surfaces.
2. Clean loading area and metallizing room.
3. Empty all waste and scrap bins.
4. Empty vacuum cleaner.
5. Clean pumping group,

The above schedule is provided to be used as a guide. Individual customers may find that they will have to modify this schedule depending upon their operational experiences.

8.10 Disposal

General Notes

After dismantling any sections of the machine, for disposal, the components and process media must be separated in the following categories:

- Process media.
- Components contaminated with process media or which had contact with process media.
- Other machine components.

	Note!
	Components and process media must be disposed of in accordance with local site regulations where the machine is operated.

Process Media

	Warning !
	Pump oil and oil mists contaminated with hazardous substances can be hazardous to your health. Skin or eye contact with pump oil or inhaling oil mists can be hazardous to your health. Wear protective clothing, oil-resistant gloves, goggles and a P3 respiratory protection mask when opening vacuum lines, vacuum valves or vacuum pumps containing harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.

	Caution !
	The process media of the vacuum pumps are harmful to the environment if not disposed of correctly. Never allow pump oil or parts dipped in pump oil to penetrate the supply of drinking water. Pump oil must be disposed of as hazardous waste.

Process media for the machine should be returned to the supplier in unopened original containers.

Process media in opened original containers and used must be disposed of as hazardous waste in accordance with local regulations.

Consumables items containing process media (filter cartridges, cleaning rags, etc.) must also be disposed of as hazardous waste.

Components

All machine components must be separated in the following classes, and recycled or disposed of, depending on the material:

- Aluminium
- Steel
- Stainless steel
- Precious metal
- Copper
- Glass
- Plastic
- Electronic or electrical parts
- Batteries

Components contaminated with operating media must be thoroughly cleaned before further processing.

Plasma Treater Target Materials

Precious metal target materials can be recycled.

Other target materials must be disposed of in accordance with the safety data sheets and local regulations.

HEALTH HAZARDS OF ALUMINIUM

By Wan Zainal Azman Wan Abdullah

Aluminium is the most widely used metal in the world after iron. Comprising about 8% of the earth's crust, it is the third most common element. It has a great affinity for oxygen and therefore occurs only in the oxidized form, mainly as alumina. It is found in almost 300 different minerals and widely distributed in rocks, clay and soils.

The sources of aluminium are largely unknown to the public. As well as occurring naturally in food and water, aluminium may be found in many processed foods, cosmetics, toothpaste, antiperspirants and adjuvants in various parenteral preparations. It could also be found in different amounts in vegetation and animals according to its presence in the environment.

Because of the widespread distribution of aluminium compounds through food and drink intake and the air that we breathe in, exposure to it is unavoidable. Daily intake of aluminium for people in the general population has been reported to range from 9 mg/day to 36 mg/day, with an average of 20 mg/day.

Normal ingestion of aluminium in the diet is about 3mg/day to 5 mg/day, of which only about 15 microgram is absorbed through the wall of the gastrointestinal tract. In humans, this small amount is usually excreted through the kidneys. However, with intakes of greater than 1000 mg/day, retention does occur.

The most common foods that contain substantial amounts of aluminium-containing additives include processed cheeses, baking powders, cake mixes, frozen dough's, pancake mixes, self-raising flours and pickled vegetables.

Some forms of aluminium, like insoluble aluminium compounds that constitute aluminium hydroxide mixture are poorly absorbed from the gastrointestinal tract. Certain organic ligands of aluminium, such as aluminium citrate, pass very rapidly from the food chain into the blood. Once absorbed, aluminium is bound extensively to protein.

Aluminium compounds can also affect absorption of other elements in the gastrointestinal tract and alter intestinal function. Aluminium inhibits fluoride absorption and may decrease the absorption of calcium and iron compounds. It possibly hinders the absorption of cholesterol too by forming an aluminium-pectin complex that binds fats to nondigestible vegetable fibers.

Aluminium salts are found in some common medicines as well. These include analgesics (buffered aspirins), antacids, anti-diarrhoeal products, anti-haemorrhoidal preparations.

Buffered aspirins may be used over prolonged periods and in high dosage by elderly arthritic patients. Three to 9gm of aspirin daily could lead to the ingestion of 126mg to 728mg aluminium daily from this source. As for antacids, the aluminium content per dose may range from 35mg to 208mg. If taken at the maximum dosage, 24 tablets could lead to a possible daily intake of between 840mg and 5,000mg, resulting in acute toxicity.

Another important route of exposure to aluminium is through the contamination of fluids used in parenteral nutrition therapy, particularly in patients with renal impairment.

Aluminium retention in the body results in its accumulation in the bone and decreased mineralisation. The principal sources of aluminium contamination in typical total parenteral nutrition (TPN) formulations are potassium phosphates, amino acids, and calcium gluconate, gluceptate or glutionate salts.

The chronic use of medicines and fruit juices containing citrates should also be avoided in patients with impaired renal function, especially if prescribed together with antacids containing aluminium hydroxide because they greatly increase absorption of aluminium.

The most serious problem of aluminium toxicity involves patients with end-stage chronic renal failure managed by long-term intermittent hemodialysis. It can cause neurological disturbance known as dialysis encephalopathy syndrome (DES).

The symptoms include memory loss, malaise, tremor, jerking movements, depression and anxiety. A toxin in the untreated tap water used for dialysis is suspected to be the cause.

Children can also be exposed to aluminium poisoning, particularly children with chronic renal failure. This could be due to the use of aluminium contaminated dialysate or orally administered aluminium-containing phosphate binders.

Milk products are high in phosphorus. Relatively large amounts of phosphate binders are required to maintain normal serum phosphate levels and they may be a major factor in raised aluminium levels.

Other clinicians have confirmed that infant formulae contain aluminium and its concentration is generally higher than that of cow or human milk.

Other contributing factors could be the immature gastrointestinal tracts of newborns and infants that could absorb an increased amount of orally administered aluminium as well as growing bones that could be especially exposed to its accumulation.

Aluminium poisoning has also been linked to Alzheimer's disease, a progressive neurodegenerative disorder associated with dementia. Higher concentrations of aluminium have been found in brain tissue of patients dying of Alzheimer's disease. The use of aluminium-based chemicals in the treatment of its treatment is believed to have caused that.

Occupational exposure also occurs in workers exposed to aluminium in the work place. Coughing, increased phlegm production and a sense of irritation are noted in this group of workers. These effects are believed to be related to increased aluminium dust concentration in the environment.

Chronic inhalation of fine aluminium dust or vapors has resulted in fibrosis of the lung, asthma, dyspnea, cough and pneumothorax.

In this aspect, prevention of toxicity is the major objective of the public health policies. As such, environmental and occupational hygienic programmes must be monitored and maintained. *The writer is a Science Officer at the National Poison Centre, Universiti Sains Malaysia, Penang.*

Source of Further Information

<http://www.world-aluminium.org/iai/publications/index.html>

The Measurement of **Dust** Exposures in the Aluminium Industry February 1989 UK: £0.60; International: \$6 per copy The Measurement of Employee Exposures in Aluminium Reduction Plants - Physical Exposures April 1992 UK: £0.60; ...

International Chemical Safety Cards

ALUMINIUM POWDER

ICSC: 0988



4



National Institute for Occupational Safety and Health



Al



Al



Atomic mass: 27.0



Aluminum powder

ICSC # 0988
CAS # 7429-90-5
RTECS # BD0330000
UN # 1396 (uncoated)
EC # 013-001-00-6

TYPES OF HAZARD/ EXPOSURE	ACUTE HAZARDS/ SYMPTOMS	PREVENTION	FIRST AID/ FIRE FIGHTING
FIRE	Flammable.	NO contact with acid(s) alcohol oxidants water	Dry sand, special powder. NO water. NO carbon dioxide, foam.

EXPLOSION	Finely dispersed particles form explosive mixtures in air. Risk of fire and explosion on contact with acid(s) alcohol oxidants water	Prevent deposition of dust; closed system, dust explosion-proof electrical equipment and lighting.	
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EXPOSURE		PREVENT DISPERSION OF DUST!	
•INHALATION		Local exhaust or breathing protection.	Fresh air, rest.
•SKIN		Protective gloves.	Rinse skin with plenty of water or shower.
•EYES		Safety goggles.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then take to a doctor.
•INGESTION		Do not eat, drink, or smoke during work.	Rinse mouth.

SPILLAGE DISPOSAL	STORAGE	PACKAGING & LABELLING
Sweep spilled substance into covered and dry containers. (Extra personal protection: P2 filter respirator for harmful particles).	Separated from strong oxidants, strong bases, strong acids water See Chemical Dangers. Dry. Well closed.	F symbol R: 15-17 S: 2-7/8-43 UN Hazard Class: 4.3 UN Packing Group: II

SEE IMPORTANT INFORMATION ON BACK

ICSC: 0988

Prepared in the context of cooperation between the International Programme on Chemical Safety & the Commission of the European Communities (C) IPCS CEC 2000. No modifications to the International version have been made except to add the OSHA PELs, NIOSH RELs and NIOSH IDLH values.

International Chemical Safety Cards

ALUMINIUM POWDER

ICSC: 0988

I M P O R T A N T D A T A	PHYSICAL STATE; APPEARANCE: WHITE GREY POWDER	ROUTES OF EXPOSURE: The substance can be absorbed into the body by inhalation	
	PHYSICAL DANGERS: Dust explosion possible if in powder or granular form, mixed with air.	INHALATION RISK: Evaporation at 20°C is negligible; a harmful concentration of airborne particles can, however, be reached quickly.	
	CHEMICAL DANGERS: Reacts with water alcohols violently with oxidants strong acids strong bases causing fire and explosion hazard.	EFFECTS OF SHORT-TERM EXPOSURE:	
	OCCUPATIONAL EXPOSURE LIMITS: TLV (as Al): 5 mg/m ³ (as TWA) (pyro powder) (ACGIH 2000). TLV: 10 mg/m ³ (as TWA) (metal dust) (ACGIH 2000). OSHA PEL: TWA 15 mg/m ³ (total) TWA 5 mg/m ³ (resp) NIOSH REL: TWA 10 mg/m ³ (total) TWA 5 mg/m ³ (resp) NIOSH IDLH: No data	EFFECTS OF LONG-TERM OR REPEATED EXPOSURE: Lungs may be affected by repeated or prolonged exposure to dust particles. The substance may have effects on the nervous system , resulting in impaired functions	
	PHYSICAL PROPERTIES	Boiling point: 2327°C Melting point: 660°C Density: 2.7 g/cm ³	Solubility in water: none reaction Auto-ignition temperature: 590°C
		ENVIRONMENTAL DATA	
		NOTES	
	Other UN number: UN1309 Aluminium powder, coated, Hazard class 4.1, Pack group II. Transport Emergency Card: TEC (R)-43G14 NFPA Code: H0; F3; R1; (uncoated)		
	ADDITIONAL INFORMATION		
ICSC: 0988		ALUMINIUM POWDER	
(C) IPCS, CEC, 2000			
IMPORTANT LEGAL NOTICE:	Neither NIOSH, the CEC or the IPCS nor any person acting on behalf of NIOSH, the CEC or the IPCS is responsible for the use which might be made of this information. This card contains the collective views of the IPCS Peer Review Committee and may not reflect in all cases all the detailed requirements included in national legislation on the subject. The user should verify compliance of the cards with the relevant legislation in the country of use. The only modifications made to produce the U.S. version is inclusion of the OSHA PELs, NIOSH RELs and NIOSH IDLH values.		

International Chemical Safety Cards

ALUMINIUM OXIDE

ICSC: 0351



National Institute for
Occupational Safety and Health







alpha-Aluminu

Aluminum trioxide
Al₂O₃
Molecular mass: 101.9

ICSC # 0351
CAS # 1344-28-1
RTECS # BD1200000

TYPES OF HAZARD/ EXPOSURE	ACUTE HAZARDS/ SYMPTOMS	PREVENTION	FIRST AID/ FIRE FIGHTING
FIRE	Not combustible.		In case of fire in the surroundings: all extinguishing agents allowed.
EXPLOSION			
EXPOSURE		PREVENT DISPERSION OF DUST!	
•INHALATION	Cough.	Local exhaust or breathing protection.	Fresh air, rest.
•SKIN		Protective gloves.	Rinse and then wash skin with water and soap.
•EYES	Redness.	Safety goggles, or eye protection in combination with breathing protection.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then take to a doctor.
•INGESTION		Do not eat, drink, or smoke during work.	Rinse mouth.
SPILLAGE DISPOSAL		STORAGE	PACKAGING & LABELLING
Sweep spilled substance into containers; if appropriate, moisten first to prevent dusting. Wash away remainder with plenty of water. (Extra personal protection: P1 filter respirator for inert particles).			R: S:
SEE IMPORTANT INFORMATION ON BACK			
ICSC: 0351		Prepared in the context of cooperation between the International Programme on Chemical Safety & the Commission of the European Communities (C) IPCS CEC 2000. No modifications to the International version have been made except to add the OSHA PELs, NIOSH RELs and NIOSH IDLH values.	

International Chemical Safety Cards

ALUMINIUM OXIDE

ICSC: 0351

I M P O R T A N T D A T A	PHYSICAL STATE; APPEARANCE: WHITE POWDER. PHYSICAL DANGERS: CHEMICAL DANGERS: OCCUPATIONAL EXPOSURE LIMITS: TLV: 10 mg/m³ (as TWA) A4, for particulate matter containing no asbestos and 1% crystalline silica (ACGIH 2000). MAK: 1.5 mg/m³; respirable fraction of the aerosol (1999) MAK: class II,2 (1999) OSHA PEL: TWA 15 mg/m³ (total) TWA 5 mg/m³ (resp) NIOSH REL: See Appendix D NIOSH IDLH: No data ROUTES OF EXPOSURE: The substance can be absorbed into the body by inhalation of its aerosol. INHALATION RISK: Evaporation at 20°C is negligible; a harmful concentration of airborne particles can, however, be reached quickly. EFFECTS OF SHORT-TERM EXPOSURE: Inhalation of high concentrations of dusts of this substance may cause eyes and upper respiratory tract irritation. EFFECTS OF LONG-TERM OR REPEATED EXPOSURE: The substance may have effects on the central nervous system
PHYSICAL PROPERTIES	Boiling point: 3000°C Melting point: 2054°C Density: 3.97 g/cm³ Solubility in water: none
ENVIRONMENTAL DATA	
NOTES	
There is a different and hard crystalline form of aluminium oxide which occurs abundantly in nature under the name corundum (CAS 1302-74-5). Other melting points: 2015°C (approx.) (corundum). Occurs also as the minerals: bauxite, bayerite, boehmite, diaspore, gibbsite.	
ADDITIONAL INFORMATION	
ICSC: 0351 ALUMINIUM OXIDE (C) IPCS, CEC, 2000	

**IMPORTANT
LEGAL
NOTICE:**

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SECTION 9

PROCESS CONSUMABLES

Section 9.1 Release Paint
Section 9.2 Graphite Tape
Section 9.3 Thermal Evaporators
Section 9.4 Aluminium Wire
Section 9.5 Lubricants
Section 9.6 Antifreeze
Section 9.7 Boat Contact Blocks
Section 9.8 Wire feed Tubes
Section 9.9 Graphite Tape – Coating Shield
Section 9.10 Gases for Plasma Process

The process consumable items required for the machine are, evaporators, aluminium wire, lubricating oils, vacuum greases, graphite tape and release paint.

There are some mechanical parts upon the machine, which will require changing periodically, and these are classed as wear parts. These include for example; bearings, bushes (bushings), boat contact posts, and wire feed tubes

Electrical items such as internal light bulbs, desk illuminated indicator bulbs, are classed as consumable items, as they are expected to "burn out" over a period of time.

9.1 Release Paint

Function

Release paint is used as a release agent to reduce the adhesion of aluminium to machine surfaces. This will save a considerable amount of time in cleaning down the machine.

Boron Nitride Water-Based Release Coating:

Directions for use:

Shake the container vigorously and apply with a brush to give a uniform coating. If necessary dilute with water or ethanol.

	Note!
	Do not allow boron nitride paint to come into contact with the evaporator clamping block contact area.

Package:

Available in polythene screw top containers. Boron Nitride is available in either a powder or liquid form.

Available from all the evaporator manufacturers.



Figure 9.1 - 1 Photographs of Release Paint Containers - Consumable

Aquadag Water-Based Graphite Coating

Aquadag is a colloidal dispersion of extremely fine, pure graphite in a water carrier that dries to form an adherent film on virtually all surfaces. It exhibits excellent parting abilities and an extremely low coefficient of friction.

Surface Preparation

Substrates must be clean and dry. No further pre-treatment is required.

Dilution

Aquadag is supplied as a concentrate and may be diluted before use.

Dilution ratios vary according to end-use requirements. Contact Acheson Colloids Company for further information.

	Note!
	Always add the water to the concentrate, never vice versa

Stir the concentrate thoroughly before diluting to break up the gel structure. Then, stirring gently, slowly add small amounts of water to the concentrate until a fluid consistency is obtained with no small gels. Stir in the remaining water to attain the desired dilution. Tightly reseal container after use.

Application

Aquadag may be applied using conventional spray, brush, or dip methods.

Curing

Aquadag will air dry to touch in 10-15 minutes, depending on room temperature and humidity.

9.2 Graphite Tape

Function

Used to provide a good electrical contact between the thermal evaporators and the water-cooled clamping blocks.

Directions for Use

Cut into strips the same width as the clamping blocks. Use a single thickness to cover the inside faces the of the evaporator clamps before inserting the evaporator.

Availability

Available in rolls 30m long x 15mm wide x 0.2mm thick. Available from all the evaporator manufactures and other suppliers.



Figure 9.2 - 1 Photograph of Graphite Tape – Consumable

9.3 Thermal Evaporators

There are a number of suppliers of thermal evaporators. When ordering new supplies refer to the evaporator information in the machine specification.

General Vacuum Equipment machines are normally designed to use 150 or 125mm long evaporators. The size to be used in individual machines is specified in the technical specification.

The above sizes are common to the metallizing industry and are available from all the suppliers.

The amount of evaporators required in a normal production environment will depend upon the deposition level during coating cycles. Normal life should be approximately 15 hours (approximately 22.5 cycles based upon 40 minute evaporation cycles). The number of evaporators that are mounted in the machine is listed in the machine's technical specification.

The machine will automatically only use a relevant number of the evaporators that is proportional to the width of the material being coated.



Figure 9.3 - 1 Photograph of Typical Thermal Evaporators

9.4 Aluminium Wire

Several different alloys of aluminium wire are available and used in the metallizing industry. We recommend alloy 1080, which has a purity of 99.80 percent.

However, less pure wire alloys, typically 1100, is extensively used in the metallizing industry.

Some users, particularly in the capacitor, industry use alloy 1199 which is 99.99 percent pure aluminium.

The alloys that are normally readily available are as follows;

Alloy	1100	1350	1075	1080	1188	1199
% purity	99.00	99.50	99.75	99.80	99.88	99.99

Table 9.4 - 1 Alloy Purity

The wire used in the metallizer will be generally 1.6 mm or 2.0mm diameter. The correct size for this machine will be listed in the machine specification.

The correct temper of wire is important. All General Vacuum Equipment machines require H18(USA) or H8(UK) temper

It is essential that the wire used in the machine is clean (i.e.: solvent degreased). Wire is generally purchased on individual spools is supplied clean.

However, some metallizing companies buy wire in bulk and wind it onto spools themselves. Bulk wire can be bought in spools of 125 lbs and can be wound onto smaller spools.

Usage of the wire will depend upon the running speed and the thickness of deposits.

The General Vacuum Equipment K4000 machine is designed to accept industrial standard 10.0kg spools of wire.

9.5 Lubricants

Ensure that only the correct lubricants recommended for the machine are used.

Use of incorrect grease within the vacuum chamber can lead to vacuum problems.

See the maintenance manual (Section 13.0) for the recommended lubricant types and equivalents.

9.6 Antifreeze

The chiller unit uses a medium made of a mixture of ethylene glycol and water. The quality of the ethylene glycol should be as follows;

Univar Ltd Pine Street South Bank Road Cargo Fleet Middlesbrough TS3 8BD United Kingdom Tel: +44 (0) 1642 227388 Fax: +44 (0) 1642 221958 E-Mail: info@univareurope.com www.univar.co.uk			
DATA SHEET			
UNIVAR BS6580 COOLANT			
Univar BS6580 Coolant is a nitrite, amine, silicate, borate and phosphate free ethylene glycol based engine coolant concentrate, which meets the requirements of British Standard BS6580:1992. A 30-50% aqueous solution will function both as antifreeze and coolant and impart very good corrosion protection to all cooling system metals and alloys found in internal combustion engines.			
Chemical Nature	Ethylene glycol with inhibitors		
Appearance	Clear Liquid Colour as specified		
Technical Data	Specific gravity 25/25oC	1.110 – 1.140	ASTM D 941
	Boiling Point	oC >150	ASTM D 1120
	Flash Point,	oC >110	DIN ISO 2592
	Water Content,	% Max.6	ASTM D 1123
	Miscibility with Water	Free miscible	
	Hard water stability	Stable no precipitation	
	Freezing Point 50% vol. in water 33.3% vol. in water	Below –34oC Below –18oC	ASTM D 1177
	pH-value	33.3% vol. in water 7.7 – 8.5	ASTM D 1287
PLEASE NOTE THESE FIGURES ARE TYPICAL VALUES			

ALUMINIUM HEAT TRANSFER CORROSION TEST

	Weight change in mg/cm ² /week
BS 6580:1992 (max.)	1.0
E&E BS6580 Coolant	0.3

COOLANT REPLACEMENT

Unlike our high performance engine coolants, it is recommended to replace BS 6580 engine coolant every year.

HANDLING

The usual precautions for handling chemicals, together with the information and advice contained in our Safety Data Sheet, should be observed.

1. Identification of the substance/preparation and company/undertaking.**NAME Mono Ethylene Glycol****S.D.S. No. M027****Synonyms** Ethylene glycol * MEG * Ethane-1,2-diol ***Supplied by: Univar Ltd**

46 Peckover Street

BRADFORD

West Yorkshire

BD1 5BD

Tel: 01274 377000

Fax: 01274 737058

Emergency No. (Office hours): 01274 377070

Emergency No. (24 hours): 01865 407333

2. Composition/information on ingredients.

Contains:

Chemical	% Conc	Classification	Exposure	CAS	EINECS
Ethylene glycol	>90	Xn;R22	10 mg/m ³	107-21-1	203-473-3

3. Hazards identification.

Ingestion of excessive doses may cause central nervous system effects, cardiopulmonary effects and kidney failure. Estimated lethal dose for an average person is 100ml. Excessive exposure may cause irritation to upper respiratory tract.

4. First Aid measures.

Exposure Route Symptom Treatment

Inhalation Irritation of breathing passages. Remove from exposure, rest and keep warm. In severe cases, or if recovery is not rapid or complete seek medical attention.

Skin Contact Mild irritation. Drench the skin with plenty of water.

Remove contaminated clothing and wash before reuse. If large areas of the skin is damaged or if irritation persists seek medical attention.

Eye Contact Mild irritation Irrigate thoroughly with water for at least 10 minutes. Obtain medical attention.

Ingestion Irritation of upper digestive tract nausea and vomiting.

Wash out mouth with water. Do not induce vomiting. If patient is conscious, give water to drink. If patient feels unwell seek medical attention.

Immediate Treatment / Antidote: Supportive care. Treatment based on judgement of the physician in response to reaction of the patient

S.D.S. No M027 Page 1 of 4**S.D.S. No M027 NAME: Mono Ethylene Glycol****5. Fire Fighting measures.****Suitable Extinguishers** Water fog, alcohol resistant foam, CO₂, dry chemical.**Hazardous Combustion Products** Oxides of carbon.

Special Equipment for Fire Fighting Self contained breathing apparatus and full protective clothing.

6. Accidental Release measures.**Safety Precautions** Wear appropriate PPE - See section 8**Environmental Precautions** Avoid entry into sewers or water courses.**Clean up Procedure** Absorb or bund spillage, transfer to suitable containers for waste disposal.**7. Handling & Storage.****Handling Ventilation** Good general ventilation.

Recommended procedures & equipment Avoid skin and eye contact. Prevent accidental ingestion.

Storage Temperature range Ambient**Keep away from** See section 10**Suitable storage Media** Store in original containers.**Precautions against static discharge****8. Exposure Control/personal protection.****Exposure Limits** 10 mg/m³ 8Hr TWA particulates 52 mg/m³ 8HrTWA vapour 104 mg/m³ STEL**Type** EH40**Monitoring Method** as Ethane-1,2-diol

Protective Measures**Respiratory:** Type approved RPE for organic vapours and mists if required.**Hand:** Impervious gloves**Eye:** Goggles**Skin:** Overalls**Hygiene Measures** Always wash thoroughly after handling chemicals.**S.D.S. No M027 Page 2 of 4****S.D.S. No M027 NAME: Mono Ethylene Glycol****9. Physical & Chemical Properties.****Appearance** Clear, colourless liquid.**PH** 6-8**Boiling Point/range** 198°C**Melting Point/range** -13°C**Flash point** 116°C closed cup**Autoignition temperature** 400°C**Explosive limits** 3.2% - 15.3% v/v**Vapour Pressure** <10 Pa @ 20°C**Relative density** 1.115**Solubility** Completely miscible in water**Partition coefficient** -1.36 (ethylene glycol)**10. Stability & Reactivity.****Stability** Stable**Known Hazardous reactions** Violent reactions with oxidizing agents.**Materials to avoid** Oxidizing agents.**Hazardous decomposition products** Oxides of carbon.**11. Toxicological Information.****Effects**

Excessive exposure may cause Central Nervous System effects. Lethal dose in humans is approximately 1.6g/Kg.

LD50 6000-13000 mg/Kg oral-rat**12. Ecological Information.****Environmental Effects****Mobility** Liquid with low volatility**Degradability** BOD28/COD = 56 - 64%; COD = 1.29 gO2/g**Aquatic Toxicity** LC50, goldfish, 24 h static >5000 mg/l**Bioaccumulative potential** Low.**S.D.S. No M027 Page 3 of 4****S.D.S. No M027 NAME: Mono Ethylene Glycol****13. Disposal considerations.****Substance** Via an authorized waste disposal contractor to an approved waste disposal site, observing all local and national regulations.**Container** As substance. Use containers must not be cut up or punctured until completely purged of product residues**14. Transport Information.****Not regulated for transport.****15. Regulatory Information.****Supply label details** CHIP 2000**Label Name** Mono Ethylene glycol.**Symbols** Harmful**Risk Phrases** 22 Harmful if swallowed.**Safety Phrases** 2 Keep out of reach of children.**E.C. No** 203-473-3**Use of this material may be governed by the following regulations:-****Users are advised to consult these regulations for further information.**

The information contained in this data sheet does not constitute an assessment of workplace risks.

16. Other Information.**This material is usually used for:**

Heat exchange medium.

It must not be used for:

Further details may be available upon request from your local Univar Ltd distribution site.

Legal Disclaimer:

The above information is based on the present state of our knowledge of the product at the time of publication. It is given in good faith, no warranty is implied with respect to the quality or the specification of the product. The user must satisfy himself that the product is entirely suitable for his purpose.

Revision No 5 Dated March 2004

Replaces S.D.S. Dated April 2003

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9.7 Boat Contact Blocks

The copper blocks that the boats are held between are considered a consumable. Please refer to sections 2 and 8 of this manual for information on why they wear out.

Live and Neutral Blocks manufactured to a GVE drawing

Details

- The copper can be purchased in rectangular lengths
- The copper is described as highly conductive, corrosion resistant and malleable.
- CU 99.90
- PB 0.005
- Bi 0.0070

Specification

High Density, Highly Conductive Electrical grade copper

Properties

Tensile Strength - 14-25-tons/sq. inch Vickers Hardness - 45-115 VH

Source of Supply

Precious Metal Dealers

<http://www.anchorbronze.com/c10100.htm>

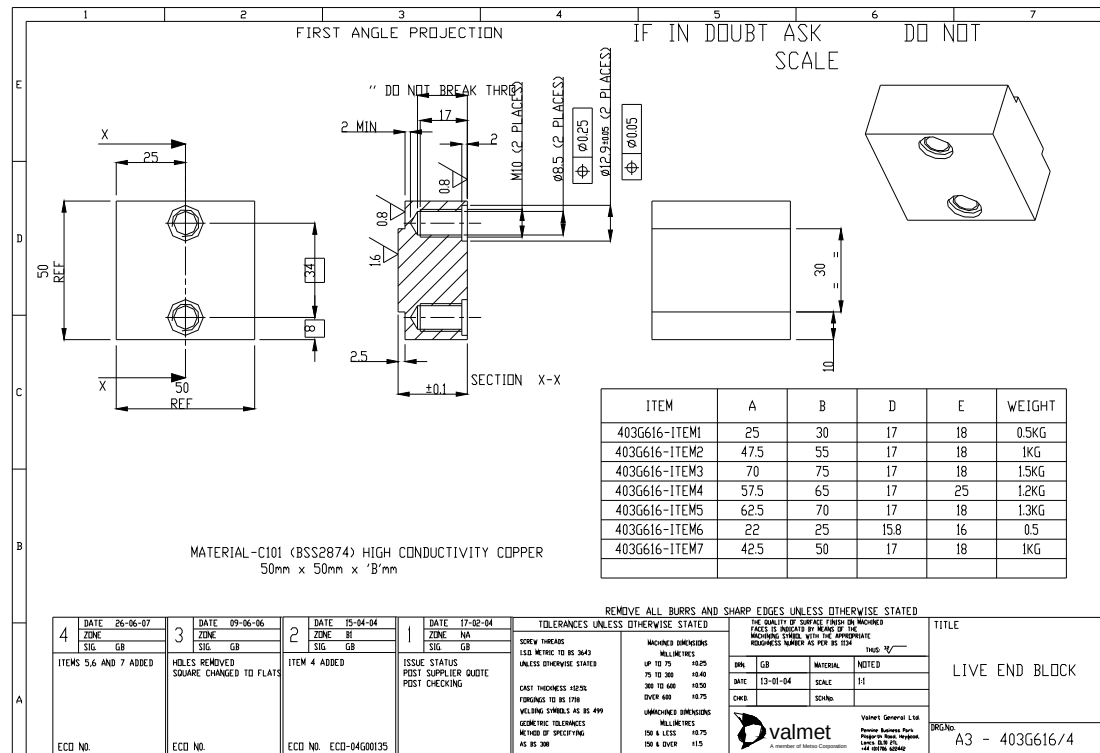


Figure 9.7 - 1 Boat Contact Block

9.8 Wire Feed Tubes

General Vacuum Equipment will supply a machine with either copper or stainless steel wire fed wear tubes. These are the tubes that pass into the source area and are coated with aluminium as it is evaporated.

It is normal to replace these tubes on a regular basis.

Copper Tube

Details

- Tube wall thickness 1mm
- Tube Outside diameter 6mm
- Tube Inside diameter 4mm

Specification

Half Hard Copper Tube- Allows easy formation of guide tube

Source of supply

Enots UK - Pneumatic Distributor

Stainless Steel Tube

Details

- Tube wall thickness >1mm
- Tube Outside diameter >6mm
- Tube Inside diameter <3mm

Specification

- Standard Stainless Tube
- More difficult formation of guide tube
- Withstands Heat

Source of supply

Steel Suppliers

9.9 Graphite tape For Coating Shield

There is a requirement to place a strip of graphite tape behind the sine wave fingers when installing them on the machine.

Available in rolls 10m long x 40mm wide x 0.5mm thick.

An explanation of how we apply the tape use can be found in section 6 of this manual.

We purchase our tape from Gee Graphite, whose web site is

www.geegraphite.com/materials/tape.htm

9.10 Gases for Plasma Process

It has been found through experimentation that the most common gas used is a pre-mixed bottle of 80/20 argon/oxygen mixture works well in most applications.

GENERAL VACUUM EQUIPMENT provide a mass flow regulator for gas control and a fix connection point to the festoon to allow a connection to be made.

Correctly rated gas regulators should be used.

	Warning !
	The following safety precautions should be adhered to at all times. This equipment is designed to run on a process gas containing less than 20% oxygen. Use of process gases containing more than 20% require specialized pumping systems for safe operation. Therefore always use the correct gas mixture

The gas purity should normally be 100%.

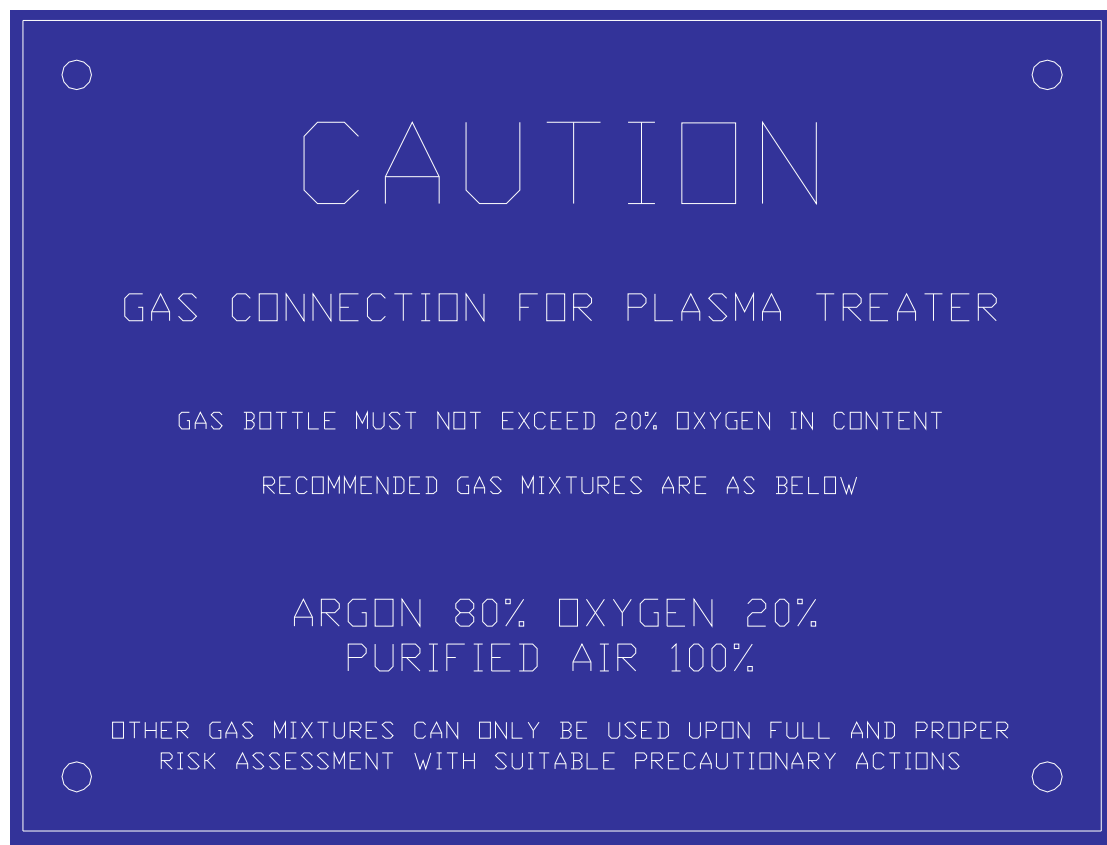


Figure 9.10 - 1 Plasma Treater Gas Mixture Warning

Contact General Vacuum Equipment for further advice.

SECTION 10

VIEWING OF OPERATION

Section 10.1 Source Area

Section 10.2 Unwind Roll Area

Section 10.3 Rewind Roll

Section 10.4 Coated Substrate

Section 10.5 CCTV System for Viewing the Substrate

The machine is designed to allow the operator clear viewing of various important areas within the chamber.

10.2 Unwind Roll Area

Two round windows are provided at the rear of the machine to allow viewing of the unwind roll.

This allows viewing of both unwind and rewind rolls from this position.

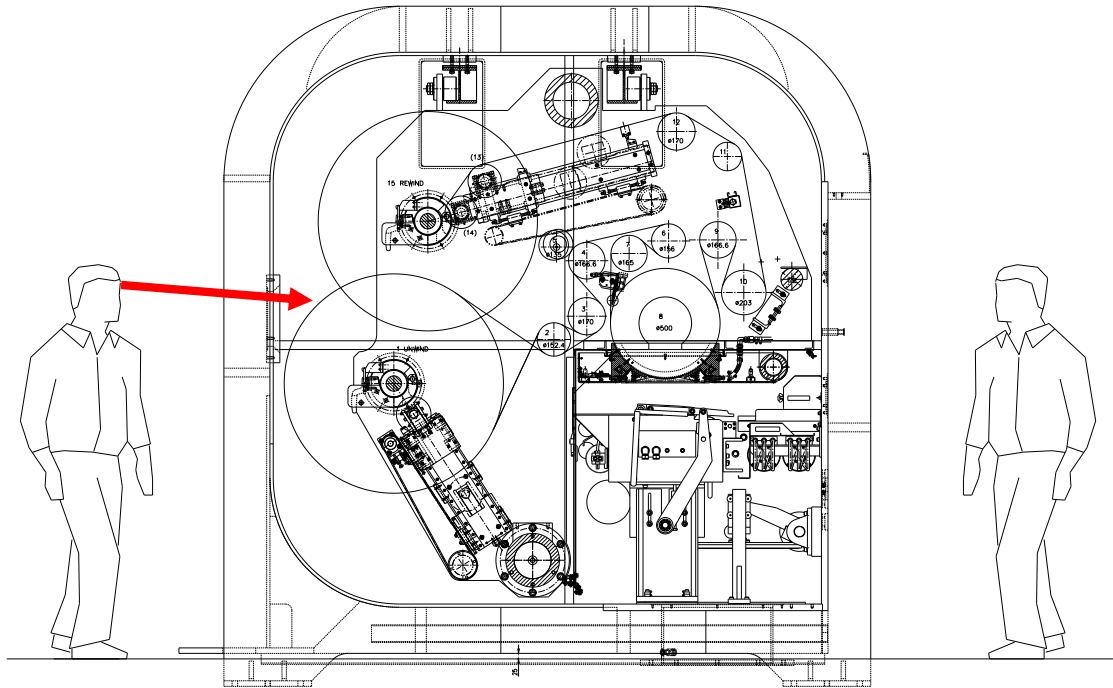


Figure 10.2 - 1 Viewing Of The Unwind Roll On A Metallizer

10.3 Rewind Roll

The rewind roll can be viewed from the machine operating position via CCTV cameras located inside the chamber.

The roll can be viewed via the CCTV monitor located in the operator location.

This monitoring position is used to watch for edge build up and also for edge slippage.

10.5 CCTV System for Viewing the Substrate

Within the vacuum chamber, there are a number of CCTV cameras. These are housed in protective domes and have been set up to give the user a remote viewing facility.



Figure 10.5 - 1 Location of CCTV cameras in the chamber

The individual display of any of the cameras is seen on the display located at the front of the machine.



Figure 10.5 - 2 Photograph of CCTV Monitor at Operator Station.

The selection controls allow the user to select one of the internal cameras or multiple cameras in a sequence.

SECTION 11 CONTROL/INTERLOCK DESCRIPTIONS

The following section describes the purpose and location of all the control push-buttons and switches on the machine.

- Section 11.1 Operator MMI Panel
- Section 11.2 Rear Local Station
- Section 11.3 Front Local Station
- Section 11.4 Lockout Isolator
- Section 11.5 MMI Interface Screens

11.1 Operator MMI Panel

The following controls are located upon the operator MMI Panel. A brief description of their function is described.



Figure 11.1 - 1 MMI Interface Screen

Mechanism Cart Close Push-button

This push-button is used to drive the winding cart into the chamber from its parked open position. Use in conjunction with the ENABLE push-button.

Mechanism Cart Open Push-button

This push-button is used to move the winding cart out of the chamber to its fully open position. Use in conjunction with the ENABLE push-button.

Vacuum Push-button

This push-button is used to start the vacuum sequence in conjunction with the ENABLE push-button. Pressing both buttons will also drive in the mechanism cart for a time limited period.

Enable Push-button

This push-button is used in conjunction with the MECHANISM CART OPEN OR CLOSED push buttons.

Emergency Stop Reset Button

When the Emergency stop circuit has been utilized, this button is used to reset the control circuit back into an operational state.

This action is also required when starting the machine.

Guard Stop Reset Button

When the guard stop circuit has been utilized, this button is used to reset the control circuit back into an operational state.

This action is also required when starting the machine

E- Stop Push-button:

This push-button will place the drive system into a controlled fast stop mode. This time period is set to be 10 seconds. After this period of time the drive system will remove power to the drive control units and allow the motors to "Coast" to a stopped position.

11.2 Rear Local Control Station

This station is located at the rear of the machine for control of the layarm system.



Figure 11.2 - 1 Photograph of Rear Local Control Station

Rewind Layarm Raise Push-button

This push-button is used to raise the layarm when the chamber is fully open.

Rewind Layarm Lower Push-button

This push-button is used to lower the layarm when the chamber is fully open.

Unwind Layarm Raise Push-button

This push-button is used to raise the layarm when the chamber is fully open.

Unwind Layarm Lower Push-button

This push-button is used to lower the layarm when the chamber is fully open.

Inch Push-button

This push-button is used to inch (Jog) the substrate through the winding mechanism.

11.3 Front Local Station

This station is located at the front of the winding cart next to the lockout isolator switch. It is provided to allow control of the coating shield raise and lower circuit.

Raise Push-button

This push-button is used to raise the coating shield when the chamber is fully open.

Lower Push-button

This push-button is used to lower the coating shield when the chamber is fully open.

Ensure the locating bolt has been removed before attempting to lower the shield.



Figure 11.3 - 1 Photograph of Operator Front Control Station

11.4 Lockout Isolator

This lockout isolator is provided to ensure that the machine is in a safe condition when it has been switched to the OFF position.



Figure 11.4 - 1 Locking Out Of Movement Isolator

	Warning ! Under no circumstances must any person enter the chamber unless they have locked out the isolator.
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11.5 MMI Interface Screens

The computer allows the user easy interfacing with the machine by either a touch screen or a keyboard and mouse pointer.

Descriptions of the functionality of the screens is given in this manual

SECTION 12 PREVENTATIVE MAINTENANCE

Section 12.1 Preventative Maintenance Schedule
Section 12.2 Preventive Maintenance Tasks
Section 12.3 Machine Trouble Shooting
Section 12.4 Major System Failures
Section 12.5 Machine Interlock Problems
Section 12.6 Component Malfunctions
Section 12.7 Fault Finding Using PLC Input - Output Racks
Section 12.8 Computer Alarm Monitoring

To cover most aspects of preventative maintenance of the third party equipment supplied with this machine, proprietary manuals are provided. These should be referred to when maintenance is undertaken on the machine. These manuals are located in the proprietary equipment binders.

	Warning !
	Before any maintenance is carried out on any part of the rotating machinery, the machine must be brought to a stationary condition and be locked out. Under no circumstances must a user attempt to clean the machine rollers when the mechanism is rotating or with tension is applied to the substrate.

Overhaul of any of the machines components must ONLY be carried out by qualified personnel who are fully conversant with all mechanical and electrical SAFETY INTERLOCK devices.

Maintenance personnel should read the instruction manuals covering the components fitted to the machine before commencing any maintenance procedures on them.

Definition of Maintenance Personnel

Maintenance or service personnel will generally have a limited knowledge of the machine operation.

	Warning !
	Reduced level of safety. Improper operation can cause serious injuries of the personnel and considerable equipment damage.

The maintenance and service personnel may operate the machine in normal operation, regularly perform necessary maintenance work and additionally perform all further service work which is necessary for proper operation. For this purpose, the machine may be operated in certain manual modes.

Maintenance of the machine comprises the following activities:

- Switching the system on and off.
- All work mentioned in the maintenance schedule

Service of the machine comprises the following activities:

- Shutting down and dismantling the machine for access to the system components.
- Repair work on mechanical and electrical components of the machine.

Maintenance and service of the machine may only be carried out by trained competent employees of the system owner.

For work on electrical components, training as an electrical technician or comparable technical training is recommended

NOTE: For the majority of machine malfunctions, various "Check Lists" are provided under Section 12.2 of this manual. This covers faults that could be encountered during normal use of the machine. Each checklist gives the probable fault, cause and suggested action to remedy the fault.

12.1 Preventative Maintenance Schedule

In an attempt to reduce downtime and consequent lost production through breakdowns, it is recommended that the checks (actions) detailed in the following schedule be fully complied with. Refer to the proprietary manuals supplied by the original equipment manufacturers, for further information.

NOTE: Failure to carry out scheduled maintenance of the equipment as described in the following pages will invalidate any warranty agreements.

Records of maintenance procedures shall be kept and be available to the supplier of this equipment, General Vacuum Equipment and their vendors.

If there is a part of the machine that requires maintenance that is not referenced in this manual, contact General Vacuum Equipment for advice.

This schedule is based upon a normal operation of 24 hours/day, 5 days/week. The frequency of preventive maintenance should be increased as the operation time increases. It is strongly suggested, that the equipment be maintained to the following schedule, even if the equipment is operated for fewer hours.

	Caution !
	Failure to carry out maintenance on the machine will result in shortened life cycles of the parts supplied.

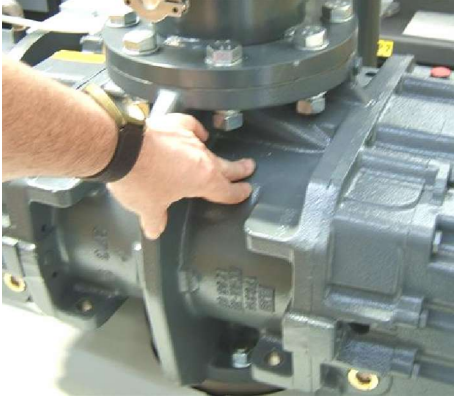
12.2 Preventive Maintenance Tasks

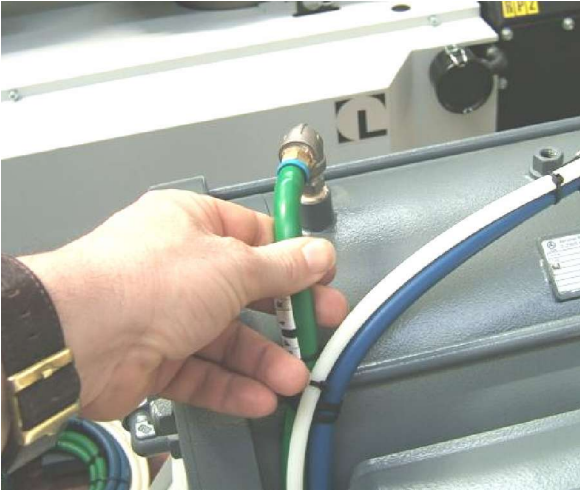
The following identified tasks should be carried out as indicated in the chart:

	Caution !
	Failure to carry out maintenance on the machine will result in shortened life cycles of the parts supplied.

Maintenance Requirement	Equipment & Task to Be Performed	
LEYBOLD ROTARY VANE PUMP		
Daily	Check Oil Condition, If Milky Colour, Check That The Automatic Gas Ballast Is Working Correctly.	
Figure 12.2 - 1 Check Oil Condition & Level		
Daily	Check the oil level (Refer to section 3.2.1 in the vendor manual)	
Figure 12.2 - 2 Check Guards Are in Place		
Weekly	Top up the oil level as required to ensure that the oil level is between the levels shown on the guidance plate on the pump.	

Maintenance Requirement	Equipment And Task To Be Performed
Figure 12.2 - 3 Top Up Oil Level As Required	
Monthly	Clean filter of gas ballast unit
Bi-Annually	Check the belt tension as recommended in the vendor manual
Figure 12.2 - 4 Check Belt Tension	
Monthly	Clean the dirt trap (Refer to section 3.5 in the vendor manual)
Quarterly	Replace the oil (Refer to section 3.3 in the vendor manual)
Quarterly	Replace the oil filter (Refer to section 3.3 in the vendor manual)
Annual	Replace the exhaust filter (Refer to section 3.4 in the vendor manual)
Annual	Check the anti-suckback valve (Refer to section 3.6 in the vendor manual)
Annual	Clean Oil Cooler
Annual	Install Major Preventative Maintenance Kit
2 Years	Overhaul The Pump Completely- Manufacturer Recommendation

Maintenance Requirement	Equipment & Task to Be Performed
MECHANICAL BOOSTER PUMPS-REFER TO THE SUPPLIER'S MANUAL	
Daily	Check for abnormal mechanical noises and temperature rise (hot spots-discolouration)
Figure 12.2 - 5 Check For Abnormalities	
Daily	Check gear case oil levels as per vendor's manual, excess losses should be investigated. Check against level shown in vendor manual, with the pump stopped. Note: Oil reservoir on top fills and empties as pump cools. For oil level indication refer to sight glass.
Figure 12.2 - 6 Check Gear Case Oil Level – Aerzen Booster pump	
Daily	Check drive oil levels as per vendor's manual, excess losses should be investigated. Check against level shown in vendor manual, with the pump stopped.

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 7 Check Drive for Oil Level – Aerzen Booster pump	
Daily	Check drive motor case oil levels as per vendor's manual, excess losses should be investigated. Check against level shown in vendor manual, with the pump stopped.
Figure 12.2 - 8 Check Condition of Water Lines– Aerzen Booster pump	
Daily	Check the drive motor case water lines for indication of leakage and repair as necessary.
Monthly	Check holding down bolts for tightness
Figure 12.2 - 9 Check Holding Down Bolts	

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Replace the oil in both sides of the pump. Ensure pipe work has been vented, both sides of the pump.
Annual	Install manufacturers preventative maintenance kit
3 Years	Overhaul the pump completely- manufacturer recommendation

DIFFUSION PUMPS

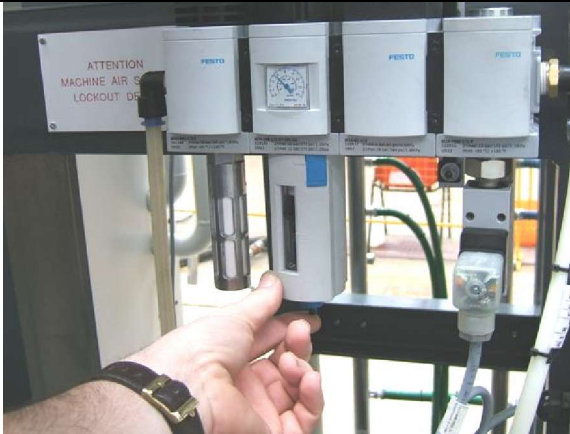
Daily	Check for water leaks
Weekly	Check the cooling water temperature leaving the pump is just less than 167 ° C.
Monthly	Check the pump fluid level
Monthly	Inspect the pump fluid
Monthly	Check that the heater currents are equal on all three phases. They should be balanced and agree to the level in the vendor manual
Figure 12.2 - 10 Check Heater Currents Are Equal On Three Phases 	
Bi-Annual	Inspect heater connections with the pump switched off and cold.
Bi-Annual	Check the pump for it's ultimate pressure
Annual	Remove pump from the machine and clean the pump, condenser and replenish the oil.

WATER SERVICES

Daily	Check for leaks on all manifolds and rubber hoses.
Weekly	Check all functions of differential pressure switches on all manifolds. Close feed and return valves. The switch should indicate a water shortage in these conditions

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 11 Check All Functions Of Pressure monitoring equipment	
Weekly	Check for correct pressure differential across the manifolds. Check the values against the water schematic drawing.
Monthly	Remove and clean all in-line filters
Figure 12.2 - 12 Remove And Clean In-Line Filters	

PNEUMATIC (AIR) SYSTEMS

Daily	Check all air lines for leaks
Daily	Discharge any water that has collected in the lubricator unit
Figure 12.2 - 13 Discharge Water Collected In the Lubricator Unit	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the operation of the regulator.
Weekly	Check all spool valves for leaks
Weekly	Check all cylinders for leaks
Weekly	Check the operation of the pressure monitoring switch by isolating the air supply
Figure 12.2 - 14 Check The Operation Of The Pressure Monitoring Switch	 A close-up photograph showing a person's hand, wearing a brown leather watch, pressing a blue button on a piece of industrial equipment. The equipment has several white cylindrical components and a red warning label that reads "ATTENTION MACHINE AIR SUPPLY LOCKOUT".
Monthly	Check all the spool valves for correct operation

HIGH VACUUM VALVE (VV7)

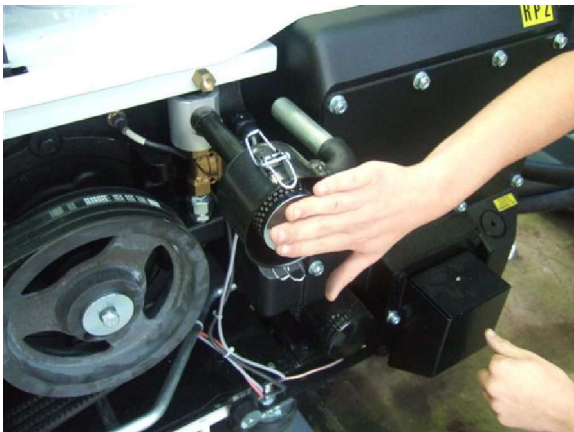
Daily	Check the gravity oil bottles for loss of oil. Investigate any loss
Figure 12.2 - 15 Check The Gravity Oil Bottle	 A photograph showing a hand holding a clear glass gravity oil bottle connected to a complex system of pipes and valves. The bottle contains a yellowish liquid. The background shows industrial machinery with various pipes and components.
Bi-Annual	Lift the valve seats with the manual override on the controlling spool valve when pump is removed to check o-ring condition. Replace as required

PIPEWORK ISOLATION VALVES

Daily	Check for air leaks
Monthly	Check the operation of the position switches
Monthly	Manual check the operation of the valves. Ensure that both sides of the valve are at atmosphere.

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 16 Manually Check The Operations Of The Valves	

GAS BALLAST VALVES

Daily	Check the operation of the valves when machine venting
Figure 12.2 - 17 Check The Operation Of The Valves	

AIR ADMITTANCE VALVES

Daily	Check the operation of the valves when the pumps are switched off.
Figure 12.2 - 18 Check The Operation Of The Valves	

VACUUM GAUGES

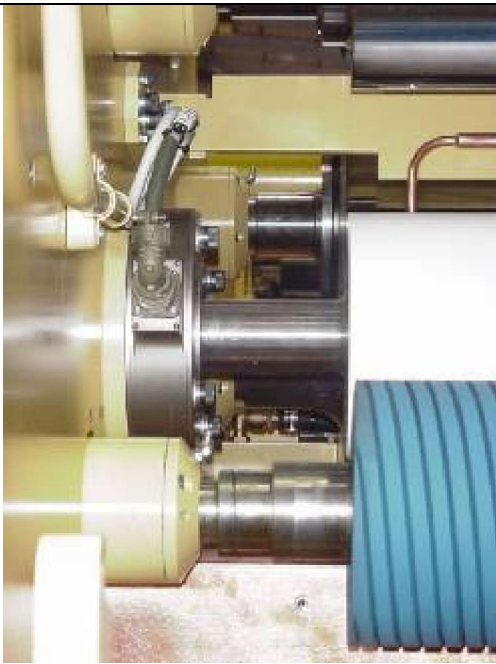
Monthly	Check the condition of the gauges and calibrate against a known standard gauge.
2 Years	Replace all vacuum gauges

Maintenance Requirement	Equipment & Task to Be Performed
WINDING MECHANISM	
Weekly	Check all metal rollers for surface damage repair as required.
Figure 12.2 - 19 Check All Metal Rollers	
Weekly	Check all cork covered rollers for surface damage. Replace the cork if required.
Figure 12.2 - 20 Check all Cork Covered Rollers	
Weekly	Check all rubber rollers for damage repair as required.
Figure 12.2 - 21 Check All Rubber Rollers	

Maintenance Requirement	Equipment & Task to Be Performed	
Figure 12.2 - 22 Check All Rubber Rollers		
Weekly	Check the internal support castor bearing and assembly for wear. Replace as required.	
Figure 12.2 - 23 Check The Internal Support Caster		
Weekly	Check all idler rollers by hand for bearing wear and stiffness.	
Figure 12.2 - 24 Check All Idler Rollers		
Weekly	Check the layarm roller assembly slide rails for debris & wear. Clean & repair as required. Grease unwind units	
Figure 12.2 - 25 Check The Layarm Roller Assembly Slide		

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check The Wear Section On The Chucks For Wear. The Reel Bar Should Fit Tightly Into The Chuck With "No Play", Once Clamped.
Figure 12.2 - 26 Check The Wear Section On The Chucks	
Weekly	Check the operation of the drum/roller zoning mechanism (s) for correct operation.
Weekly	Check that the load cell transducer cables are correctly fitted in their sockets.
Figure 12.2 - 27 Check The Load Cell Transducer Cables	
Weekly	Examine and replace worn or damaged belts in the tendency drive enclosure.
Figure 12.2 - 28 Examine Tendency Drive Belts	

Maintenance Requirement	Equipment & Task to Be Performed	
Weekly	Check the tightness of all taper-lock units used on the tendency drive system.	
Weekly	Check for bearing wear of the bearings used in the tendency drive system.	
Figure 12.2 - 29 Check Tendency Drive Bearings		
Monthly	Check reel shafts bearings for wear. Grease lightly with the correct grease.	
Figure 12.2 - 30 Check The Reel Shaft Bearings		
Monthly	Extend The Engagement Sections On The Mechanical Reel Shafts, Check For Wear And Free Movement	
Figure 12.2 - 31 Check The Engagement Sections On The Mechanical Reel Shafts		
Monthly	Check External Drive Belts And Pulleys For Wear And Correct Alignment.	

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 32 Check External Drive Belts And Pulleys	
Monthly	Check load cell calibration as described in this manual.
Monthly	Grease the bearings, where possible on the rollers.
Figure 12.2 - 33 Grease The Bearings On The Rollers	
Figure 12.2 - 34 Grease The Bearings On The Rollers	

Maintenance Requirement	Equipment & Task to Be Performed	
Figure 12.2 - 35 Grease The Bearings On The Rollers		
Figure 12.2 - 36 Grease The Bearings On The Rollers		
Monthly	Layarm roller assemblies should be checked for alignment to the rewind shaft.	
2 YEARS	Replace lip seals on all leadthroughs	
2 YEARS	Replace bearings on all rollers	

Maintenance Requirement	Equipment & Task to Be Performed
CABLE COILER	
Weekly	Check the cable coiler. Repair where necessary.
Figure 12.2 - 37 Check The Cable Coiler	

WINDING CART TRAVERSE DRIVE MOTOR & GEARBOX UNIT	
Monthly	Check the chain for slackness. Tighten as required and grease lightly
Figure 12.2 - 38 Check the Chain	
Monthly	Check the condition of the cart castors (under the cart). Grease with suitable grease.
Figure 12.2 - 39 Check the Cart Castors	

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 40 Check The Cart Castors	
Monthly	Check for excess wear on tracks. Alignment of castors maybe required
Figure 12.2 - 41 Check Tracks	
Weekly	Check traverse limit switches work correctly on stopping drive and slowing it down.
Annual	Change the oil in the traverse gearbox.

NON CONTACT RESISTANCE PROBE SYSTEM

Monthly	Check the distance between the probes and the roller
Figure 12.2 - 42 Check The Distance Between The Probes and The Roller	

Maintenance Requirement	Equipment & Task to Be Performed	
Weekly	Check the cables for damage. These cables run between the DCM system and the buffer box. Also between the buffer box and the beam.	
Figure 12.2 - 43 Check The Cables		
Monthly	Calibrate the system.	

ELECTRICAL CABINETS

Monthly	Check that all the communication cable termination's are fully inserted	
Figure 12.2 - 44 Check Communication Cable Terminations		
Monthly	With the electrical cubicles isolated, use a vacuum cleaner to remove dust etc. Do not allow build ups	
Bi-Annual	Check tightness of all connections.	
Bi-Annual	Check that all interconnecting cables between plc input/output racks are inserted correctly.	
Bi-Annual	Check the condition of all the contactors used.	
Bi-Annual	Check the operation of all isolator switches	

Maintenance Requirement	Equipment & Task to Be Performed
ROTARY UNIONS	
Monthly	Lubricate the union and check for leaks.
Figure 12.2 - 45 Lubricate The Union	
2 Years	Replace internal seal

VACUUM CHAMBER	
Weekly	Clean and lubricate the winding mechanism joint face o-ring with a suitable grease.
Weekly	Check all joint faces for damage
Figure 12.2 - 46 Check Joint Faces	
Weekly	Clean the interior walls as per the instructions given in the operation manual.

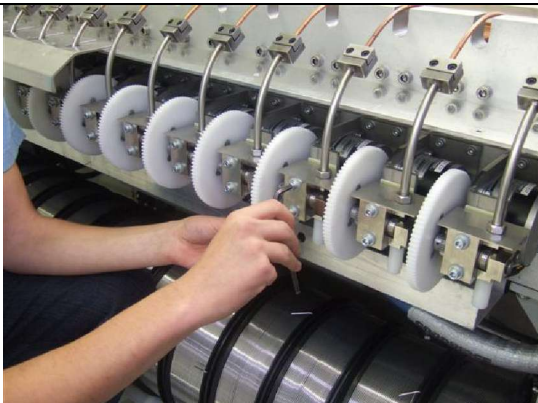
VACUUM PIPELINE RUBBER ISOLATION BELLOWS	
Weekly	Check for wear or damage to the bellows. Replace as necessary.
Figure 12.2 - 47 Check Bellows	

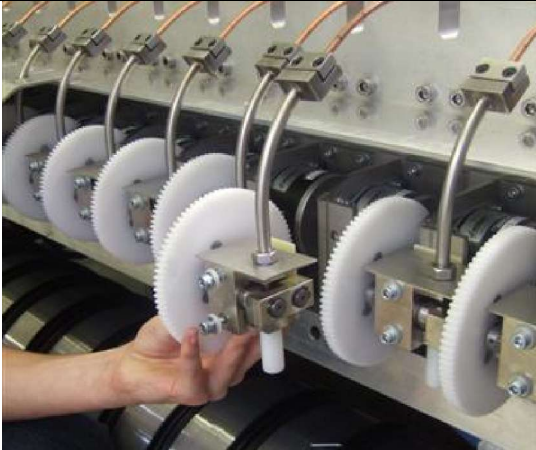
Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check for excess movement - this can lead to air leaks
Monthly	Check the tightness of the hose clamps

WINDING CART ROTARY LEAD THROUGHS

Weekly	Check the gravity oil bottles for loss of oil. Major losses will indicate seal failure.
Figure 12.2 - 48 Check Gravity Oil Bottles	
Bi-Annual	Check for bearing play in all lead-throughs

WIRE FEEDER UNITS

Monthly	Check for wear on drive gear sprockets.
Figure 12.2 - 49 Check Drive Gear Sprockets	
Quarterly	Remove the feeders and check for wear.
Figure 12.2 - 50 Remove And Check Feeders	

Maintenance Requirement	Equipment And Task To Be Performed	
Figure 12.2 - 51 Remove And Check Feeders		
Bi-Annual	Strip the feeder completely, and check for wear. Remedy any defects and reassemble.	
Figure 12.2 - 52 Strip The Feeder		
Annual	Replace all internal bushes	
Figure 12.2 - 53 Replace Internal Bushes		

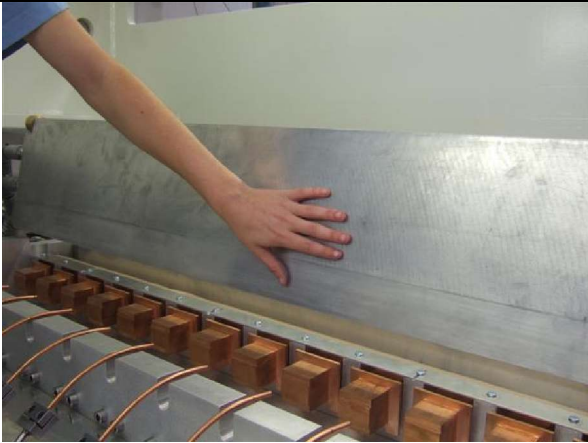
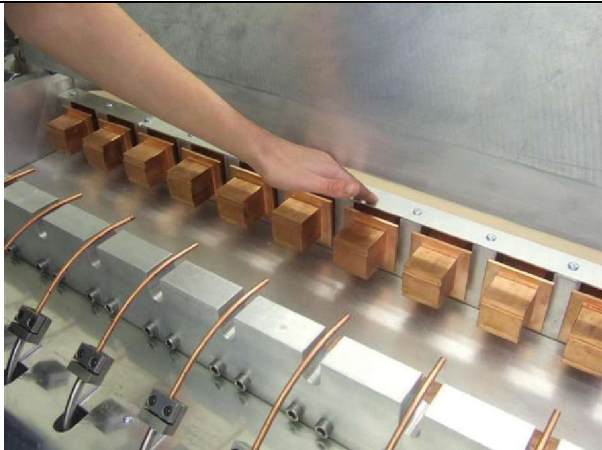
SOURCE ELECTRICAL BUSBAR SYSTEMS

Monthly	Check for debris in the transformer enclosure. Remove any found.
Quarterly	Check the condition of the insulation on the external flexible cables between the busbars and transformers.

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 54 Check The External Cables	
Quarterly	Check cables tightened
Figure 12.2 - 55 Check The External Cables	
Bi-Annual	Check that all the connections on between busbars are tight. Tighten bolts as required.

VACUUM VENT VALVES

Monthly	Check the condition of any filters fitted. Clean or replace as required
Figure 12.2 - 56 Check The Filters	
Annual	Replace The Filter

Maintenance Requirement	Equipment & Task to Be Performed
EVAPORATOR & SOURCE AREA	
Weekly	Grease The Shutter Movement Mechanism Bearings.
Figure 12.2 - 57 Grease The Shutter Mechanism Bearings	
Weekly	Check The Shutter For Mechanical Damage.
Figure 12.2 - 58 Check The Shutter	
Weekly	Check the source area shielding for damage. Repair or replace as required.
Figure 12.2 - 59 Check The Source Area Shielding	
Monthly	Check the insulators around the source area and wire feeders for electrical shorts or damage. Replace as required.

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 60 Check The Insulators	
Quarterly	Check the condition of all the ceramic insulators. Replace as required.
Figure 12.2 - 61 Check The Ceramic Insulators	
5 Years	Replace all insulators

GEARBOXES

Weekly	Check all gearboxes for excessive noise (where possible)
Annual	All gearboxes should have their oil or grease changed

ZONE SEALING

Weekly	Paint the seals with release paint
Weekly	Check the condition of the zone sealing strips. Renew as required.
Figure 12.2 - 62 Check The Zones Sealing Strips	

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 63 Check The Zones Sealing Strips	
Figure 12.2 - 64 Check The Zones Sealing Strips	
Annual	Replace the seals

AUTOMATIC CONTROL SYSTEMS

Weekly	Check all interlocks are in place and operate correctly.
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SAFETY EQUIPMENT

Daily	Check the operation of the emergency stop circuit
Daily	Check the operation of the audible alarm
Daily	Check the operation of the visual alarms

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 65 Check The Visual Alarms	
Weekly	Check all local electrical isolators operate correctly (if fitted)
Figure 12.2 - 66 Check Local Electrical Isolators	

VIEWING WINDOWS

Daily	Check that all windows for damage. Replace as required.
Figure 12.2 - 67 Check Windows	

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 68 Check Windows	

ELECTRICAL LEAD THROUGHS

Weekly	Check for damage to cables.
Figure 12.2 - 69 Check Cables	
Monthly	Check all lead-throughs for tightness
Figure 12.2 - 70 Check Leadthroughs	

OPERATOR INTERFACE UNITS

Monthly	Check for correct calibration of touch segments. Re calibrate when possible or replace the unit.
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VACUUM SYSTEM

Quarterly	Leak check the chamber, pumps and pipe work for air leaks
Bi-Annual	Check the ultimate pressure overnight

Maintenance Requirement	Equipment & Task to Be Performed
AC VECTOR MOTORS	
Monthly	Check and clean the cooling fan
Figure 12.2 - 71 Check And Clean Cooling Fins	
Quarterly	Check The Electrical Connections For Tightness
Figure 12.2 - 72 Check The Electrical Connections	
Quarterly	Check encoder to motor connection is ok

SIEMENS COMBIMASTER MOTOR/GEARBOX

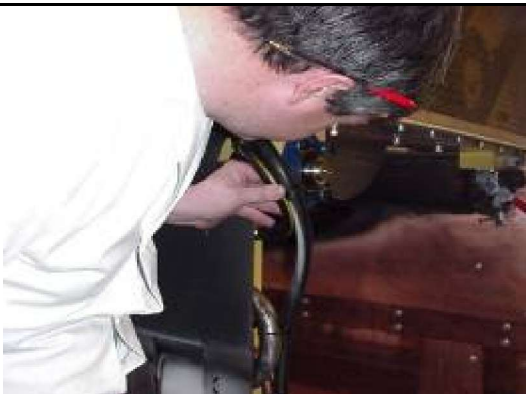
Annual	Change The Oil In The Gearbox
Quarterly	Clean The External Body Of The Motor/Gearbox
3 Years	Change The Bearings As Recommended By The Vendor

LOCKOUT ISOLATOR

Weekly	Check That The Non Operation Of The Switch Makes The Siren Sound To Warn All Users
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Maintenance Requirement	Equipment & Task to Be Performed
BELT GUARDS	
Weekly	Check That All Guards Are In Position And That Screws Are In Place
Figure 12.2 - 73 Check All Guards and Screw Are In Place	
Weekly	Check That The Safety Warning Signs Have Not Been Removed

BOWED SPREADER ROLLER	
Weekly	Check The Rubber Sleeve For Damage
Figure 12.2 - 74 Check Rubber Sleeve	
2 Years	Send Roller Back To Manufacturer For Roller Overhaul.

Maintenance Requirement	Equipment And Task To Be Performed
COATING SHIELD	
Weekly	Check The Shield For Sealing Surfaces To Drum Area Zoning
Figure 12.2 - 75 Check The Zones Seals	
Weekly	Check That The Sealing Sections To The Drum Are Not Damaged
Figure 12.2 - 76 Check The Sealing Sections To The Drum	
Daily	Check The Condition Of The Hoses
Figure 12.2 - 77 Check The Hoses	
Weekly	Check The Condition Of The Shields For Mechanical Damage

Maintenance Requirement	Equipment And Task To Be Performed
Figure 12.2 - 78 Check The Shields	
Weekly	Check That The Wear Plates Are Good Condition And That The Reel bar Does Not "Rock" When Moved From Side To Side
Figure 12.2 - 79 Check The Wear Plates And Reel Bar	
Figure 12.2 - 80 Check The Wear Plates And Reel Bar	
Weekly	Ensure That the Clamp Is Holding the Bearing
Weekly	Check The Manual Clamps Engage Over The Outside Of The Chuck OK.

Maintenance Requirement	Equipment And Task To Be Performed
O-RINGS	
Daily	Check Main O-Ring For Damage
	 Figure 12.2 - 81 Check Main O-Ring
Annual	Replace The O-Ring In The Right Angle Valve Seat
5 Years	Replace O-Rings On Machine As Components Are Removed For Maintenance.

NIP ROLLER	
Daily	Check The Roller For Damage
Weekly	Check For Correct Contact Between The Rollers
	 Figure 12.2 - 82 Check For Correct Contact Between The Rollers
Weekly	Check That The Mechanical Stop Stops The Roller From Being Crushed

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2 - 83 Check The Mechanical Stop	
2 Years	Replace The Roller

SOURCE ROTARY WINDOW

Weekly	Grease Bearings At Either End Of The Assembly
Weekly	Check Coupling Is Engaged Correctly
Weekly	Check Drive Belt Is Tight
Figure 12.2 - 84 Check Drive Belt	
Weekly	Grease Gearbox
Annual	Change Bearings

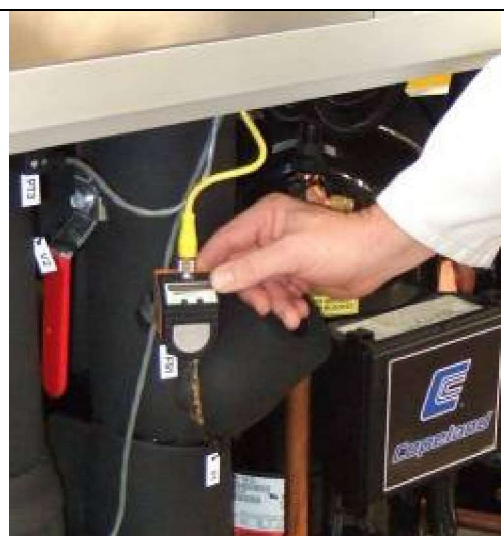
Maintenance Requirement	Equipment & Task to Be Performed
CRYO-GENERATOR - POLYCOLD	
Monthly	Check suction and discharge pressures
Figure 12.2 - 85 Check Suction And Discharge Pressures	
Monthly	Check Discharge Line Temperature
Monthly	Check Liquid Line Temperature
Monthly	Check #1/2 Coil In Temperature
Monthly	Check #1/2 Coil Out Temperature
Bi-Annual	Check Noise Level and For Abnormal Sounds
Bi-Annual	Check Refrigerant Line For Ice Or Water
Bi-Annual	Check For Water In Valve Boxes
Bi-Annual	Check Balance Pressure

GAS WEDGE	
Daily	Check gas wedge valve
Figure 12.2 - 86 Check Gas Wedge Valve	
Bi-Annual	Check the gas wedge is not blocked by feeding compressed air into it. Note: This task may need to be carried out more often if the machine is located in a dusty atmosphere.

B & R STEPPER DRIVES	
Weekly	Check the fans located below the output modules are running.

Maintenance Requirement	Equipment & Task to Be Performed
TES CHILLER UNIT	
Daily	Check externally for water leaks
Figure 12.2 - 87 Removing Chiller Access Panel	
Figure 12.2 - 88 Removing Chiller Access Panel to Electrical	
Daily	Check for unusual operating noises
Weekly	Check externally for water leaks
Figure 12.2 - 89 Check For Water And Refrigeration Oil Leaks	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check all electrical connections are tight
Figure 12.2 - 90 Check Electrical connections	
Monthly	Check external fasteners
Figure 12.2 - 91 Check the panel external fasteners for operation.	
Quarterly	Check suction and discharge pressures are correct on Interface screen
Quarterly	Check sight glass for dryness and bubbles to verify that there are no leaks
Figure 12.2 - 92 Check the refrigerant sight glass for	

Maintenance Requirement	Equipment & Task to Be Performed	
Quarterly	Check compressor oil levels	
Figure 12.2 - 93 Check The Oil Level In Compressor		
Quarterly	Check correct water temperatures are maintained	
Quarterly	Check water flow is correct using FS1	
Figure 12.2 - 94 Check water flow is correct using FS1		
Quarterly	Check correct operation of the pump and flow switch	
Quarterly	Check voltage and current characteristics of components	
Figure 12.2 - 1 Check The Pump And Compressor Motor Currents		

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Clean evaporator water strainer
Figure 12.2 - 95 Clean evaporator water strainer	
Quarterly	Check that pipe work clamps are secure
Figure 12.2 - 96 Check that pipe work clamps are secure	
Quarterly	Check that compressor mounts are in good condition and are tight
Figure 12.2 - 97 Check that compressor mounts are in good condition and are tight	
Quarterly	Visually Check The Pressure Relief Valves For Signs On Operation (There Should Not Normally Be Any)
Quarterly	Check The Coolant Concentration (Glycol)

Maintenance Requirement	Equipment And Task To Be Performed
Figure 12.2 - 98 Check The Coolant Concentration by accessing the hot tank from the top of the unit.	
Quarterly	Check That The Temperature Measuring Devices Are In Good Contact With The Source Of Contact
Figure 12.2 - 99 Check The Temperature Measuring Devices	
Quarterly	Check that the compressor heater is operational
Figure 12.2 - 100 Check that the compressor heater is operational	

Maintenance Requirement	Equipment And Task To Be Performed
Quarterly	Check That All The Capillary Tubing Is In A Good Condition
Figure 12.2 - 101 Check The Temperature Measuring Devices	
Quarterly	Check That The Pump And Compressor Motor Currents Do Not Exceed The Rating On The Motor Plates Under Normal Running Conditions
Quarterly	Check The Operation Of The Flow Switch by restricting the flow at the outlet valve.
Annually	An annual full service by a qualified refrigeration engineer is recommended.

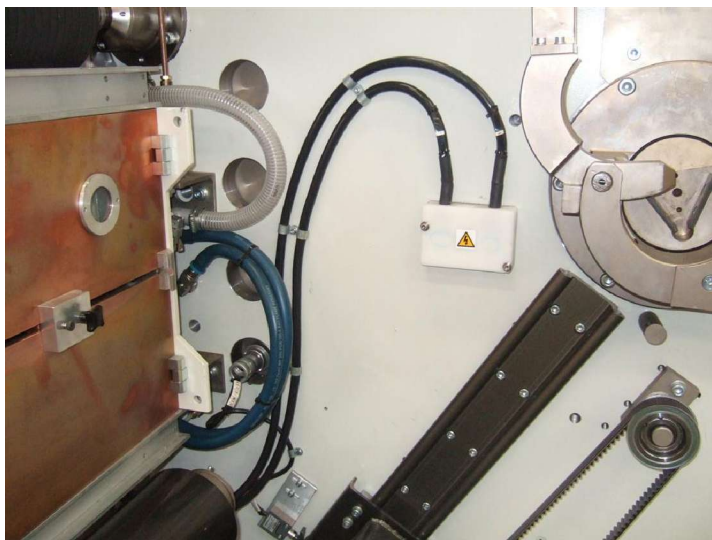
AE PLASMA POWER SUPPLY

Quarterly	Check Connections for tightness
Quarterly	Check for indication of corrosion damage

PLASMA HIGH VOLTAGE INTERCONNECTION CABLES

Weekly	Check For Damage To The Conduit Containing The High Voltage Cables.
Monthly	Check The Connections For Tightness And Damage

Figure 12.2 - 102
Check The Connections



Maintenance Requirement	Equipment And Task To Be Performed
MKS MASS FLOW CONTROLLERS	
Monthly	Check For Physical Damage And Replace As Necessary
Quarterly	Check That The Shut Off Valve Is Operating Correctly
Quarterly	Check Calibration Of Gauge
MKS BARATRON PRESSURE TRANSDUCER	
Monthly	Check For Physical Damage And Replace As Necessary
Quarterly	Check The Zero Value On The Gauge
Quarterly	Check That The Interconnect Cable Is Installed Correctly
Quarterly	Check Calibration Of Gauge
PLASMA TREATER	
Monthly	Check Insulating Materials For Indication of Over Heating
Monthly	Check Tightness Of All Fixing Components.
Monthly	Check For Wear On Plates By Physical Erosion.
Annual	Replace O-Rings
2 Years	Have Magnetic Field Checked by General Vacuum Equipment Technician

Table 12.2 - 1 Preventative Maintenance Tasks

12.3 Machine Trouble Shooting

In the event that any major section of the machine or the whole machine fails to operate, or functions unsatisfactory, this section of the manual attempts to suggest possible reasons, and methods of correcting them.

NOTE: If any part of the proprietary equipment supplied with this machine should develop a fault that is not discussed in this manual, refer to the manufacture's manuals provided in the propriety manual binders.

12.4 Major System Failures

If the machine or any major subsystems are totally inoperative, preliminary checks may discover the problem (see the lists on the following pages). If the initial checks do not succeed in locating the problem, more detailed checks will be necessary (see Section 12.4 & 12.5).

FAULT	REASONS & POSSIBLE REMEDIES
Machine Totally Inoperative	Check for incoming power supply. Check that the main isolator is switched on. Check that the I/O Fault indicator on the network diagnostic screen is not highlighted.
Winding System Inoperative	Check that power supplies are available. Check if the drive system control circuit is operational (see troubleshooting section). Check if the Emergency Stop circuit has been reset. Check for control power supply. Check that the safety interlocks circuit is functioning correctly.
No Linear Movement (Cart)	Local Lockout Switch is disabled. Mechanical obstruction. Circuit breaker tripped in cubicle. Check that safety interlock circuit is functioning correctly. Certain interlocks not made.

Table 12.4 - 1 Major System Failures

12.5 Machine Interlock Problems

The following section of this manual has been written as a guide to explain the sequence interlocks that are required to allow operation of certain parts of the machine. These interlocking components are controlled from within the computer.

The machine has been manufactured with a number of various screens to assist the user when an operation does not occur. These are presented on the MMI screen in a flow chart.

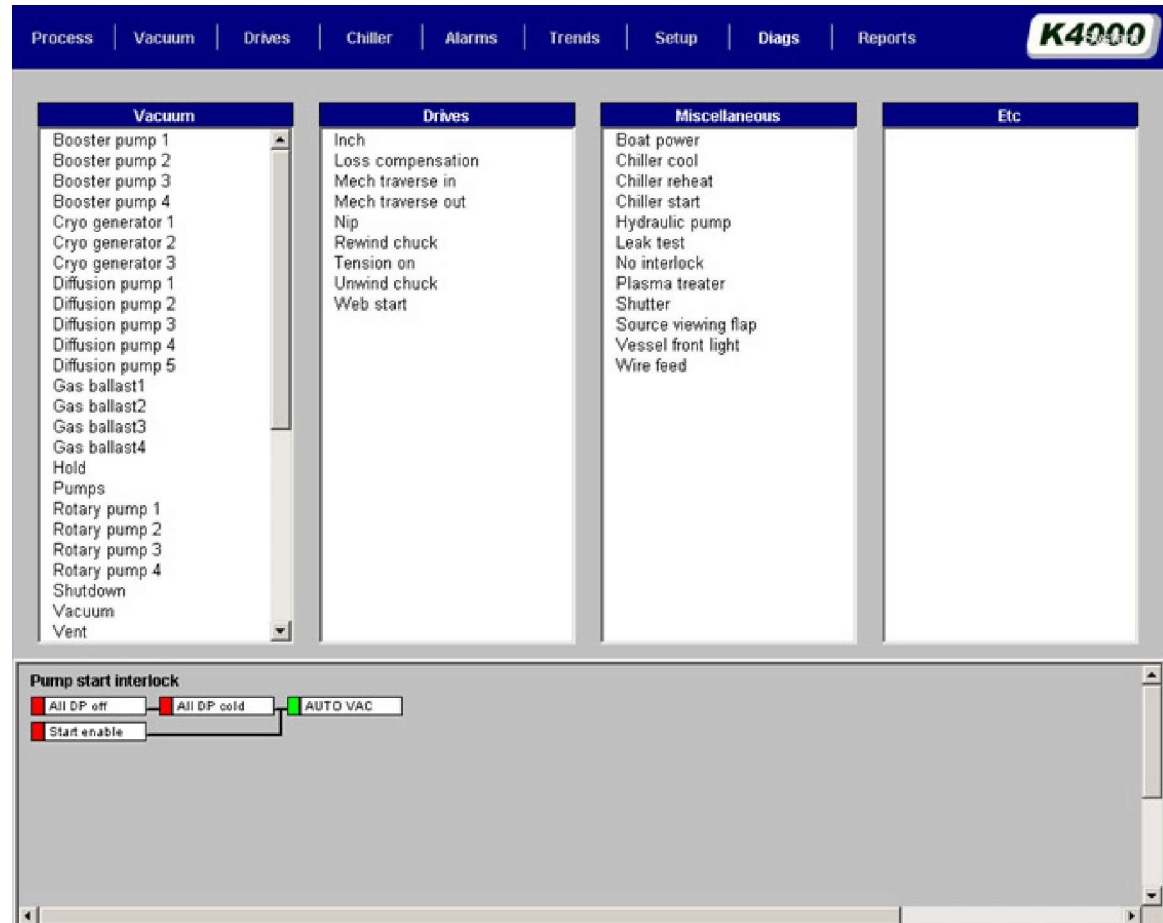


Figure 12.5 - 1 Interlock Diagnostic Screen

Where the block has a back slash in front of the wording, as shown below, this signifies that the interlock is a Not condition.



Figure 12.5 - 2 Interlock Diagnostic Screen – Not Function

When attempting to operate one of the functions, if the correct interlocks are not present, the flow chart will be shown.

The green or red blocks show the condition of the individual interlock. There must be a continual line of green blocks to allow the function to take place.

Also if there is an alarm state, this will be indicated.

On this machine there are interlock charts for the following functions.

Vacuum

- Pumps Function
- Shutdown Function
- Vacuum Function
- Vent Function
- Rotary Pump 1 Operation
- Rotary Pump 2 Operation
- Rotary Pump 3 Operation
- Rotary Pump 4 Operation (Option – Width Dependant)
- VV1
- VV1B (Option – Width Dependant)
- VV2
- VV2B
- VV4 Operation
- VV6 A/B/C/D/E Operation (Option – Width Dependant)
- VV7 Operation
- VV8 Operation (Option – Width Dependant >1650mm)
- VV10A/B Operation
- Booster Pump 1 Operation
- Booster Pump 2 Operation
- Booster Pump 3 Operation
- Booster Pump 4 Operation (Option – Width Dependant >1650mm)
- Cryogenerator 1 Operation
- Cryogenerator 2 Operation (Option)
- Diffusion Pump 1 Operation
- Diffusion Pump 2 Operation
- Diffusion Pump 3 Operation
- Diffusion Pump 4 Operation (Option – Width Dependant >1650mm)
- Diffusion Pump 5 Operation (Option – Width Dependant >2050mm)
- Gas Ballast – Rotary Pump 1 Operation
- Gas Ballast – Rotary Pump 2 Operation
- Gas Ballast – Rotary Pump 3 Operation (Option – Width Dependant)
- Gas Ballast – Rotary Pump 4 Operation (Option – Width Dependant)

Drives

- Inch Drive Function
- Loss Compensation Function
- Mechanism Cart Traverse in
- Mechanism Cart Traverse Out
- Reset Layarms
- Web Start Function
- Tension on Start

Miscellaneous

- Boat Power
- Chiller Cool
- Chiller reheat
- Coating Shield lower
- Coating Shield raise
- Shutter Open/Close
- Source Viewing Flap
- Vessel Lights
- Wire Feeder Operation
- Plasma Treater Operation

12.6 Component Malfunctions

This section provides details on possible faults in the major components used on the machine. Equipment provided by outside vendors will generally provide proprietary component manuals with their equipment. Please refer to these manuals for troubleshooting of the equipment. This equipment will include the following;

- Vacuum Pumps
- Drive Systems and motors
- Chiller Units
- Cryo-Generator Units
- Vacuum Gauging Equipment
- Computers
- Control Systems
- Gearboxes.

If further information or help is required, please contact General Vacuum Equipment.

The following list is provided for convenience in locating the fault find procedures. These are appropriate to the major components found on General Vacuum Equipment machines. The majority of these suggestions are based on general and sometimes specific guides written by vendors' engineers and General Vacuum Equipment engineers.

Equipment	Applicable to this machine
Air Admittance Valve Malfunctions	YES
AC Contactors Malfunctions	YES
Busbar System Malfunctions	YES
Rubber Bellows Malfunctions - Pipelines	YES
Drive Belts Malfunctions - Drive Belts	YES
Drive Belts Malfunctions - Vee Belts	YES
Rotary Blowers And Booster Pumps Malfunctions	YES
Diffusion Pump Malfunctions	YES
Roller / Drum Low Conductance Seals Malfunctions	YES
Electrical Vacuum Leadthroughs Malfunctions	YES
Drive System Malfunctions	YES
Flexible Zone Seals Malfunctions	YES
Gauge Malfunctions, - Edwards Active Gauge, Pirani Gauge	YES
Vacuum Dial Gauge Malfunctions	YES
Viewing Window Malfunctions	YES
Keystone Isolation Valves Malfunctions	YES
Source Power Supply Cable Malfunctions	YES
Layarm System Malfunctions	YES
Internal Chamber Lighting Malfunctions	YES
'Bow' Spreader Roller Malfunctions	YES
AC Motor Electrical Malfunctions	YES
Push-Button Malfunctions	YES
Cryo-Generator System Malfunctions	YES
PLC System Malfunctions	YES
Pneumatic Valves And Cylinders Malfunctions	YES
High Vacuum Valve	YES
Reed Switches Malfunctions - Located On Cylinders	YES
Winding Mechanism Roller Malfunctions	YES
Liquid Rotary Glands Malfunctions	YES
Shutter (Moving Shield) Malfunctions	YES
Solenoid Malfunctions	YES

Equipment	Applicable to this machine
Window - Anti Coating Devices - Malfunctions	YES
Vacuum Pump Motor Control Switchgear Malfunctions	YES
Evaporator Problems Thyristors TU 1471	YES
Computer To Thyristor Communications Related	YES
Power Supply Related	YES
Operator Related Problems	YES
Power Transformers Malfunctions	YES
Winding Cart Linear Drive Malfunctions	YES
Water Pressure Switches Malfunctions - Differential Type	YES
Software System Malfunctions	YES
Rittal Air Conditioning Unit Malfunctions	YES
Siemens Combimaster Drive Malfunctions - Traverse Drive	YES
TES Chiller Unit	YES
Kinertrol Pneumatic Actuators Malfunctions	YES
Siemens Asi Modules Malfunctions	YES
AFM SI Interface Module Malfunctions	YES
Omni-Phase Surge Protector Malfunctions	YES
Machine System Communications Malfunctions	YES
B&R Wire Feeder Drive System	YES
Plasma Power Supply	YES
KEB Inverter Units	YES
Eurotherm TE10P Thyristor Units	YES

Table 12.6 - 1 Component Malfunctions Table

Air Admittance Valve Malfunctions		
Edwards IPVA 10EK Valve		
Symptoms	Probable Cause	Possible Remedies
FAULT: Valve is not sealing against atmospheric pressure	No power supply to the solenoid	- Check the power supply is healthy - Check the PLC output is on - Check the wiring
	Seal damaged	Inspect the seal, repair or replace valve
FAULT: Valve does not de-energize after power is removed.	Valve seal is sticking	Remove the valve and check for debris.

Table 12.6 - 2 Air Admittance Valve Malfunctions

AC Contactors Malfunctions		
AC contactors are used in the control circuits of motors used on the machine.		
Symptoms	Probable Cause	Possible Remedies
FAULT: Noisy - humming	Low Control Voltage	Check the line voltage and make modifications to supply.
	Defective Coil	Repair- replace the coil
	Contamination	Clean magnet faces
FAULT: Frequent operating coil failure	Over voltage	Check coil is correctly rated for supply voltage
	High environmental temperature	Ensure that cooling fans/system are operating correctly.
	Magnet not sealing in	Remove any debris and check spring tension is not excessive
	Excessive cycling of contactor	Reduce the cycling factor
	Moisture	Remove all traces of moisture. Use electrical contact cleaner

Table 12.6 - 3 AC Contactors Malfunctions

Busbar System Malfunctions		
The busbar system carries power from the evaporator supply transformers to the evaporator posts via water-cooled leadthroughs. The system is manufactured from high-grade copper.		
Symptoms	Probable Cause	Possible Remedies
FAULT: Intermittent loss of power to evaporators	Loose connections at busbar joints	- Check the connections at all joints - Check connections at the braid connection between the evaporator post and busbar.
FAULT: Vacuum leak at busbar leadthrough.	O-ring failure	This is a rare fault and is usually caused by overheating of the busbar. Replace the o-ring. Seek advise from General Vacuum Equipment
FAULT: Electrical short at leadthrough.	Failed insulator.	Fault is indicated by local arc marks between busbar and wall of chamber. Replace the insulator.

Table 12.6 - 4 Busbar System Malfunctions

Rubber Bellows Malfunctions - Pipelines		
These bellows are provided to isolate vibration from the pumping group passing to the chamber via the pipework.		
Symptoms	Probable Cause	Possible Remedies
FAULT: Vacuum leaks at bellows	Hose clips loose	Check the tightness of the clips
	Bellows damaged	Check the bellows for damage, repair if possible, or replace.
FAULT: Bellows move during pump down and eventually leak	Excess grease between bellows and pipework	Remove excess vacuum grease

Table 12.6 - 5 Rubber Bellows Malfunctions - Pipelines

Drive Belts Malfunctions - Drive Belts		
Symptoms	Probable Cause	Possible Remedies
FAULT: Breakage of belts	Rubbing of Belts on Guards	Check the guards/covers for "rub" marks
	Rubbing of Belts on drive pulleys	Check alignment of pulleys and align as required.
FAULT: Excess noise from belts	Loose Belt	Check the belt for the correct tightness
	Pulley Moved on shaft	Check that the drive pulleys used have not moved

Table 12.6 - 6 Drive Belts Malfunctions - Drive Belts

Drive Belts Malfunctions - Vee Belts		
Symptoms	Probable Cause	Possible Remedies
FAULT: Belt slipping	- Not enough tension - Drive overload - Oily drive	- Increase tension - Increase drive capacity - Correct oil leakage
FAULT: Belt Squealing	- Drive overload - Heavy Starting load - Small arc of contact	- Increase drive capacity - Increase tension - Lengthen centre distance
FAULT: Turned over belts	- Broken cords from running belts onto pulleys - Not enough tension - Impulse loads	- Reduce pulley centres when installing belts - Increase tension - Provide pivoted motor base or spring loaded tensioner roller
FAULT: Repeated belt breakage	- Shock loads - Broken cords from running belts onto pulleys - Foreign objects	- Increase tension or increase drive capacity - Reduce pulley centres
FAULT: Rapid wear	- Worn pulley grooves - Pulley too small - Misalignment - Drive overloaded - Belts rubbing guards	
FAULT: Underside cracked or belt hardened	Excessive heat	

Symptoms	Probable Cause	Possible Remedies
FAULT: Excessive whipping	• Drive centres too long • Pulsating load	

Table 12.6 - 7 Drive Belts Malfunctions - Vee Belts

Rotary Blowers And Booster Pumps Malfunctions		
This is a general guide for particular faults refer to the manufactures manual		
Symptoms	Probable Cause	Possible Remedies
FAULT: Abnormal Noises Emitted From Machine When Running	• Excessive backlash between flanks of timing gear wheel teeth.	• Remove gear casing and measure amount of backlash all the way round the entire circum of gear wheels, first making sure that the wheels are dry.
		• Should backlash be found to exceed the mean clearance value engraved on gear wheels by more than 30-50%, fit new wheels.
	• Excessive play in bearings.	• Measure clearances, using a dial type indicator gauge, and compare readings with SKF bearing catalogue. If bearing play is too great, fit new bearings
	• Rotary pistons fouling cylinder walls and rubbing against one another.	• Measure operating pressures and temperatures, and compare readings obtained with the data supplied
	• Rotary pistons rubbing against one another or against cylinder walls due to dirt.	• Clean out cylinder and examine for possible cracks.
FAULT: Blower Overheating	• Excessively high compression heat resulting from a too high-pressure differential.	• Check inlet pressure to the pump to be less than 50 mbar while running
	• Rotary pistons coming into contact with side plates, due to an excessive heating-up of pistons when running, caused by overloading. • Rotary pistons suffering from imbalance as the result of a build-up of dirt deposits.	• Clean components and check for possible shaft deflection
FAULT: Blower Overheating	• An accumulation of dirt in the inlet.	• Check inlet of pump for dirt.
	• Lubrication oil.	• Check oil level is correct and oil type is of the type specified

Symptoms	Probable Cause	Possible Remedies
	Rotary pump "backing" the booster is not operating correctly	- Check the air admittance valve is closed. - Check that the outlet from the pump is not blocked
	Excessively large rotary piston clearances	Check clearances and refer the matter back to the supplier
FAULT: Leakage of Oil Into The Conveying Chamber	Oil level too high.	Reduce to correct level and flush out conveying chamber
	Internal seal failure.	Replace inner seals.
FAULT: Excessively Low Volumetric Flow At Inlet	Layout of pumping system is incorrect	- Check running speed and blower pressures. - Check the layout with the provided information.
	Excessively large internal clearances, due to wear.	Check clearances, record readings and refer the matter back the supplier.
FAULT: Excessively High Power Consumption	Actual operating conditions differ from original use.	Check with supplier.
	Oil level too high.	Reduce to correct level
FAULT: Loss of oil from gravity bottle supplying the shaft seal	Shaft Seal worn	If oil is seen escaping down the outside of the pump, replace the outer shaft seal.
	Pressure building up inside gear case cavity and forcing the oil out	Monitor for this action when pump running. Consult with manufacturer if seen.
	External leak from oil bottle.	Replace the oil bottle
FAULT: Oil levels in pump gear cases increase	Oil from gravity bottle leaking into pump due to seal failure.	Replace the inner shaft seal.
FAULT: Excessive noise on starting to rotate pump	Belt slip	Check the tightness of the drive belt.
	Misalignment of pulleys	Check that external drive pulleys are in-line and not allowing the drive belts to touch the guard.

Table 12.6 - 8 Rotary Blowers and Booster Pumps Malfunctions

Diffusion Pump Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Poor Ultimate Pressure	Air leak in vacuum system, virtual or real.	Locate and repair
	Excess cooling of pump.	Check that the cooling water returning to the manifold is approximately 130 ° F
	Machine contaminated	Clean machine

Symptoms	Probable Cause	Possible Remedies
	Low heat input	- Check supply voltage is correct - Check current being taken by the heaters using and amp meter. Check actual readings against vendor manual. - Check for burned out heater elements, or poor thermal contact.
	Inadequate cooling water	- Check water pressure. - Check tubing for obstructions and backpressure.
	High foreline pressure.	- Check for air leak in foreline - Poor mechanical pump performance. Monitor the foreline pressure gauge. - Breakdown of pump fluid.
	Contaminated pump fluid.	Examine for clear appearance, if dark, replace it.
FAULT: Low Speed (Prolonged Cycle)	Low heat input.	Check current being taken by the heaters using and amp meter. Check actual readings against vendor manual.
	Low fluid level.	Add the correct fluid to recharge the pump
	- Malfunctioning pump assembly. - Improperly located jets. - Damaged jet system.	Check and repair/replace
FAULT: Inlet Pressure Surges	Incorrect heater input.	Check and correct
	Fluid outgassing	Condition fluid
	Leak in system ahead of pump inlet.	Check and correct.
FAULT: Chamber contamination by oil. (back streaming)	High fore pressure.	- Check for leak in foreline - Poor mechanical pump performance - Breakdown of pump fluid
FAULT: Indication of Over-temp by monitoring system.	Low oil level	Check the oil level and fill as required.
	Inadequate cooling from water supply	The cooling water returning from the pump should be approximately 67 ° C. Open the valves to ensure excess temperatures are avoided.
FAULT : Pump switches off when chamber is open	Vacuum level rises in pump and trips pressure switch	- Check for debris between valve plate and plate mating surface. Clean - Valve VV6 stays closed and oil-outgassing pressure increases. Check pumping sequence for correct operation and valve operation.
	Power source circuit breaker tripped.	Check the circuit breaker is made.

Symptoms	Probable Cause	Possible Remedies
	Loss of oil activates the oil "over-temp" thermostat	Check the oil level.
	Cooling water temperature is too high and water "over-temp" thermostat is tripped.	The cooling water returning from the pump should be approximately 67 ° C. Open the valves to ensure excess temperatures are avoided.

Table 12.6 - 9 Diffusion Pump Malfunctions

Drum Low Conductance Seals Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Scratching substrate	Aluminium Debris in gap between seal and drum	Lower coating shield and remove debris.
FAULT: Vacuum pressure levels breakdown between zones poor.	Conductance seals have been lowered	Close seals to mechanical stop and check for clearance to be approximately 2mm.
FAULT: Debris flaking off the drum	Aluminium oxide build up on drum	Adjust the drum edge shields to cover the edge of the substrate.

Table 12.6 - 10 Roller / Drum Low Conductance Seals Malfunctions

Electrical Vacuum Leadthroughs Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Loss of power/control signal	Loose connection	Check tightness of plug into socket section
	Open circuit on pins	Check pins for correct termination
	Dry soldered joint	Re-solder the suspect joint
FAULT: Electrical short circuit	Incorrect Installation	Replace and install correctly
	High pressure region around leadthrough	Replace and ensure leadthrough is leak tight
	Excess Power	Ensure that leadthrough is rated for required use

Table 12.6 - 11 Electrical Vacuum Leadthroughs Malfunctions

Drive System Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: When inching, the drive system pulls the substrate off the rewind core.	Inch direction mode selected to reverse	Check that the inch direction mode is selected to the forward direction.
	Rewind direction incorrect to underwind/overwind mode selection	Check that the correct mode is selected

Symptoms	Probable Cause	Possible Remedies
FAULT: The drum will not move when the inch function is selected.	Mechanical binding	Check for rotation by hand.
	Drive system fault	Check the drive for faults.
FAULT: When tension is applied, the substrate is pulled off the rewind core.	Rewind direction incorrect to underwind/overwind mode selection	Check that the correct mode is selected
FAULT: When the tension is applied, no tension is taken up in the substrate.		Check that the correct modes are selected
FAULT: When tension is applied, the substrate on the unwind roll is not applied and the roll starts to pay off.	Incorrect web path used	Check that the substrate has been threaded correctly through the winding mechanism
	Incorrect under / overwind mode selected.	Check that the correct mode is selected
	Zero tension set-point selected	Ensure the tension set point is more than zero.
	Fault with load cell system	Check the load cell system for correct calibration.
FAULT: The drive system will not allow "line Start" to be active.	System in "Auto slowdown mode"	Check that the rewind and unwind diameters are correct.
	Rotation direction selected is reverse	Check that the direction of rotation is selected forward. If in reverse, the drive system will think that the rewind core is empty and therefore place the drive system in a slowdown mode. No visual or audible alarm is given
	Drive interlock circuit not energized	The machine will only allow the running of the drive system at speed when the chamber is closed.
FAULT: On starting the acceleration of the drive system, large drop off in tension is seen	Incorrect roll diameter set point in drive system.	Check that the pre-start diameter is correct.
FAULT: On starting the acceleration of the drive system, high tension is seen to be pulled.	Incorrect roll diameter set point in drive system.	Check that the pre-start diameter is correct.

Table 12.6 - 12 Drive System Malfunctions

Flexible Zone Seals Malfunctions		
These are the seals that run along the side of the winding mechanism and chamber.		
Symptoms	Probable Cause	Possible Remedies
FAULT: Zone seal not sealing correctly - evaporant leaking past seal	Poor figment or movement of seal	Check the seal between the seal with the surrounding contact faces
	Seal damage	- Inspect seals and replace as required - Check the joint surface for sharp edges and repair
	Seal turning as the chamber is closed	Lubricate or replace as required
FAULT: Seal turns over as winding mechanism is driven into the chamber	Seal to sealing face too close	Check that the distance is not too close. Adjust as required.
FAULT: Damage to seal	Seal catching on external surface.	Replace the seal

Table 12.6 - 13 Flexible Zone Seals Malfunctions

Gauge Malfunctions Edwards Active Gauges Pirani Gauge		
Symptoms	Probable Cause	Possible Remedies
FAULT: Gauge Gives False or No Reading	Gauge Dirty	Clean as per vendor manual instructions
	Power supply Fault	- Check that the plug is engaged at the gauge - Check that the wiring is terminated correctly at the PLC rack. - Check that the 24 volt supply to the gauges is on
FAULT: No Readings upon the VACUUM Screen.	Signal Missing	- Check that the connectors are not loose - Check all the wiring

Table 12.6 - 14 Gauge Malfunctions Edwards Active Gauges Pirani Gauge

Vacuum Dial Gauge Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Gauge Gives False Vacuum Levels and Sticks	Dust contamination	Clean the gauge carefully by removing the front from it
FAULT: Offset in reading with the chamber open.	Machine location not at sea level.	None.
FAULT: Offset in reading with chamber evacuated.	Faulty gauge	Replace the gauge. The gauge cannot be re-calibrated.

Table 12.6 - 15 Vacuum Dial Gauge Malfunctions

Viewing Window Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Source Windows become coated with Aluminium	Aluminium Contamination, which is normal over a long period of time.	Clean the glass as described in the operating manual Use supplied shields, etc. to reduce the coating level.
FAULT: Vacuum leak on window		Consult with GVE before attempting to work on the window.
FAULT: Window becomes scratched or foggy	Incorrect method of cleaning used.	Replace the window. A replacement is available from GVE.

Table 12.6 - 16 Viewing Window Malfunctions

Keystone Isolation Valves Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Valve sticking	Incorrect Air Pressure	Check that the air supply to the machine is 80 psi
	Contaminated Spool Valve	Strip the valve and check for debris, clean as required
	Drive shaft Problem	Remove actuator and check rotation of shaft
FAULT: Valve not closing	Contaminated Spool Valve	Strip the valve and check for debris, clean as required
	Faulty Actuator	Remove, and check operation off valve
	Drive shaft Problem	Remove actuator and check rotation of shaft
FAULT: Open/close indication not correct	Actuator position switches not being activated	Check operation of reed switches and alter as required
	Valve not opening or closing correctly	- Check that the air supply to the machine is 80 psi. - Strip the valve and check for debris, clean as required
	Electrical wiring Fault	- Check for correct - Check open/close signal at ASI controller
FAULT: Vacuum leak across seal when closed	Actuator not closing valve correctly	Check operation of actuator. Note: The valve should be normally closed in most cases
	Valve shaft sheared	Remove the valve from the pipework and check for damage
	Seal Damaged	Remove the valve from the pipework and check for damage, replace where required
FAULT: Solenoid Chatters	Intermittent power supply to solenoid	Check the power supply for constant feed

Table 12.6 - 17 Keystone Isolation Valves Malfunctions

Source Power Supply Cable Malfunctions		
These are the cables that run between the busbars and transformers.		
Symptoms	Probable Cause	Possible Remedies
FAULT: Cables become warm	Loose connections	Check the terminals are tight
FAULT: Loss of power supply	Loose connections.	- Check the terminals are tight - Check all connections to the transformers.

Table 12.6 - 18 Source Power Supply Cable Malfunctions

Layarm System Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Layarm does not operate	No Power supply	- Check the layarm motor controller drive circuit breaker. - Check the motor control drive unit for the correct power up indicators.
	Interlock Stopping Operation	Check that the correct interlocks are active to allow the arm to operate.
	Mechanical failure	- Check the layarm belts for damage - Check the slide rails for damage - Check the vacuum leadthrough. - Check the gearbox/coupling to be in good condition. - Check that the brake unit s not engaged.
FAULT: Layarm assembly out of calibration	Position feedback not Operational	Check calibration as per section 14 of this manual.
FAULT: Layarm assembly moves slowly	Layarm Drive problem	Check fault condition against vendors manual
	Layarm mechanical friction increased	Check the layarm slide area for friction
	Brake not disengaged (if fitted)	Check that the brake is disengaging fully.
FAULT: Layarm will not move into roll when chamber is closed	Interlocks Incorrect	Check the interlocks to be correct for this operation to occur.
	Mechanical Obstruction	Check that there is no visual obstructions around the layarm assembly.
FAULT: Lay arm roller assembly out of alignment.	Can cause wrinkles in substrate.	Check alignment between last roller and rewind shaft as described in the maintenance manual
FAULT: Layarm will not layon to the roll in the "Lay on" mode	Friction binding of arm	Check that the assembly will move easily down the slide assembly.
	Incorrect balancing of assembly	Check that the configuration is set up correctly. Changes in rollers weight will affect this balance weight. Consult General Vacuum Equipment for advise.

Table 12.6 - 19 Layarm System Malfunctions

Internal Chamber Lighting Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Internal lighting does not work	Ballast unit failure	Check ballast unit for correct operation. replace as required
	No power supply	Check the incoming supply via the PLC or desk switch for correct value in accordance to the electrical drawings.
	Loose Electrical Leadthrough	Check the tightness of the electrical leadthrough.
	Wiring or connection Fault	Check the internal and external wiring to the lighting circuit for continuity.

Table 12.6 - 20 Internal Chamber Lighting Malfunctions

'Bow' Spreader Roller Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Roller will not rotate	Roller bearings seized	Remove roller and send to repair agency for overhaul.
FAULT: Roller will not rotate at the correct line speed	The roller is driven by a tendency driven device.	Check that the device has been set up correctly as per the instructions in the maintenance manual.
FAULT: Roller sleeve becomes Un-attached from the driven pulley	Breakdown of adhesive layer between pulley and sleeve	Roller has to be replaced
FAULT: Roller sleeve become hard and cracks	Roller sleeve has aged or has been attached by solvent	Replace roller
FAULT: Apex position motor does not operate	Motor Failure	Check motor – electrical continuity
	Positioning potentiometer problem	Check for feedback and reset as required

Table 12.6 - 21 'Bow' Spreader Roller Malfunctions

AC Motor Electrical Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Motor Does Not Start Without Load Or Doesn't Attain Full Speed	Bad contact between electrical supply and motor	Check all connections on wiring, terminals and contactors.
	Phase failure	Check fuses and supply voltage
	Low supply voltage	Check supply is to specification
FAULT: Motor Speed Drops When Load Is Applied	Incorrectly connected	Change connections from "star" to "delta" arrangement
	Bad contacts	Check all connections on wiring, terminals and contactors.
	Phase failure	Check fuses and supply voltages
	Overload	Check for and remove cause of overload
	Supply voltage too low	Check electrical supply and connections
FAULT: Starting Windings Overheat And Trips Protection Switch	Incorrect connections	Check the connections and correct if incorrect

Symptoms	Probable Cause	Possible Remedies
	Overload of motor	Check and remove the cause of the overload
	Supply voltage is too low	Check electrical supply and connections.
FAULT: Motor "Hums"	Phase failure	Check for three phases and repair
FAULT: Circuit Breaker Or Fuses Fail On Switching On The Motor	Overloaded motor	Check for and remove cause of overload

Table 12.6 - 22 AC Motor Electrical Malfunctions

Control Push-Button Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Occasional sticking of device	Contamination	Clean/replace device
	Worn Components	Repair/replace components
	Improper adjustments	Correct adjustment
	Mechanical Binding	Check for free operation
FAULT: Mechanical failure	Installed Incorrectly	Install Correctly or replace
	Damaged during operation	Replace and change operating procedures
FAULT: No operation, either make or break action	Loose connections	Check all terminations
	No control voltage present	Check the supply voltage is present
FAULT: Illuminated push-button does not illuminate	Bulb failed	Check the bulb for failure
	No voltage present	Check that the supply is present at the push-button

Table 12.6 - 23 Control Push-Button Malfunctions

Cryo-Generator System Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: The unit has no main power supply	The power supply from the relative power source has been removed	- Check the circuit breaker/fuses/overload to be OK. See the electrical drawing - Check that the water supply to the unit has not been reduced. - Check that the unit isolator switch is in the ON position
FAULT: The cryogenic system does not change into "cool" mode	The machine PLC interlocks are not in the correct mode to allow the unit to cool	- The vacuum sequence is not in the in the "Automatic" mode - Check the wiring for 24 v dc
	Machine "Cool" relay not being operated.	- Check the operation of the relay. - Check the fused output from the PLC output card.

Symptoms	Probable Cause	Possible Remedies
FAULT: The cryogenic system does not change into "defrost" mode	The machine PLC interlocks are not in the correct mode to allow the unit to cool	- The vacuum sequence is not in the in the "Automatic "mode - Check the wiring for 24 v dc
	Machine "Defrost" relay not being operated.	Check the operation of the relay.
FAULT: The cryogenic system does not change into "standby" mode	The machine PLC interlocks are not in the correct mode	Check the operation of the control circuitry. The "Cool relay should only be energized when the "vacuum" mode is requested. The "Defrost" relay should only be energized for a period of three minutes after the chamber has been vented and opened.
	"Cool or defrost" relays are not disengaged.	Check the operation of the relays.
	Winding cart closed switch not operating correctly.	Check the operation of the fully closed limit switch.
FAULT: The Cryo-Generator unit shuts off during normal operation	The unit may have shut down internally	Refer to the proprietary manual supplied

Table 12.6 - 24 Cryo-Generator System Malfunctions

Refer to the detailed manual supplied with the Cryo-Generator unit for further troubleshooting ideas.

PLC System Malfunctions		
This section is written as a general statement. Specific details will be found in the relevant manuals supplied by the vendors.		
Symptoms	Probable Cause	Possible Remedies
FAULT: The plc's outputs do not operate	There is a hardware/software Fault	Check that there is no fault lights on the processor module
	The power supply to the outputs is not present	Check for the power supply at the rack
	Output fuse failed	Check and rectify fault. Refer to the unit's manual

Table 12.6 - 25 PLC System Malfunctions

Refer to the proprietary manuals supplied with the machine for the PLC system for further troubleshooting methods.

Pneumatic Valves And Cylinders Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Pneumatic Isolation Valves Not Operating Or Sticking	No compressed air supply	- Check main air supply is switched on - Manual isolation valve closed on the manifold panel. - Pneumatic regulator closed, reset to 80 PSI - Pneumatic lines contaminated
	Solenoid valves faulty \ contaminated	- Check for power to solenoid valve - Test the valve for manual operation - Test the coil's integrity electrically - Strip the valve and inspect for debris
	Faulty valve actuator	- Pneumatic leak on actuator body - Excessive backlash due to worn piston rack or pinion, remove, examine and replace worn parts. - Manual override binding due to compounded grease in piston teeth. Remove, examine and clean and remove any worn parts.
	Valve installed incorrectly	Remove the assembly and examine for damage.
FAULT: Cylinders Inoperative	Worn piston lip seal	Remove piston and change seal. Inspect for damage to seal and inner bore of the cylinder.
	No air Supply	- Check that the air supply is available - Check that the solenoid spool valve is operating correctly.
	Mechanical Obstruction	- Check for any possible obstruction - Check that the coupling between the plate and cylinder has not parted
FAULT: Spool Valves not operating correctly -	Moisture in valves	Strip the valve and clean, check performance of customer system dryer
	Strip the valve and clean, check performance of system dryer	- Strip the valve and clean. - Investigate the source of the debris
FAULT: Magnetic Reed switches do not indicate action of pneumatic cylinders	Switch not being activated by internal magnetic strip.	Use a paper clip on the outside of the cylinder to locate the magnetic strip and position the switch for indication.

Table 12.6 - 26 Pneumatic Valves and Cylinders Malfunctions

High Vacuum Valve Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Valve does not appear to open correctly in sequence	Valve not connected to cylinder/assembly	Visually check the connection between the valve assembly and cylinder
FAULT: Valve opens and closes rapidly as the pump down sequence takes place	Vacuum pressure in chamber too high	Compare operation pressures with standard pump down pressures. Investigate for reasons of high pressure, i.e.; water leak

Symptoms	Probable Cause	Possible Remedies
	Foreline pressure too high	Check if the foreline pressure is rising above 5 E-1 mbar when the right angle valve opens. If it does, check the performance of the roughing pumps.
FAULT: Leadthrough Gravity fill bottle empties	Vacuum seal failed	Check for oil on the top of the valve, which has leaked, replace the seal arrangement.
	Oil bottle has external leak	Check if the bottle is leaking externally. Replace if required
FAULT: When valve opens, there is no indication on the vacuum mimic screen.	Incorrect position of reed switch on cylinder	Reset the switch to the correct position
	Air gap between switch and cylinder	Reset the switch to contact the cylinder.
	Wiring Fault	Check the continuity of the control circuit.
FAULT: Leakage across seal	Debris between valve plate and mating surface	Vent the pump, lift the valve plate and check for debris.
	O-ring Failure	Vent the pump, lift the valve plate and check for damage.
FAULT: Oil contamination in valve area	Back streaming of diffusion pump oil.	Refer to diffusion pump chart

Table 12.6 - 27 High Vacuum Valve Malfunctions

Reed Switches Malfunctions - Located On Cylinders		
Symptoms	Probable Cause	Possible Remedies
FAULT: Switch does not indicate	Incorrect position on cylinder	Reset the switch to the correct position
	Air gap between switch and cylinder	Reset the switch to contact the cylinder.
	Wiring Fault	Check the continuity of the control circuit.

Table 12.6 - 28 Reed Switches Malfunctions - Located On Cylinders

Winding Mechanism Roller Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Roller does not rotate	Bearing seized?	Remove the bearing and check the bearing
FAULT: Roller at incorrect speed	- Roller bearing tight - Tendency drive system failed	- Check the bearing for wear - Check that the tendency drive system is in good condition.
FAULT: Roller out of balance	Roller weights loose	Check that the balance weights have not become loose
FAULT: Roller damaging substrate due to roller surface damage	Substrate can be scratched	Polish the surface of the roller until the surface is flat
FAULT: Wrinkles appearing between rollers	Misalignment of rollers	Check the alignment using " outside micrometers between rollers.

Symptoms	Probable Cause	Possible Remedies
	Bearing Wear	Check the bearings for wear
FAULT: Arcostretcher Rollers guide substrate in left or right direction	Depths of machined cut is different from side to side	Check the depth of the machined cut
FAULT: Wrinkles appear at Archostretcher roller	Wear on roller causes grooving	Replace roller

Table 12.6 - 29 Winding Mechanism Roller Malfunctions

Liquid Rotary Glands Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Cooling Medium Leaking	Seal failure	Check seal tightness and replace as required
FAULT: Mechanical Grinding Noise	Seal Assembly dry	Lubricate with correct grease
	Bearing wear	Drain the cooling system and inspect the gland assembly, replace parts as required.
	Insert tube has worked loose	Drain the cooling system and inspect the gland assembly and insert tube, replace parts where required.
FAULT: Excessive movement of assembly when rotated	Poorly fitted union.	Replace. Excessive movement can lead to vibration.
FAULT: Rotary union "sweats" water onto the surrounding equipment.	Ambient temperature and humidity in surrounding atmosphere is > 45 ° C	Add extra insulation or lower the ambient atmosphere conditions.
	Insulation material has been removed or degraded	Replace the insulation material.

Table 12.6 - 30 Liquid Rotary Glands Malfunctions

Shutter (Moving Shield) Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Shutter does not operate	Mechanical obstruction or failure	Check the alignment of the shutter assembly and replace/realign as required
	Enable circuit not energized	- Check that the track enable limit switch is not binding - Test the circuit wiring and operational control circuit. - Check the spool valve
	Pneumatic fault	Check for pneumatic leaks on the pipework, if there is, repair them. There may be some bleeding of air from the regulated valves and this is normal.
		It is possible that a spool valve may become seized because of a build up of debris in the valve, clean the valve and reassemble it.

Symptoms	Probable Cause	Possible Remedies
Fault: Shutter "Bows" while metallizing takes place.	This is not a fault, but is a reaction of the difference in temperature between the lower and upper surfaces.	
Fault: Shutter opens or closes erratically	Mechanical interference.	- Push the shutter open and closed and check for binding to other sections of the machine. - Check shutter wheels turn. Release where required.

Table 12.6 - 31 Shutter (Moving Shield) Malfunctions

Solenoid Malfunctions		
Air Admittance Valves - Gas Ballast Valves - Refrigeration Valves - Keystone Valves - Pneumatic Spool Valves - Hydraulic Valves		
Symptoms	Probable Cause	Possible Remedies
FAULT: Noisy operation	Low supply voltage	Check the supply voltage for correct supply voltage.
	Mechanical binding	Repair -replace source of binding.
	Defective Coil	Repair/replace coil
FAULT: Coil Failure	High ambient temperature	Ventilate the area
	Supply voltage too high	Check supply voltage and take action to reduce voltage
	Incorrect duty cycle	Use continuous related coils for correct duty conditions.
FAULT: No operation	Mechanical Overload	Reduce the mechanical loading.
	Incorrect Coil	Investigate and change as required
	Incorrect supply voltage	Check the power supply
FAULT: Coil/device will not de-energize	Voltage still applied to the coil	Check why the power supply is still present
	Mechanical Binding	Check the device for debris and clean.
FAULT: Coil melted - mechanically damaged	Over voltage Condition	Check that the coil is rated for the voltage applied.
	Coil removed from device	Replace the coil, as it will have expanded.

Table 12.6 - 32 Solenoid Malfunctions

Window - Anti Coating Devices - Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Windows get coated during process run	Anti coating Device not installed.	Install the device supplied
FAULT: Window scratched	Incorrect materials used to clean windows.	Replace window

Table 12.6 - 33 Window - Anti Coating Devices - Malfunctions

Vacuum Pump Motor Control Switchgear Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Contactor electrical contacts chatter	Poor contact in control circuit.	Improve contact or use holding contact
	Low voltage	Check voltage condition. Check momentary voltage dip during starting of equipment.
	Dirty contacts	Disassemble and clean.
FAULT: Contacts Welded Together	Abnormal inrush of current	- Ensure the contactor is rated for the application. - Check for shorts in wiring or motor
	Rapid switching of device	Check control circuit Install a larger device
	Short circuits	Remove the short and check that the protection device is correctly sized
	Loose connection	Clean and tighten
FAULT: Open Circuit	Sustained overload	Install larger device Check for excessive motor load current
	Mechanical damage to the coil	Store the coil carefully
FAULT: Coil Overheating	Over Voltage	Check if correct equipment has been purchased
	High ambient temperature	Ensure that the cubicle cooling device is operating
FAULT: Overload Tripping Constantly	Sustained overload	- Check for short circuit to earth - Check for short circuit between phases - Check for excessive motor current
	Loose connection on load wiring	Clean and tighten
	Incorrect current setting of device	Reset to the correct level

Table 12.6 - 34 Vacuum Pump Motor Control Switchgear Malfunctions

Evaporator Problems Thyristors TU 1471		
Mechanical Related		
Symptoms	Probable Cause	Possible Remedies
FAULT: Volts & amps screen displays only voltage and no current for 1 evaporator.	Loss of conductance through flexible joint	Check the condition of the flexible joint and the fasteners used to attach it.
	Loss of electrical conductance through busbar system	Check the condition of the joints in the copper busbar system and the clamps onto the water cooled busbars
FAULT: Intermittent heating of evaporator	Temporary loss of conductance through flexible joint	Check the condition of the flexible joint and the fasteners used to attach it.
	Temporary loss of electrical conductance through busbar system	Check the condition of the joints in the copper busbar system and the clamps onto the water cooled busbars

Symptoms	Probable Cause	Possible Remedies
FAULT: Volts & amps screen displays only voltage and no current for a number of evaporators. The evaporators are not common to a single stack	Loss of electrical conductance through neutral busbar system	Check the condition of the joints in the copper busbar system and the clamps onto the water cooled neutral busbar
FAULT: Volts & amps screen displays only voltage and no current for one or more evaporators	Loss of conductance between evaporators and contact blocks.	Check the contact between the evaporators and the contact blocks. Examine for poor contact and re-install with graphite tape.
FAULT: Tripping of circuit breakers as the power is slowly increased to the evaporators or during a coating cycle.	- Short between surrounding metalwork and live post or evaporator - Excessive amperage is drawn and the fuse is blown.	- Visually examine the area between the evaporators for "burn or arcing" marks. - Check the condition of the insulating spacers for cracks and damage. replace as required.
FAULT: Tripping of circuit breakers as the power is slowly increased to the evaporators or during a coating cycle. The evaporators maybe next to each other physically.	- Short circuit of evaporator power supply between separate phases. - Excessive amperage is drawn and the fuse is blown	- Visually examine the area between the evaporators for "burn or arcing" marks. - Check between fastening screws in the post assembly.
FAULT: After a period of running the evaporators, one or more evaporators cool down and there is no amperage displayed on the volts/amps screen	Local heating of tensioning springs, causing loss of torque required to clamp the evaporators.	Check that the springs have not lost the retained tension levels as described in the attached notes. If they have replace the springs.

Table 12.6 - 35 Evaporator Problems Thyristors TU 1471

Computer To Thyristor Communications Related		
Symptoms	Probable Cause	Possible Remedies
FAULT: No Evaporators will heat up when the "Heaters On" button is touched on the Process Screen	Loss of communications to stack by disconnection of wiring.	- Check communications is taking place as described in the machine maintenance manual. - Check that the wiring between the computer and the thyristors has not been damaged. Check for continuity using a low voltage multimeter. Ensure that the machine is de-energized before carrying out this test. - Check the wiring for a short circuit, as described above. - Ensure that all the plugs to the thyristor stacks are located in the correct place. - Ensure that the "jumper" ST9 is set to the correct position for digital control. - Check that the communications boards

	• Short between control signal wires between thyristor stacks and computer. • Plug not connected to top of communications board. • More than one unit set to the same communication address	do not have the same addresses. check that the "jumpers are correct as described in the GENERAL VACUUM EQUIPMENT maintenance manual supplied and electrical drawings.
	• Loss of internal power supply module.	• Check that the internal power supplies are present using the Eurotherm 260 diagnostic unit. If they are not present and the power supply is not being supplied, change the power supply module. • Alternatively, check that the green "power on" led is illuminated
FAULT: Lower power level applied to evaporators after ramp up	• Base wire speed level is too low	• Ensure that the base wire feed speed level is set at initialise stage

Table 12.6 - 36 Computer to Thyristor Communications Related

Power Supply Related		
Symptoms	Probable Cause	Possible Remedies
FAULT: The volts and amps screen displays a fixed level for all selected channels	• Loss of auxiliary power to thyristor unit.	• Check that the green led is lit on the communications board. If is not check that the auxiliary power supply is being fed to the unit.
FAULT: There will be a voltage displayed on the DCM volts & amps screen, but no current for one of the evaporators.	• Loss of main power to one of the channels 2,3,4	• With the main power removed from the C4 contactor, check the conductance through each fuse or circuit breaker. This will locate the problem. If the fuse/CB continues to blow, refer to other sections of these notes.
FAULT: Loss of current display for 4 evaporators on volts & amps screen. The evaporators will not be next to each other, i.e.: 1,4,7,10	• Loss of power to channel 1	• With the main power removed from the C4 contactor, check the conductance through each fuse or circuit breaker. This will locate the problem. If the fuse/CB continues to blow, refer to other sections of these notes.
FAULT: On supplying power to the thyristor units, a evaporator heats up instantly and stays at the same power level. This level cannot be increased or decreased by the operator.	• Short-circuiting of failed SCR unit.	• Check the SCR with power off to the machine. Use a multimeter to check that there is no short circuit between the supply and output terminal.

Symptoms	Probable Cause	Possible Remedies
FAULT: On pressing heater, an evaporator will heat up very quickly and the supply fuses to the channel will blow. (circuit breakers will trip) There is no fault lights illuminated.	- Faulty driver board. - Poor connection to internal current transformer	- Replace the suspect driver board as described in the attached notes. Ensure that addressing of board is correct. Refer to the maintenance manual supplied. - Check the current transformer connections to the driver board
FAULT: The heater supply contactor will not switch on when the "Heaters On" button is touched	All the required interlocks are not present.	Check that the required interlock signals are present.
FAULT: As described in the prior fault, and all the required interlocks are present. The output from the PLC is supplying power to the slave contactor	Fault on slave contactor	- Check the slave contactor HC is being supplied 110 volts ? - Check that the coil is OK
FAULT: The slave contactor is energized after touching the "Boats On" -button, but the main contactor C4 is not energized.	- Fault on main contactor coil - No power supply to coil - Mechanical failure of coil	- Use a multimeter to check the coil - Check that the power supply is being fed to the contactor coil - Inspect the contactor for mechanical damage
FAULT: The contactor C4 does not hold in after the push-button is released	Latch circuit not being engaged.	Check the auxiliary contact power supply and engagement signal to the PLC.
FAULT: The contactor C4 is engaged, but no power is supplied to the thyristors	No main power supply to this part of the machine	Check the incoming main supply to the C4 contactor
FAULT: On touching the "Boats Off" button the evaporators do not cool down.	Faulty Contactor	Check the operation of the contactor.
FAULT: Volts and amps display does not agree with actual values.	Incorrect calibration of output signals from thyristor units. Especially when new communication boards have been fitted, or primary transformer tapping are changed.	Communications board calibration must be carried out as per Eurotherm manual using 260 diagnostic tool

Symptoms	Probable Cause	Possible Remedies
FAULT: After the voltage has increased to about 30 % of the required level, it will shut off if no current is being drawn. Version 612 software only	No current being drawn initially - reasons can be :- poor connections, high resistivity evaporators	- Check the tests as described elsewhere in this document. - Check the resistivity of the evaporator is less than 2800 Hot Resistivity
FAULT: No output from all four thyristors during cycle. appears to occur at a similar time in each coating cycle.	- Cooling fan not rotating - Ambient temperature too high - Enclosure temperature too high - Electronic fault failure	- Check cooling fan rotating - Ambient temperature too be less than 45 degrees C - Check force cooling fan is OK and filters are not blocked - Examine the components for signs of over-temperature. Send thyristor stack back to supplier for check out.
FAULT: The evaporator does not receive the correct power level, although the voltage displays a correct voltage. All other tests carried out would indicate a correct situation.	- Possible faulty component. - Power Board - Microprocessor Board - Half wave firing	Swap components as instructed to locate fault component. These instructions are given in the thyristor maintenance manual.
FAULT: Evaporator will not heat up	Loss of power supply fuse or circuit breaker tripped	Check voltage slowly increases at transformer primary terminals. Do not check a single phase to a ground reference, because feedback through the transformer can occur
FAULT: No power outputs to evaporators from a single thyristor unit	No enable signal	Check enable link is fitted at the top of the thyristor unit
FAULT: Erratic pulsing noise coming from transformers and then blowing of fuses or tripping of circuit breakers	- Incorrect firing mode selected. The correct mode of operation should be phase firing not fast cycle. - Fluctuations on incoming power supply	- A updated software mod has been made to the EPROM on the thyristor stack. Check with GENERAL VACUUM EQUIPMENT EPROM needs to be replaced. - Use power supply monitoring equipment if this to diagnose problems with waveforms, voltage levels, etc. - Seek advise from power supply company

Table 12.6 - 37 Power Supply Related

Operator Related Problems		
Symptoms	Probable Cause	Possible Remedies
FAULT: Evaporators outside the coating width are heated up.	Incorrect substrate width selected	The user should enter the correct width that the substrate will be coated . This is carried out in the ROLL SETUP Screen.
FAULT: Not enough evaporators are heated up to allow uniform coating	Incorrect substrate width selected	The user should type in the correct width that the substrate will be coated .
FAULT: Evaporator power cannot be increased to provide enough evaporation, causing poor uniformity and spitting.	Old evaporators being used, which cannot be heated to a higher rate.	Change evaporators.
FAULT: Evaporators become very hot in one area, usually away from the wire feed.	Evaporators not cleaned between coating cycles. Therefore, aluminium does not flow down the full length	Clean oxide coatings off between cycles.
FAULT: After ramp period, evaporators are very cool and wire does not melt on them.	Low set-point caused by:- Initialised Low wire feed set point or running the power down to a low level the previous coating cycle, which is remembered for the next cycle.	Ensure that the wire feed set-point is correct for the operating value to be used during the cycle
FAULT: "New Boats" ramp up time is too fast	The "New Boats" ramp curve is too quick.	There is a curve in the ROLL SET UP Screen that can be adjusted.

Table 12.6 - 38 Operator Related Problems

Power Transformers Malfunctions		
These transformers are generally used to reduce the main incoming voltage to either a 240 or 110 volts ac supply.		
Symptoms	Probable Cause	Possible Remedies
FAULT: Excessive heat generated from transformer	Improper rated unit	Check that the device is rated correctly. Replace as required.
FAULT: Corrosion of transformer	Excess moisture	Dry the transformer, remove source of moisture.
	Corrosive fumes	Relocate the transformer or vent the fumes away.
FAULT: Insulation failure leading to short circuit condition	Overheating	- Lower ambient temperature. - Lower load demand
FAULT: Tripping circuit breaker	Inrush current from equipment supplied from transformer	Install breaker rated for inrush conditions
	Loose connection	Check all connections

Symptoms	Probable Cause	Possible Remedies
	Excessive current	Check design loading
	High ambient temperature	Lower ambient temperature

Table 12.6 - 39 Power Transformers Malfunctions

Winding Cart Linear Drive Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: The cart will not move.	Check that the safety lockout isolator is in the correct mode.	Place the switch in the correct position
	Missing interlock in control logic.	Check that the required interlocks are energized.
	Mechanical obstruction	Investigate and rectify.
	Chamber closed under vacuum condition	Vent the chamber to atmosphere
	Circuit breaker tripped	Check condition of circuit breakers. There are two used in the circuit.
	Faulty contactor	Check the operation
	Mechanical Problem with assembly	- Check the torque restrictor for correct operation. - Check the chain for slackness. - Ensure that the drive is rotating the drive wheels
FAULT: Cart movement does not slow down as it reaches the fully open or closed positions.	Limit switch not operating	Check the operation of the slow limit switches.
	Faulty interconnection wiring	Check the wiring for continuity.
FAULT: The warning siren does not operate when movement is taking place.	Faulty Unit	Check the operation of the unit. Replace where required
	Missing interlock in control logic.	- Check that the required interlocks are energized. - These maybe different between machines.
	Faulty interconnection wiring	Check the wiring for continuity

Table 12.6 - 40 Winding Cart Linear Drive Malfunctions

Water Pressure Device Switches Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Pressure switches sensing low pressure and shutting down machine systems	Loss of water pressure	Check the water supply to the machine. It should meet the minimum levels shown on the water schematic drawing.
	Return Valve Closed	Check if the return valve on the manifold is closed and open it

Symptoms	Probable Cause	Possible Remedies
	Feed Valve is closed	Check if the feed valves are fully open
	Loss of control system power	Check the power supply to the sensor
	Fault on interconnection wiring	Check the continuity of the wiring.
	Strainer blocked with debris.	Remove and clean the strainer feeding the water manifold in question.
FAULT: Removing water from machine does not produce alarm statement on Alarm Screen	Software is not configured correctly to indicate failure	Consult with General Vacuum Equipment
FAULT: Removing water from machine does not switch equipment off.	Faulty pressure monitoring equipment.	- Check the operation of the device - Replace where required.
	Incorrect PLC logic control	Check that the PLC logic will shut down the equipment that the switch is protecting.

Table 12.6 - 41 Water Pressure Switches Malfunctions

Software System Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: No bar charts shown on DEPOSITION display on PROCESS Screen.	- Metallized film with heavy deposit in front of probes. - Monitor hardware failure. - System out of calibration	- Remove film - See section on hardware fault message - See below.
FAULT: Out of calibration (Symptoms: Lighter or heavier deposit measurements when read off machine.	- Calibration sequence carried out incorrectly - Calibration slip damaged - Self-Test taking place with lightly coated film in front of probes.	- Follow calibration sequence as instructed in the manual provided. - Replace calibration slip. - Ensure that there is no coated film in front of the probes at the time of Self Test.
FAULT: Film has light edges.	- Initialise page not set to the correct width - Web not cantered	- Check coating window size against that set in the ROLL SETUP Screen. - Re-centre. If web is telescoping fit side cheeks or increase unwind tension
FAULT: MMI monitor does not power up.	- No mains supply. - Loose connection - Switched off.	- Check that power is available to it - Check the cabling. - On some monitor's on/off switches may be included.

Symptoms	Probable Cause	Possible Remedies
FAULT: Controller not controlling as well as previously.	- Incorrect grade of aluminium wire being used. - Vacuum level not below 8E-4 mbar - Wrong side of web being metallized. - Dyne level too low	- Try a different batch of wire. - Check metallizing zone pressure on gauge - Only one side of the web is treated to make the metal stick to it. - Surface treatment is not sufficient for good adhesion.
FAULT: In the warm-up mode, no power outputs are displayed on the output screen.	- Controller in stop mode. - Power ramp level not set-up	- Ensure the controller power button is indicating on position. - Ensure that the ROLL SETUP screen has been down-loaded
FAULT: MMI Monitor powers up but no display appears	- No mains to PC. - Loose connection - PC switched off	- Check that power is available at the PC power connection. - Check the cabling - Check on/off switch.
FAULT: MMI Monitor powers up but program does not load.	- Fault in PC. - Software corrupted	- Consult with GVE - Install back up disc.
FAULT: Deposition Display is "frozen "	No power to buffer box	Ensure power is supplied to the interface box located inside chamber
	Faulty wiring connection	Check the following are correctly located in sockets/plugs.
		Beam to leadthrough ribbon cable (vacuum side)
		- Leadthrough to buffer box cable). - Buffer box to beam
	Broken cable	Switch the power off. Check the above, connection to connection with an ohmmeter.
FAULT: Loss of control to sources	Loss of control signals.	Check cables to thyristor stacks.
FAULT: Disc drives are re-booting during normal working cycle.	Voltage drop.	Check if any new additions to the control circuitry are causing a volt drop.
FAULT: PC shuts down.	Overheating	Check that the fans are running and that the filters are clean.

Table 12.6 - 42 Software System Malfunctions

Rittal Air Conditioning Unit Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: The temperature in the electrical enclosures is excessive	- Loss of Refrigerant gas - Loss of power to air conditioner - Blockage of intake filter - Incorrect setting of internal thermostat	- Replace Unit - Check for the correct power supply - Clean the inlet filter - Check the setting of the thermostat. refer to vendor manual supplied
FAULT: Water coming out of the bottom of the unit	Condensed water	The units have been supplied with tubes to allow the condensed water to be drained.

Table 12.6 - 43 Rittal Air Conditioning Unit Malfunctions

Siemens Combimaster Drive Malfunctions - Traverse Drive		
Symptoms	Probable Cause	Possible Remedies
FAULT: Sirens does not sound and traverse drive does not move.	Interlocks not in the correct state.	Check that all the required interlocks are in the correct state to allow the winding cart to enter the chamber
FAULT: The siren sounds , but the winding cart does not move.	- Lockout Isolator in OFF position - No power supply - Mechanical Obstruction	- Check that the switch is in the correct ON position. - Check that the supply circuit breaker has not tripped. - Check that there are no mechanical obstructions.
FAULT: The winding cart does not slow down at either the fully open or closed ends of the travel.	The slowdown limit switches are not operating correctly	Check that the striker plates are operating the switches.
Fault: Traverse motor does not attempt to operate	Investigate fault diagnostic LEDs on side of unit	See section 2 of this manual

Table 12.6 - 44 Siemens Combimaster Drive Malfunctions - Traverse Drive

TES Chiller Unit Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: System not operating – Power Off	Main / local isolator off	Check all isolators from mains to unit.
	Mains fuse(s) failed.	- Check all mains fuses. - Replace after correcting fault. Check for loose wires.
FAULT: System not operating – Power On	Unit not switched on	Switch on.
	Control MCB tripped.	Re-set after investigating and correcting fault
	Loose wire in control circuit.	Investigate and tighten connector.
	Safety device or internal relay switch open circuit.	- Check through circuitry starting at control MCB. - Check primary and secondary voltage.
FAULT: High Temperature Chilled Water	Compressor not operating	See unit not operating power off – power on
	Compressor not operating efficiently	Fit gauges and investigate. Use refrigeration section of troubleshooting.
	Load too great for system.	High ambient affecting condenser performance. Investigate design and loads.
	Controls incorrectly set or faulty	Reset controls.
FAULT: Low chilled water temperature	Compressor operating too long or loaded unnecessarily	Check operation of control thermostat
	Chilled water stat incorrectly set or faulty.	Reset, repair or replace.

Symptoms	Probable Cause	Possible Remedies
	Low water flow rate	Check pumps, flow switches and strainers
FAULT: Compressor Not Operating	Seized compressor possibly due to poor lubrication.	- Replace compressor if faulty. - Check system operating conditions
	Defective compressor motor.	Check windings to each other – replace compressor.
	Klixon out does not reset	Sometimes it takes up to 4 hours to reset. Replace compressor if necessary.
	Low pressure switch operated (large or complete loss of refrigerant charge).	Reclaim refrigerant, repair leak and recharge system. Evacuate before charging using good refrigeration
FAULT: Noisy Compressor	Check refrigerant charge.	If low, repair leak and charge to correct design conditions adding oil if necessary.
	Scroll damaged due to lack of oil.	Replace compressor.
	Low superheat allowing liquid into compressor.	Replace EEV driver.
FAULT: Head Pressure too High	Condenser clogged or dirty	Clean condenser.
	Air or other non condensable gas in system	Not possible to reclaim complete charge.
	Over charge of refrigerant.	Reclaim excess from system.
FAULT: Compressor Short Cycling or LP Cut-out Operated.	Dirty filters / drier	Replace
	LP switch operating too high	Check, recalibrate if possible or replace.
	Liquid line isolating valve not fully open.	Check and adjust as necessary.
	Lack of refrigerant.	Repair leak and recharge system
FAULT: Suction Pressure too Low.	Low water flow.	- Check filters and pipework for obstructions. - Check operation of pump and flow switch
	Flash Gas at Expansion Valve	Check for leaks. Repair. Recharge System.
	Clogged filter drier.	Replace
	Obstruction in expansion valve	Pump down and clean or replace
	Superheat too high.	Replace EEV driver.
FAULT: Suction Pressure too High.	Expansion valve passing too much.	Check superheat.

Symptoms	Probable Cause	Possible Remedies
FAULT: Freeze Thermostat is tripped.	- Incorrect temperature set-point - Thermostat open circuit - Fluid temperature too low - Low process liquid flow	- Reset to the correct level, ensuring that the anti-freeze mix is correct - Check the operation - Switch is operating correctly. examine the thermostat for correct operation - See low flow fault
FAULT: Flow Switch does not activate	- Process pump not running - Air cavitating in pump - System not filled. - Blocked in-line filter - Valve closed	- Check why pump is not running. - Bleed the air from the pump. - Ensure that the system is fully filled - Check that if a filter is fitted, it is clean. - Ensure that the relevant valve is open
FAULT: Cooled roller does not heat UP	- Loss of power to heater banks - Incorrect thermostat setting - In-line pump failure - Faulty Level switch	- Check the heater contactor and power supply - Check the thermostat setting in the heaters - Check the operation and control circuit of the in-line pumps - Check that the level switches are operating correctly
FAULT: Loss of cooling medium	- Evaporation of liquid in storage tank - Leakage from a fitting	- Keep lids on tanks at all times - Check for leakage and rectify
FAULT: Heating storage tank is getting too hot	Thermostat set incorrectly	Check the setting of the thermostat and rectify where necessary

Table 12.6 - 45 TES Chiller Unit Malfunctions

Kinertrol Pneumatic Actuators Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Indication of operation not occurring.	- Internal limit switches not being activated. - Faulty wiring	- Check the operation of the switches - Check the wiring
FAULT: Full movement of unit not occurring	- Mechanical limitation by external fixture. - Mechanical restriction by in-built limiting screws	- Check for obstructions - Reset the limiting screws
FAULT: Operation is sluggish	Low Air pressure	Ensure the air supply is at 80 psi

Table 12.6 - 46 Kinertrol Pneumatic Actuators Malfunctions

Siemens Asi Modules Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Power Led is not illuminated	- No power supply - Incorrect cabling	- Check that power supply unit is powered up - Check that the interface cable has been installed correctly
FAULT: Power Led is illuminated, but signals not being transmitted back to interface unit.	- Incorrect addressing - Faulty Unit	- Check addressing at interface unit - Replace unit

Table 12.6 - 47 Siemens Asi Modules Malfunctions

AFM SI Interface Module Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: CTRL led not lit	Unit not in run condition	Change operating mode
FAULT: ASI PWR led not lit	- No power to Unit - Unit Failed	- Check power supply - Replace unit
FAULT: Message E-50 or E-59	Internal Error	Check configurations

Table 12.6 - 48 AFM SI Interface Module Malfunctions

Refer to vendor manual supplied.

Omni-Phase Surge Protector Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Led light are not lit	- No Power Supply - Unit failed	- Ensure the circuit breaker is switched on - Replace the unit

Table 12.6 - 49 Omni-Phase Surge Protector Malfunctions

Machine System Communications Malfunctions		
Symptoms	Probable Cause	Possible Remedies
FAULT: Vacuum system not operable	Lost communications link between in Profibus network	- Check that all relevant units on the Profibus network are powered up. - Check for loose wiring between units - Check that addresses of modules have not been changed - Check for fault indications as referred to in section 2 of this manual.

Table 12.6 - 50 Machine System Communications Malfunctions

B&R Wire feeder System		
Symptoms	Probable Cause	Possible Remedies
FAULT: No wire feeders operating	Loss of power to module. - System shut down - Faulty component	Check for power - Check that system is operating - Check as instructed in vendor manual
FAULT: Communications fault	- Profibus plug loose - Profibus connector end resistor engaged	- Check - Check

Table 12.6 - 51 B&R Wire Feeder Drive System

Plasma Power Supply		
Symptoms	Probable Cause	Possible Remedies
FAULT: Fault Message Displayed		Check vendor manual

Table 12.6 - 52 Plasma Power Supply

KEB Inverter Units		
Symptoms	Probable Cause	Possible Remedies
FAULT: Fault LED Indicated		Check vendor manual

Table 12.6 - 53 KEB Inverter Units

Eurotherm TE10P Thyristor Units		
Symptoms	Probable Cause	Possible Remedies
FAULT: Unit does not Power Up	Loss of fuses or circuit breaker	Check fuses and circuit breaker in panel E3.0
FAULT: Fault Code Indicated		Check vendor manual

Table 12.6 - 54 Eurotherm TE10P Thyristor Units

12.7 Fault Finding Using PLC Input - Output Racks

The PLC input and output racks supplied with this machine can be used for faultfinding.

If a certain part of the equipment is suspected to be faulty (i.e. a limit switch, we can look at the corresponding input indicator on the PLC system). If the indicator is illuminated, the switch is made, if there is a fault at the switch the light will be extinguished.

On outputs, if the indicator is illuminated but the corresponding function does not take place, check the output fuse and external wiring.

For more detailed faultfinding, use a relevant programming/investigating device to status inputs, internal timers etc.

To allow for overcoming potential plc interlock problems refer to section 12.2 of this manual.

12.8 Computer MMI Generated Alarms

The computer will report to the user, alarm conditions that occur with the machine hardware and services supplied to the machine.

An alarm condition is relayed to the machine user by two visual indicators. The ALARM tab at the top of the screen will flash and also there is a colour indicator at the bottom of the screen as shown below.



Figure 12.8 - 1 Alarm Indicator

The colour codes indicate the following:

- Green Cleared Fault
- Yellow Acknowledged Fault Present
- Red New Alarm

Pressing the ALARM tab at the top of the screen will bring up the following screen

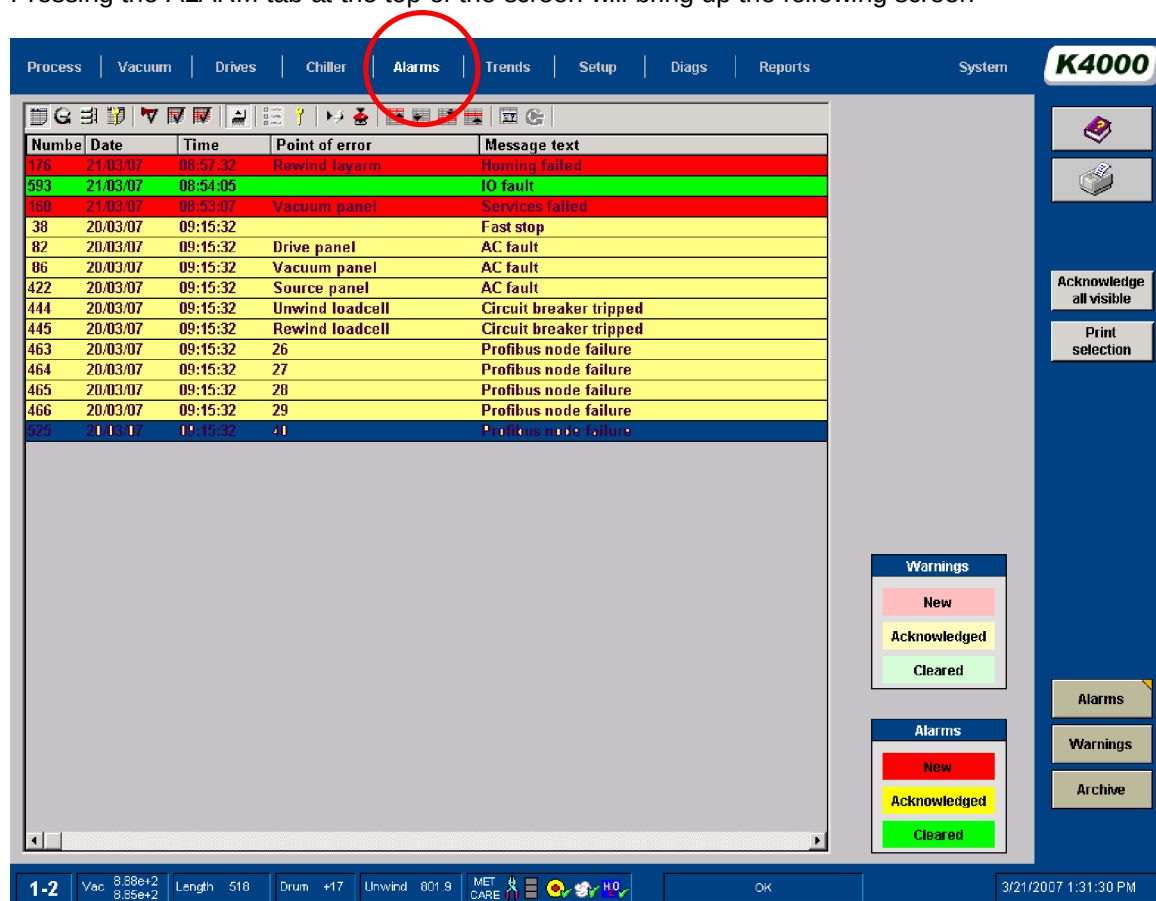


Figure 12.8 - 2 Alarm Screen

Messages displayed in this box indicate the time and date that they occurred and the type of failure.

Alarms that have been acknowledged will be highlighted a different colour from their original.

Only the alarms that can be viewed can be acknowledged. Once the fault has been overcome, the message will be removed after acknowledgement.

Alarm Screen Options:

Acknowledge all visible	Pressing this button on the screen will acknowledge all those visible on the screen. It will be recorded by the computer that this acknowledgement has taken place.
Print selection	This button can be pressed to print out an individual alarm message.
Alarms	Touching this button will display the current alarms
Archive	Touching this button will display the historic alarms
Warnings	Touching this button will display the current warnings

Figure 12.8 - 3 Alarm Options Table

There are other functions, but these are not used. These are standard WINCC functions.

If the Archive button is then highlighted, the following screen or similar will be displayed.

Index	Date	Time	Alarm
04	07/09/04	04:35:20 PM	Source water pressure low
04	07/09/04	04:35:28 PM	Source water pressure low
04	07/09/04	04:35:28 PM	Source water pressure low
04	07/09/04	04:35:31 PM	Source water pressure low
04	07/09/04	04:35:39 PM	Source water pressure low
04	07/09/04	04:35:39 PM	Source water pressure low
04	07/09/04	04:35:41 PM	Source water pressure low
04	07/09/04	04:35:48 PM	Source water pressure low
04	07/09/04	04:35:49 PM	Source water pressure low
04	07/09/04	04:35:51 PM	Source water pressure low
04	07/09/04	04:35:58 PM	Source water pressure low
04	07/09/04	04:35:58 PM	Source water pressure low
04	07/09/04	04:36:01 PM	Source water pressure low
04	07/09/04	04:36:08 PM	Source water pressure low
04	07/09/04	04:36:08 PM	Source water pressure low
04	07/09/04	04:36:11 PM	Source water pressure low
04	07/09/04	04:36:18 PM	Source water pressure low
04	07/09/04	04:36:18 PM	Source water pressure low
04	07/09/04	04:36:22 PM	Source water pressure low
04	07/09/04	04:36:28 PM	Source water pressure low
04	07/09/04	04:36:28 PM	Source water pressure low
04	07/09/04	04:36:29 PM	Source water pressure low
04	07/09/04	04:36:30 PM	Source water pressure low
04	07/09/04	04:36:38 PM	Source water pressure low
04	07/09/04	04:36:33 PM	Source water pressure low
04	07/09/04	04:36:40 PM	Source water pressure low
04	07/09/04	04:36:49 PM	Source water pressure low
04	07/09/04	04:36:43 PM	Source water pressure low
04	07/09/04	04:36:50 PM	Source water pressure low
04	07/09/04	04:36:50 PM	Source water pressure low
04	07/09/04	04:36:53 PM	Source water pressure low
04	07/09/04	04:37:01 PM	Source water pressure low
04	07/09/04	04:37:01 PM	Source water pressure low
04	07/09/04	04:37:04 PM	Source water pressure low
04	07/09/04	04:37:11 PM	Source water pressure low

Figure 12.8 - 4 Alarm History Screen

Warnings Screen

This screen shows the current warning on the machine. Warnings highlight potential problems or reminders and do not result in the stopping of the machine or equipment on it.

The colour codes indicate the following:

- Light Green Cleared Fault
- Light Yellow Acknowledged Fault Present
- Light Red New Warning

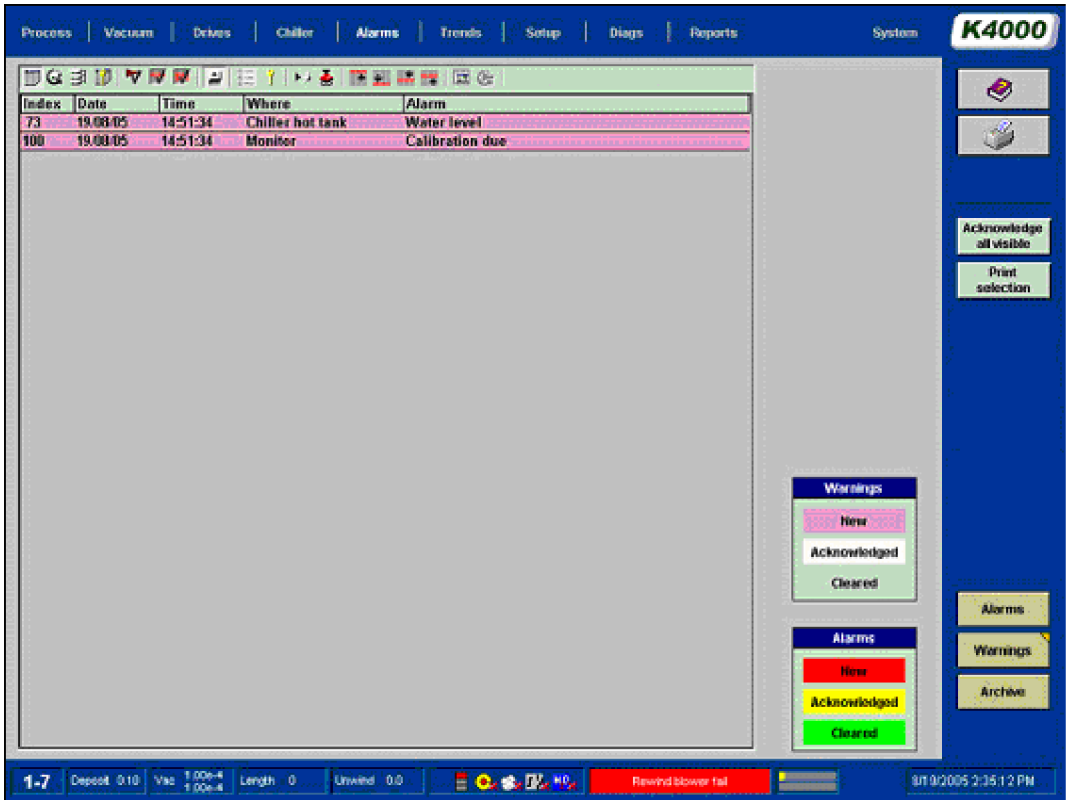


Figure 12.8 - 5 Warnings Screen

SECTION 13

MACHINE LUBRICATION

The following pages contain reference tables of the oils and greases used in the various components used on General Vacuum Equipment machines.

Section 13.1	Preventative Maintenance Lubrication Tasks
Section 13.2	Lubrication Points on Machine
Section 13.3	Equipment Oil /Grease Types and Capacities
Section 13.4	Gearbox Capacities and Oil Level Indication
Section 13.5	Grease and Oil Specifications
Section 13.6	Grease and Oil MSDS Sheets

	Note!
	Contaminated components and operating media must be disposed of in accordance with local government regulations where the machine is operated.
	Warning !
	Pump oil and oil mists contaminated with hazardous substances can be hazardous to your health. Skin or eye contact with pump oil and inhaling oil mists can be hazardous to your health. Wear relevant personal protective equipment when draining oil from vacuum pumps, which can potentially contain harmful gases or other harmful substances, and when disposing of exhaust condensation or filters.
	Caution !
	The lubricating oils from the vacuum pumps are potentially harmful to the environment. Never allow pump oil or parts dipped in pump oil to penetrate the supply of drinking water. Pump oil (even small amounts!) must be disposed of as hazardous waste.

13.1 Lubrication Tasks

The following table shows the suggested grease lubrication period on various equipment found on the machine. Photographs showing the actual greasing points are shown in the preventative maintenance schedule.

	Caution !
	Failure to follow recommended maintenance procedures may lead to machinery damage

Grease Lubrication

Equipment	Grease Type	D	W	M	Q	BA	A	R	S
Filton Rotary Gland	B		XX					XX	
Face-plate O-ring	C							XX	
Load cell bearings	A							XX	
Cart Castors	A			XX				XX	
Tendency Drive bearings	A		XX					XX	XX
Winding Mechanism Bearings	A		XX					XX	XX
Shutter Leadthrough	A							XX	
Reel Shaft Bearings	A				XX			XX	
Shutter Drive Bearings	A							XX	
Vacuum Test Ports	C							XX	XX
Pipework O-rings	C							XX	XX
Window O-rings	C							XX	XX
Isolation Valve Seals	C							XX	XX
Rubber Bellows	C							XX	XX
Grease Filled Gearboxes- int-chamber	A							XX	XX
Layarm Slides (grease type)	A		XX						
Grease Filled Gearboxes external	see plates								
Baldor Motors (Kinney vacuum pumps)	E			XX					
Strobe Window O-ring	A		XX					XX	XX
Transformer to Steel Connections	F								XX
Strobe Cylinder Bearings	A			XX				XX	XX

Table 13.1 - 1 Lubrication Schedule - Greases

Key to above chart:

D	Daily	W	Weekly
M	Monthly	Q	Quarterly
BA	Bi-annual	A	Annual
R	Removal Of Equipment For Repair	S	Strip Out and replacement

Notes:

- Please note that cross-reference of the greases and the codes used by suppliers may change from country to country.
- If users propose to use a grease type not listed, it is suggested that before the grease is purchased, the specification of the grease is to be sent to the equipment manufacture for their recommendations.
- The greases used within the machine are selected for the required working conditions and low outgassing properties.

	Caution !
	Do not use grease types that are not on the recommended list, as this could lead to failure of components upon the machine.

Oil Lubrication

Equipment	Oil Type	D	W	M	Q	BA	A	R	S
Kinney KT505 LP Pumps	Section 13.3				XX			XX	XX
Leybold Sogevac Pumps	Section 13.3				XX				
Aerzen Mechanical Booster Pumps	Section 13.3					XX	XX	XX	
Diffusion pumps	Section 13.3						XX		XX
Unwind Chuck Gravity Bottle	Section 13.3							XX	XX
Rewind Chuck Gravity Bottle	Section 13.3							XX	XX
Drum Leadthrough Gravity Bottle	Section 13.3							XX	XX
DTR1 Leadthrough Gravity Bottle	Section 13.3							XX	XX
DTR 2 Leadthrough Gravity Bottle	Section 13.3							XX	XX
VV7 Leadthrough Valve Gravity Bottle	Section 13.3							XX	XX
Flap Valve Leadthrough Gravity Bottle	Section 13.3							XX	XX
Chiller Compressor	Section 13.3							XX	XX
Polycold Compressor	Section 13.3							XX	XX
Lauda Refrigeration System	Section 13.3								XX
Winding Cart Movement Gearbox	Section 13.3						XX	XX	XX
Traverse Drive Chain	Chain Guard				XX			XX	XX

Table 13.1 - 2 Lubrication Schedule – Oil

Key to above chart:

D	Daily	W	Weekly
M	Monthly	Q	Quarterly
BA	Bi-annual	A	Annual
R	Removal Of Equipment For Repair	S	Strip Out and replacement

Notes:

- In the case of the oil in the rotary pumps, it maybe found that the oil has to be changed on a more regular basis, depending upon the moisture in the surrounding atmosphere, etc.
- Top up gravity bottles and booster pumps as required. High degree of oil loss indicates a problem that should be investigated.

- Please note that cross-reference of the oils and the codes used by suppliers may change from country to country.
- If users propose to use an oil type not listed, it is suggested that before the oil is purchased, the specification of the oil is to be sent to the equipment manufacture for their recommendations.

	Caution !
	Do not use oil types that are not on the recommended list, as this could lead to failure of components upon the machine.

Replacement Parts

Any replacement parts such as load cell transducer bearings must be cleaned with a degreaser and then acetone before being re-greased with the approved grease type.

13.2 Lubrication Points On Machine

Refer to the preventative maintenance section of the manual for photographs of lubrication tasks taking place.

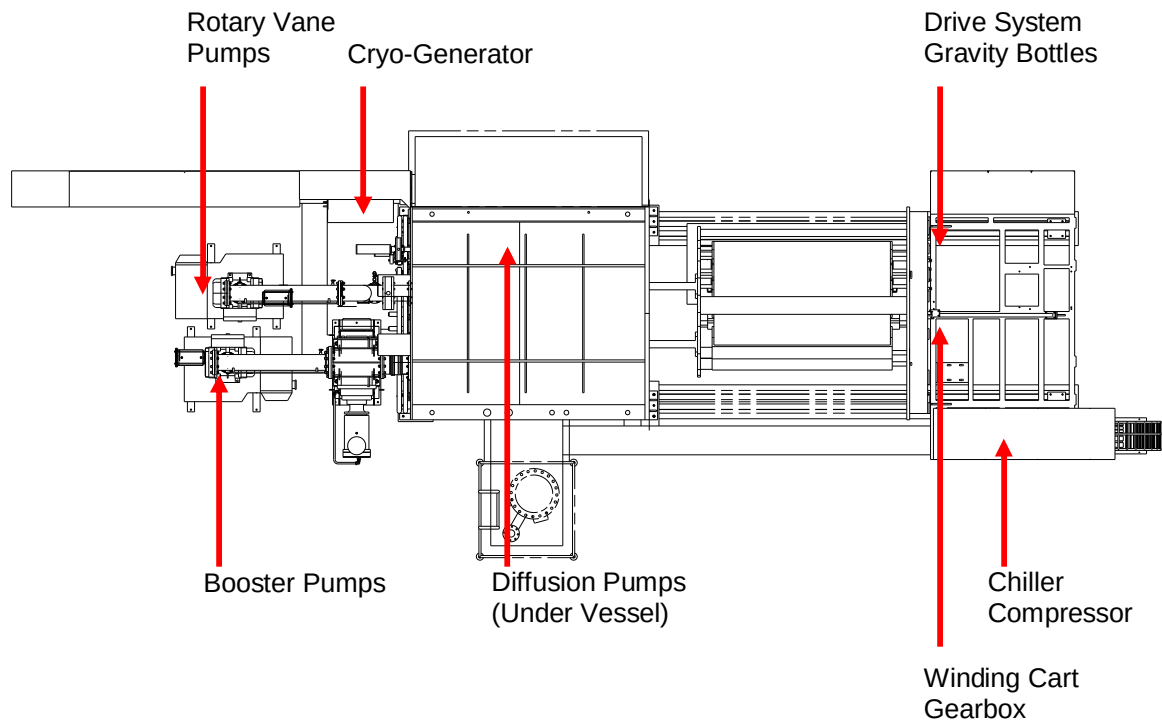


Figure 13.2 - 1 Location of Equipment Requiring Oil Lubrication

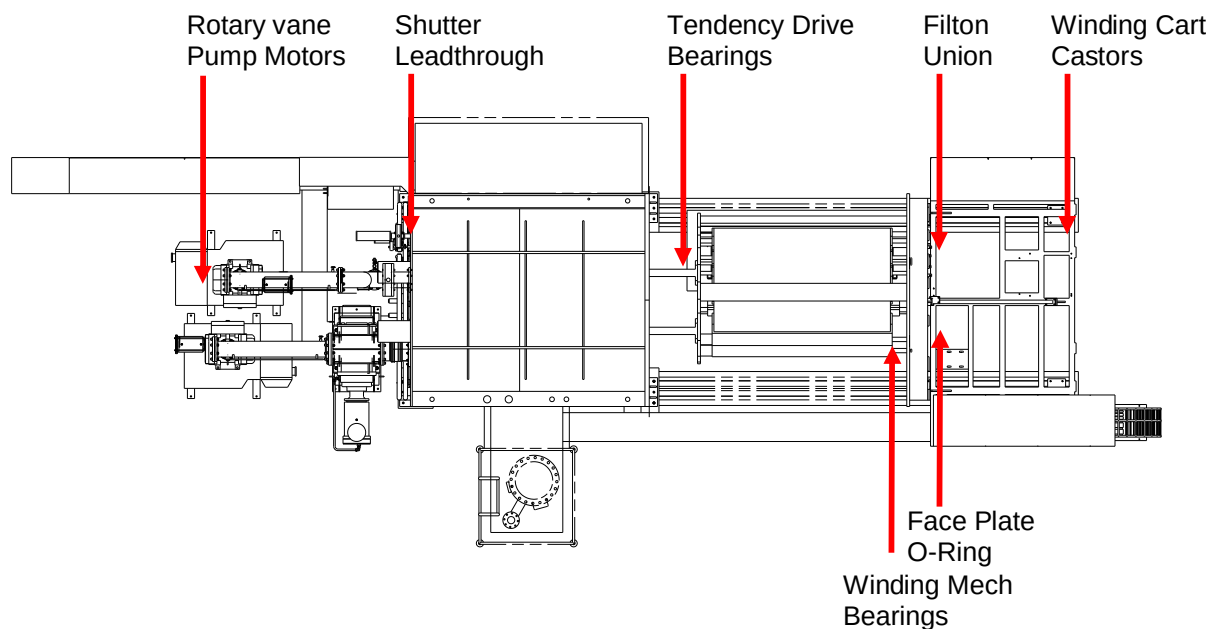


Figure 13.2 - 2 Location of Equipment Requiring Grease Lubrication

13.3 Equipment Oil /Grease Types and Capacities

Refer to section 13.1 for periodic lubrication time scales.

	Caution !
	Do not use oil types that are not on the recommended list, as this could lead to failure of components upon the machine.

Equipment	Leybold 630 F Sogevac Rotary Vane Pumps
Capacity:	35/38 litres
Special Notes:	
Recommended Lubricant Type from vendor	Leybold GS777 Oil
Equivalent Type 1	
Equivalent Type 2	
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 1 Leybold Rotary Piston Pumps

Equipment	Aerzen Mechanical Booster Pumps - Gear Case
Capacity:	Size 14.9 20 litres / 16.13 11 litres
Special Notes:	
Recommended Lubricant Type from vendor	See vendor manual
Equivalent Type 1	Shell (UK) Terreso 100
Equivalent Type 2	Mobil (UK) DTE Extra Heavy
Equivalent Type 3	Esso Terreso 100
Equivalent Type 4	BP (UK) Enercol 100
Equivalent Type 5	

Table 13.3 - 2 Aerzen Mechanical Booster Pumps - Gear Case

Equipment	Varian Diffusion Pumps
Capacity:	HS 20 - 4.73 litres HS35- 11.35 litres
Special Notes:	
Recommended Lubricant Type from vendor	Dow Corning 704
Equivalent Type 1	CVC Silicone 4
Equivalent Type 2	
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 3 Varian Diffusion Pumps

Equipment	Unwind Chuck Oil Lubrication Gravity Feed Bottle
Capacity:	Dependent upon chuck size
Special Notes:	Fill to within 12 mm of the top of the gravity bottle.
Recommended Lubricant Type from vendor	Mobil (UK)DTE Heavy Medium
Equivalent Type 1	Crown (UK) GV10
Equivalent Type 2	Grahams (UK) GV10
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 4 Unwind Chuck Oil Lubrication Gravity Feed Bottle

Equipment	Cooled Drum/rollers Oil Lubrication Gravity Feed Bottle
Capacity:	Dependent upon leadthrough size
Special Notes:	Fill to within 12 mm of the top of the gravity bottle. Only to be used on leadthroughs that are cooled i.e. drum leadthrough
Recommended Lubricant Type from vendor	Castrol(UK) Hyspin AWS 10
Equivalent Type 1	
Equivalent Type 2	
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 5 Cooled Drum/rollers Oil Lubrication Gravity Feed Bottle

Equipment	High vacuum valve (VV7) Oil Lubrication Gravity Feed Bottle
Capacity:	Dependent upon leadthrough size
Special Notes:	Fill to within 12 mm of the top of the gravity bottle.
Recommended Lubricant Type from vendor	Mobil(UK) DTE heavy medium
Equivalent Type 1	Crown(UK) GV10
Equivalent Type 2	
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 6 Right Angle Valve (VV7) Oil Lubrication Gravity Feed Bottle

Equipment	Process Chiller Compressor Oil
Capacity:	See Vendor manual
Special Notes:	For R404 Gas
Recommended Lubricant Type from vendor	Dependent upon refrigerant gas charge type

Table 13.3 - 7 Process Chiller Compressor Oil

Equipment	Cryo-generator Compressor Oil
Capacity:	3.9 litres
Special Notes:	see vendor manual
Recommended Lubricant Type from vendor	Zerol 150

Table 13.3 - 8 Cryo-generator Compressor Oil

Equipment	Traverse Movement Gearbox Oil
Capacity:	Varies , see vendor manual
Special Notes:	
Recommended Lubricant Type from vendor	Shell(UK) Omala 320
Equivalent Type 1	Mobil(UK) Gear 632
Equivalent Type 2	Castrol(UK) Alpha SP320 K310
Equivalent Type 3	Esso Spartan EP320
Equivalent Type 4	BP(UK) Energol GR-XP320
Equivalent Type 5	

Table 13.3 - 9 Traverse Movement Gearbox Oil

Equipment	Lauda Compressor Oil
Capacity:	Varies , see vendor manual
Special Notes:	
Recommended Lubricant Type from vendor	Reniso Triton SEZ22
Equivalent Type 1	
Equivalent Type 2	
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 10 Lauda Compressor Oil

	Caution !
	Do not use grease types that are not on the recommended list, as this could lead to failure of components upon the machine.

Grease Type:	A
Capacity:	Unknown
Special Notes:	
Recommended Lubricant Type from vendor	Mobil (US) Grease 28
Equivalent Type 1	Mobil (US) Supertemp SHC32
Equivalent Type 2	Castrol (UK) Longtime PD2
Equivalent Type 3	Esso(UK) Nova 275
Equivalent Type 4	Shell(US) Retina AX

Table 13.3 - 11 Grease Type A

Grease Type:	B
Capacity:	Unknown
Special Notes:	
Recommended Lubricant Type from vendor	Dow Corning Molyoke 44M
Equivalent Type 1	Shell(UK) Alvania RA
Equivalent Type 2	Mobil(US) Mobilplex 47
Equivalent Type 3	Esso(UK) Beacon 2
Equivalent Type 4	BP(UK) Energrease LS2
Equivalent Type 5	Texaco(UK) Multifak AFB2

Table 13.3 - 12 Grease Type B

Grease Type:	C
Capacity:	Unknown
Special Notes:	
Recommended Lubricant Type from vendor	Apizon M Grease
Equivalent Type 1	Dow Corning High Vacuum Grease
Equivalent Type 2	Dupont Krytox LVP2
Equivalent Type 3	Varian Vacuum Grease
Equivalent Type 4	

Table 13.3 - 13 Grease Type C

Grease Type:	D
Capacity:	Unknown
Special Notes:	
Recommended Lubricant Type from vendor	Shell(UK) Aero Shell #7
Equivalent Type 1	Esso(UK) Temperex N3
Equivalent Type 2	SKF LGHQ3
Equivalent Type 3	

Table 13.3 - 14 Grease Type D

Grease Type:	E
Capacity:	Dependant upon unit fitted
Special Notes:	
Recommended Lubricant Type from vendor	Shell Dolium R
Equivalent Type 1	Chevron SRI
Equivalent Type 2	
Equivalent Type 3	

Table 13.3 - 15 Grease Type E

Grease Type:	F
Capacity:	
Special Notes:	
Recommended Lubricant Type from vendor	Tecknit Conductive Silver Filled silicone Grease
Equivalent Type 1	
Equivalent Type 2	
Equivalent Type 3	
Equivalent Type 4	
Equivalent Type 5	

Table 13.3 - 16 Grease Type F

Notes:

1. The chiller compressor oil should be suitable for the type of gas contained within the system. Consult a local refrigeration component supplier for suitable oil type
2. There may be other gearboxes fitted to components on the equipment. They are normally labelled with their requirements. If no details are available, please contact General Vacuum Equipment
3. The code/names given by lubricant manufacturers may change between continents. Please forward a data sheet on alternative products before using, to General Vacuum Equipment

	Caution ! The lubricating oils from the vacuum pumps are potentially harmful to the environment. Never allow pump oil or parts dipped in pump oil to penetrate the supply of drinking water. Pump oil (even small amounts!) must be disposed of as hazardous waste.
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13.4 Gearbox Capacities and Oil Level Indication

Capacities

See documentation for individual capacities in the vendor binders.

Oil Levels

The equipment supplied has some vendor guidelines in regards to oil levels. Please refer to the vendor manuals supplied.

In the case of the gravity bottles, the oil level should be above the half way part of it.

13.5 Grease and Oil Specifications

The following sheets are brief specifications on some of the common lubricants that are used in the machine.

- Mobil DTE Series of Oils
- Castrol Hyspin Range of Oils
- Shell Omala Range of Oils
- Shell Turbo Range of Oils
- Shell AW Hydraulic Oils
- Nu-Calgon Zerol Refrigerant Oils
- Dow Corning DC704 Oil
- Reniso Triton SEZ 22 Oil

- Mobil Grease 28
- Dow Corning 55 O-ring Grease
- Mobil Temp SHC 32 Grease
- Shell Dolium Grease
- Chevron SRI Grease

13.6 Grease and Oil MSDS Sheets

MSDS sheets are provided for the lubricants commonly used on the machine.

- Mobil DTE Heavy Medium Oil
 - Hyspin AWS 10 Oil
 - Shell Omala Oil
 - Shell AW Hydraulic Oil 32
 - Nu-Calgon Zerol 150 Oil
 - Reniso Triton SEZ 22 Oil
-
- Tecknit Conductive Silver Filled Silicone Grease.
 - Mobiltemp SHC32 Grease
 - Dow Corning 55 O-Ring Grease
 - Dow Corning High Vacuum Grease
 - Chevron SRI Grease



Castrol (U.K.) Limited

The Leading Lubrication Specialist



SAFETY DATA SHEET

1: IDENTIFICATION OF THE SUBSTANCE / PREPARATION AND OF THE COMPANY / UNDERTAKING

Product Name: **Hyspin AWS 10** Code: **6015-UK**
Application: Hydraulic systems
Company: **Castrol (U.K.) Limited**
Address: Burmah Castrol House, Pipers Way, Swindon, Wiltshire, SN3 1RE
Telephone (24 hours): **01793 512712** Fax: **01793 491442**

2: COMPOSITION/INFORMATION ON INGREDIENTS

Composition: Highly refined mineral oil and additives

Hazardous Ingredient(s)	Symbol	Risk Phrases	Other Information	%
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This product contains ingredients classified as hazardous. However, they are NOT present in sufficient quantities to warrant classifying the product as hazardous

All constituents of this product are listed in EINECS (European Inventory of Existing Commercial Chemical Substances) or ELINCS (European List of Notified Chemical Substances) or are exempt.

Refer to Section 8 for Occupational Exposure Limits.

3: HAZARDS IDENTIFICATION

This product is NOT classified as hazardous

4: FIRST AID MEASURES

Eyes: Irrigate immediately with copious quantities of water for several minutes
Skin: Wash thoroughly with soap and water or suitable skin cleanser as soon as possible
Inhalation: Remove from exposure
Ingestion: Obtain medical attention. Do NOT induce vomiting.

5: FIRE FIGHTING MEASURES

Suitable Extinguishing Media: Carbon dioxide, powder, foam or water fog - Do not use water jets
Special Exposure Hazards: None
Special Protective Equipment: None

6: ACCIDENTAL RELEASE MEASURES

Personal Precautions: Spilt product presents a significant slip hazard
Environmental Precautions: Prevent entry into drains, sewers and water courses
Decontamination Procedures: Soak up with inert absorbent or contain and remove by best available means

7: HANDLING AND STORAGE

Handling: When used in high pressure systems, leakage may result in mist formation so presenting a hazard
To avoid the possibility of skin disorders, repeated or prolonged contact with products of this type must be avoided. It is essential to maintain a high standard of personal hygiene

Storage: No special precautions

8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Occupational Exposure Limits:-

Substance	8 Hr.TWA	STEL	Source/Other Information
Mineral oil (see Oil mist, mineral)	5mg/m ³	10mg/m ³	EH40 (OES)

Engineering Control Measures: Mechanical methods to minimise exposure must take precedence over personal protective measures

Personal Protective Equipment: Avoid skin and eye contact. Wear impervious gloves (eg of PVC), in case of repeated or prolonged contact. Change contaminated clothing and clean before re-use

9: PHYSICAL AND CHEMICAL PROPERTIES

Physical State:	Liquid
Colour:	Amber
Odour:	Mild
Boiling Point/Range (°C):	Above 250
Pour Point: (°C):	Below minus 39
Kinematic Viscosity @ 40°C (cSt):	10
Flash Point (closed, °C):	Above 140
Autoignition (°C):	Above 250
Explosive Properties (%):	Not determined
Relative Density (at 20°C):	Below 1.0
Water Solubility:	Insoluble
Fat Solubility:	Not determined

10: STABILITY AND REACTIVITY

Stability:	Stable, will not polymerise
Conditions to Avoid:	Temperatures (°C) above 90
Materials to Avoid:	Strong oxidising agents
Hazardous Decomposition Products:	None

11: TOXICOLOGICAL INFORMATION

The following toxicological assessment is based on a knowledge of the toxicity of the product's components
Expected oral LD₅₀, rat > 2g/kg

Health Effects

On Eyes: May cause transient irritation

On Skin: Unlikely to cause harm on brief or occasional contact

By Inhalation: Mist and vapours may cause irritation to nose and respiratory tract

By Ingestion: May cause nausea, vomiting and diarrhoea.

Chronic: Repeated and prolonged skin contact may lead to skin disorders

Other: None known

12: ECOLOGICAL INFORMATION

Environmental Assessment: When used and disposed of as intended, no adverse environmental effects are foreseen

Mobility: Mobile liquid. Insoluble in water. Non-volatile.

Persistence and Degradability: Inherently biodegradable

Bioaccumulative Potential: Bioaccumulative based on logP values of constituents

Ecotoxicity: Not expected to be toxic to aquatic organisms
Not expected to be inhibitory to sewage bacteria

13: DISPOSAL CONSIDERATIONS

Disposal must be in accordance with local and national legislation.

Unused Product: May be sent for reclamation

Used/Contaminated Product: Dispose of through an authorised waste contractor to a licensed site
May be incinerated

Packaging: Must be disposed of through an authorised waste contractor
May be steam cleaned and recycled

14: TRANSPORT INFORMATION

This product is NOT classified as dangerous for transport

15: REGULATORY INFORMATION

This product is NOT classified as dangerous for supply in the UK

Hazard Label Data:-

EC Directives: Waste Oil Directive, 87/101/EEC
Framework Waste Directive, 91/156/EEC

Statutory Instruments: Health & Safety at Work, etc. Act 1974
Consumer Protection Act 1987
Environmental Protection Act 1990

Codes of Practice: Waste Management. The Duty of Care

Guidance Notes: Occupational exposure limits (EH 40)
Carcinogenicity of mineral oils (EH 58)
Skin cancer caused by oil [MS(B)5]
Save your skin! - Occupational Contact Dermatitis [MS(B)6]
Dermatitis - cautionary notice [SHW 367]
Effects of mineral oil on the skin [SHW 397]

The above publications are available from HMSO or HSE

16: OTHER INFORMATION

The data and advice given apply when the product is sold for the stated application or applications. The product is not sold as suitable for any other application. Use of the product for applications other than as stated in this sheet may give rise to risks not mentioned in this sheet. You should not use the product other than for the stated application or applications without seeking advice from us.

If you have purchased the product for supply to a third party for use at work, it is your duty to take all necessary steps to secure that any person handling or using the product is provided with the information in this sheet.

If you are an employer, it is your duty to tell your employees and others who may be affected of any hazards described in this sheet and of any precautions which should be taken.

Further copies of this Safety Data Sheet may be obtained from Castrol (U.K.) Limited.

Antiwear Hydraulic Oil**DESCRIPTION**

The Castrol Hyspin AWS hydraulic oil range is based upon highly refined mineral oil with a low zinc-containing anti-wear system.

APPLICATION

Castrol Hyspin AWS has been specially formulated to provide outstanding anti-wear and thermal stability performance using the very latest low zinc additive technology. The careful blend of additives with a high quality basestock ensures that Castrol Hyspin AWS has excellent hydrolytic and oxidative stability while exhibiting a minimal tendency to produce sludge and deposits. In addition, Castrol Hyspin AWS provides protection from corrosion to the ferrous and yellow metal components found within a hydraulic system. Castrol Hyspin AWS is designed for use in most industrial hydraulic systems which require anti-wear protection, DIN Classification HLP. Castrol Hyspin AWS meets and exceeds the Denison HF-O requirements.

FEATURES

- ◆ Outstanding thermal and oxidative stability
- ◆ Excellent anti-wear performance
- ◆ Outstanding filterability characteristics, both wet and dry
- ◆ Superior protection against corrosion
- ◆ Excellent water separation

BENEFITS

- ◆ Longer operating life
- ◆ Overall reduction in lubricant costs and used oil disposal costs
- ◆ Reduced downtime from unscheduled maintenance. Replacement part costs reductions
- ◆ Cost savings due to increased filter life and reduced maintenance
- ◆ Minimises deposit formation giving a cleaner system
- ◆ Reduced down-time through prolonged lubricant life